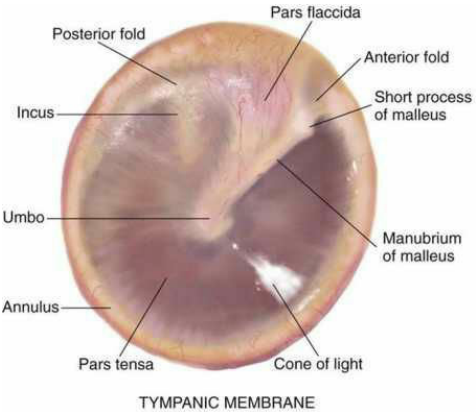
45 NOTES TO PG-IMAGES

—Dr. Ankit Yadavendra, M.B.B.S., Dr.V.M.G.M.C., Solapur

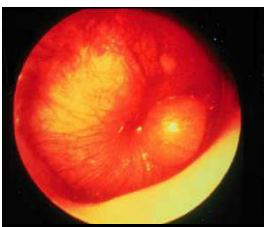
08–ENT

normal tympanic membrane



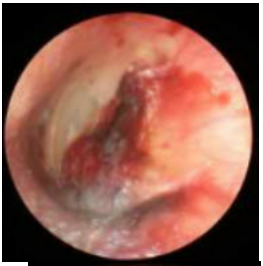
TYMPANIC MEMBRANE

ASOM-presuppurative-cartwheel



barotrauma-congested&retracted, air bubble, hgic effusion

healed myringitis bullosa-sagograin



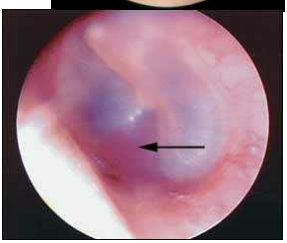
hemotympanum

glue ear

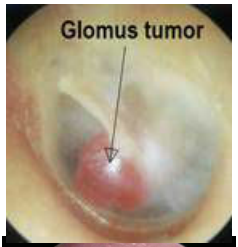
keratin deposit, osmium tetroxide-snakelike

myringitis bullosa(influenza virus)-hgic bleb

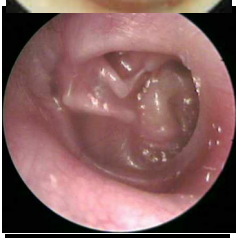
otosclerosis-norm(90%)-translucent&pearly gray, active ds-flamingo tint(pink spot)



glomus tm

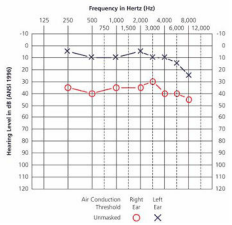


retracted-dull lustreless



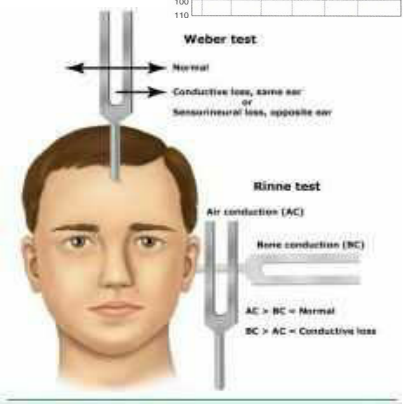
tympanosclerosis-chalky white plaque

audiometry



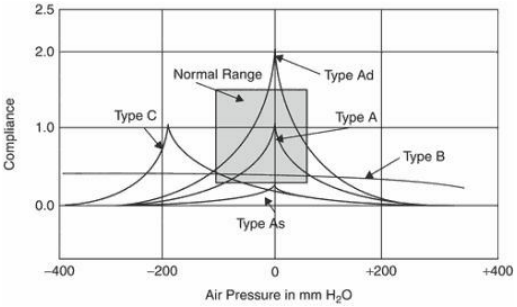
Carhart notch=2000Hz

Weber test, Rinne test

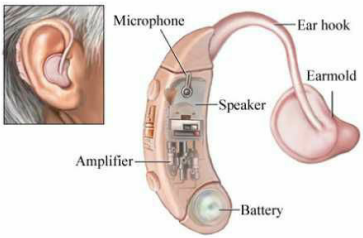


tympanometry

A-norm, AS-otoSclerosis, AD-ossicular Disruption, B-Perforation, C-EustaChian tube dysfn, Flat-Fluid/glue ear

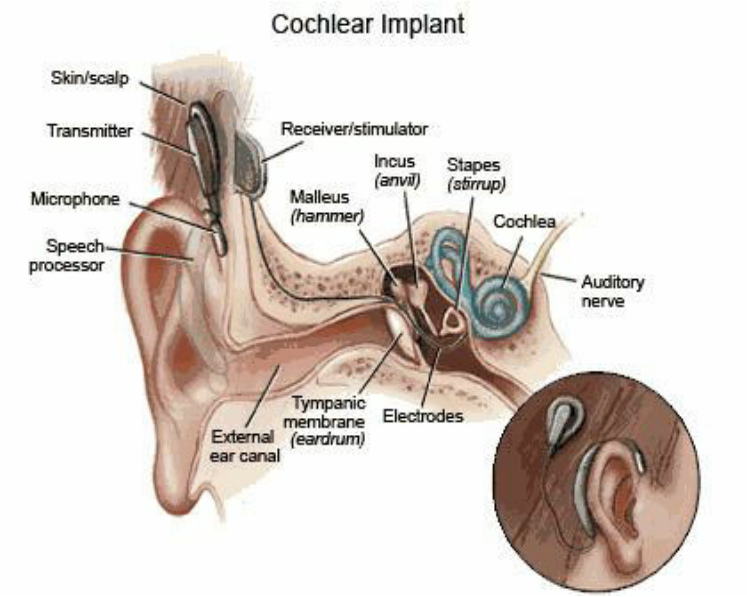


hearing aid

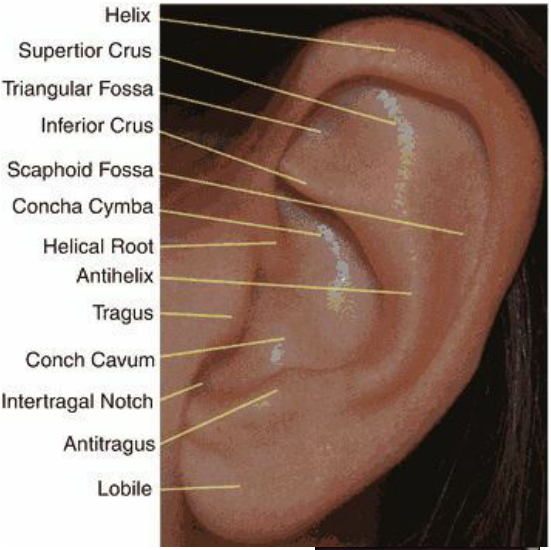


BAHA-Bone Anchored Hearing Aid(TES)-Titanium implant, Ext abutment, Sound processor

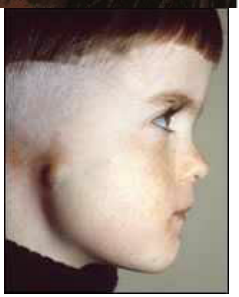
cochlear implant



normal anatomy pinna

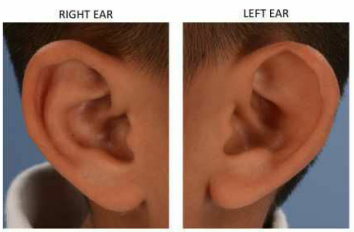


Anotia-cong Absence pinna



Loss Of the Antihelical Fold

bAt-no Antihelix

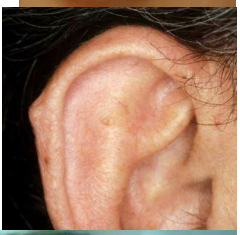


cleft pinna-cong fissure pinna

collaural fistula-b/n EAC&neck, 1st pharyngeal cleft

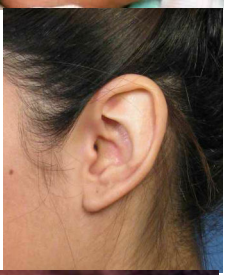
coloboma lobuli-cong fissure earlobe

Darwin/auricular tubercle-thick helix(jn up - mid)



low set ear-cong low displaced pinna

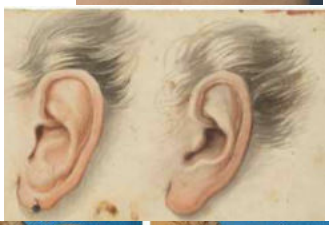
macrotia-cong large pinna



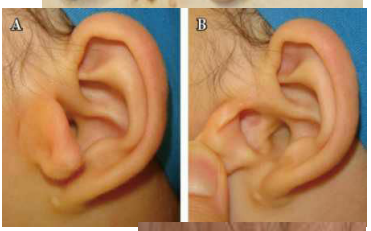
MElotia-cong displaceMEnt pinna

microtia-cong small pinna

Mozart-Mixing of antihelix&helix



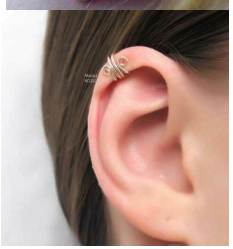
polyotia-additional pinna



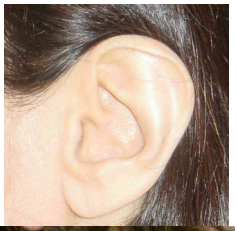
preauric sinus-1st arch anomaly

preauric tag-small appendage ant to pinna

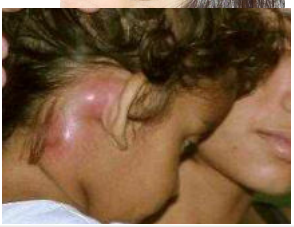
scroll ear-rim(helix) roll forward,inward



Wildermuth-no helix

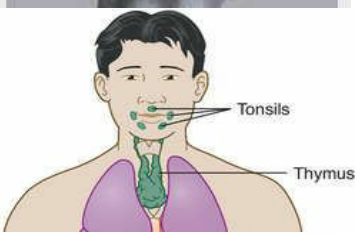


Bezold-SCM sheath

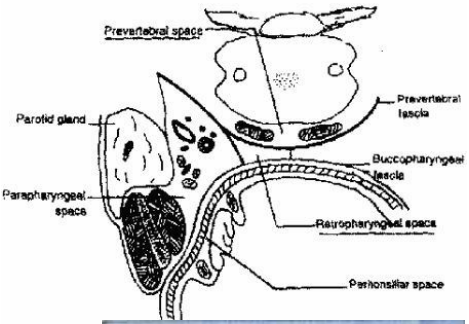


Citelle-digastric triangle

DuboiS-thymuS(SyphiliS)

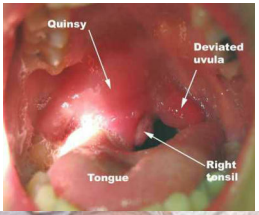


Gillete-retropharyngeal(b/n pharynx& prevertebral fascia)



Luc-temporal bone(roof of EAC)

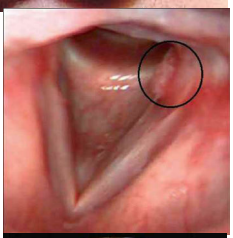
peritonsillar(quinsy)-tonsillar capsule& sup constrictor



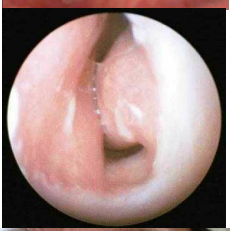
mulberry like

nasal polyP-rhinosPoridiosis

vocal corD-rhinosporiDiosis



nasal mucosa-inf turb hypertrophy

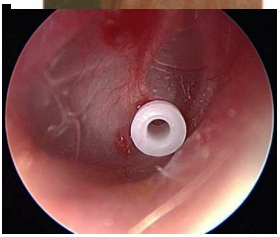


potato nose-rhinophyma

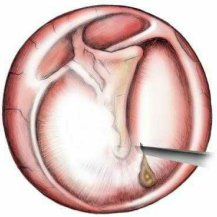
tapiR nose-Rhinosclerosis

Strawberry nose-Sarcoidosis

grommet insertion



incision of myringotomy



Pure Tone Audiometry

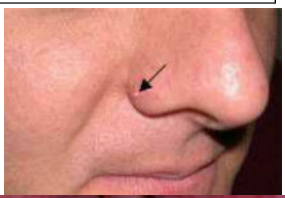
Rt ear-Red

Lt ear-bLue

Key to symbols on an audiogram:

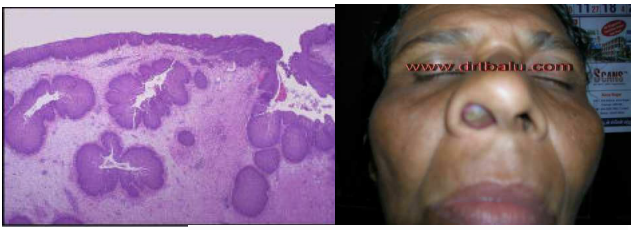
	Right	Left
Air unmasked	O	X
Air masked	△	□
Bone unmasked	<	>
Bone masked	[	]
Sound field	S	S
Aided	A	A

angiofibroma

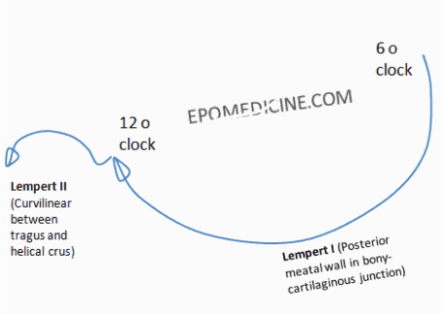


juvenile pappilomatosis-multiple, spont resolve, recur

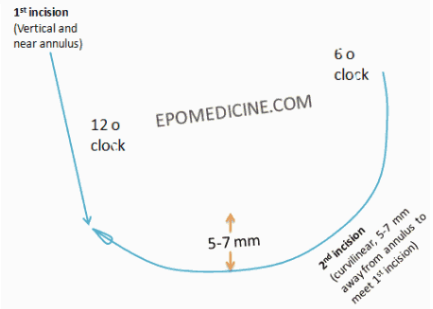
inverted papilloma



Lempert endaural incision-above tragus, incisura terminalis

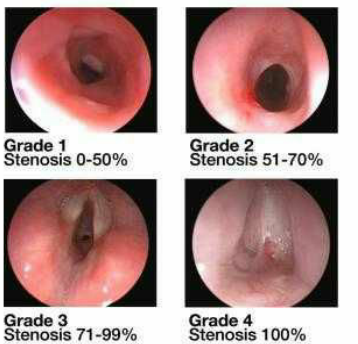


Rosen incision(post wall EAC)-stapedectomy

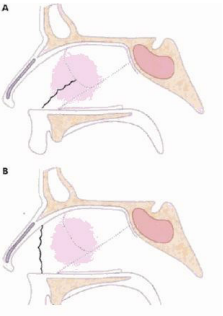


c/c hypertrophic candidiasis/candidial leukoplakia

laryng web

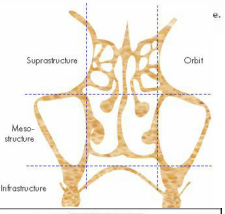


CheValley #-fr bel-nasal septum, E, orbit spare



Jarjaway #-fr front-E, orbit spare

Lederman classificat-ca larynx

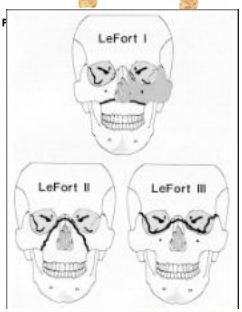


leForte-facial#

I-line through floor M-low max#/floating palate

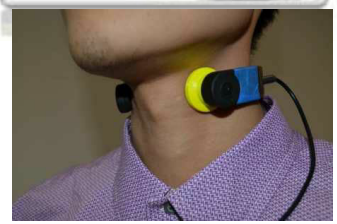
II-line across nasolabial fold-pyramidal#

III-line through orbit-craniofacial dysjn



Dunda grant apparatus-cold caloric test

handfree electrolarynx



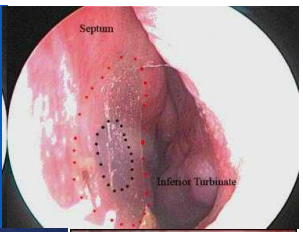
Blom Singer tracheoesoph prosthesis/
electrolarynx

head mirror-concave mirror, focal lth=25cm, diam=89mm, hole=19mm

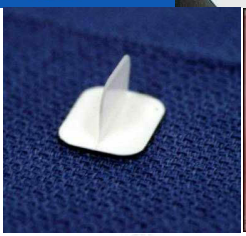
Menitt device

Siegel speculum

silastic button-2cm nasal perforation



silastic keel

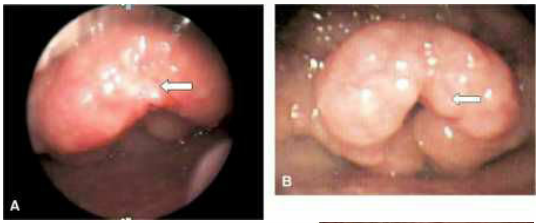


Walsham forceps-#nasal bone

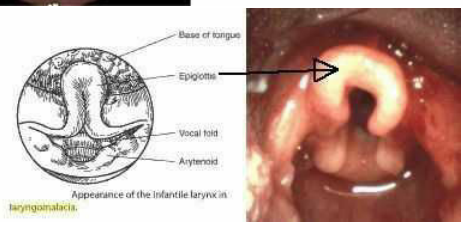
Toynbee speculum

otoscope

mousebeaten VC, turban epiglottis, pseudoedema epiglottis-TB larynx



oMeGA epiglottis-larynGoMAlacia



Cottle test-DNS

-(N.B.-THESE NOTES ARE ONLY FOR THE PURPOSE OF GUIDANCE AND HELP TO PG ASPIRANTS, NOT FOR COMMERCIAL OR OTHER PURPOSE. IMAGES AND REFERENCE HAVE BEEN TAKEN FROM INTERNET AND VARIOUS STANDARD TEXTBOOKS. FOR ANY FEEDBACK/QUERY PLEASE CONTACT- ankit.yadavendra@facebook.com or dr.ankityadavendra@gmail.com)

for theory please refer-

21-ENT https://db.tt/iYhQYbKy ,

DVT 2017
ENT (OTOLOGY)

DAMS
BEST PG MEDICAL ENTRANCE COACHING

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THREE SECTIONS

1. EAR IMPLANTS
2. CLINICAL IMAGES
3. AUDIOLOGY

DAMS
BEST PG MEDICAL ENTRANCE COACHING

COCHLEAR IMPLANT
(EXTERNALCOMPONENT)

DAMS
BEST PG MEDICAL ENTRANCE COACHING

COCHLEAR IMPLANT (INTERNAL COMPONENT)

DAMS
BEST PG MEDICAL ENTRANCE COACHING

RECIEVER-STIMULATOR

OR

ELECTRODE

COCHLEAR IMPLANT
(INTERNAL COMPONENT)

DAMS
BEST PG MEDICAL ENTRANCE COACHING

AUDITORY BRAINSTEM IMPLANT

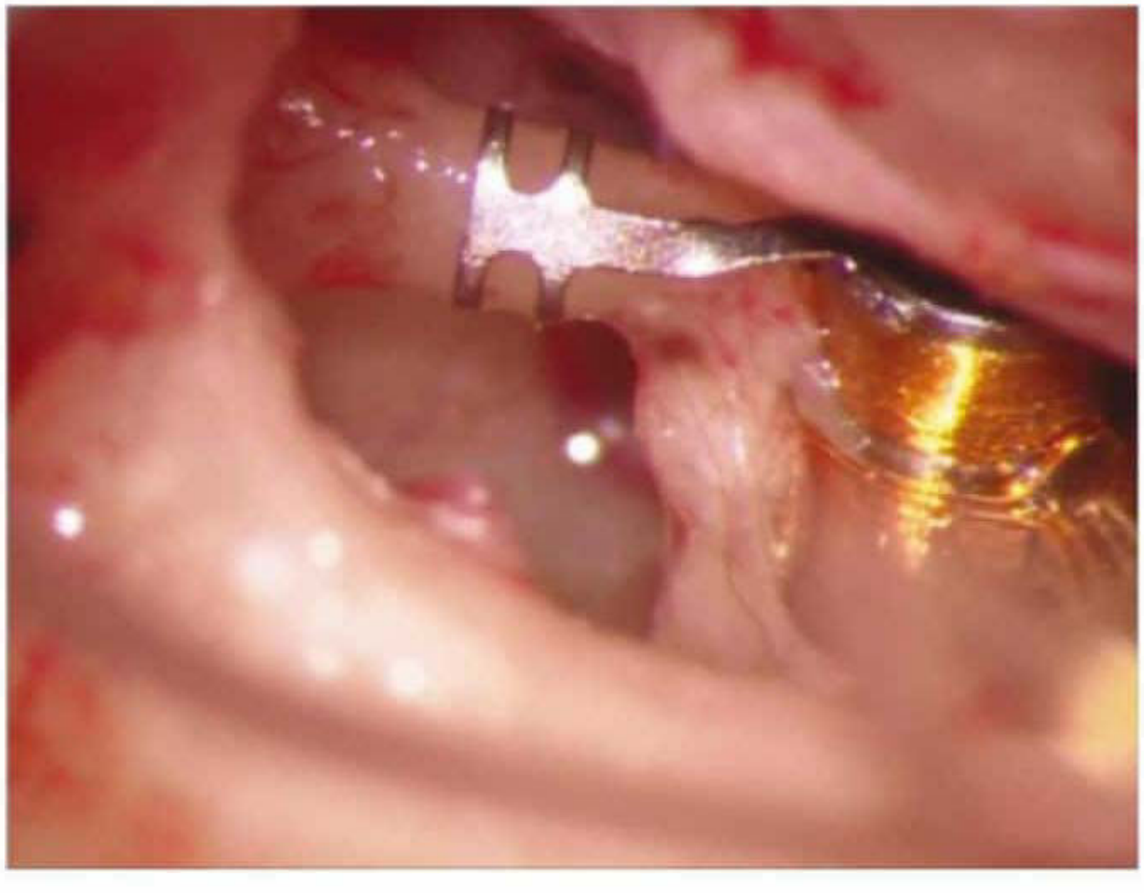
DAMS
BEST PG MEDICAL ENTRANCE COACHING

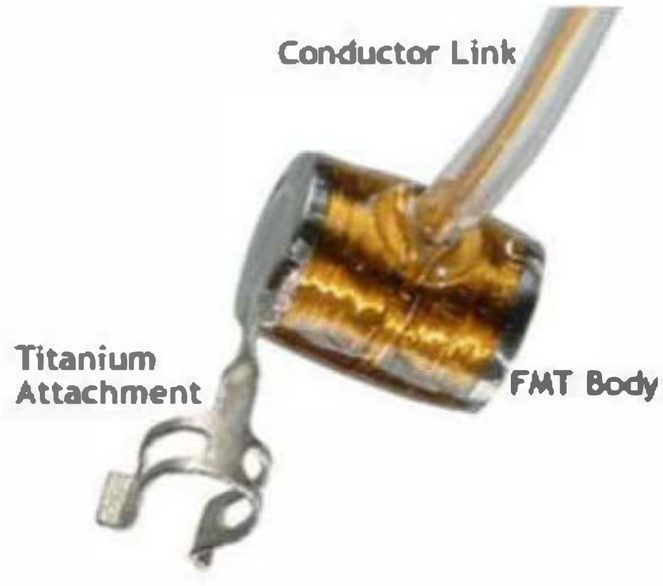
COMPARE TWO ELECTRODES

VIBRANT SOUND BRIGE USED IN SNHL

DAMS
BEST PG MEDICAL ENTRANCE COACHING

ALTERNATIVE TO HEARING AID



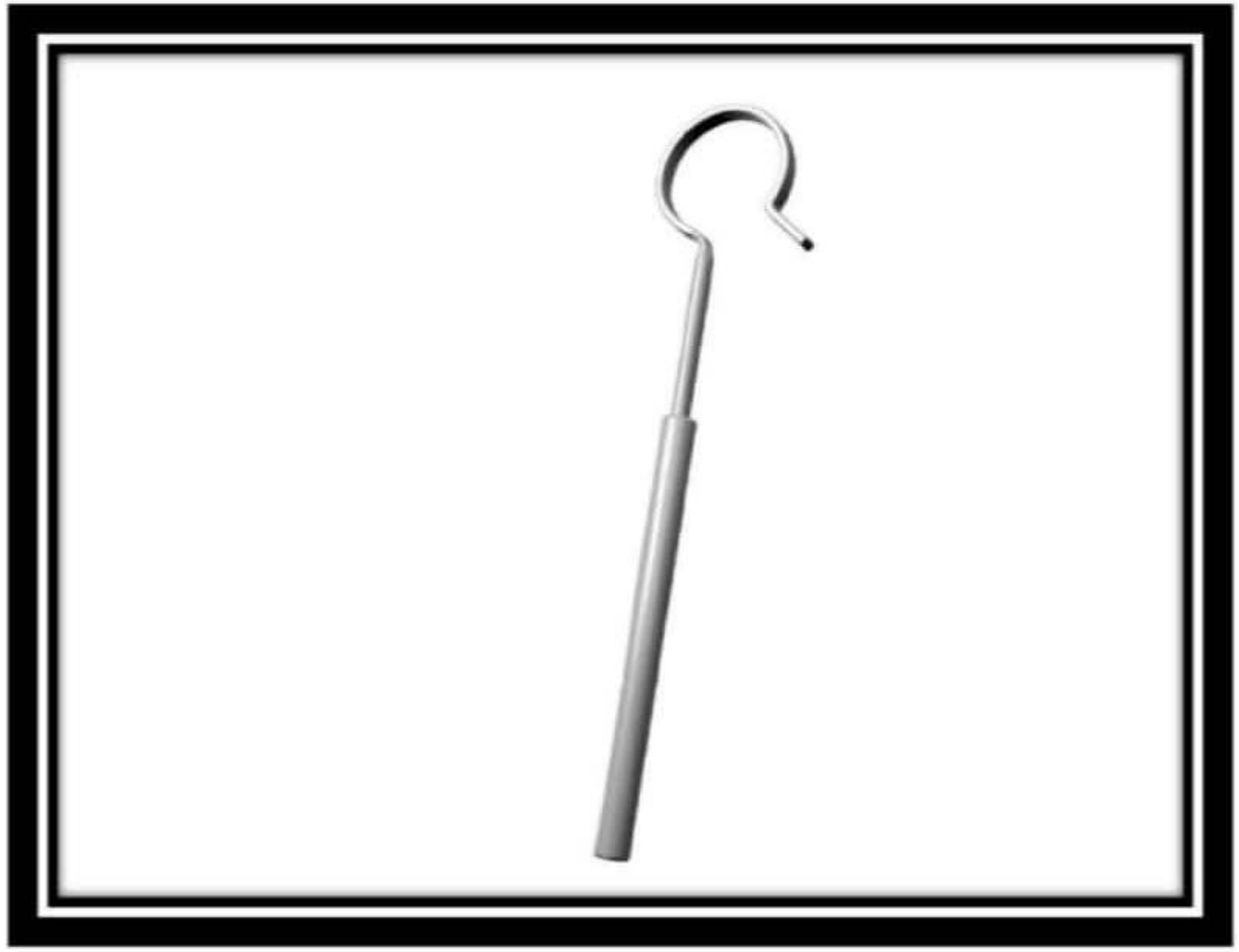


BONE ANCHORED HEARING AID

BONE ANCHORED HEARING AID

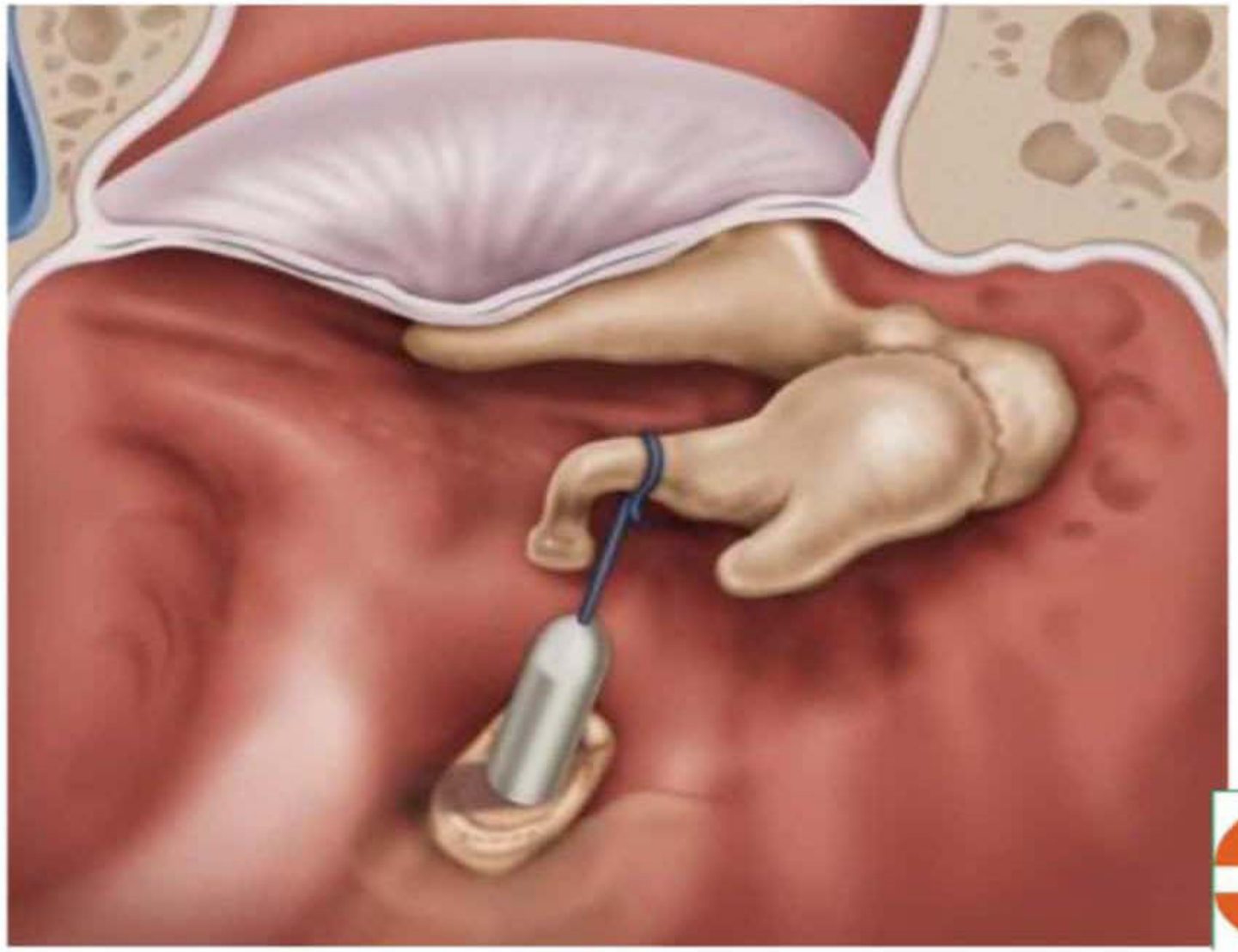
DAMS

STAPES PISTON PROSTHESIS

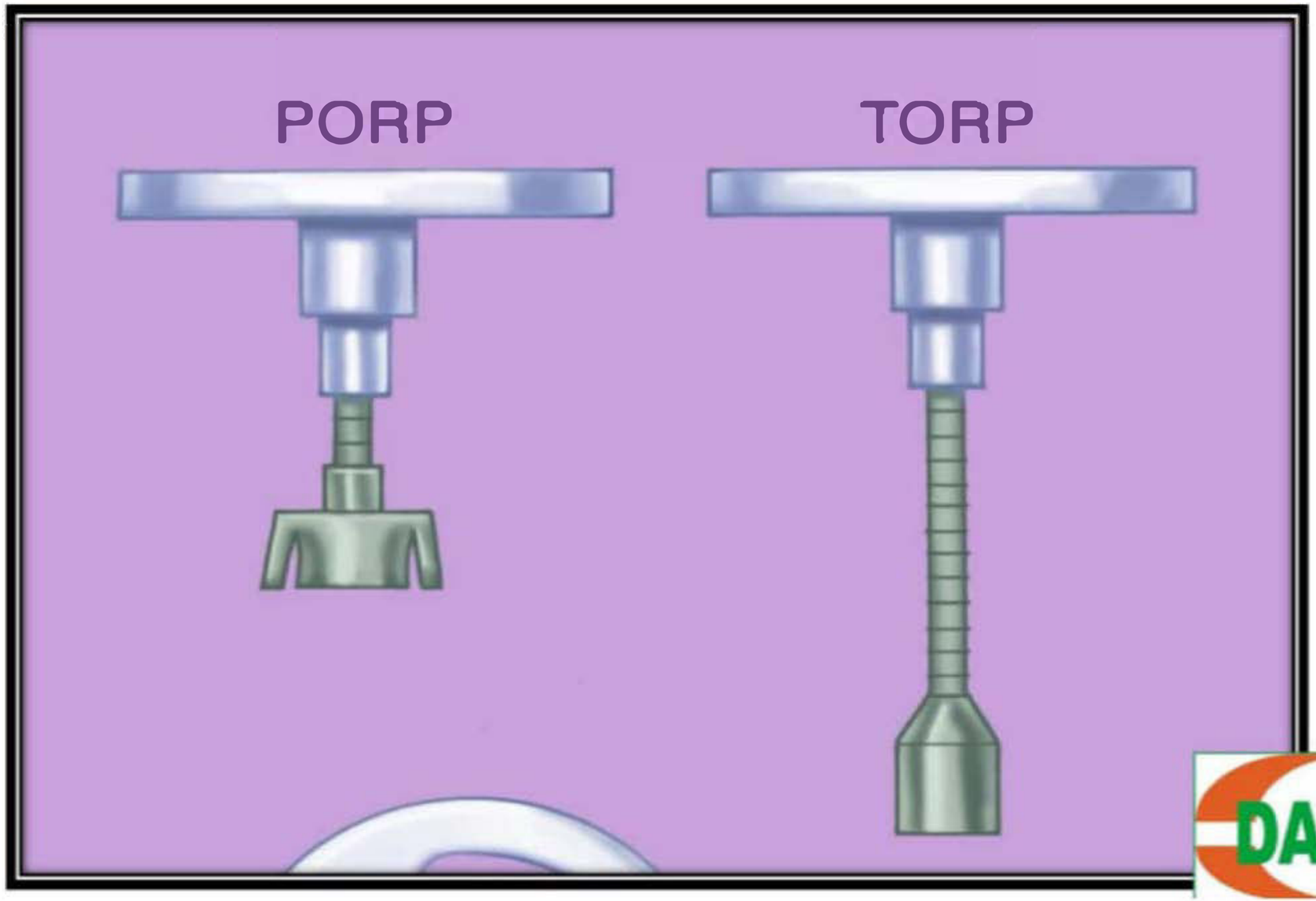


DAMS

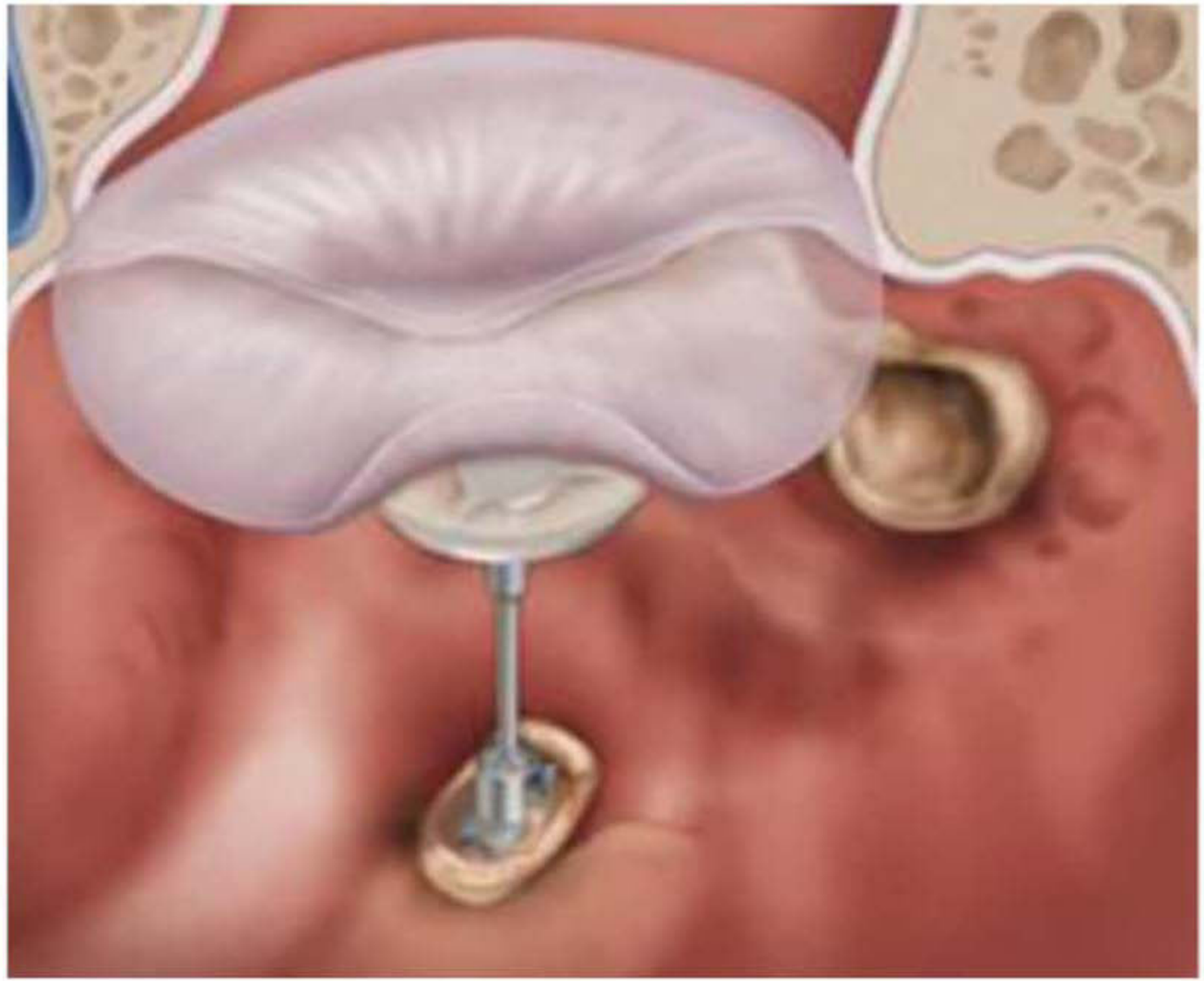
STAPES PISTON PROSTHESIS



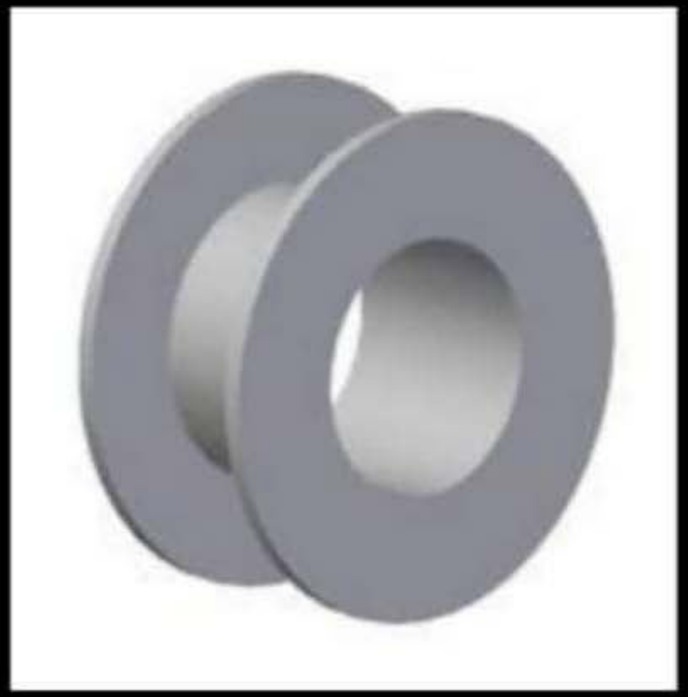
PORP TORP

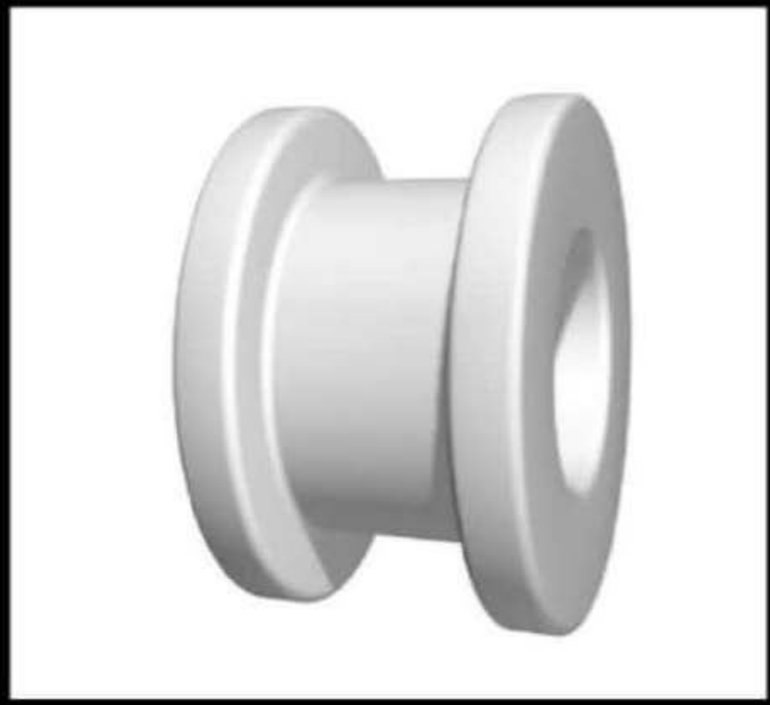


TORP



DAMS



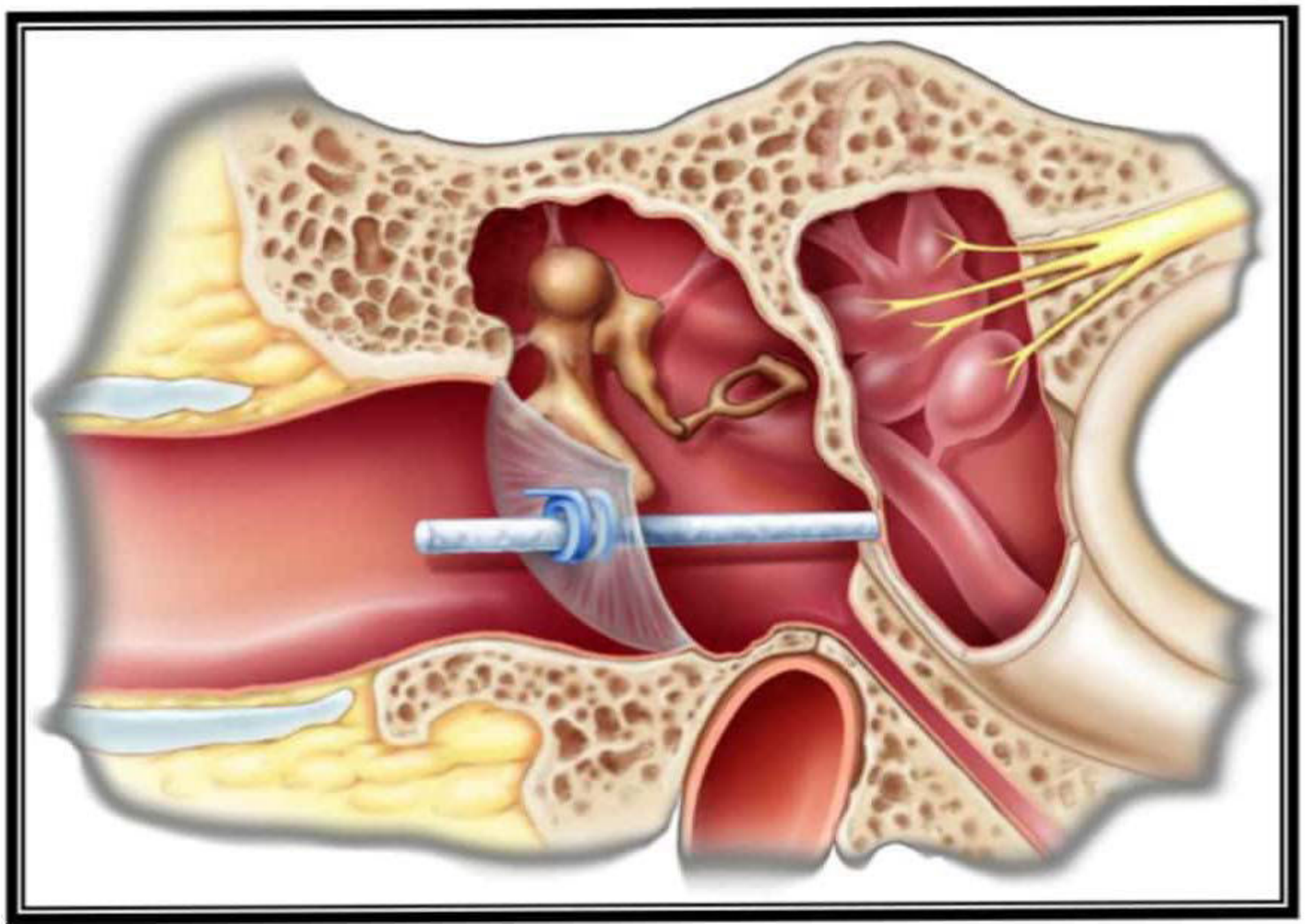


GROMMET

DAMS

GROMMET=
MIDDLE EAR VENTILATION TUBE

MICROWICK



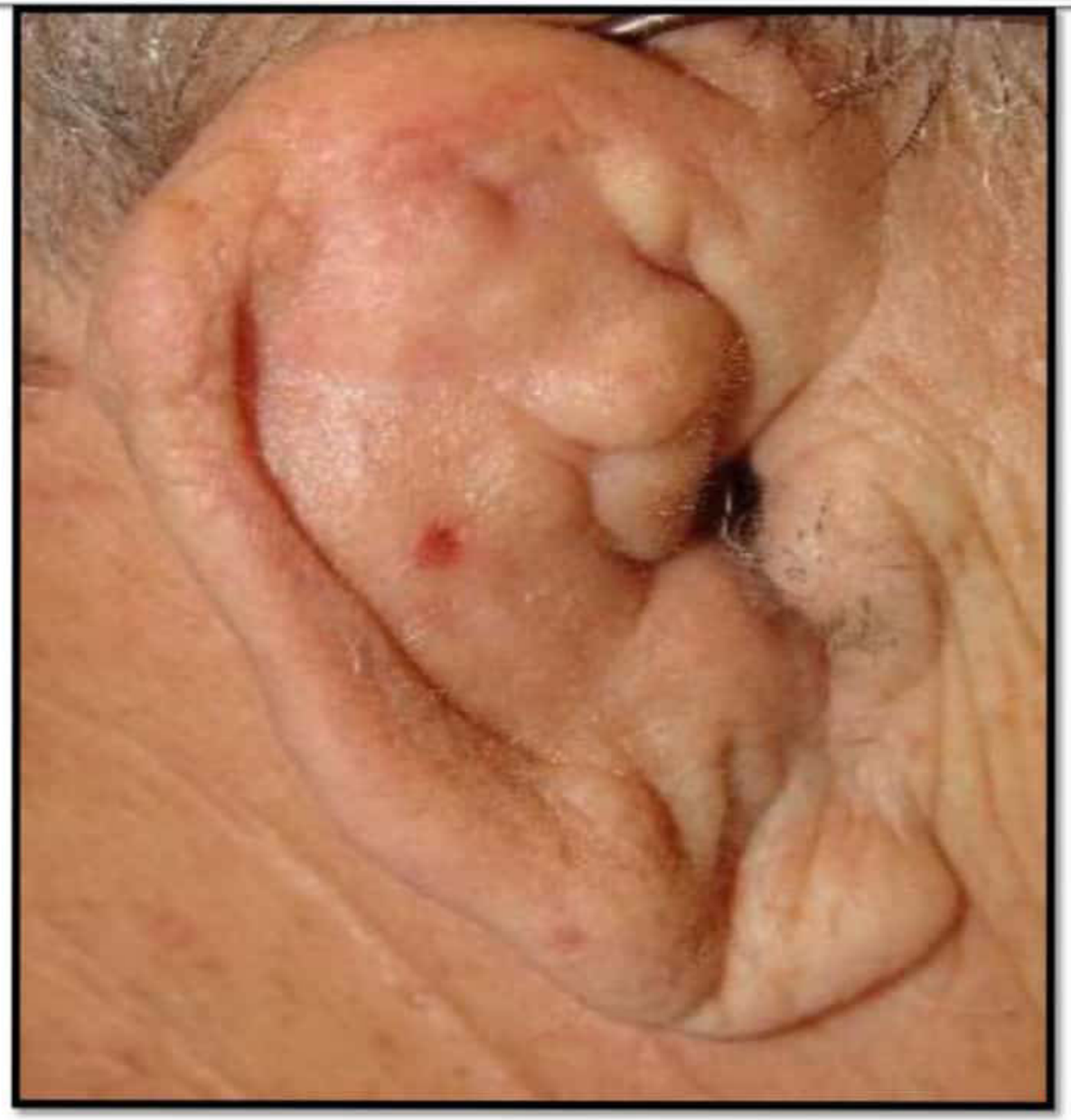
DAMS
BEST PG MEDICAL ENTRANCE COACHING

2. CLINICAL IMAGES

PINNA HEAMATOMA

DAMS
BEST PG MEDICAL ENTRANCE COACHING

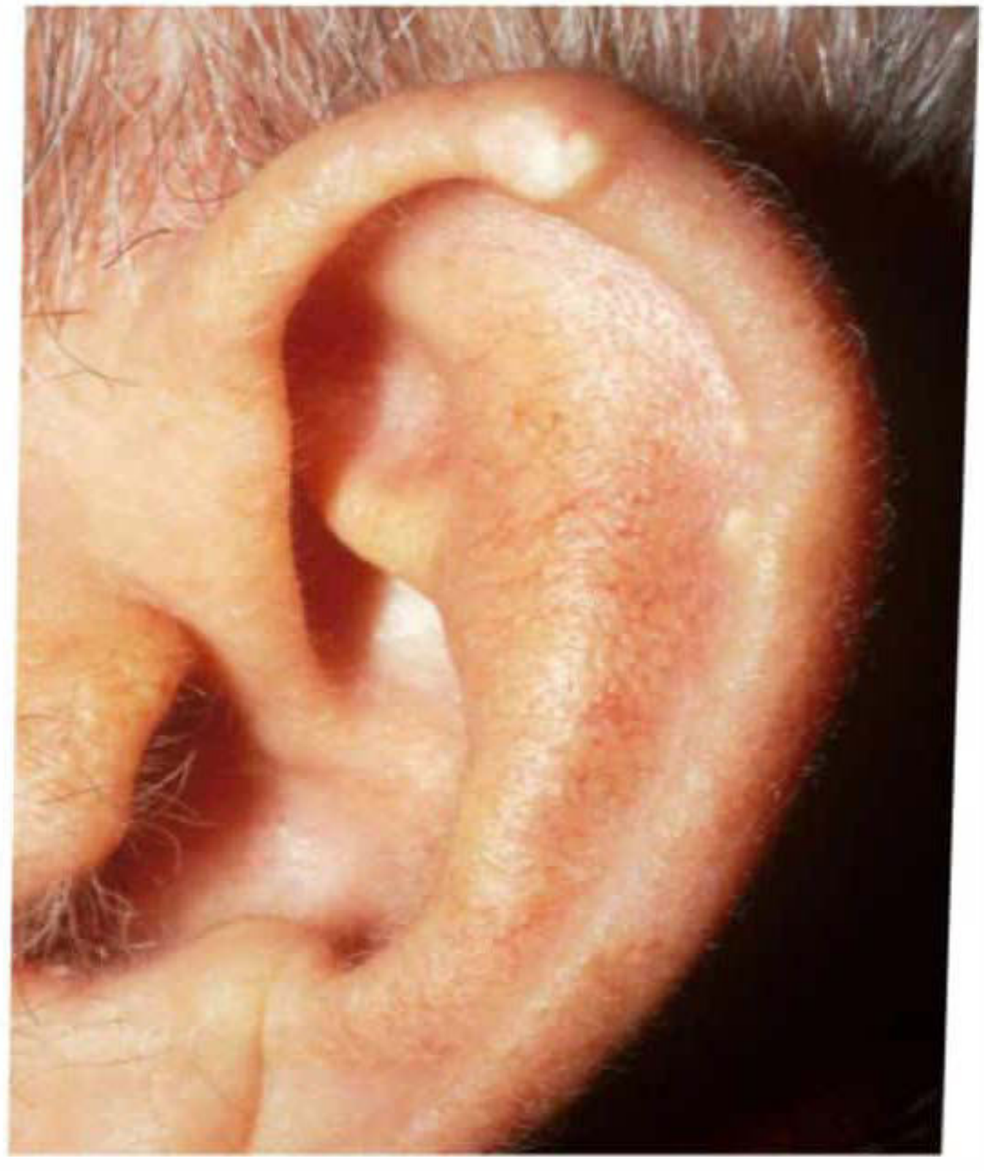
BOXER EAR= CAULIFLOWER EAR



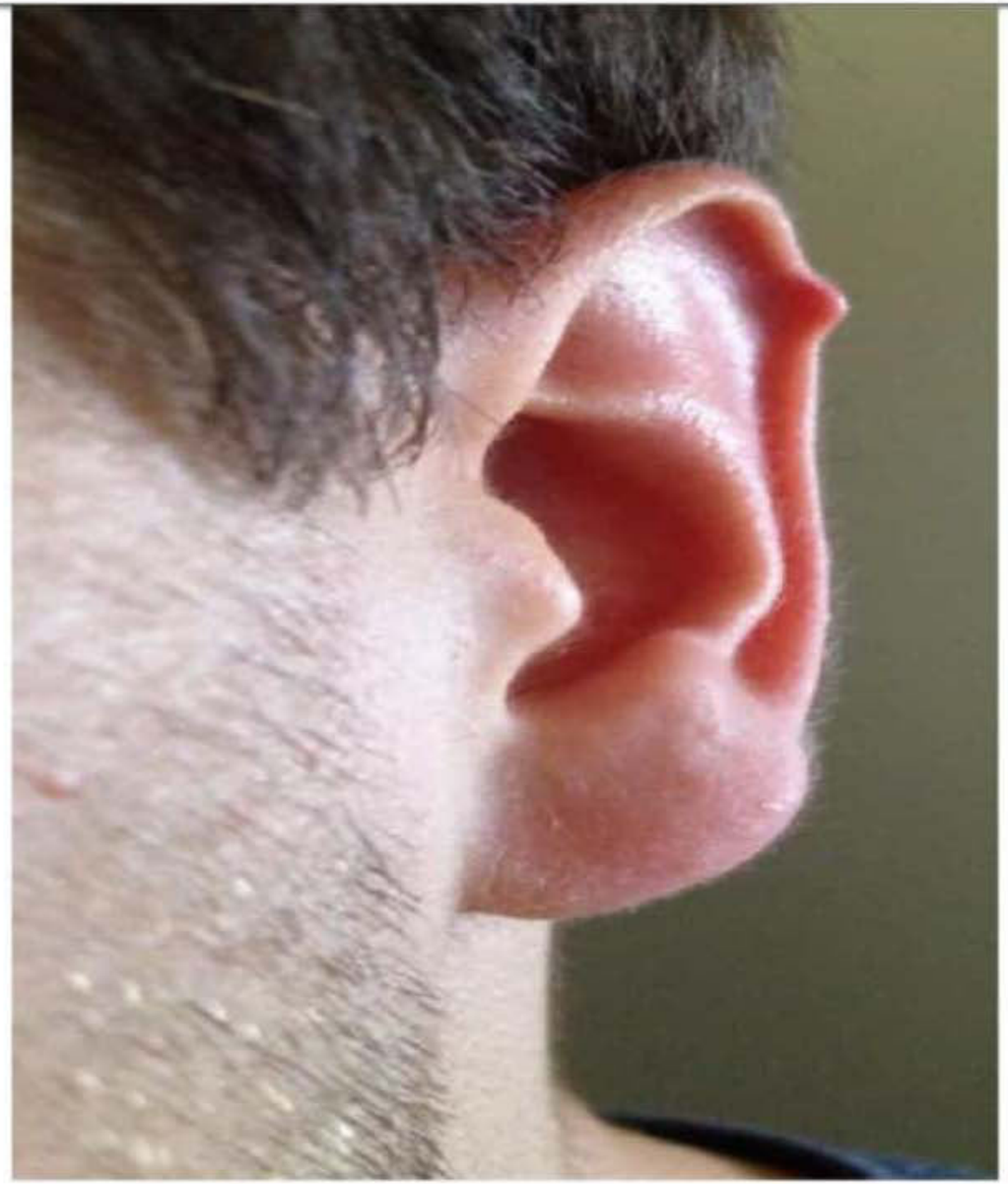
DAMS
BEST PG MEDICAL ENTRANCE COACHING

BAT EAR

GOUTY TOPHI ON PINNA



DARWIN'S TUBERCLE

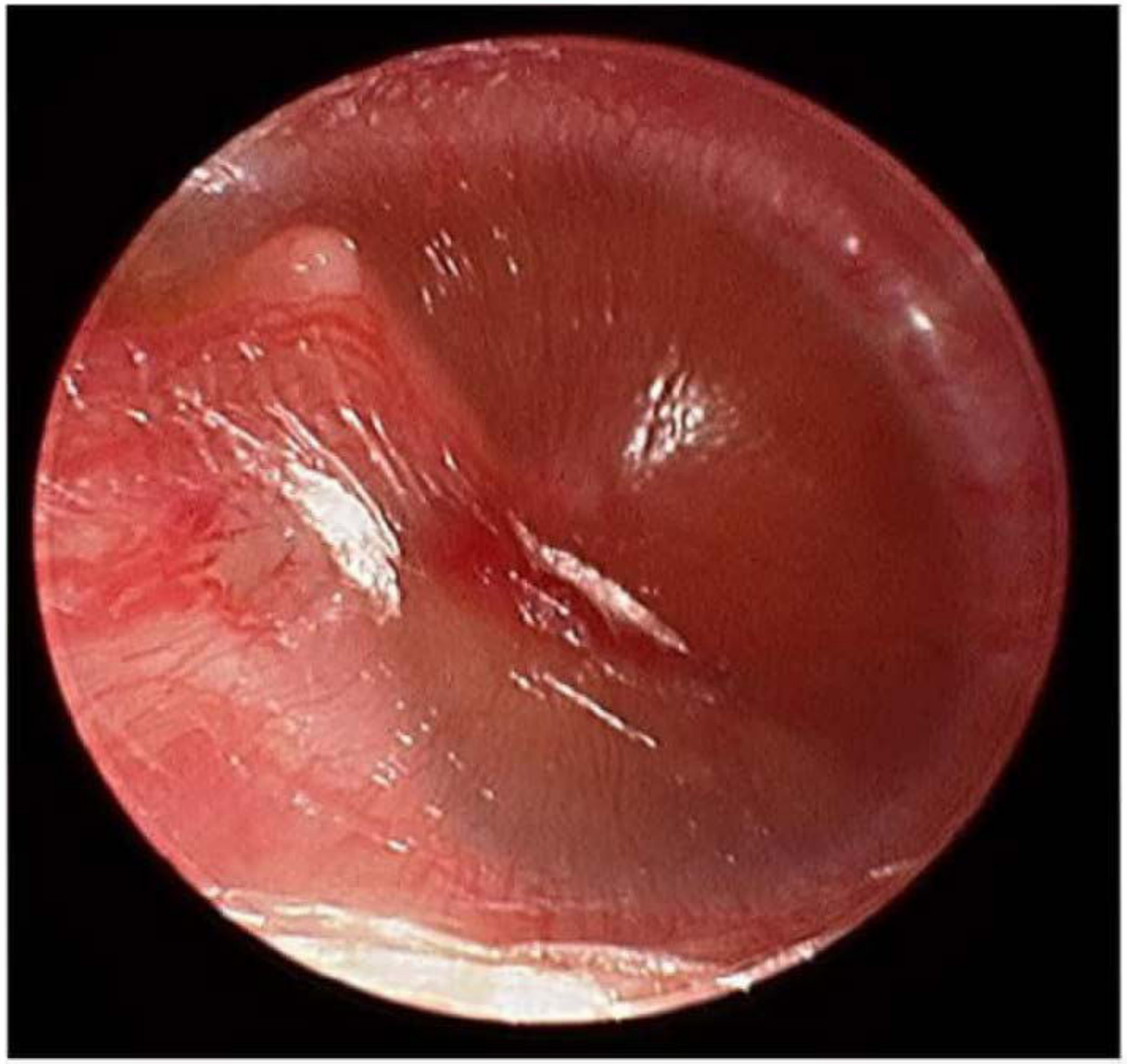


HERPES ZOSTER OTICUS= RAMSAY HUNT SYNDROME

NORMAL TYMPANIC MEMBRANE

DAMS

TYMPANIC MEMBRANE IN ASOM

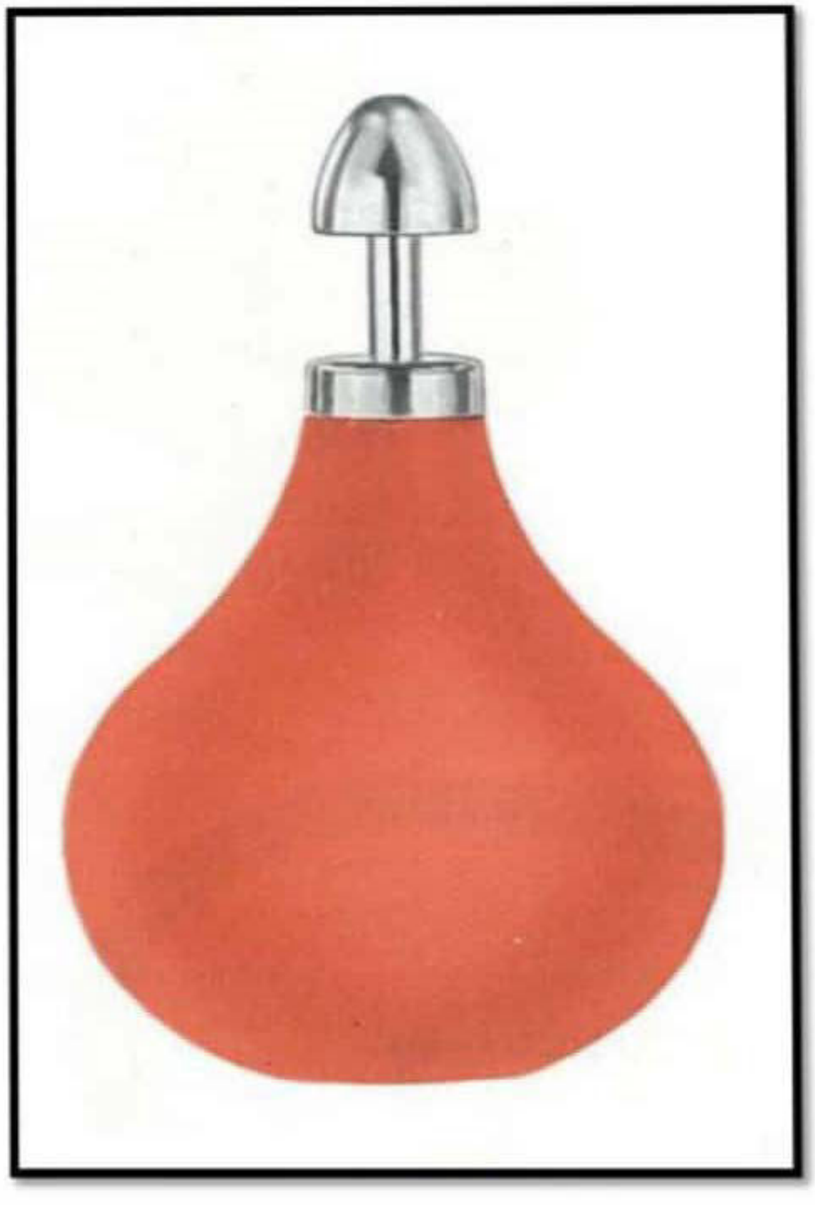


TYMPANIC MEMBRANE IN GLUE EAR

DAMS
BEST PG MEDICAL ENTRANCE COACHING

SEIGEL SPECULUM

POLITZER BAG

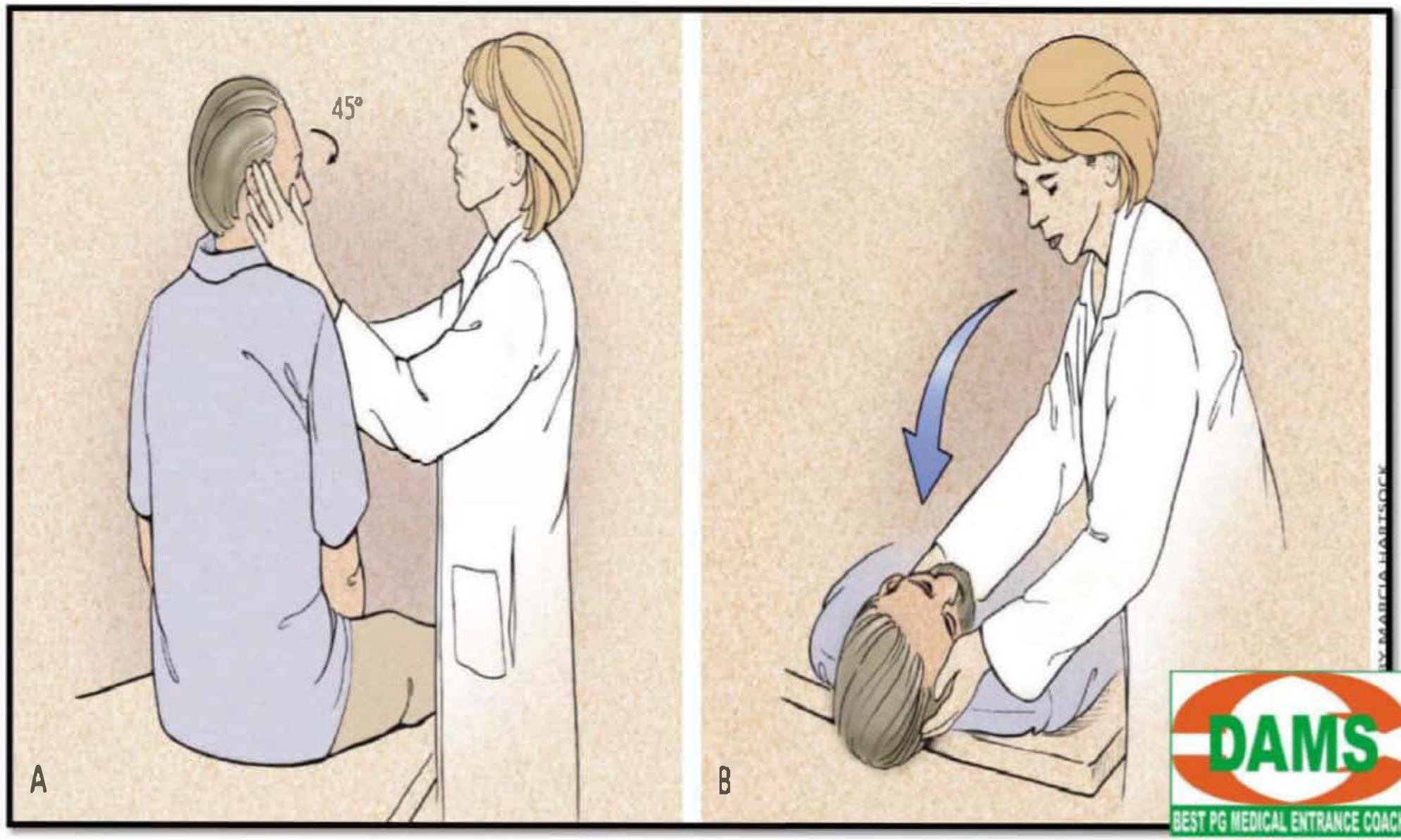


DAMS
BEST PG MEDICAL ENTRANCE COACHING

FRENZEL GLASSES

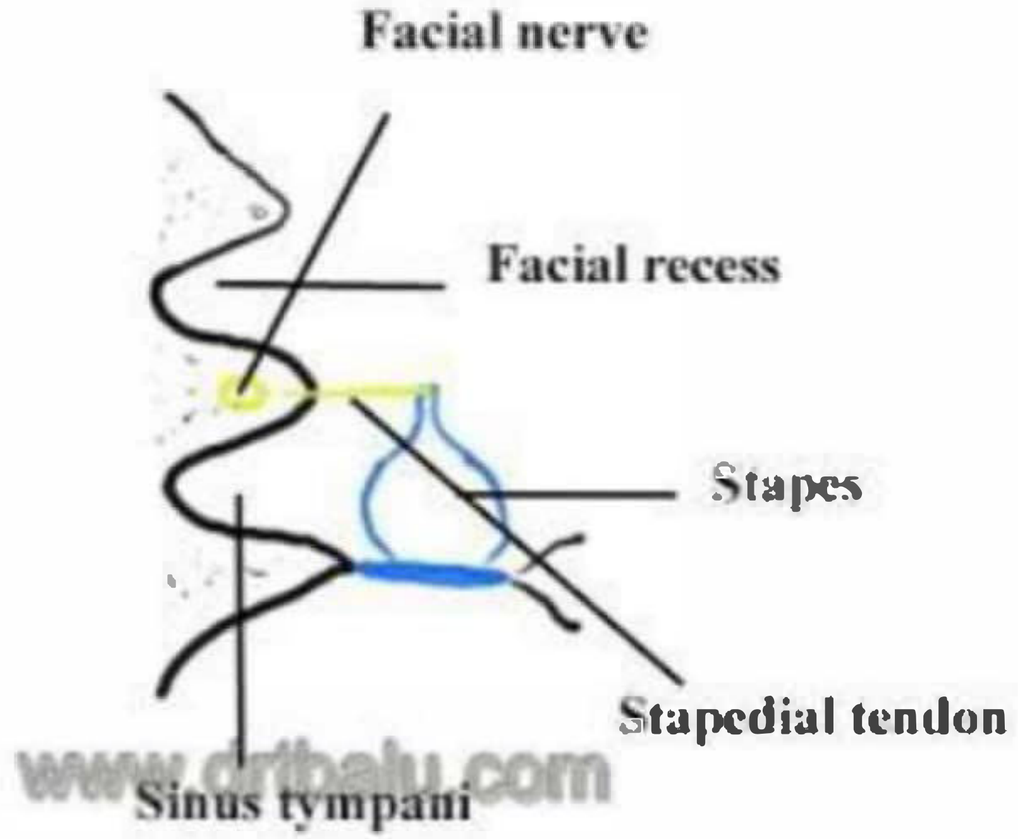
DAMS
BEST PG MEDICAL ENTRANCE COACHING

DIX HALLPIKE'S MANEUVER



DAMS
BEST PG MEDICAL ENTRANCE COACHING

FACIAL RECESS/SINUS TYMPANI



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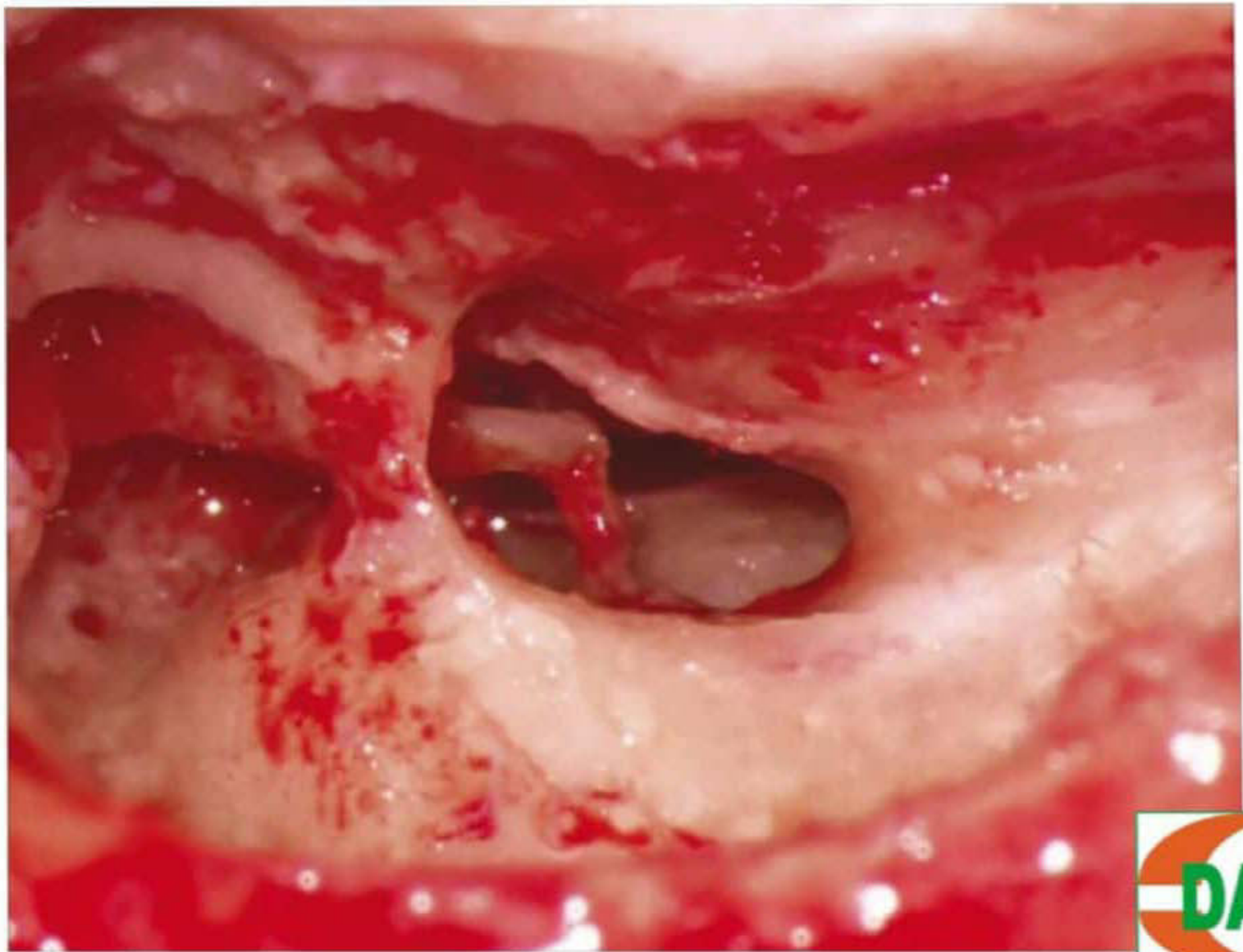
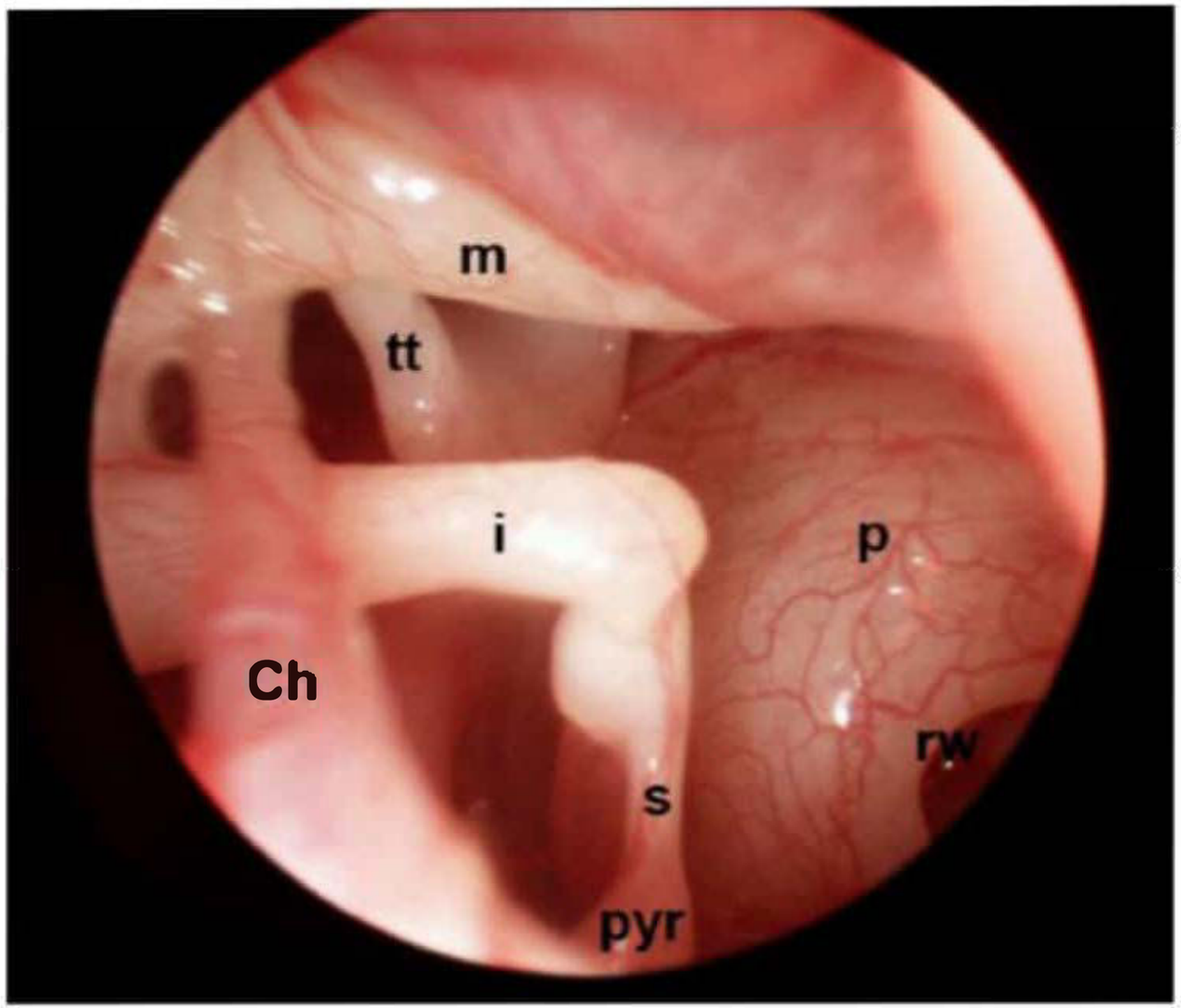
DAMS
BEST PG MEDICAL ENTRANCE COACHING

POSTERIOR TYMPANOTOMY WINDOW



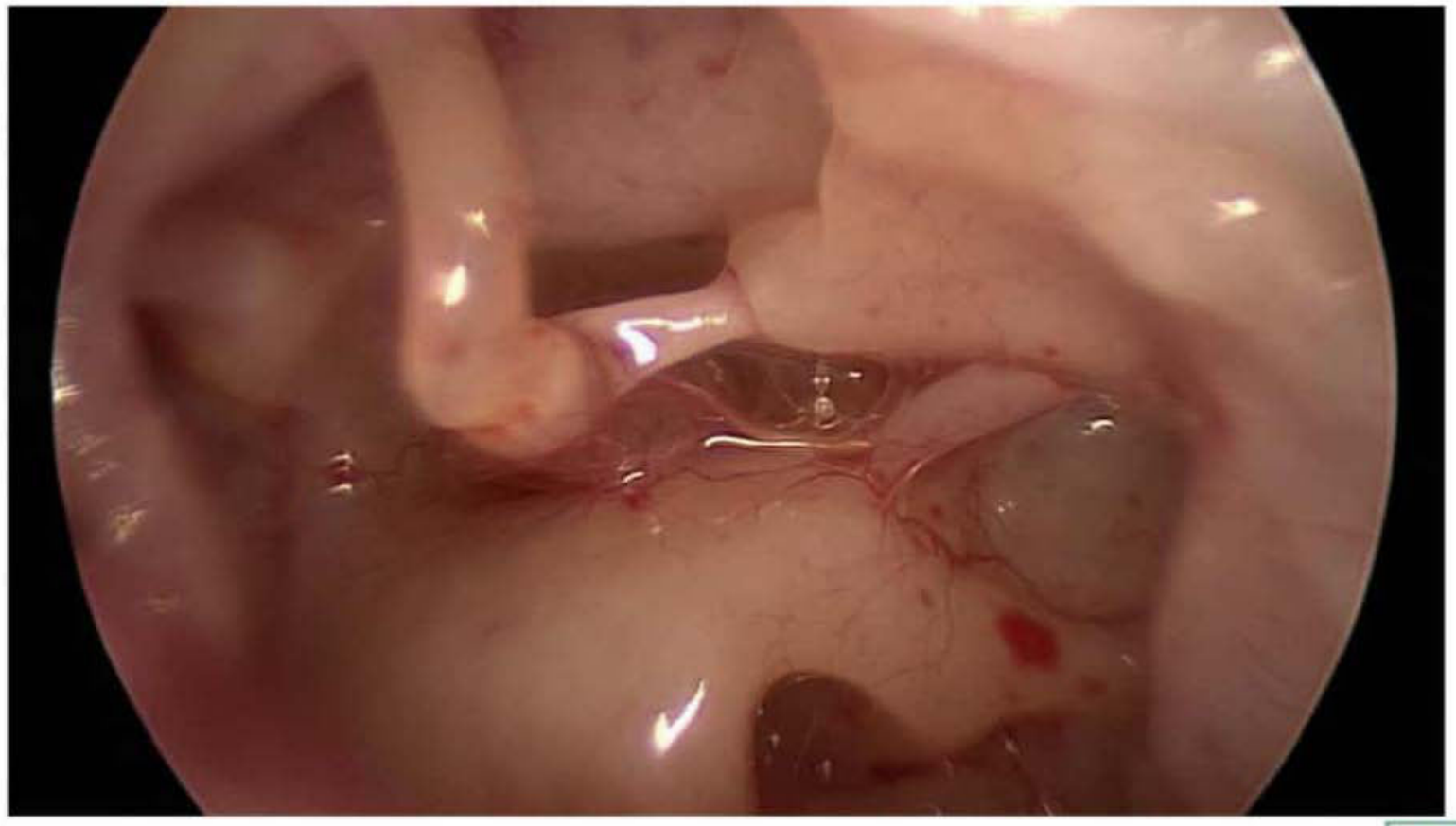
DAMS
BEST PG MEDICAL ENTRANCE COACHING

MIDDLE EAR ANATOMY



DAMS
BEST PG MEDICAL ENTRANCE COACHING

MIDDLE EAR



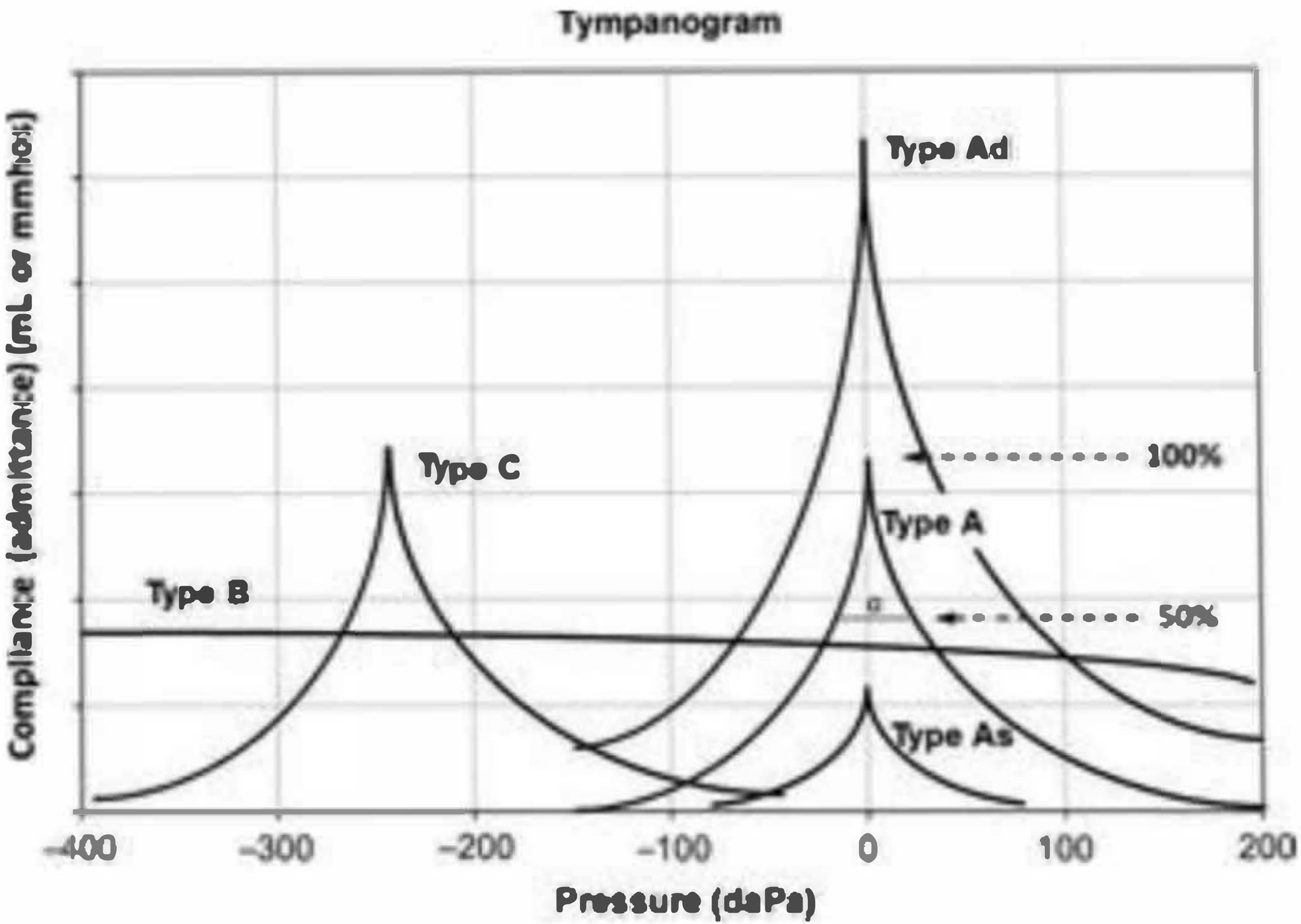
3.AUDIOLOGY

DAMS

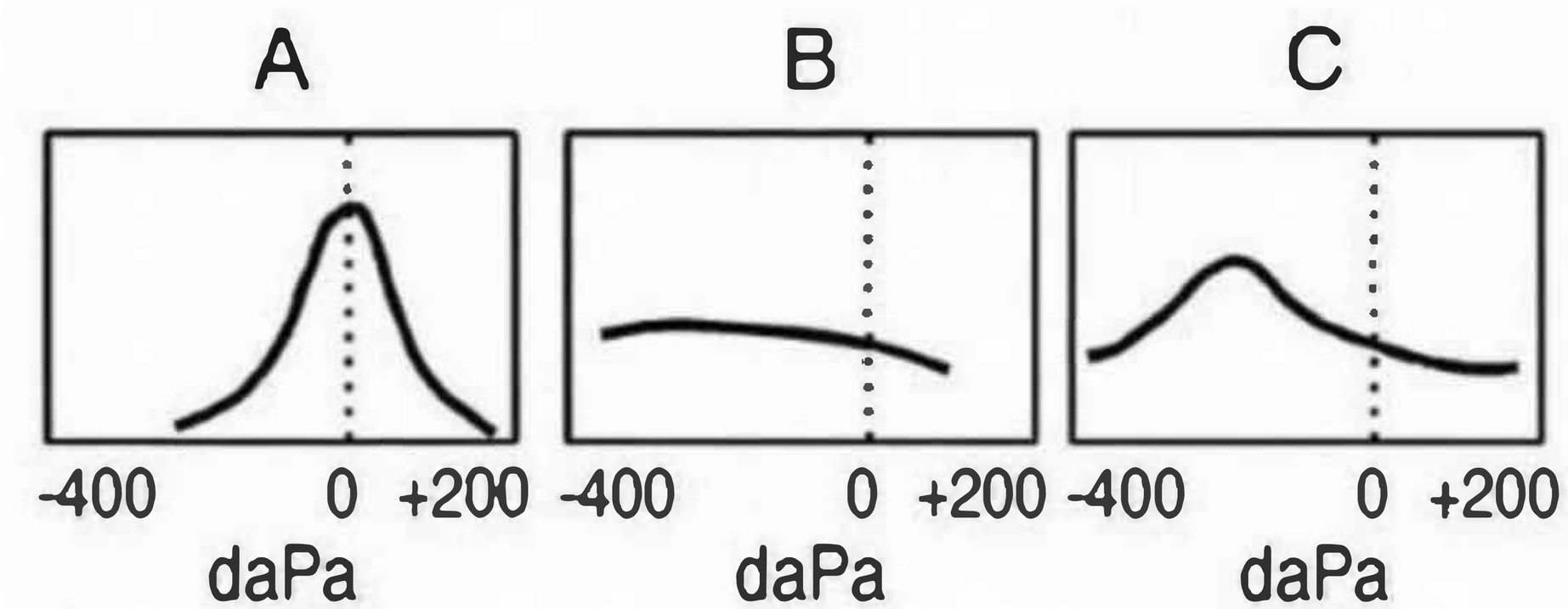
TYMPANOMETRY

DAMS
BEST PG MEDICAL ENTRANCE COACHING

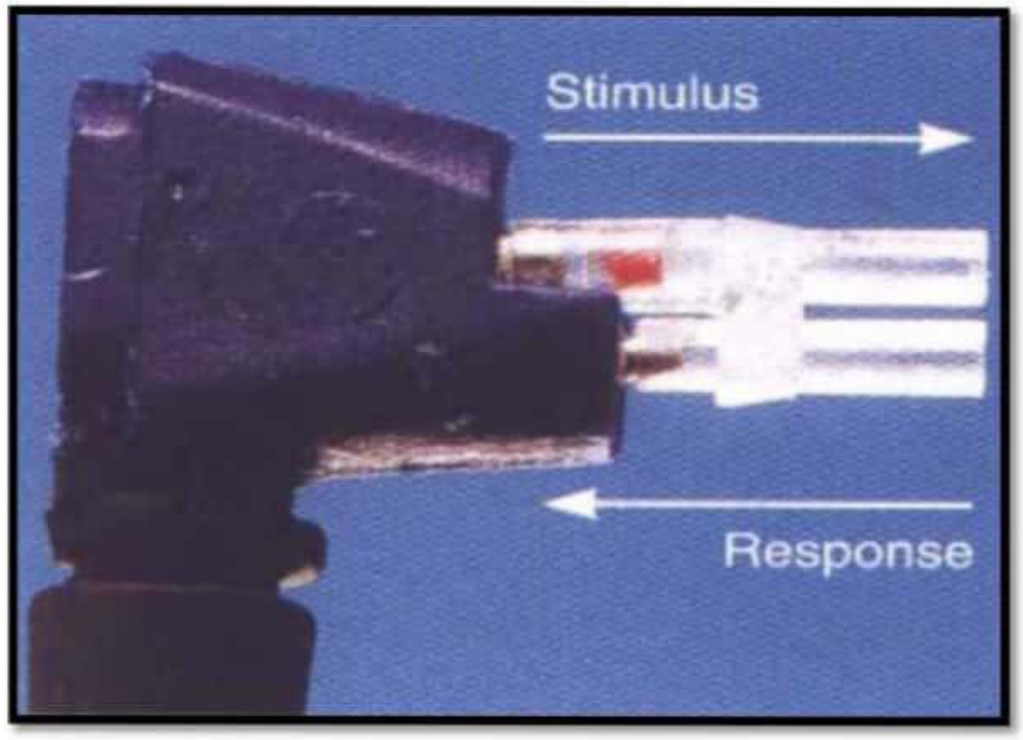
TYMPANOMETRY



TYMPANOMETRY

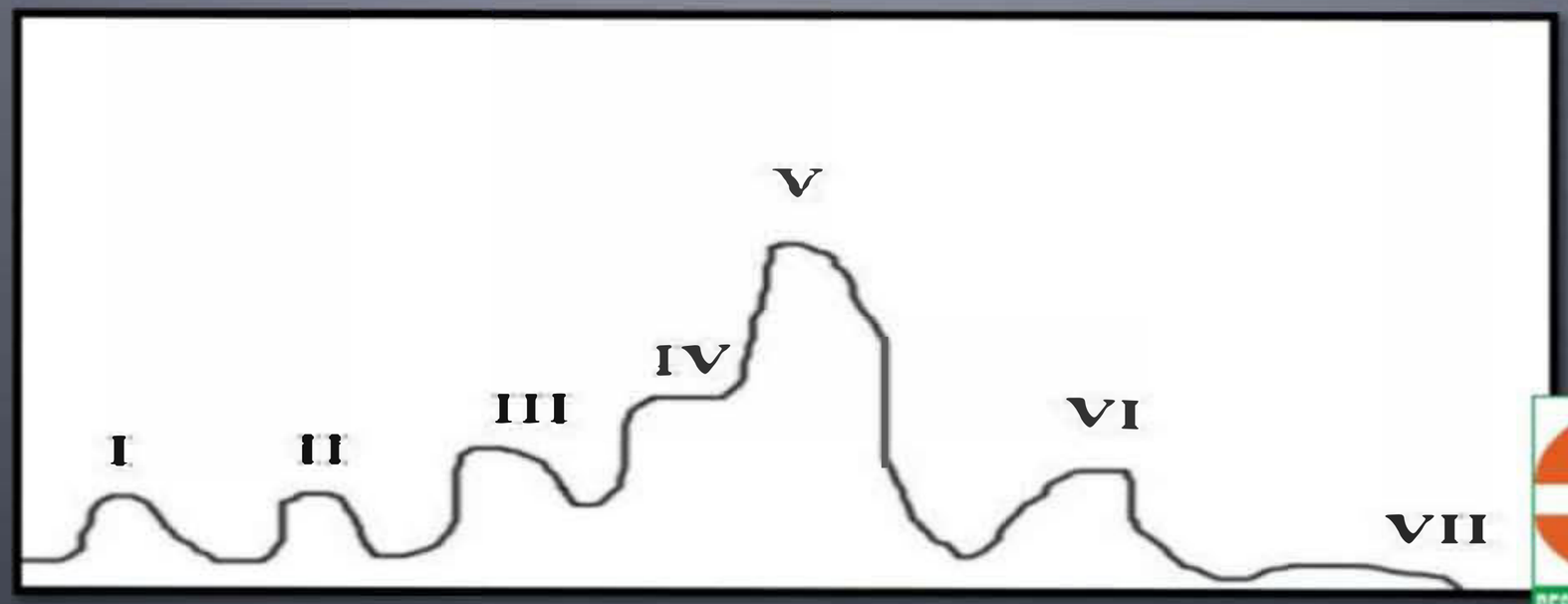


OTOACOUSTIC EMISSIONS



DAMS
BEST PG MEDICAL ENTRANCE COACHING

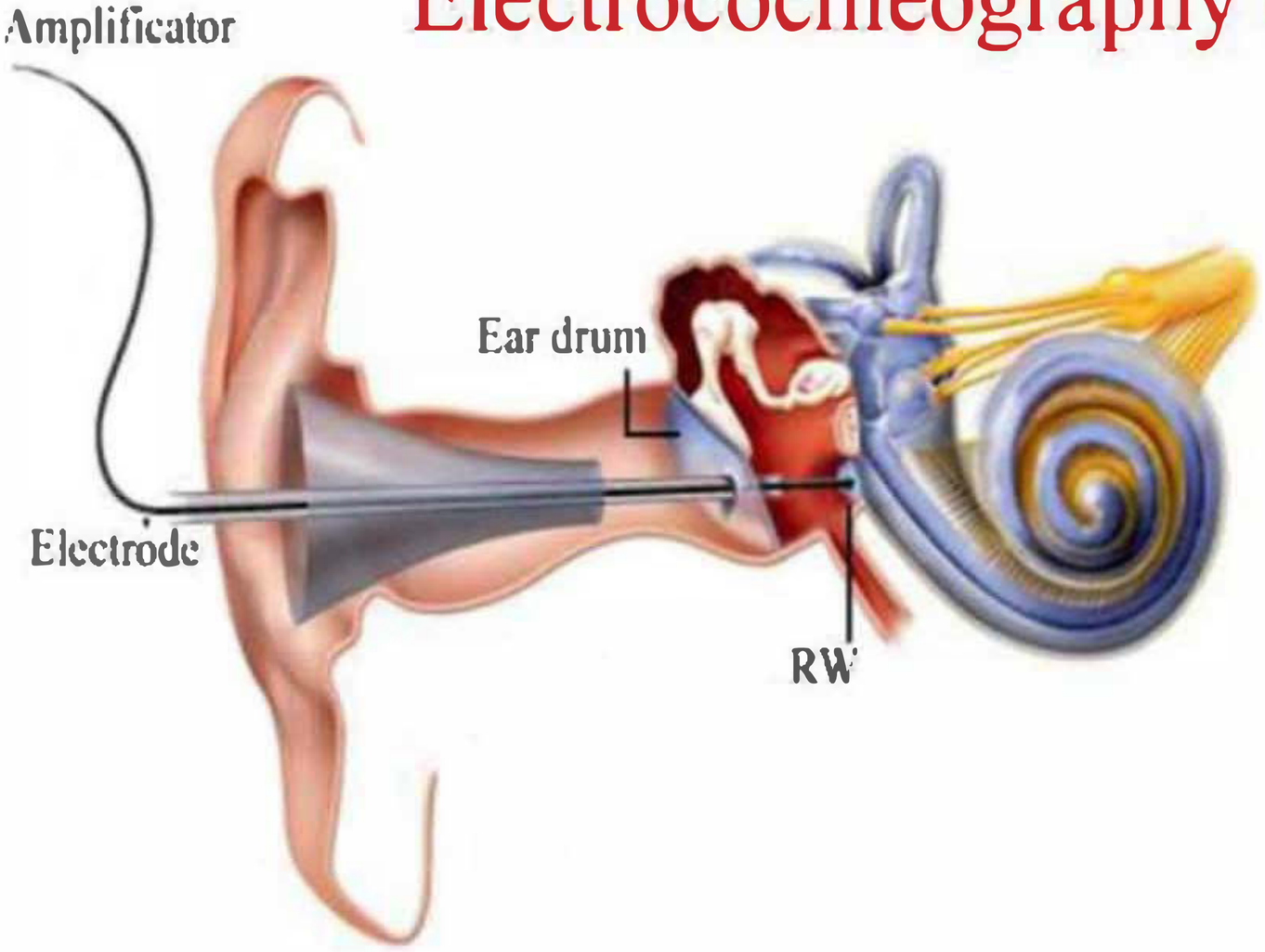
BERA
=ABR

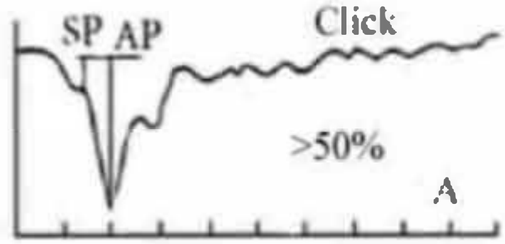


DAMS
BEST PG MEDICAL ENTRANCE COACHING

ELECTROCOCHLEOGRAPHY

Electrocochleography (EcochG)

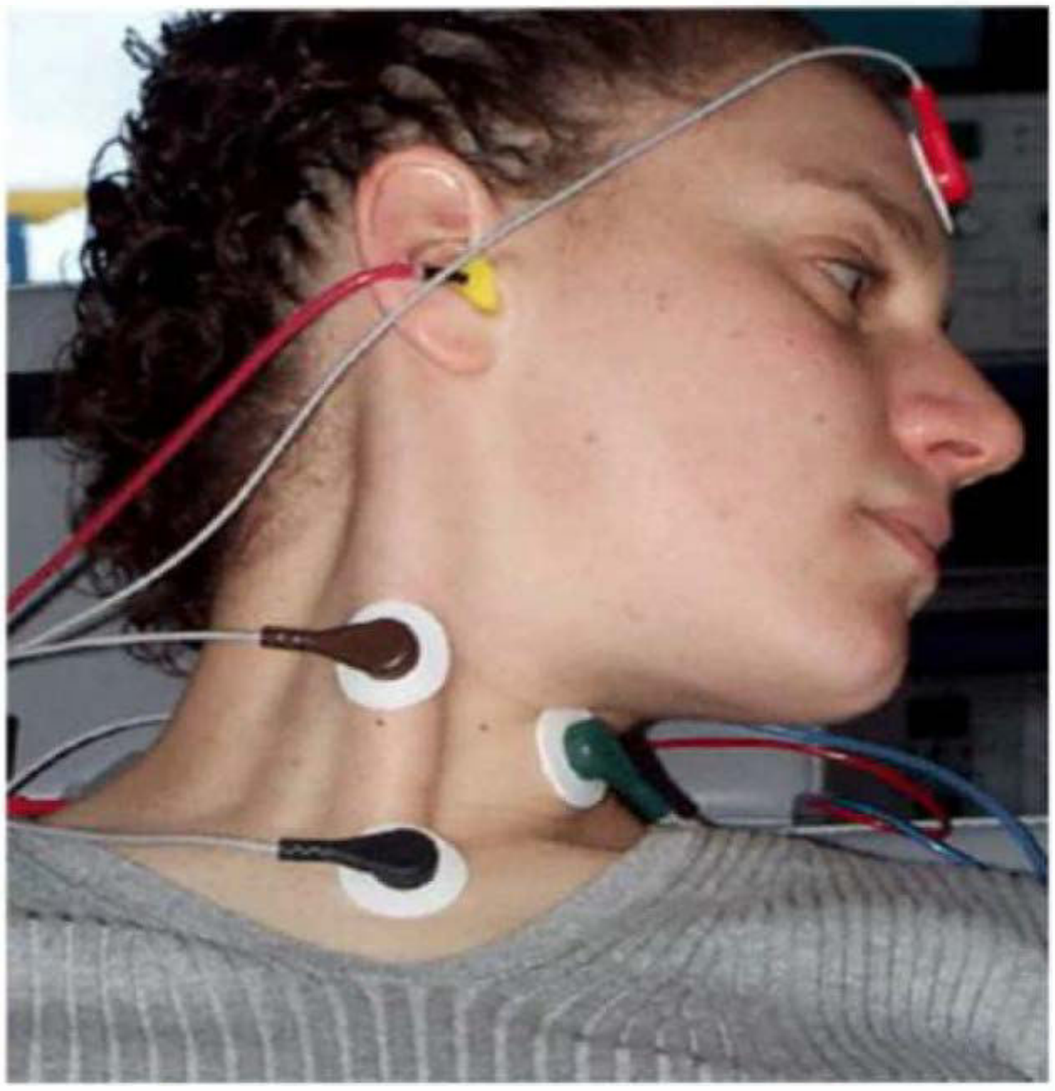




Action potension(AP)

DAMS
BEST PG MEDICAL ENTRANCE COACHING

VEMP (VESTIBULAR EVOKED MYOGENIC POTENTIAL)



DAMS
BEST PG MEDICAL ENTRANCE COACHING

AUDIOGRAM= PTA

DAMS

AUDIOGRAM BASICS

AC IS COMPLETE CHECK UP OF HEARING

BC CHECKS COCHLEA ONLY

BC IS POOR IN SNHL ONLY

BC LINE IS ALWAYS ABOVE THE AC LINE

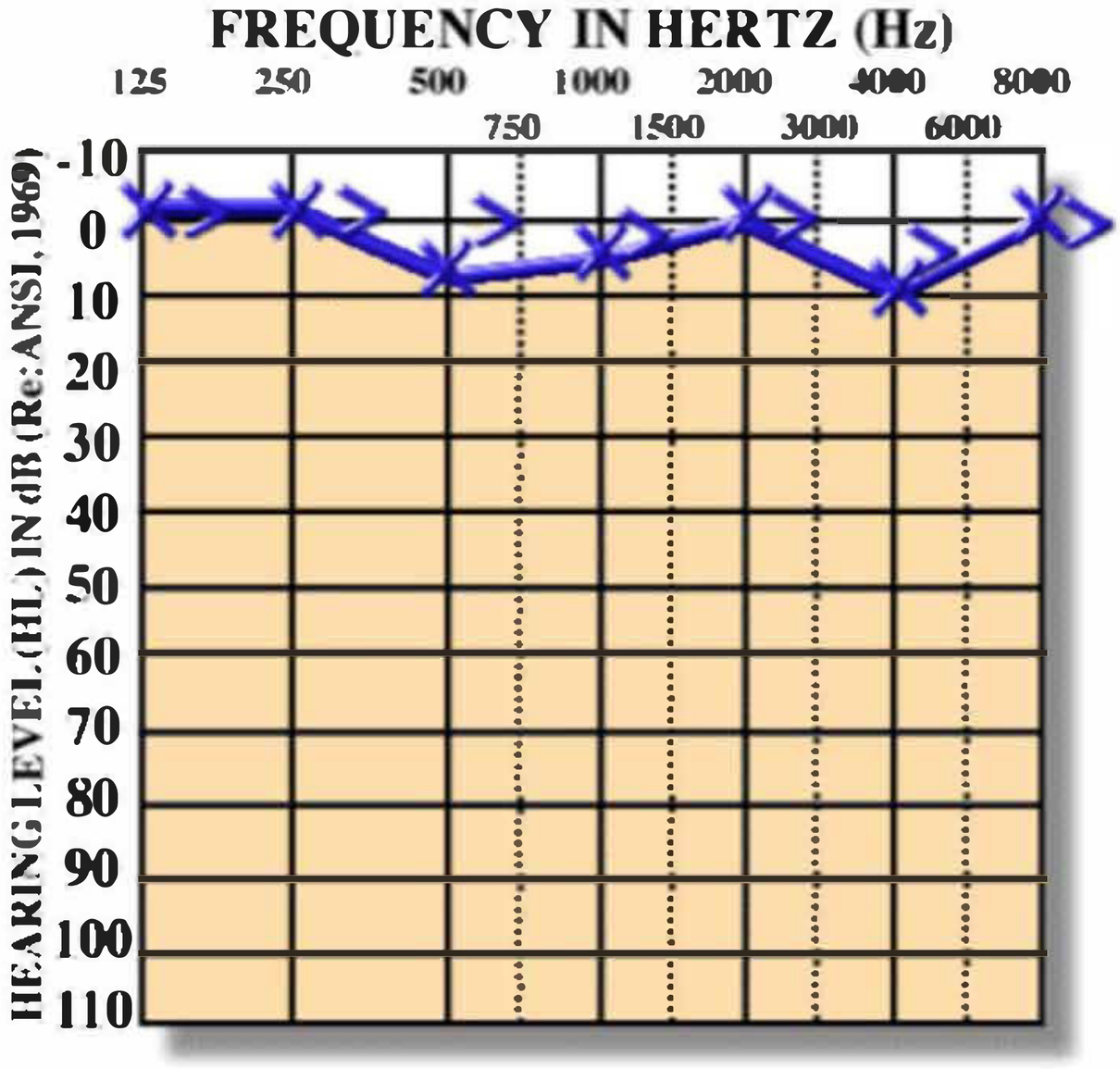
DAMS

AUDIOGRAM SYMBOLS

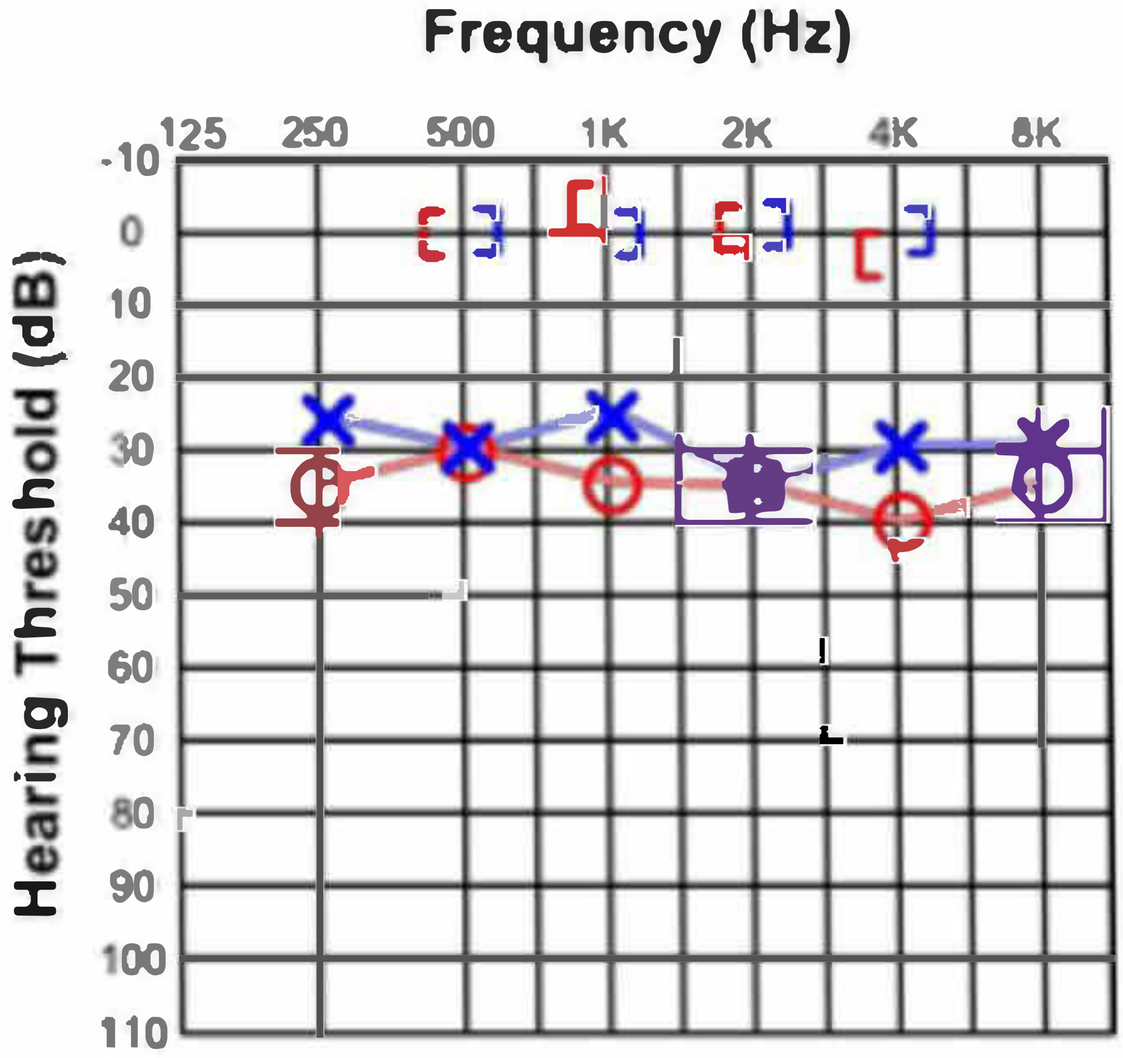
Modality	Ear	
	Right	Left
AC unmasked	○	×
AC masked	△	□
BC unmasked	<	>
BC masked	⊏	⊐

DAMS

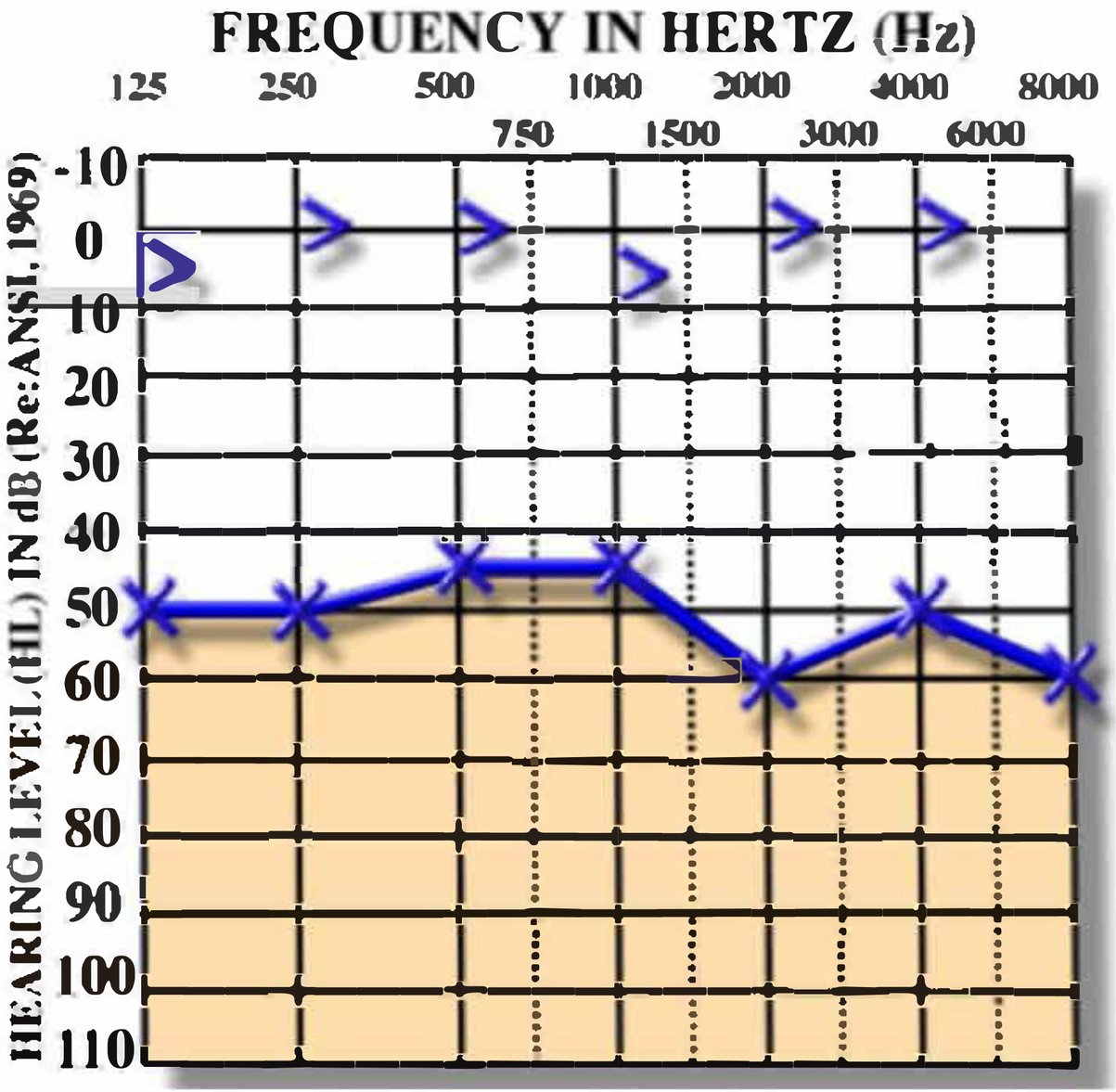
NORMAL HEARING AUDIOGRAM



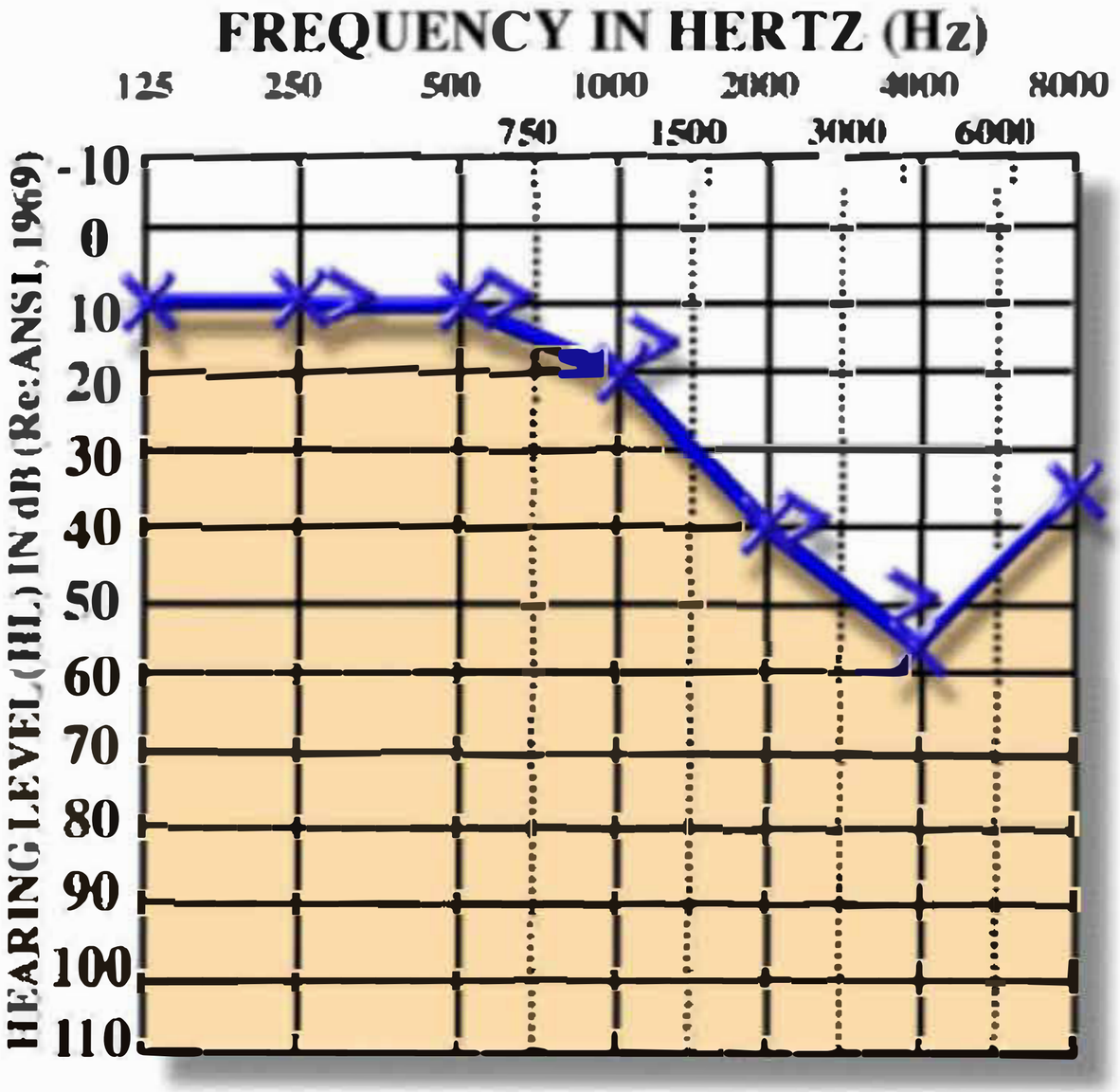
BILATERAL
CONDUCTIVE HEARING LOSS



CONDUCTIVE HEARING LOSS



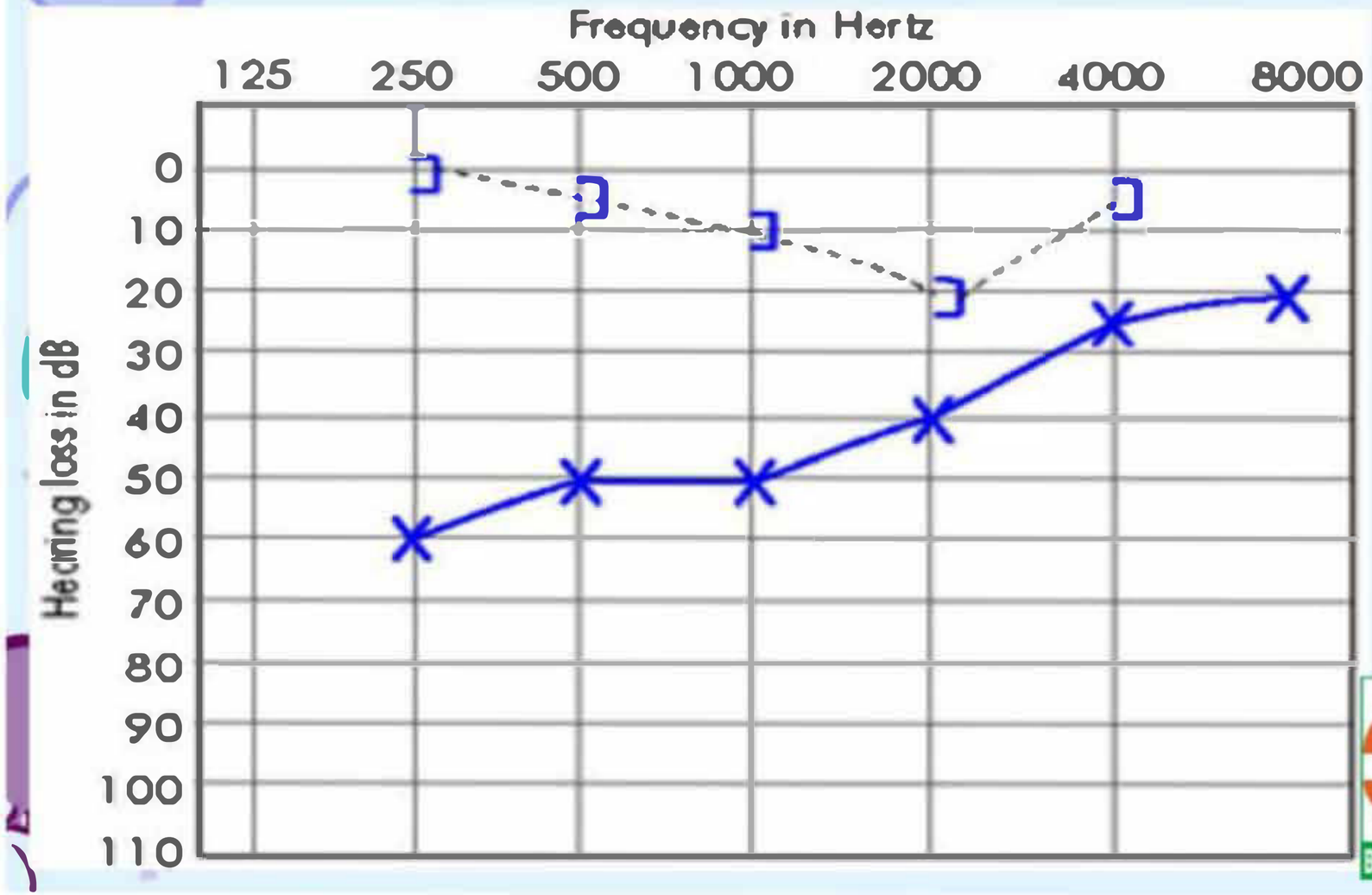
SNHL



DAMS
BEST PG MEDICAL ENTRANCE COACHING

OTOSCLEROSIS

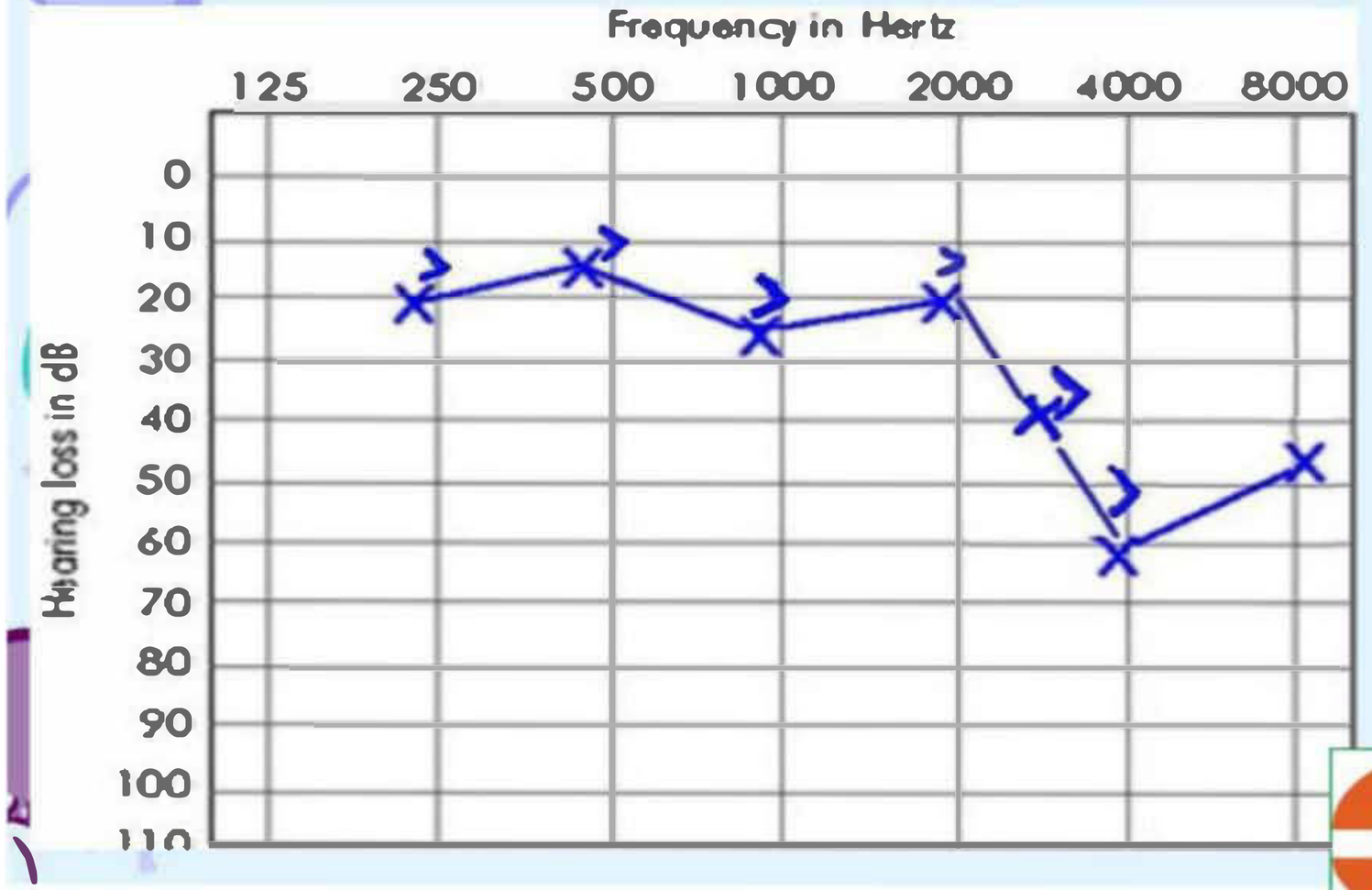
Carhart's notch (otosclerosis)



DAMS
BEST PG MEDICAL ENTRANCE COACHING

NOISE INDUCED HEARING LOSS

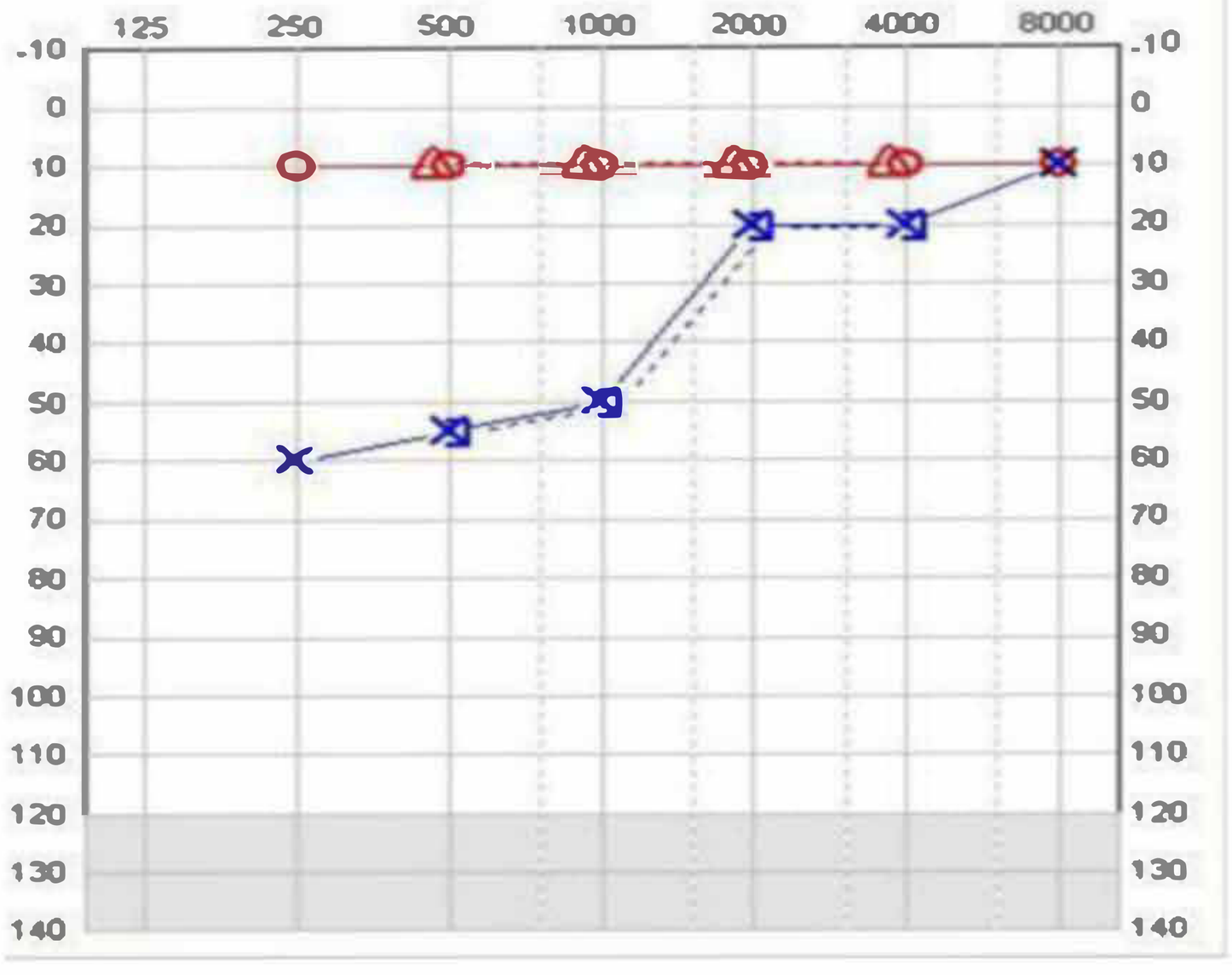
Acoustic dip (Noise deafness)



DAMS
BEST PG MEDICAL ENTRANCE COACHING

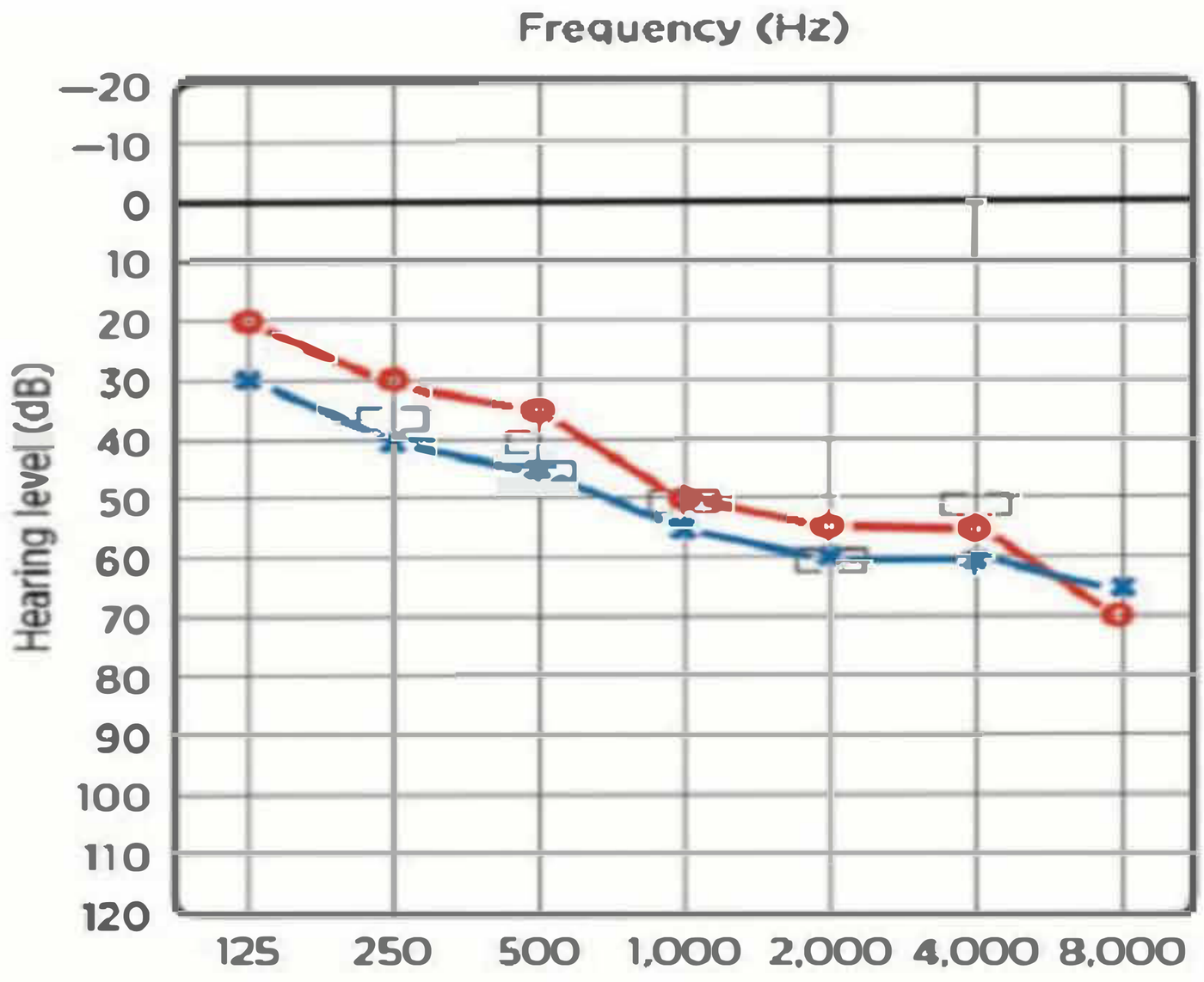
MENIERE'S DISEASE.....
RISING CURVE

DAMS
BEST PG MEDICAL ENTRANCE COACHING



PRESBYACUSIS...
SLOPING CURVE

DAMS
BEST PG MEDICAL ENTRANCE COACHING



THANK YOU

DR RAJIV DHAWAN

#RajivDhawan

Rajiv Dhawan - ENT Discussion

DAMS

ALL THE BEST FROM ALL DAMS
FACULTY MEMBERS & BEST WISHES

DAMS

BEST PG MEDICAL ENTRANCE COACHING

UPSIT

University of Pennsylvania Identification Test (UPSIT)

“scratch and sniff” booklets, each containing 10 odorants

Cross-Cultural Smell Identification Test (CC-SIT).

This test is a variant of the UPSIT. It comprises 12 items

Parkinson disease and Alzheimer disease

Septal perforations most commonly found in the septum

Anterior septum

Most commonly 1 to 2 cm

Septal button

Large perforation (> 2 cm)

DAMS

Front of the nose filter upto 3 µm

Nasal mucosa trap 0.5-3.0 µm

Nasal septum and turbinates filter particles 10 to 30 µm

Nasal vibrissae filter particles 20 to 30 µm

Particles 0.2 to 0.5 µm in diameter tend to remain suspended and are exhaled.

What are the three infundibular cells that are anterior ethmoid air cells

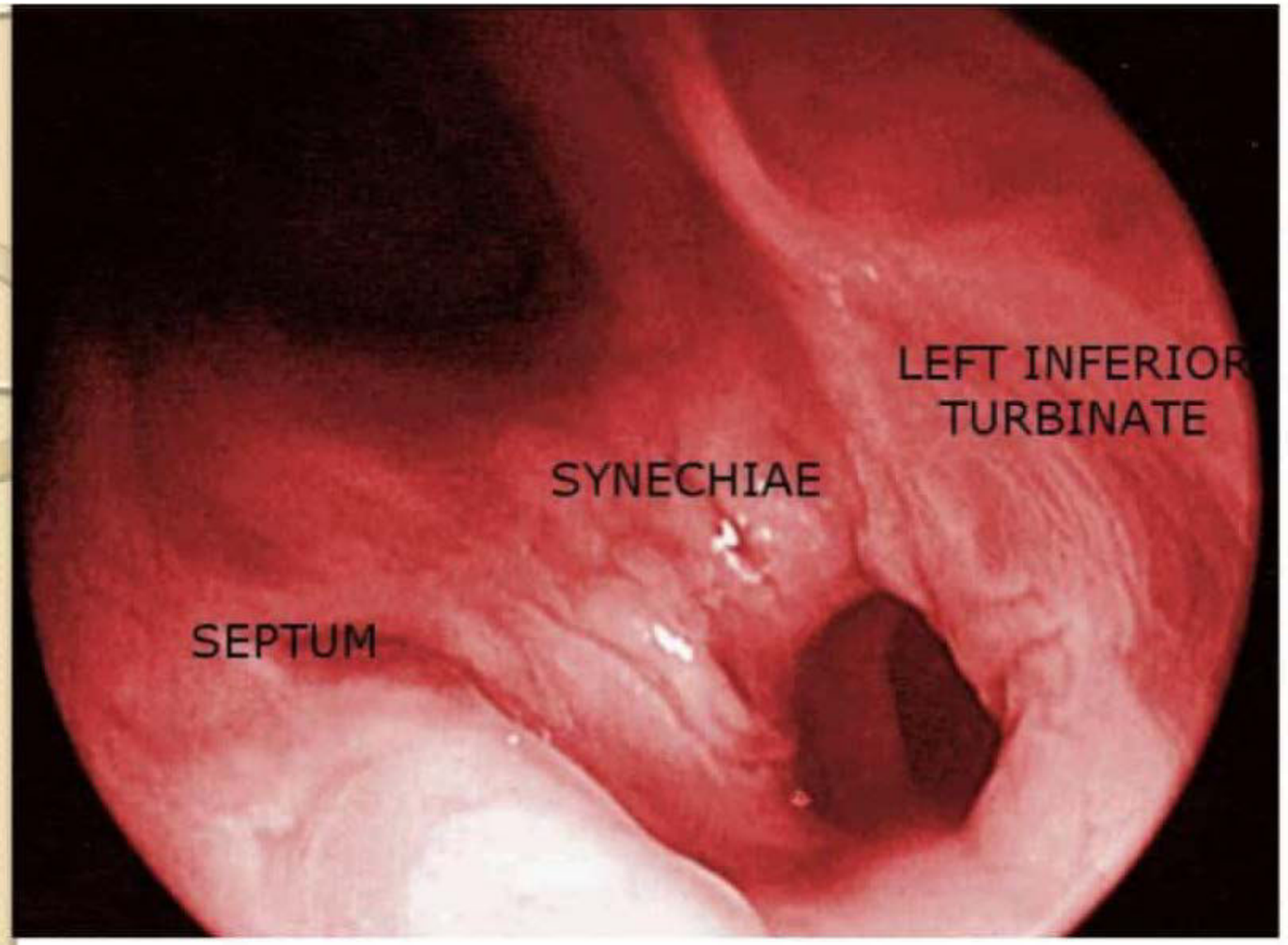
- Agger nasi cells (MOST ANT)
- Terminal cell (recessus terminalis)
- Suprainfundibular cell

What structure separates the anterior and posterior ethmoid sinuses

Ground or basal lamella

Nose and the voice

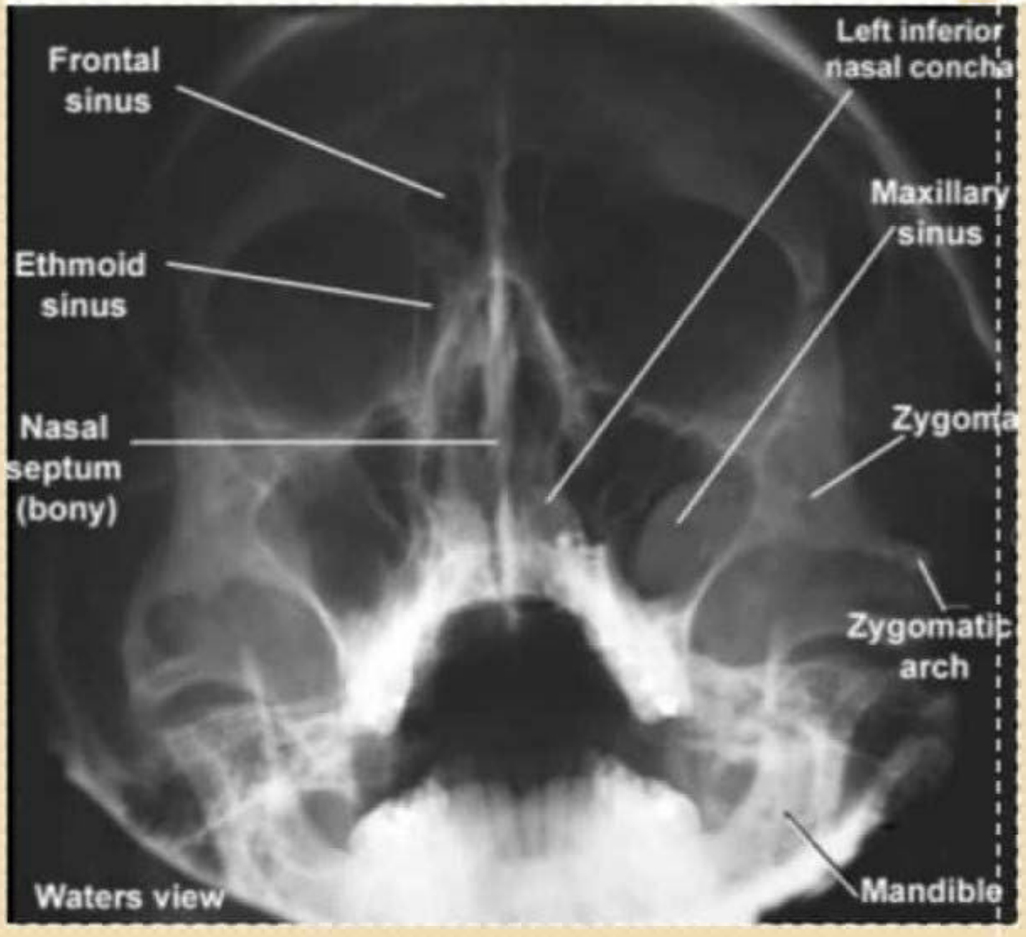
- Nose adds quality by allowing air to escape
- Rhinolalaia clausa too little air escapes
- Rhinolalia aperta too much escapes



Treatment of choice for nasal synechiae-

- Surgical removal of adhesions
- Topical mitomycin
- Nasal stent

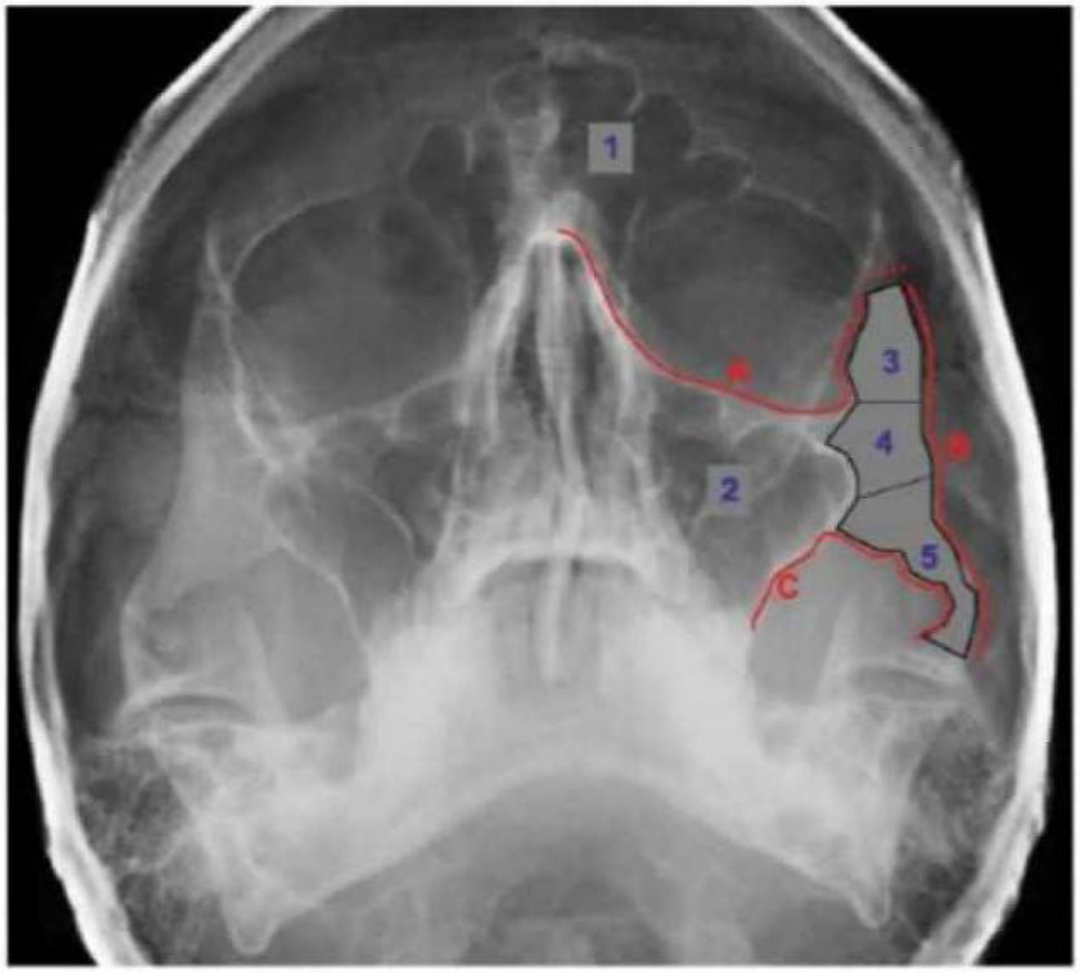
Occipitomental View



Nose Chin Position

- Frontal process of maxilla
- Superior orbital fissure
- Infratemporal fossa
- Sphenoid sinus ?

WATER VIEW



Water's View

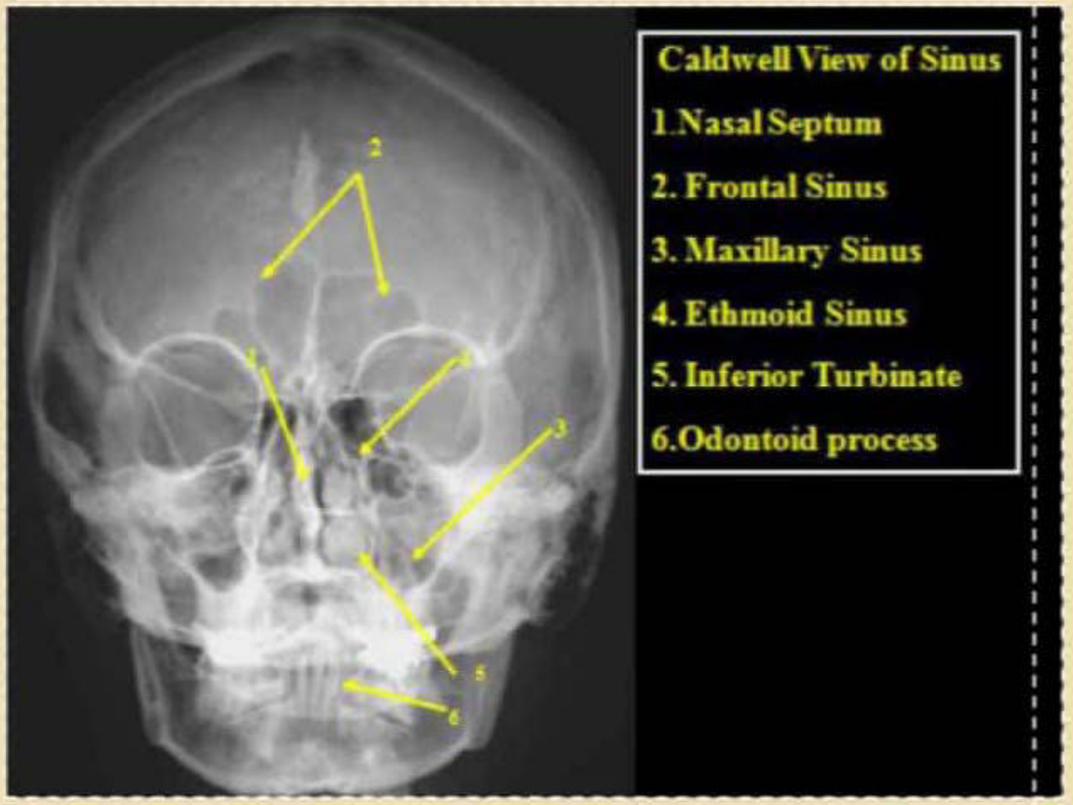
Key structures
1 = Frontal sinus
2 = Maxillary sinus
3 = Frontal process of Zygoma
4 = Body of Zygoma (malar eminence)
5 = Temporal process of Zygoma

Dotted line = zygomatico-frontal suture

Dolan's lines of reference
Line A, B, C

Rule:
smooth, non-disrupted, same contour on both sides

Occipitofrontal

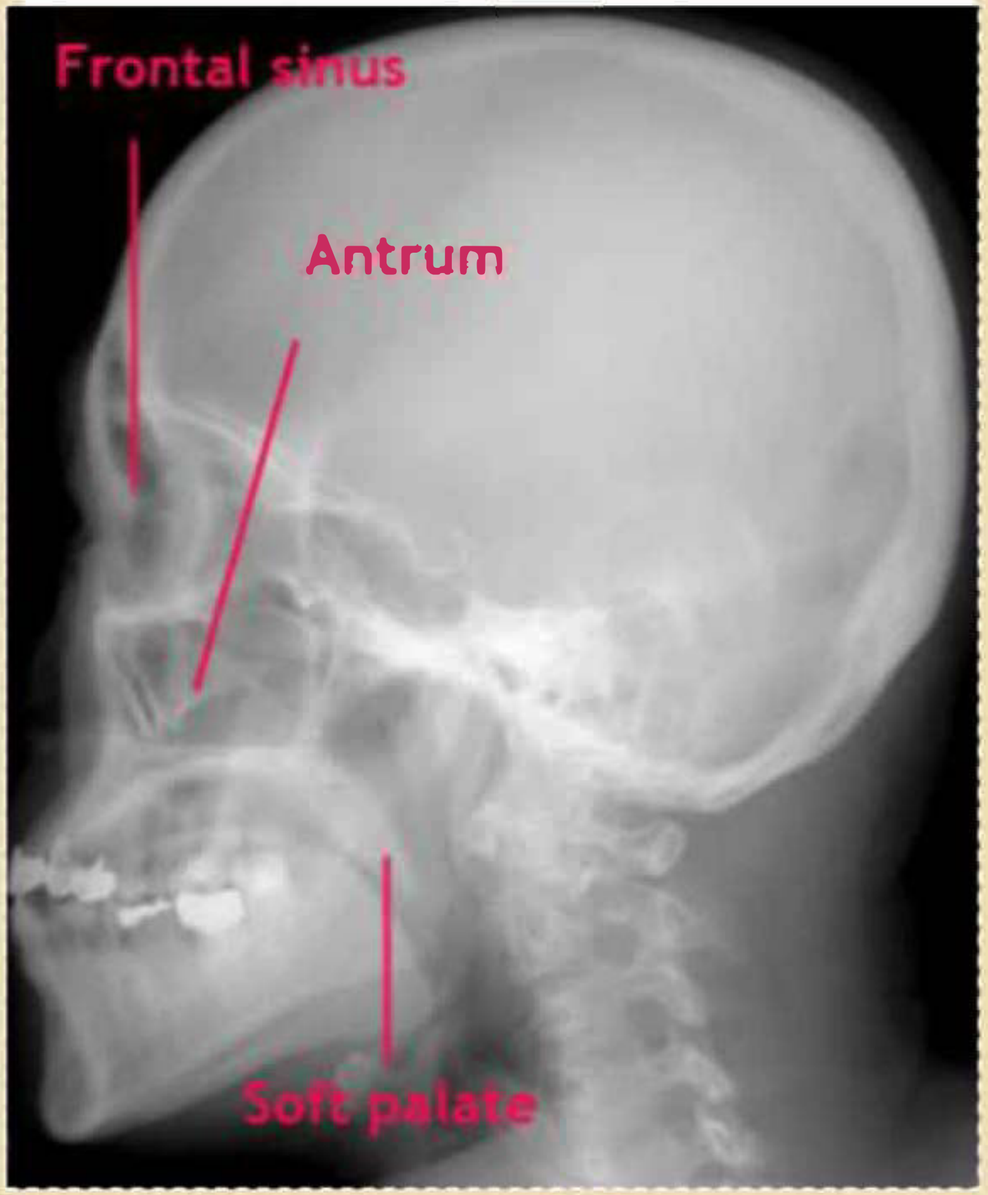


Nose –Forehead position

- Frontal process of zygoma and zygomatic process of maxilla
- Superior margin of orbit
- Superior orbital fissure
- Foramen rotundum

CALDWELL VIEW

DAMS



- Anterior and posterior extent of sphenoid,frontal and maxillary sinuses
- Sella turcica
- Ethmoidal sinuses
- Alveolar process
- Condyle and neck of mandible

LATERAL VIEW

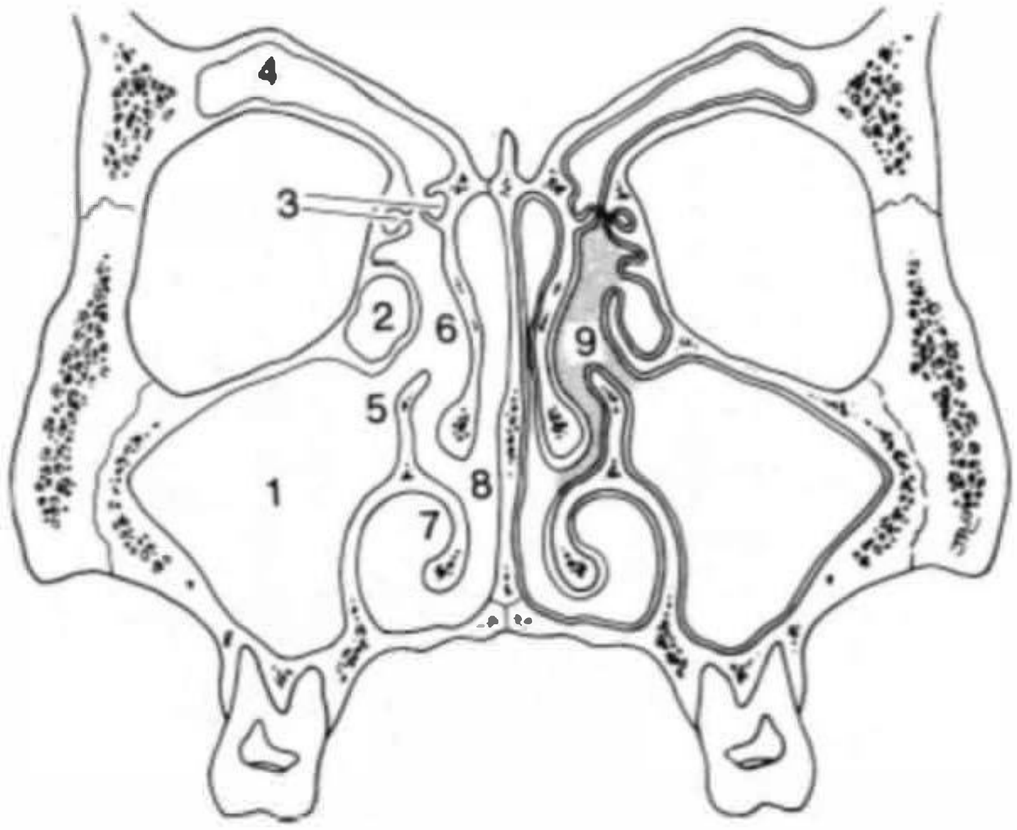
DAMS

Submentovertical View

- Sphenoid > posterior ethmoid >maxillary sinuses

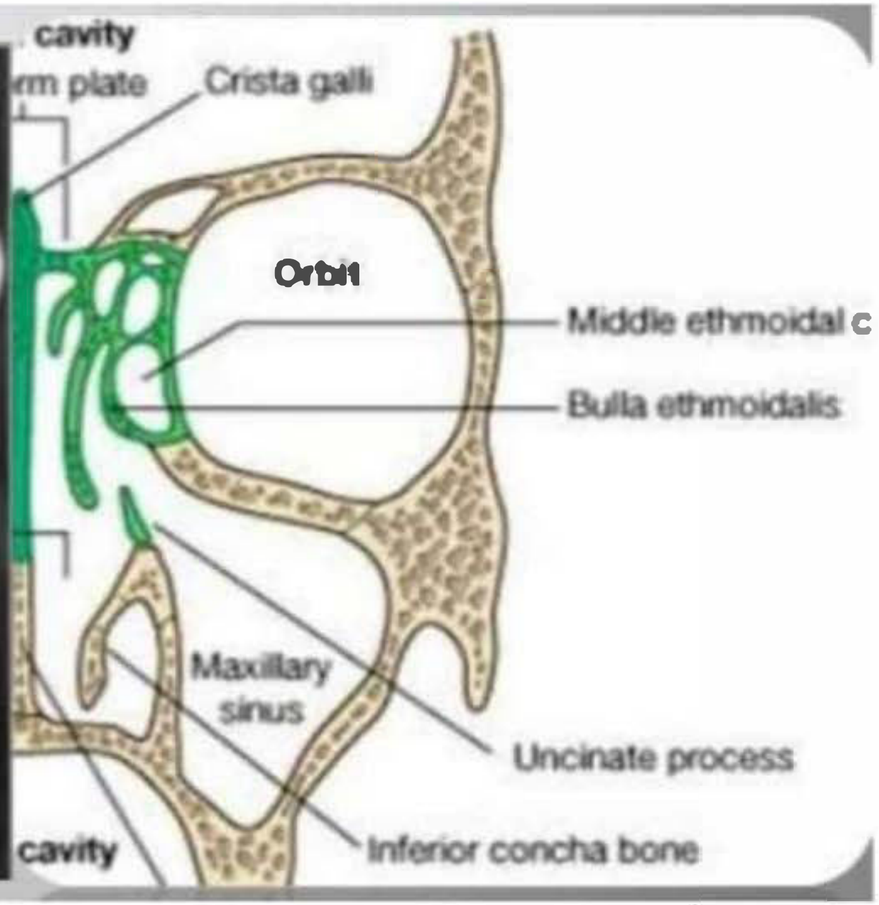
Basal view

Figure 1. Coronal View Of The Paranasal Sinuses



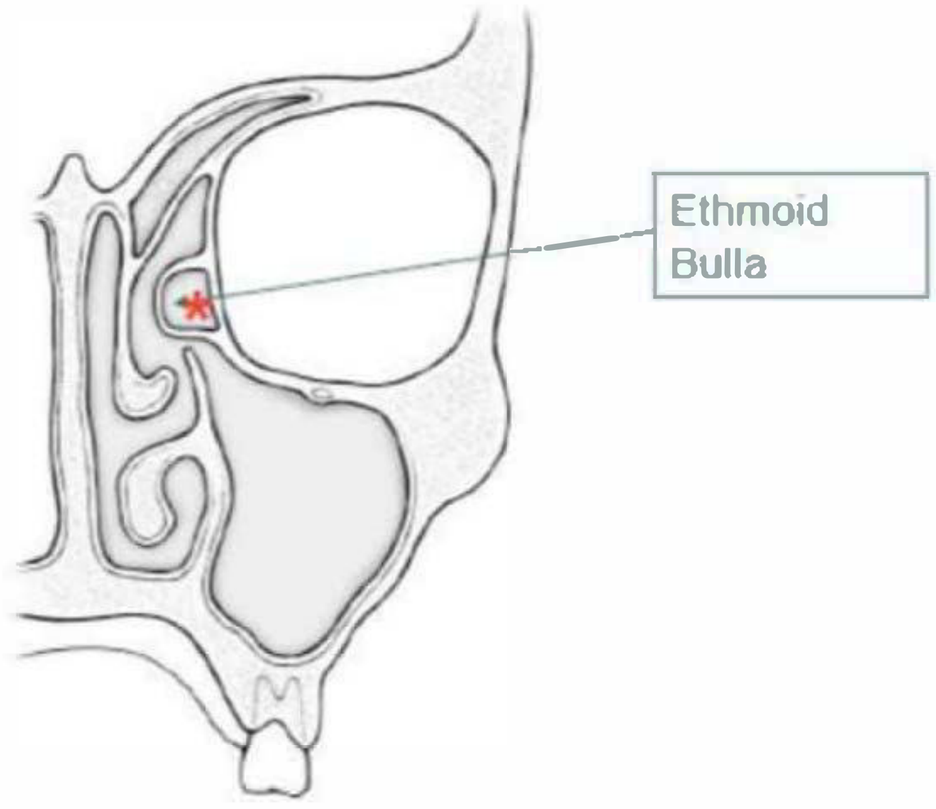
1. Maxillary sinus
2. Ethmoidal bulla
3. Ethmoidal cells
4. Frontal sinus
5. Uncinate process
6. Middle turbinate
7. Inferior turbinate
8. Nasal septum
9. Osteomeatal complex

MO
MT
MS
Uncinate


cavity
rm plate
Crista galli
Orbit
Middle ethmoidal c
Bulla ethmoidalis
Maxillary sinus
Uncinate process
cavity
Inferior concha bone

Ethmoid Bulla

- Bulla = hollow, thin-walled bony prominence
- Most consistent and well pneumatized anterior ethmoid air cell
- Makes up posterior boarder of frontal recess

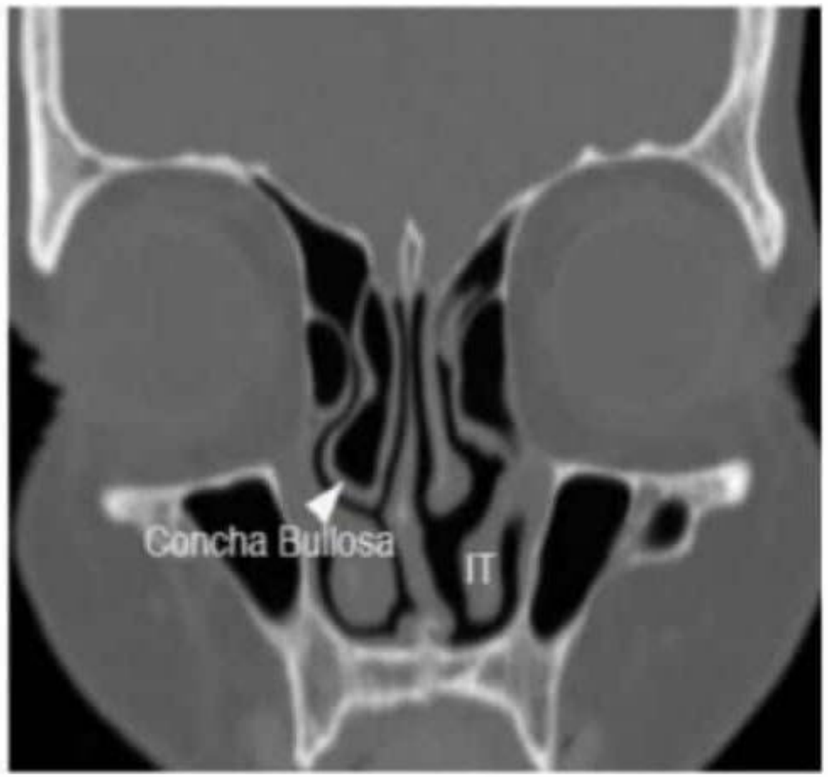


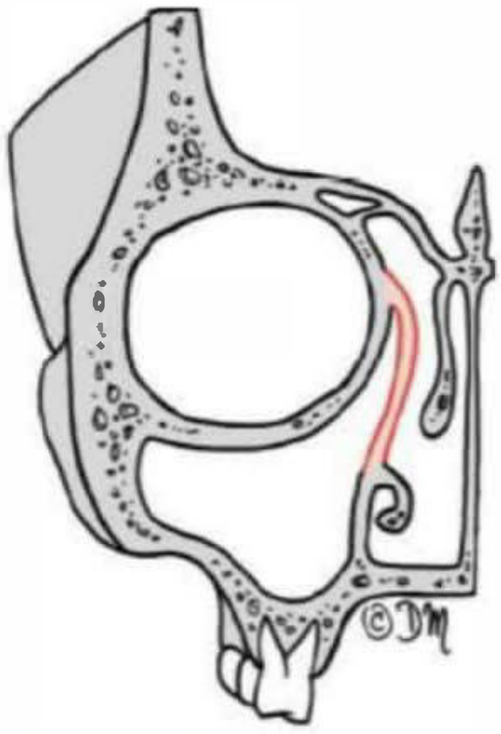
DAMS

Agger Nasi Cell
AE
PE
SpS

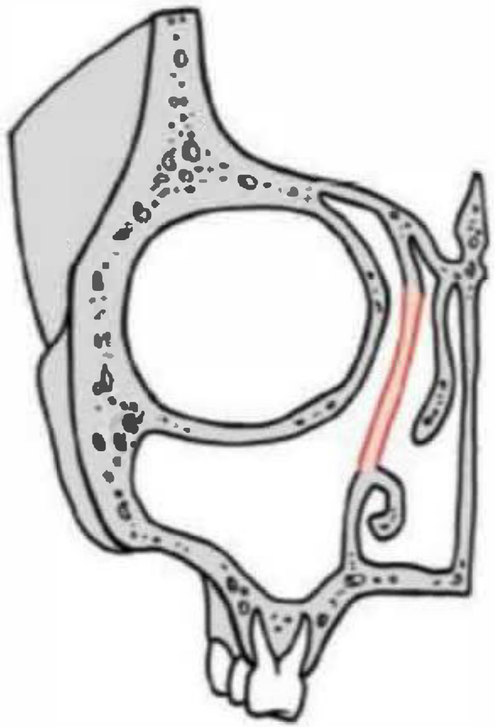
Concha Bullosa

- pneumatization of the bulbous portion of the middle turbinate
- An enlarged concha bullosa may impede drainage from the middle meatus

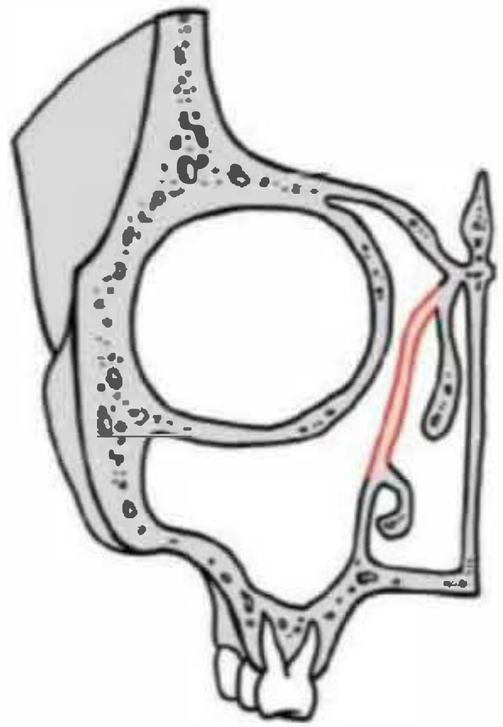




TYPE 1



TYPE II

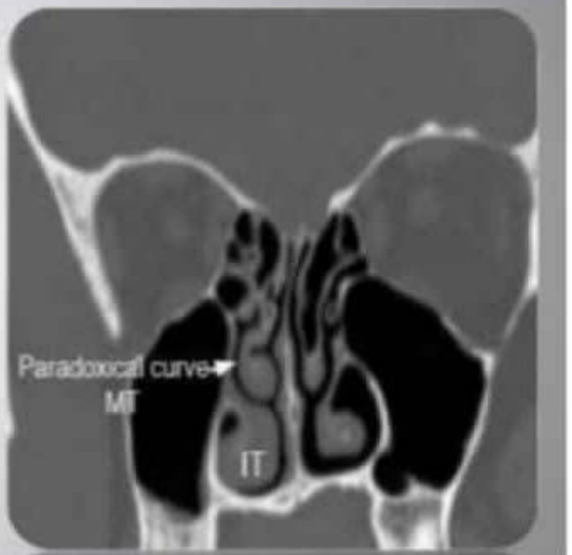


TYPE III

- Sometimes the free edge of the uncinate process adheres to the orbital floor, or inferior aspect of the lamina papyracea. This is referred to as an atelectatic uncinate process

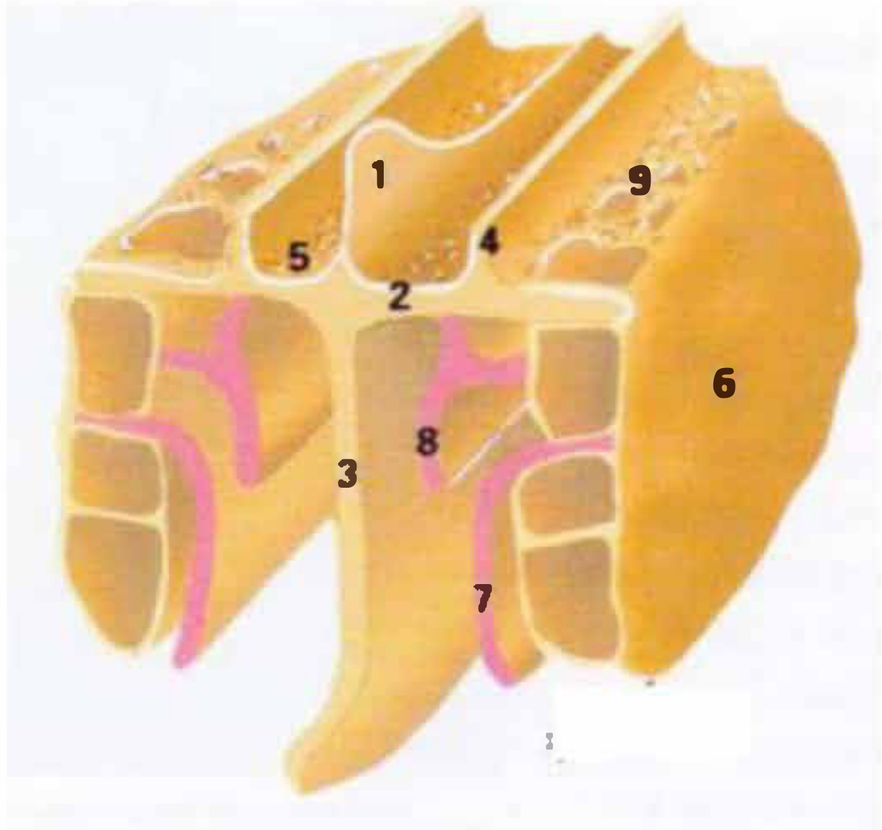
Variations of Middle turbinate

- *Paradoxic Curvature*
- Normally, the convexity of the middle turbinate bone is directed medially, toward the nasal septum.
- When paradoxically curved, the convexity of the bone is directed laterally toward the lateral sinus wall.
- The inferior edge of the middle turbinate may assume various shapes, which may narrow and/or obstruct the nasal cavity, infundibulum, and middle meatus.

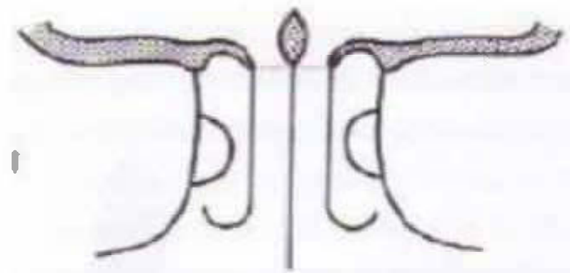


DAMS

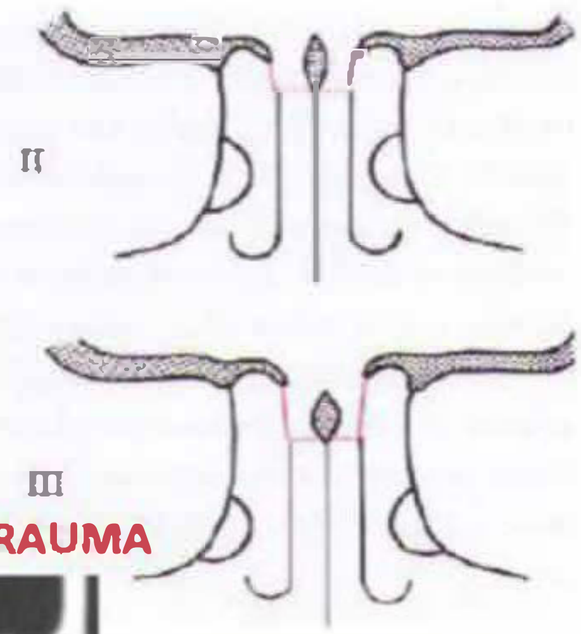
1) Crista Galli
2) Cribiform (lamina cribosa)
3) Nasal Septum
4) Cribiform (lateral lamella)
5) Olfactory Fossa
6) Lamina Papyracea
9) Fovea Ethmoidalis



Keros Classification

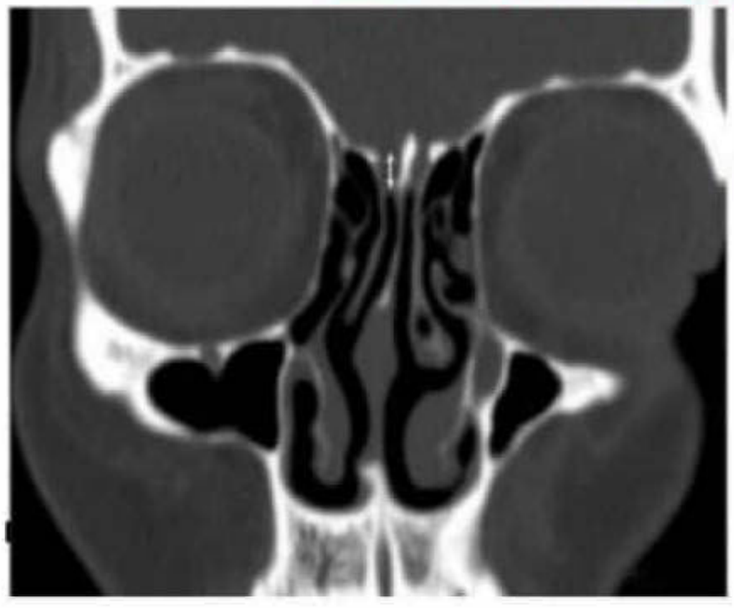


- Type 1: 1-3mm
- Type 2: 4- 7mm **MC**
- Type 3: 8-16mm



III

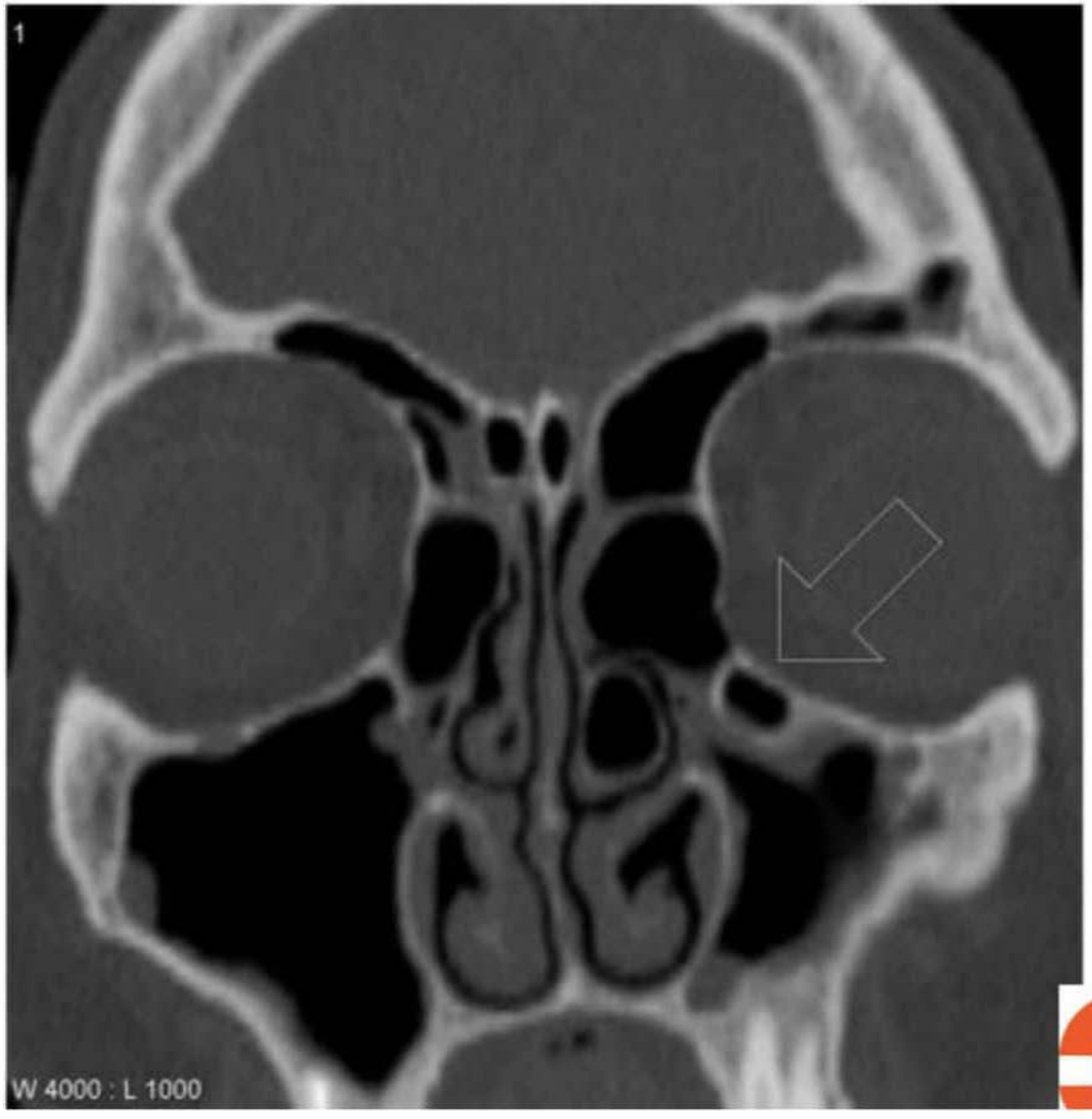
IATROGENIC TRAUMA



OLFACTORY FOSSA

HEIGHT OF LATERAL LAMINA OF CP

DAMS


1
W 4000 : L 1000

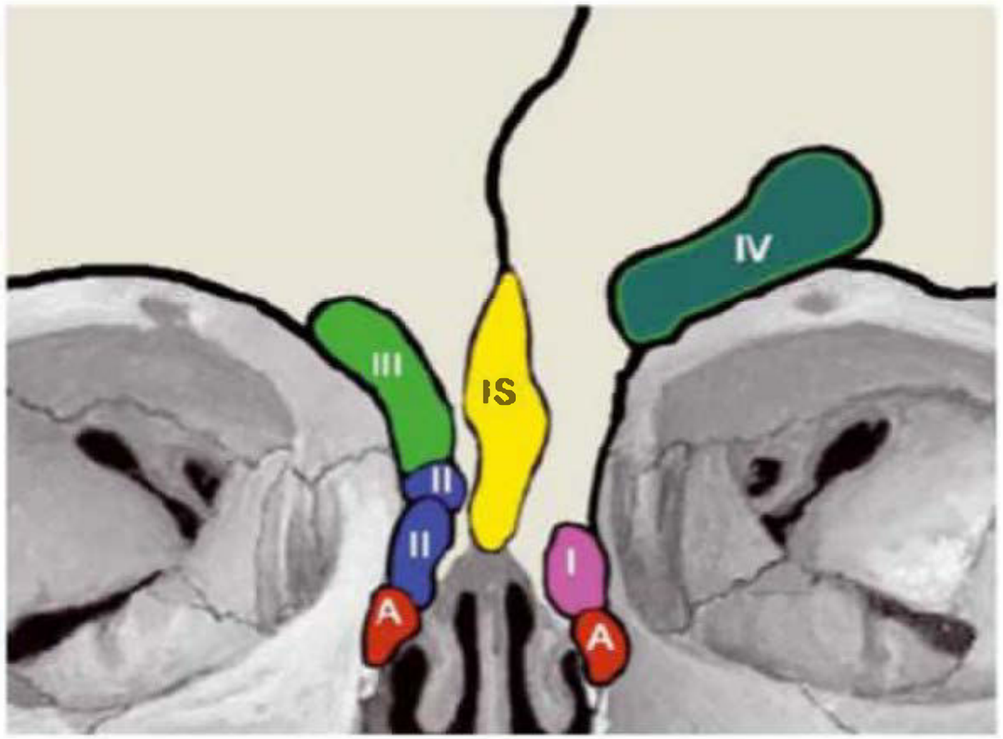
Sphenoethmoid cell

- Aka Onodi Cell
- Posterior ethmoid cell that extends over sphenoid sinus
- Close relationship to CN II

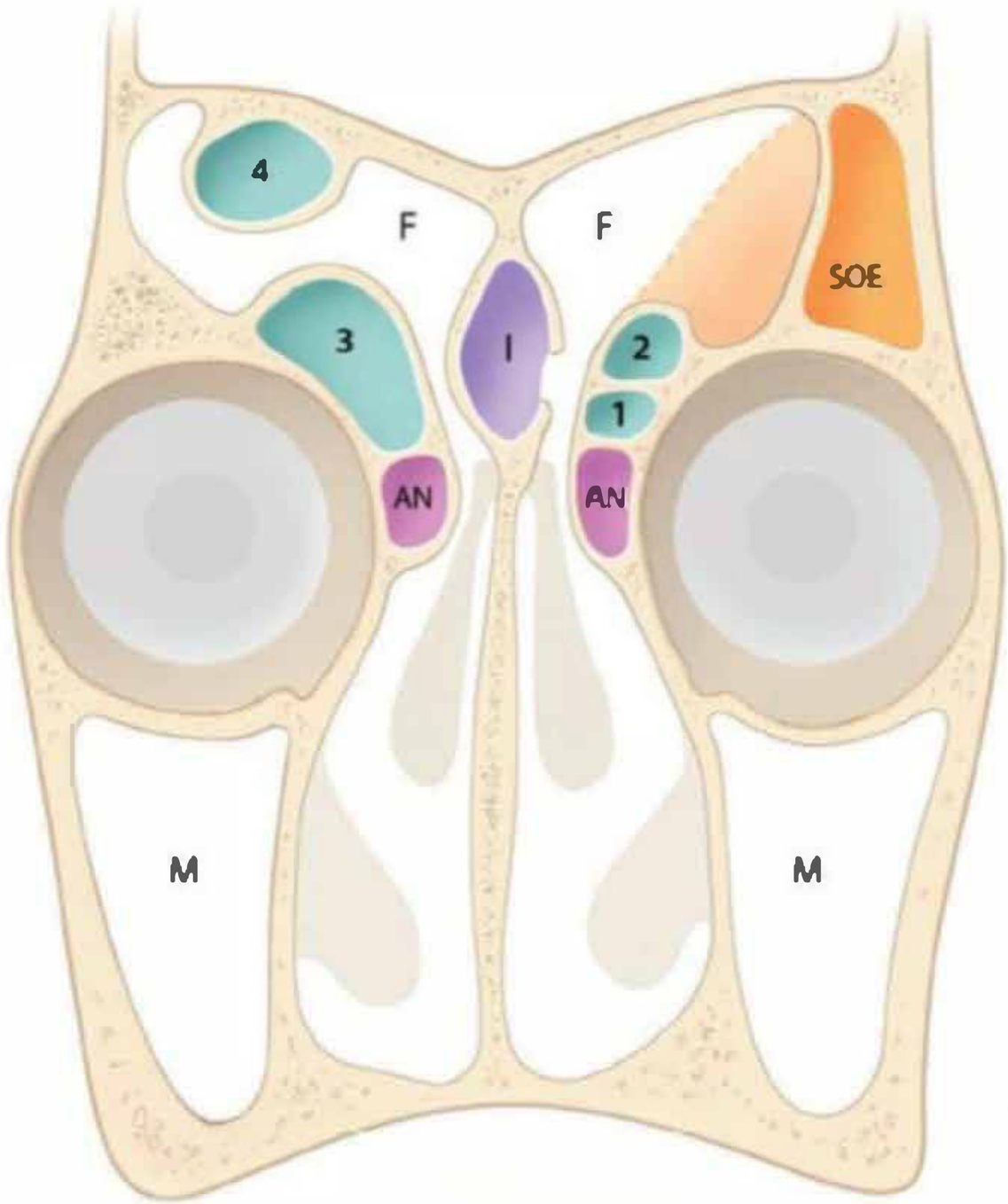
Sinuses	Status at Birth	First Radiological evidence	Reaches Adult size by
Maxillary sinus	Present at birth	4-5 months after birth	15 years
Ethmoid sinus	Present at birth	1 year	12 years
Sphenoid sinus	Not Present	4 years	15 years – adult age
Frontal Sinus	Not Present	6 years	Size increases until teens

Frontal Sinus Cells

- Type I: Single cell superior to the agger nasi but not extending into the frontal sinus
- Type II: Tier of two or more cells above the agger nasi but below the orbital roof
- Type III: Single cell extending from the agger nasi into the frontal sinus
- Type IV: Isolated cell within the frontal sinus



DAMS



recess. M, maxillary sinus; F, frontal sinus; AN, agger nasi cell; 1–4, type 1–4 frontal cells; SOE, supraorbital ethmoid cell; I, frontal intersinus septal cell.

Sphenoid Sinus

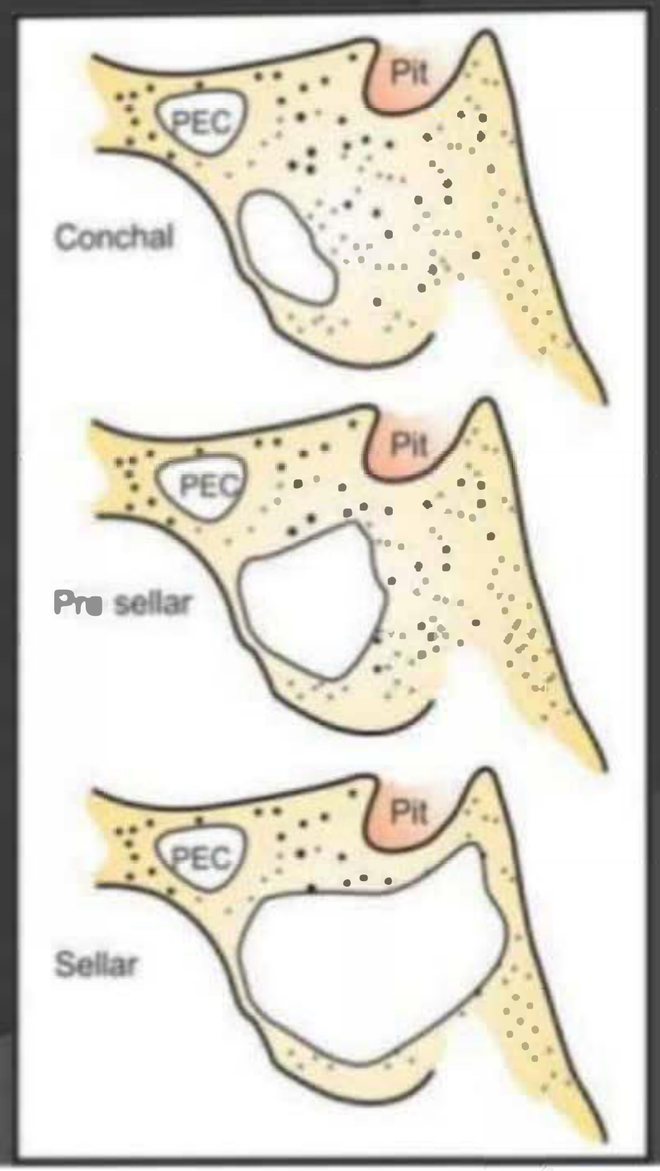
The sphenoid sinus shows great variations in pneumatization.

- It may be present as a small pit in a predominantly non-pneumatized sphenoid bone -- Conchal type.

- It may extend upto the anterior wall of the sella turcica —Presellar type.

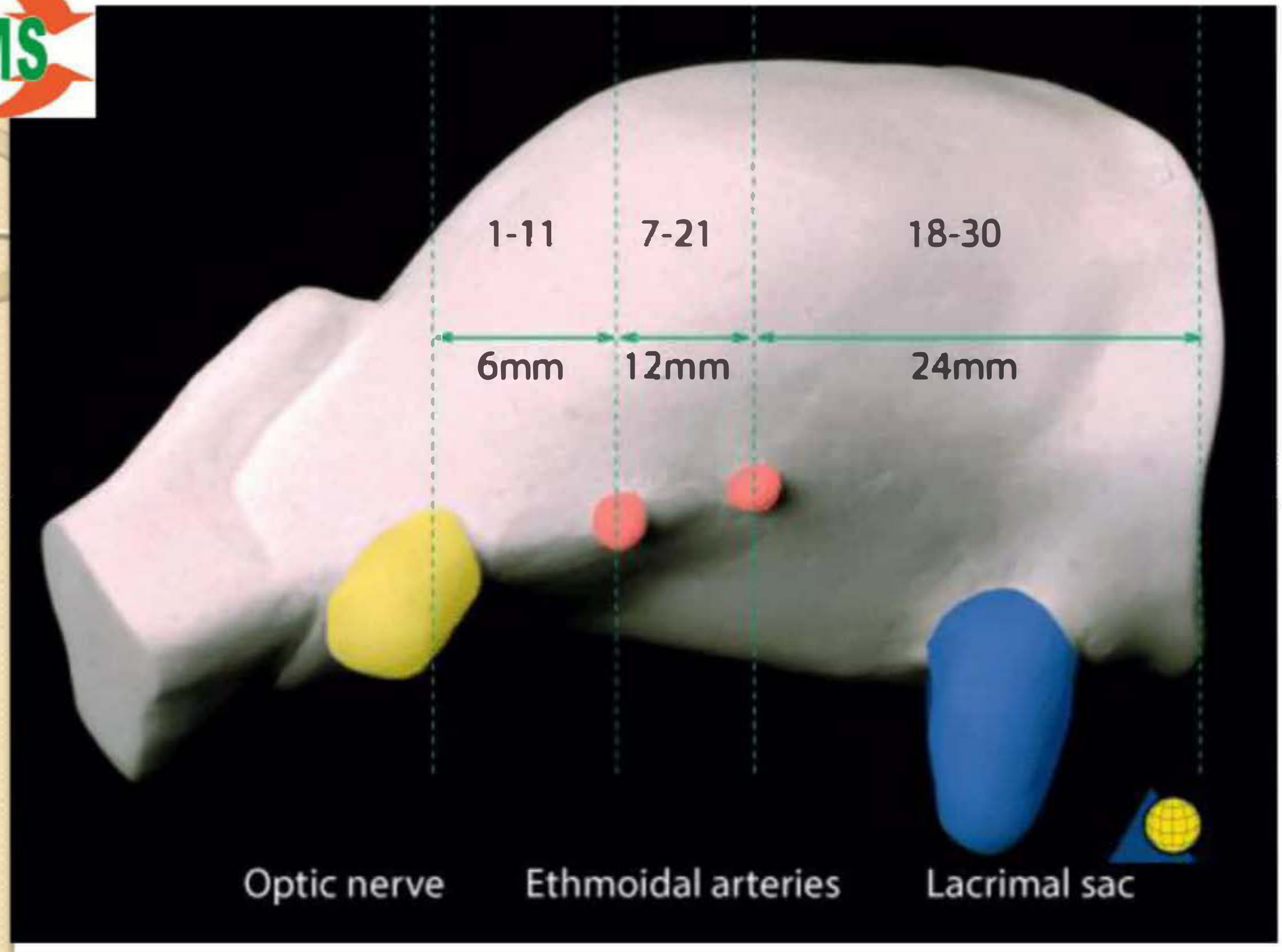
HAMBERGER CLASSIFICATION

- It may pneumatize the entire sphenoid body below and behind the sella turcica, so that the pituitary forms a distinct bulge in its posterosuperior wall—Sellar type.



A) Sellar Type 86%
B) Pre Sellar 11%
C) Conchal 3%

DAMS


1-11
7-21
18-30
6mm
12mm
24mm
Optic nerve
Ethmoidal arteries
Lacrimal sac

Most common location for the maxillary ostium within the infundibulum

Inferior third (65%)

What arterial structure typically runs through the roof of the ethmoid bulla

Anterior ethmoid artery

When removing the intersinus septum within a sphenoid sinus, attachment of this septation to what critical structure must be considered:

Internal carotid artery

The carotid artery is reported to be dehiscent in the sphenoid sinus in what percent of patients

15%

STANKIEWICK'S SIGN - indicate orbital injury during FESS. fat protrudes into nasal cavity on compression of eye ball from ouside

COTTLE MANEUVER

(a) (b)

FIGURE 1: Demonstration of the modified Cottle nasal valve maneuvers ((a) = external valve; (b) = internal valve), with the curette j externally only to demonstrate the area to be supported intranasally.

Normal Sinuses

Acute Maxillary Sinusitis

What pathogens are most commonly involved in ABRS?

- Streptococcus pneumoniae (30%)
- Haemophilus influenzae (20 to 30%)
- Moraxella catarrhalis (10 to 20%)

Two most common pathogens associated with viral rhinosinusitis

Rhinovirus and Influenza virus

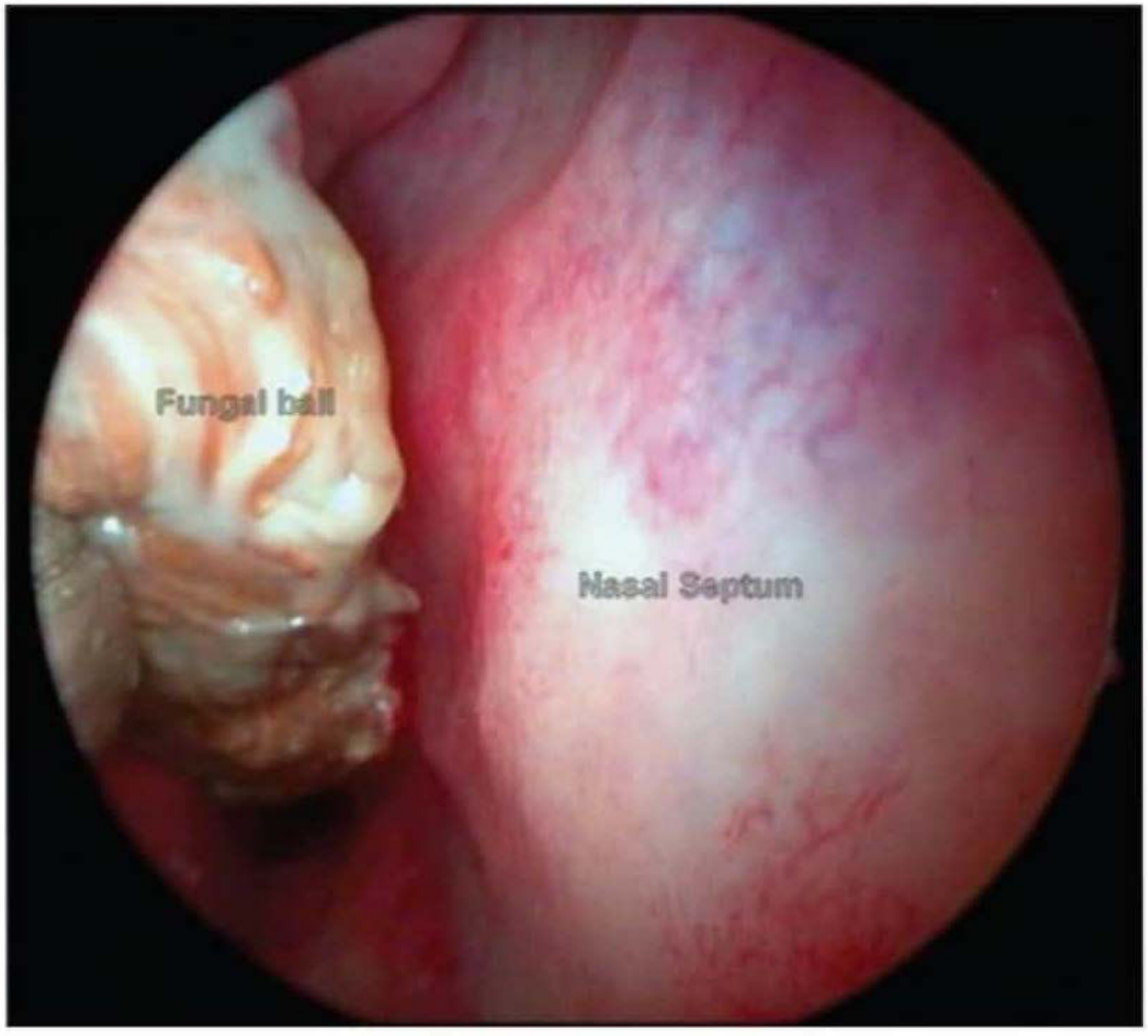
Nonallergic rhinitis of eosinophilia syndrome (NARES)

DAMS

- X ray – Water's view & caldwell view

- Ct – gold standard. Coronal & axial sections

- MRI is predominantly used for pre and post operative management of naso sinus malignancy
- The chief disadvantage of MRI is its inability to show the bony


Fungal ball
Nasal Septum

ALLERGIC FUNGAL SINUSITIS

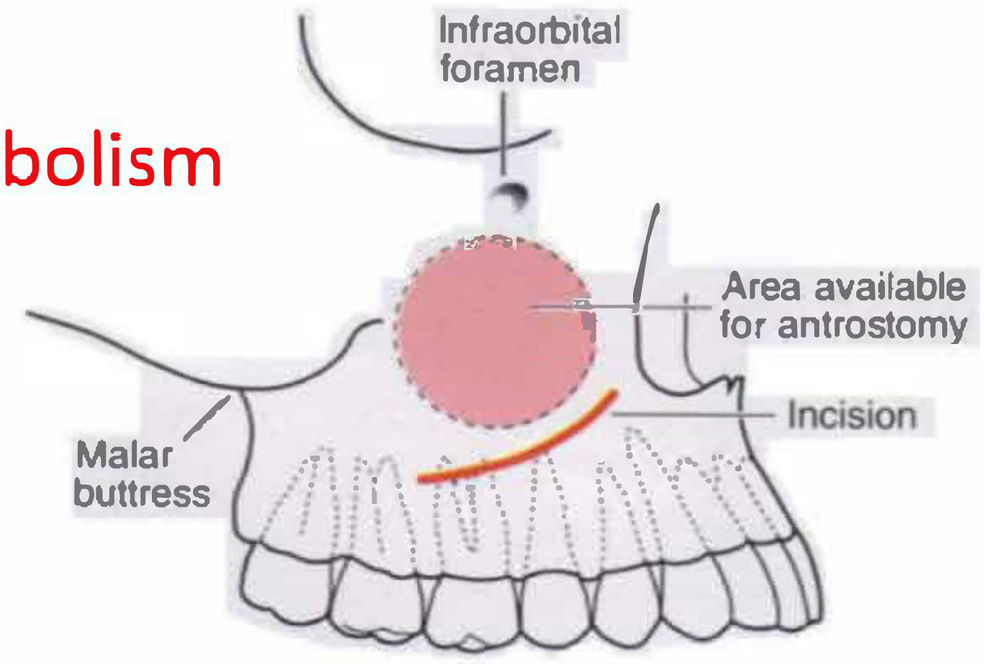
Bent and Kuhn Diagnostic Criteria

Major	Minor
Type I hypersensitivity	Asthma
Nasal polyposis	Unilateral disease
Characteristic CT findings	Bone erosion
Eosinophilic mucin without invasion	Fungal cultures
Positive fungal stain	Charcot-Leyden crystals
	Serum eosinophilia

CT - Mucosal thickening with hyperdense areas
- Bone erosion
- Expansion of sinus but no fungal invasion into mucosa.

DAMS

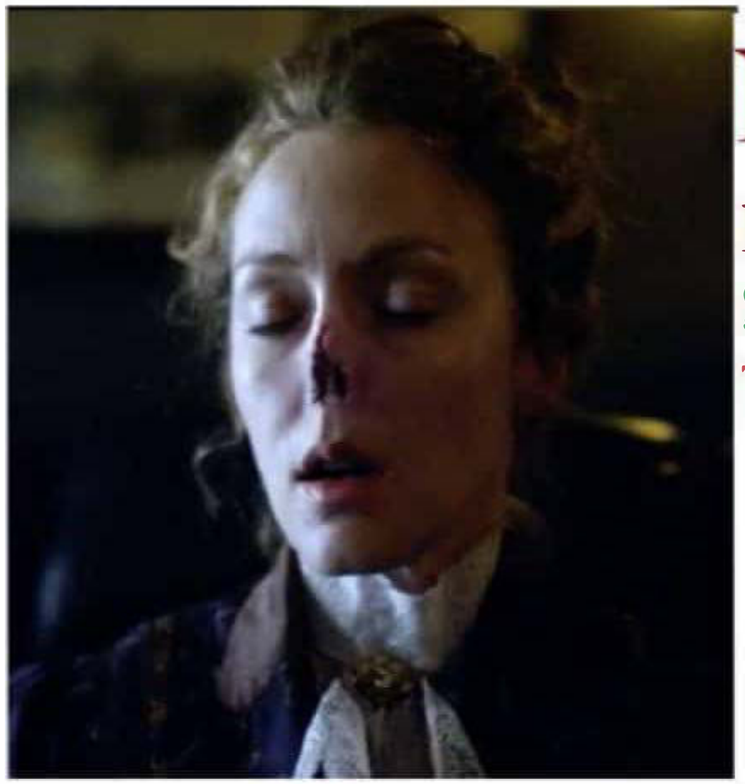
Air embolism



The *Caldwell-Luc* Operation

MUCOCELE

DAMS



NASAL SYPHILIS

Primary:Chancre of Vestibule..Rare

Secondary:Crust/Fissure ..Rare

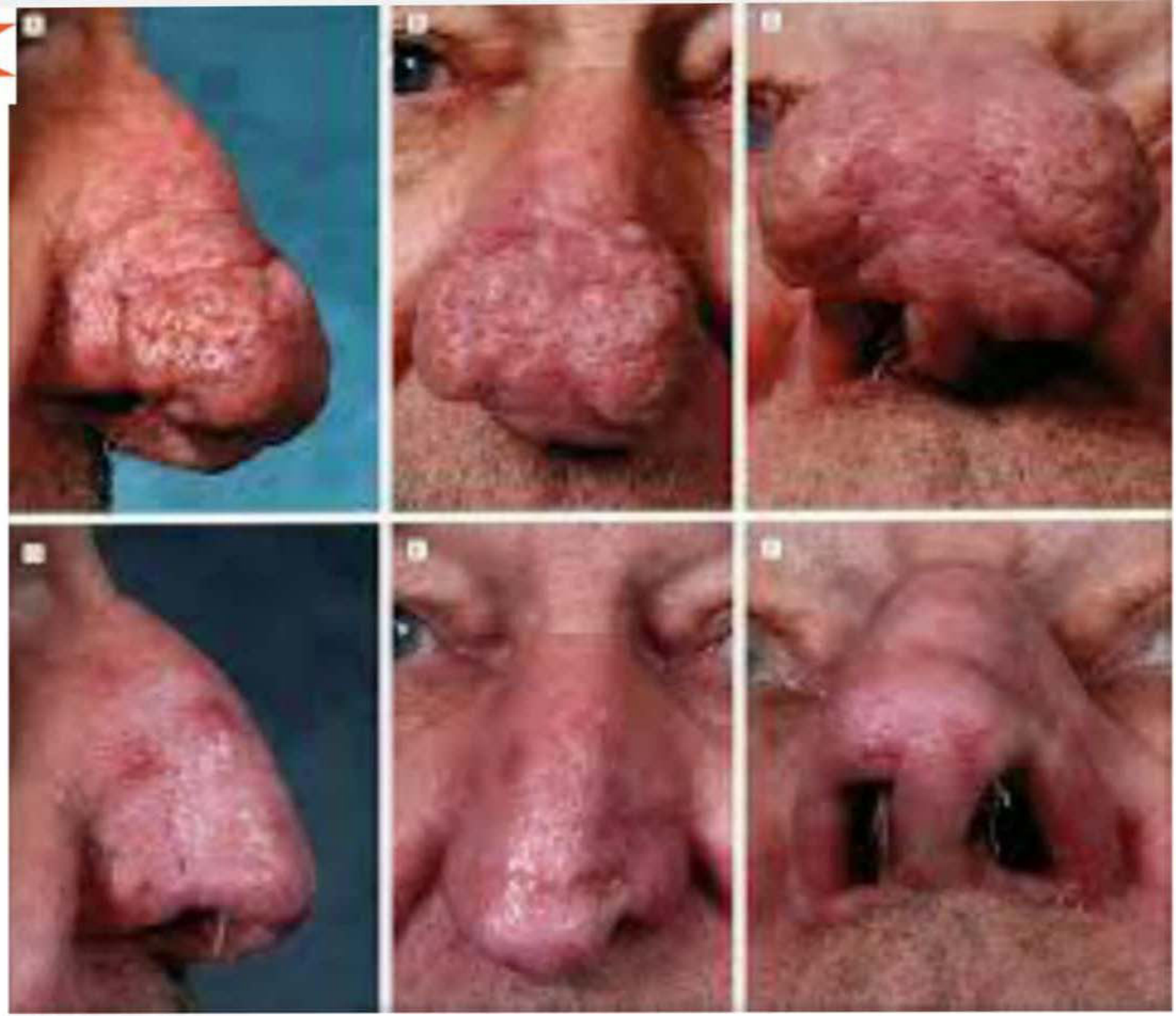
Tertiary : Most commom

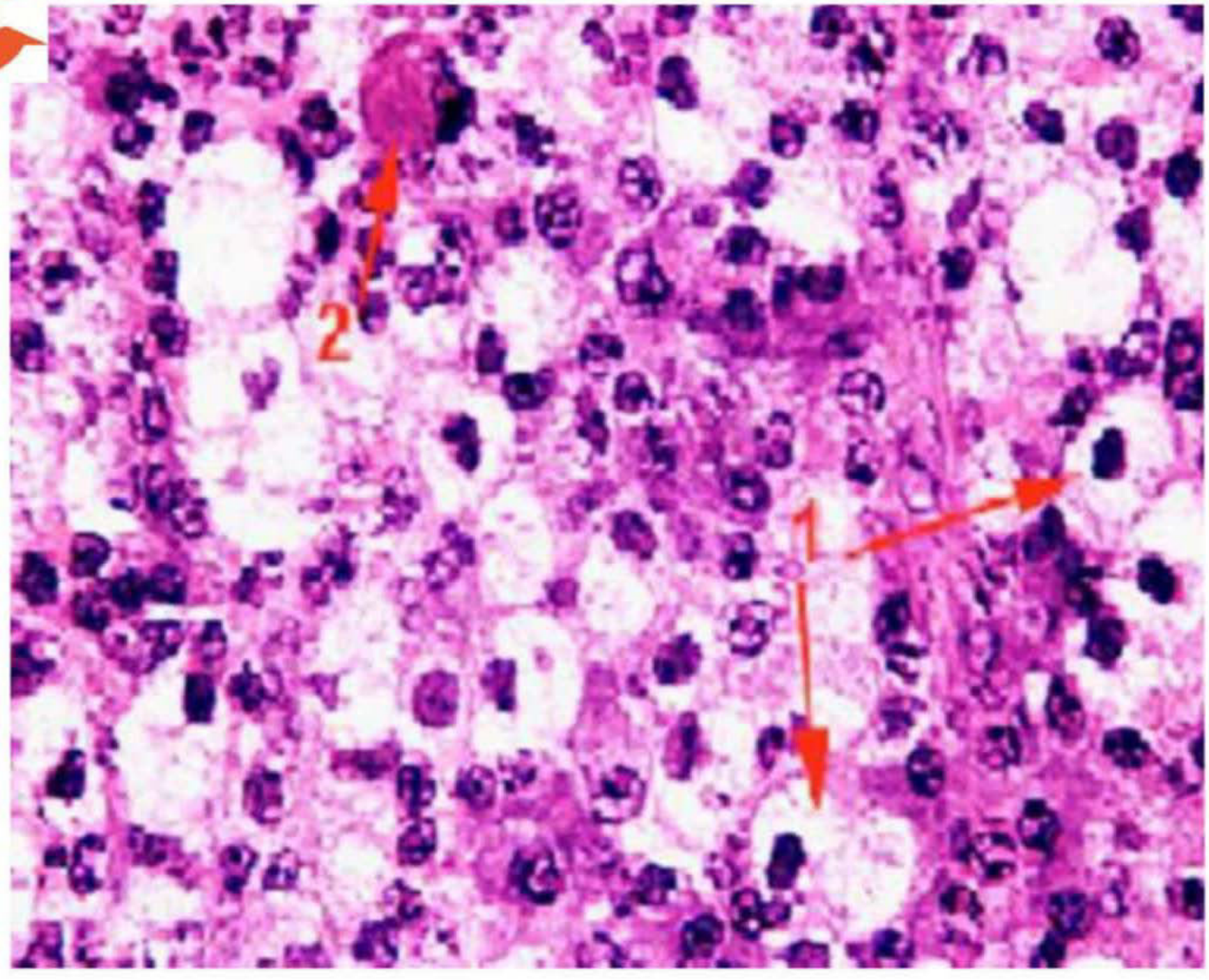
Gumma-----Saddle Nose

ANTERIOR PART OF NOSE

TB LUPUS LEPROSY

DAMS




2
1

Nasal involvement staging

1. Catarrhal Stage: foul smelling purulent nasal discharge (carpenter's glue)
2. Adhesions & stenosis coarse & distorted external nose (Tapir nose)
3. Non-ulcerative, painless nodules which widen lower nose (Hebra nose)

Nasopharyngoscopy

- Obliteration of nasopharynx due to adhesions between deformed V-shaped soft palate & posterior pharyngeal wall (Gothic sign)

RHINOSPORODIOSIS

MAGGOTS/NASAL MYIASIS

DAMS

ONE LINERS

CHRYSOMIA

200 EGGS WITHIN 24 HRS

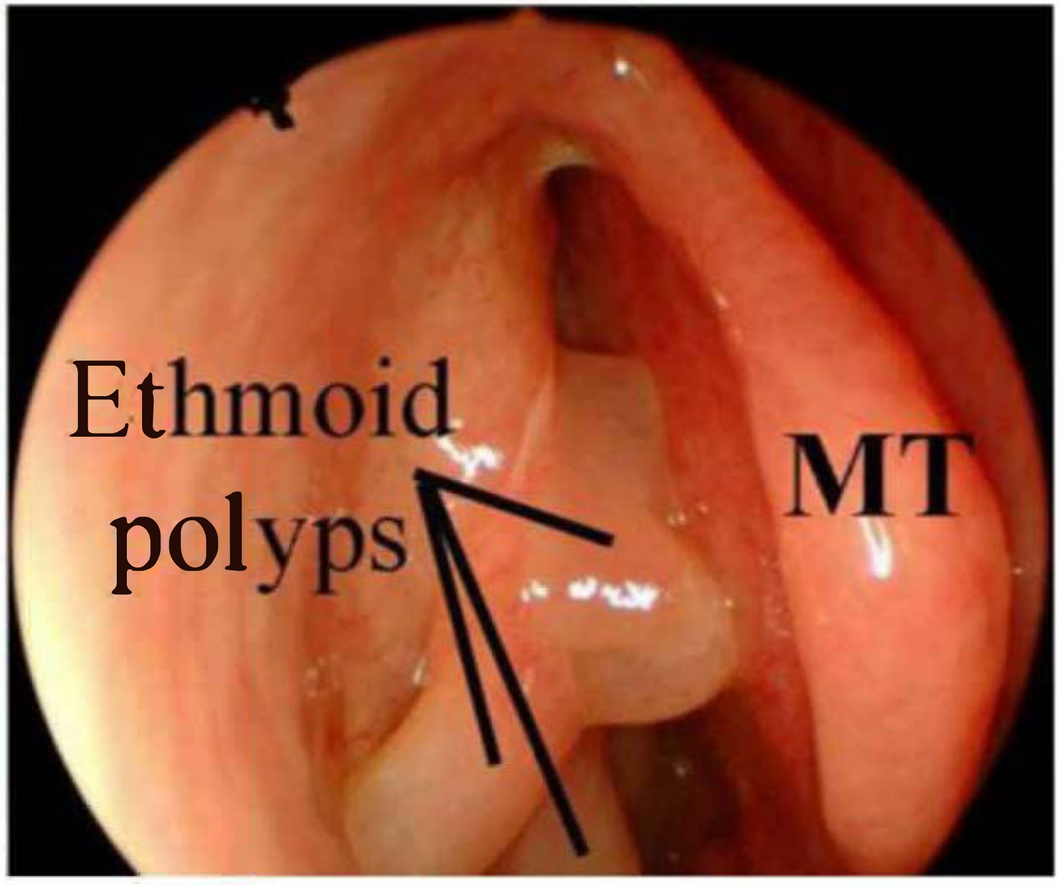
ON ¾ DAY THEY CRAWL OUT

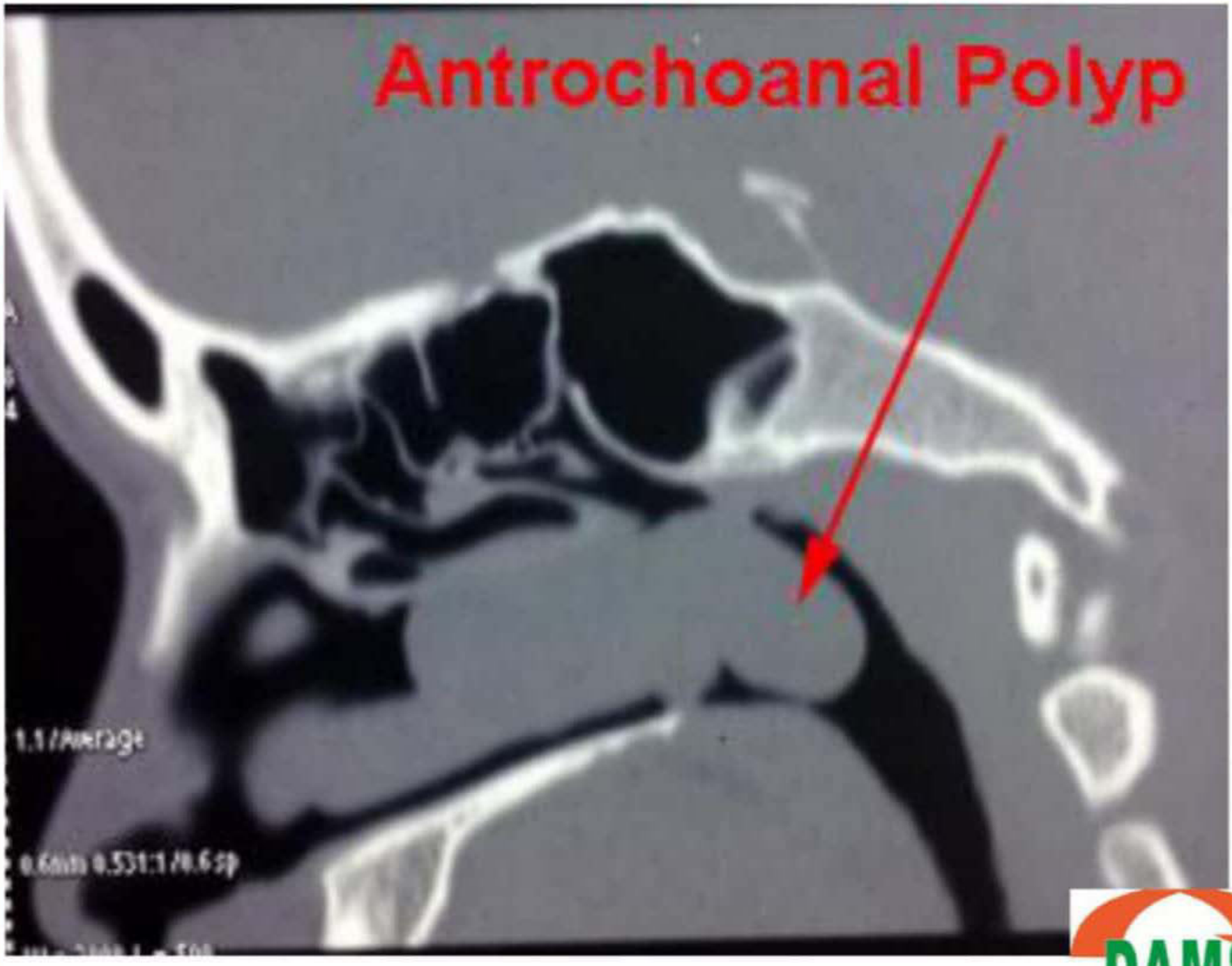
CHLOROFORM WATER

Ethmoidal Polyps

Anterior Rhinoscopy

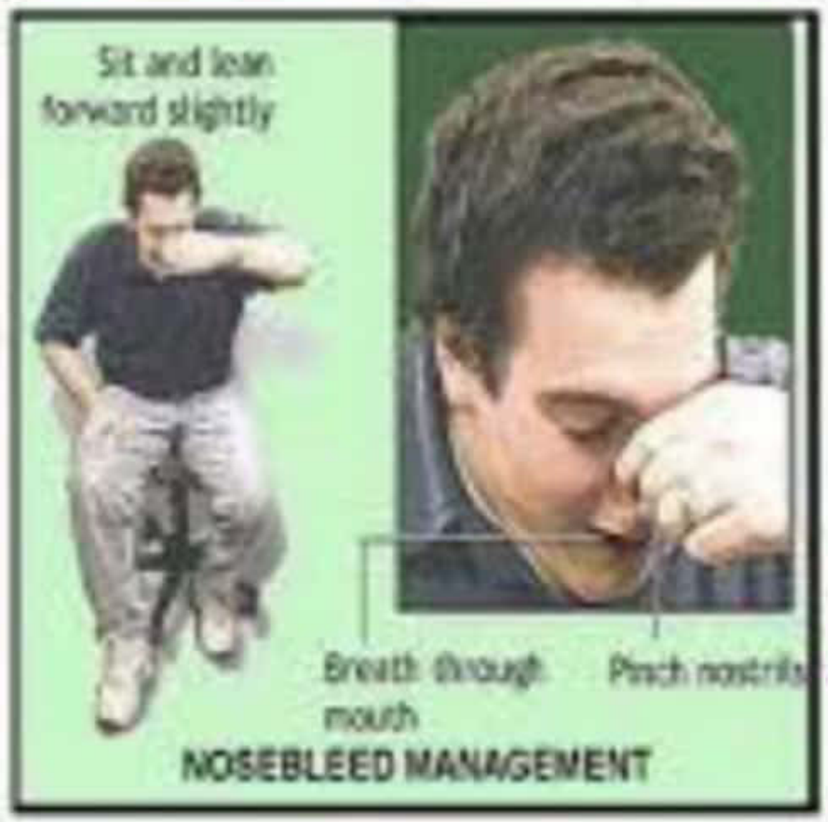
Nasal Endoscopy



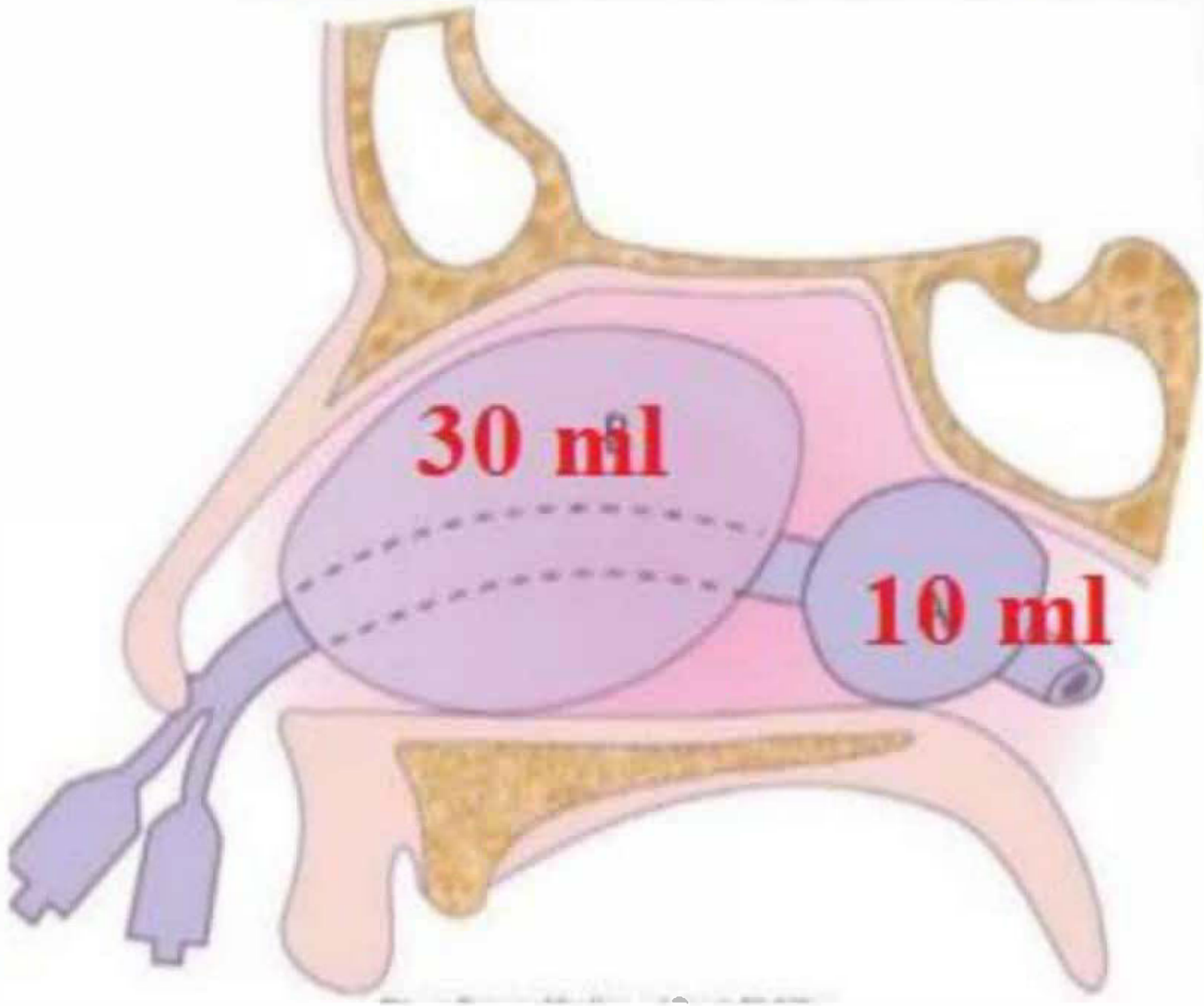

Antrochoanal Polyp

TREATMENT OF EPISTAXIS

- First aid
  - . ABC
  - . Trotter's method- Make patient sit up, pinch the nose for 5-10 minutes. Head bent forward. Open mouth and breathe
  - . Ice packs



EPISTAXIS BALLOON



Tespal: (Trans nasal endoscopic sphenopalatine artery ligation)

Location of the sphenopalatine artery for endoscopic ligation.

Posterior to the inferior attachment of the middle turbinate, submucosal on the lateral nasal sidewall

External ligation of the anterior ethmoid artery

Lynch incision

Approximately 24 mm posterior to the anterior lacrimal crest

DAMS

Osler-Weber-Rendu disease

Recurrent epistaxis

Bevacizumab targets a cancer cell protein called vascular endothelial growth factor (VEGF)

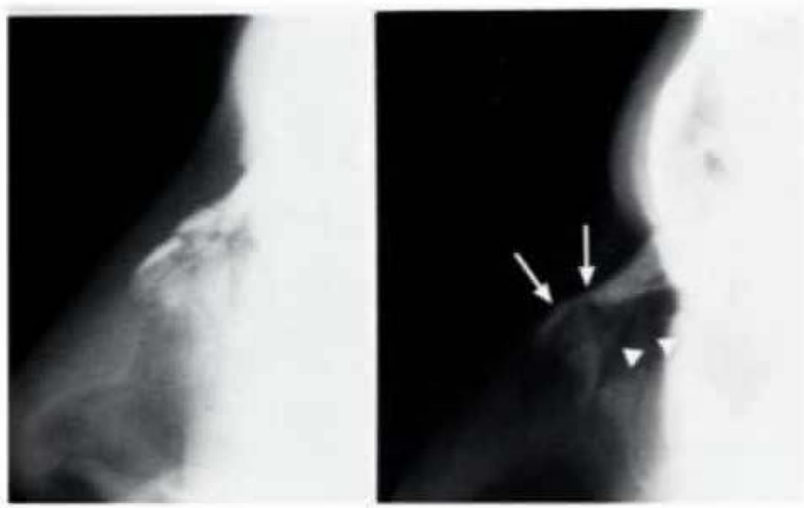
Surgical management:

- Potassium-titanyl-phosphate (KTP) laser ablation of lesions
- Injection of bevacizumab
- Septodermoplasty
- Young's procedure

Saunder's septodermoplasty.

Denuding of nasal mucosa affected by telangiectasias and coverage of denuded area with a split-thickness skin grafts

DAMS



Walsham Forceps

Ash forceps

I

II

III

- GUERIN sign: ecchymosis of palate , bilateral greater palatine foramen.

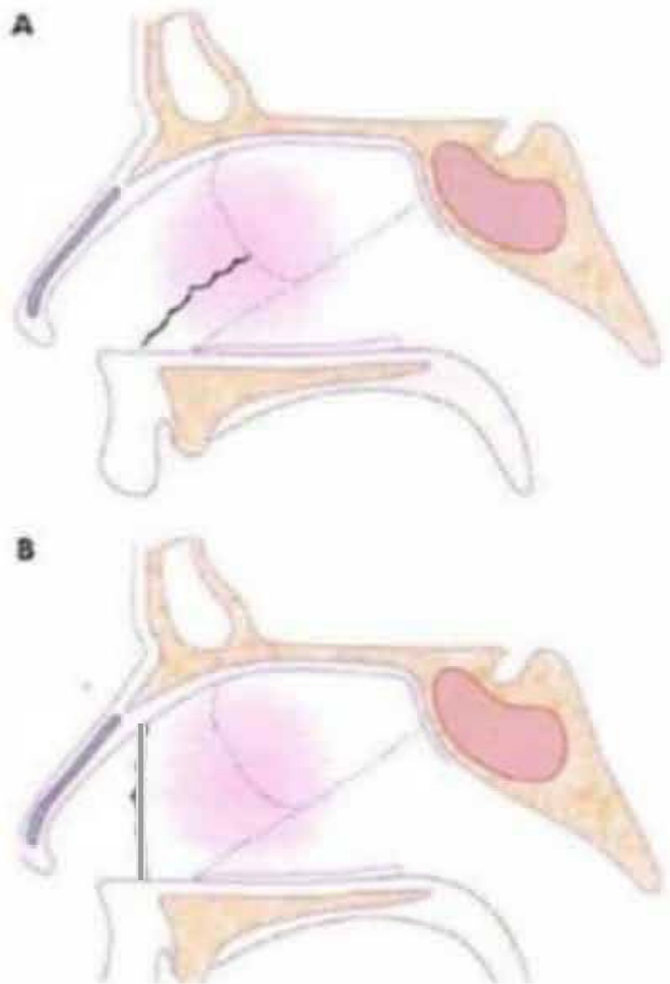
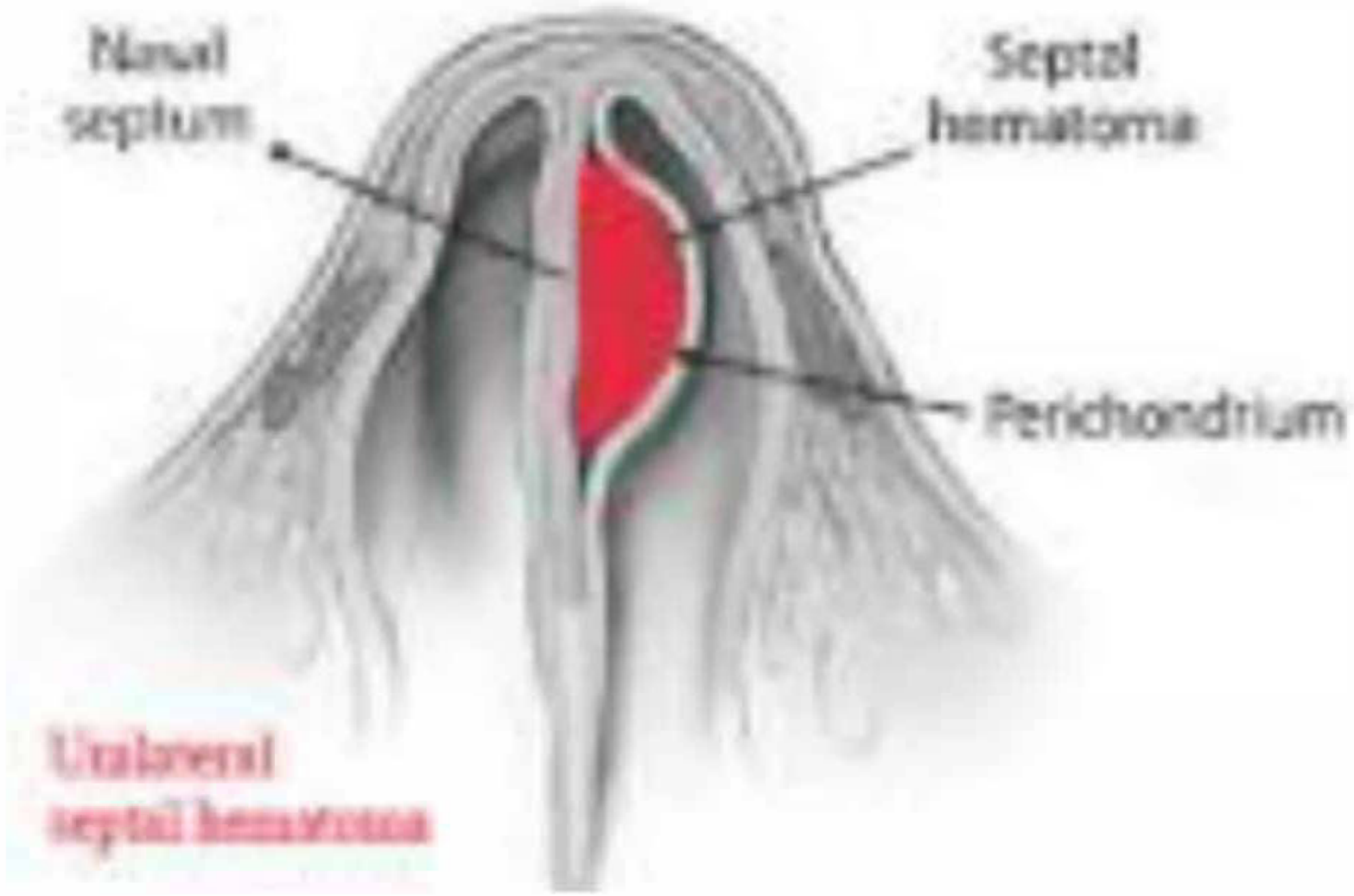


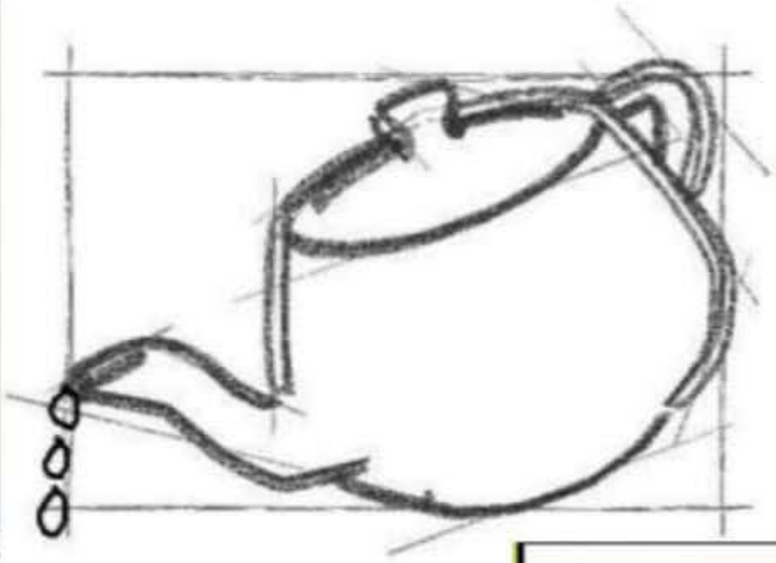
Fig. 26.2 Septal fracture showing: (A) Jansweg type. B) Chevallet type.

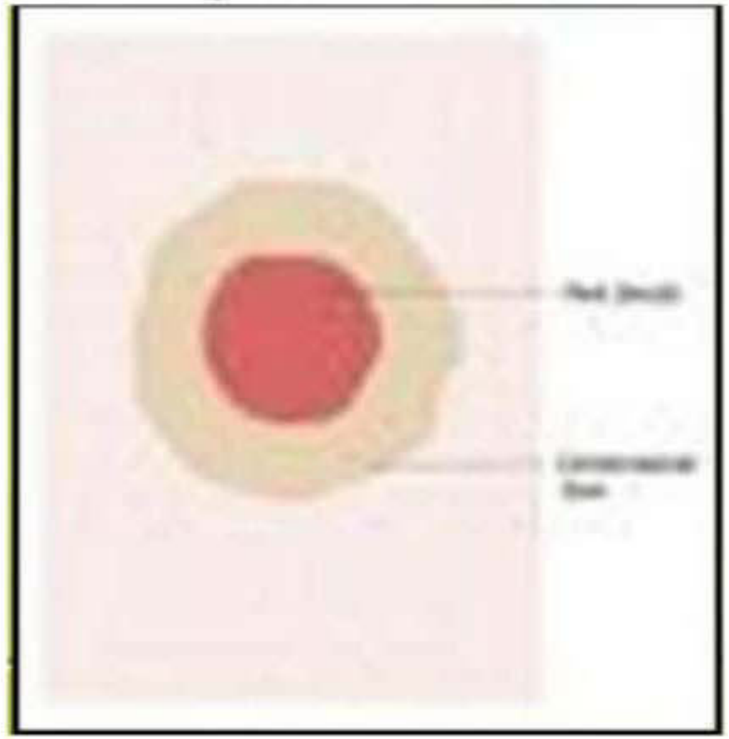
Zygoma # TRIPOD

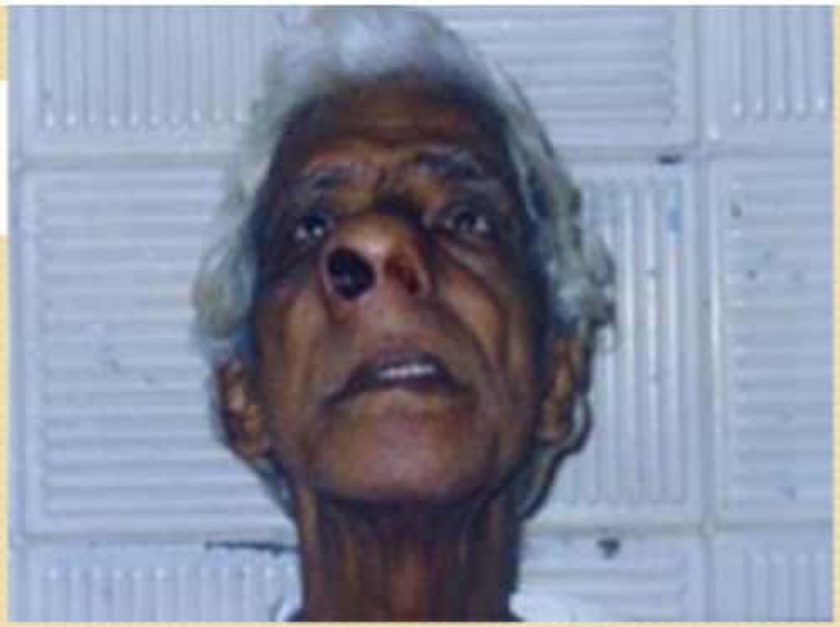
DAMS


Nasal septum
Septal hematoma
Perichondrium
Unilateral septal hematoma

CSF RHINORRHOEA



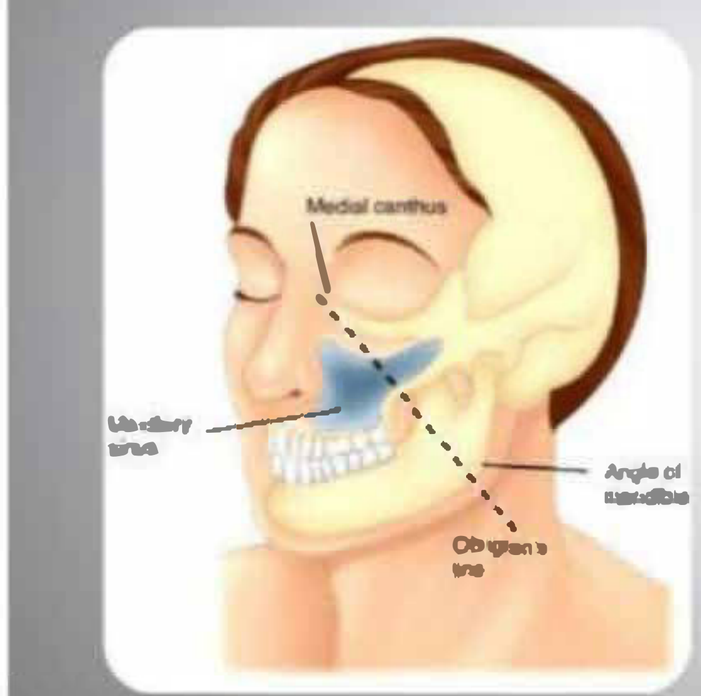


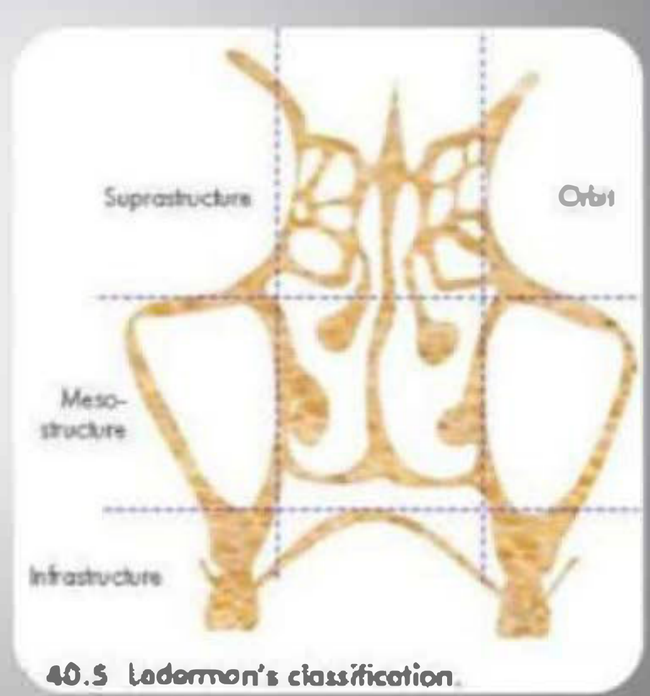


Clinical photogram showing mass in right nasal cavity on septum

PAPILLOMA/MM/CAPILLARY H

Maxillary Sinus - (Antrum of Highmore)





40.5 Ladermon's classification.

HEMOSTASIS (UPTO 1.5 MM)

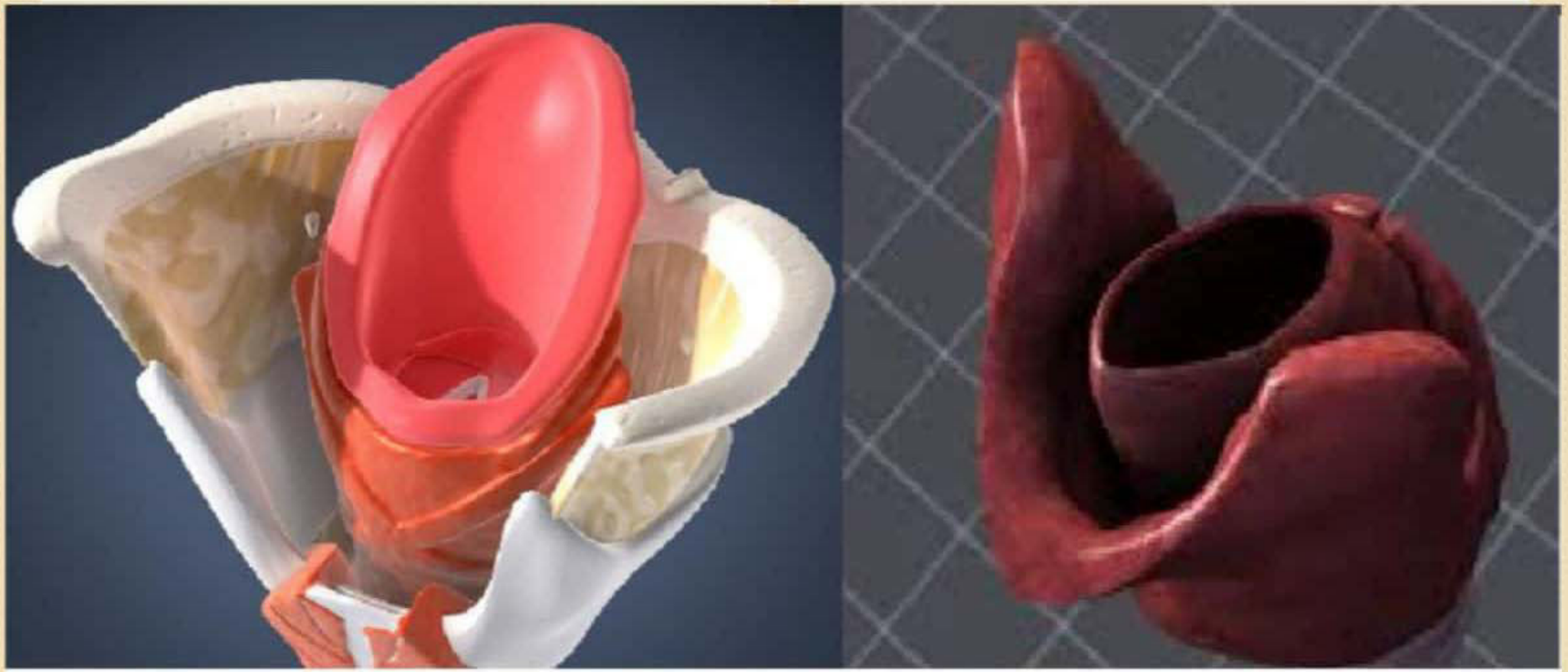
|

KOI TUMHE TAB TAK
NAHIN HARA SAKTA
JAB TAK TUM
KHUD SE
NA HAAR JAO

DAMS

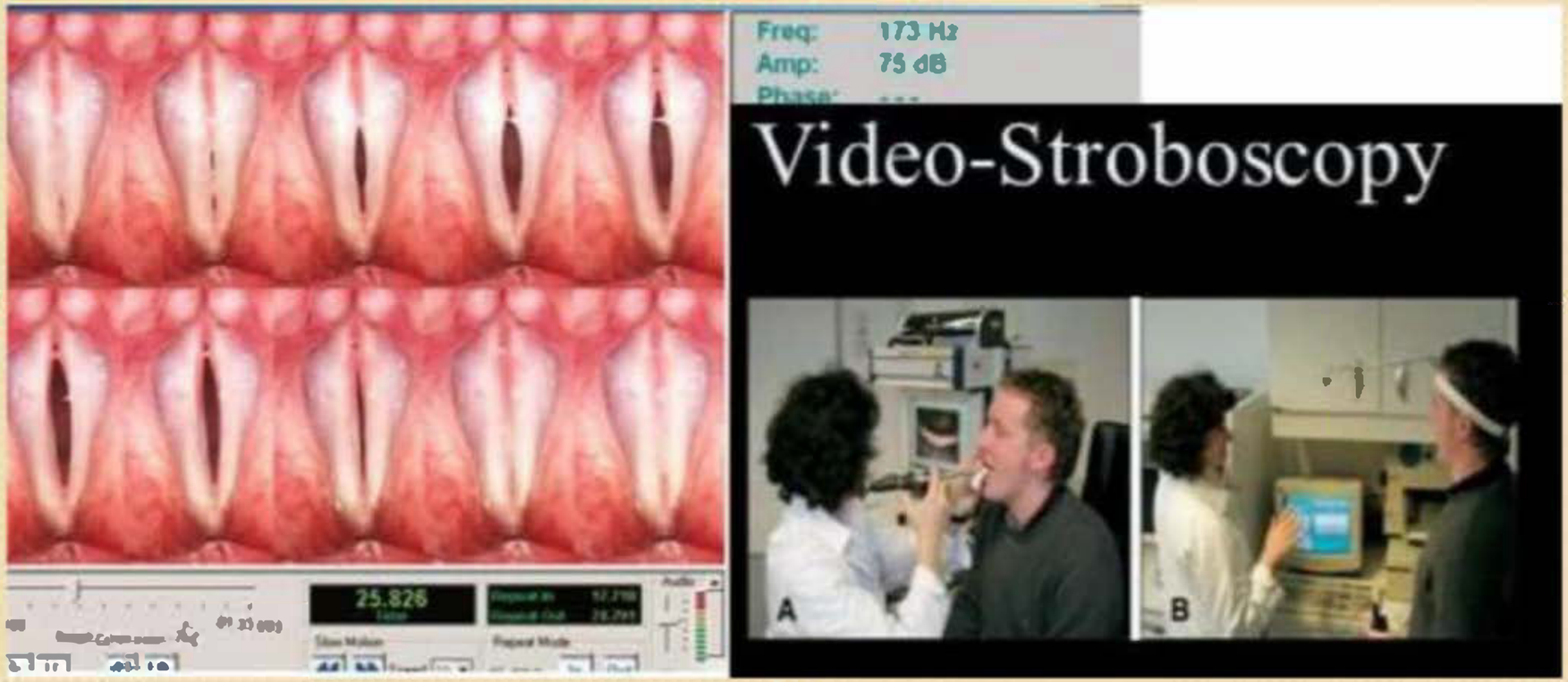
HEAD MIRROR

- CONCAVE MIRROR
- FOCAL LENGTH
  25 CMS
- DIAMETER 89 MM
- HOLE D 19 MM



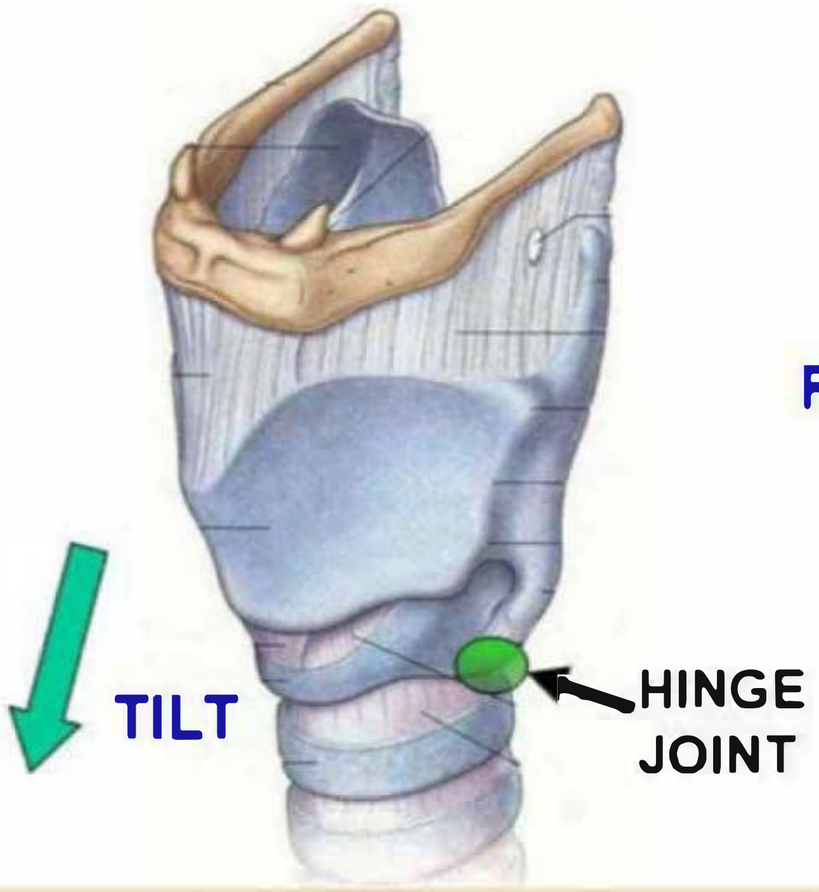
LARYNGEAL SPACES

DAMS

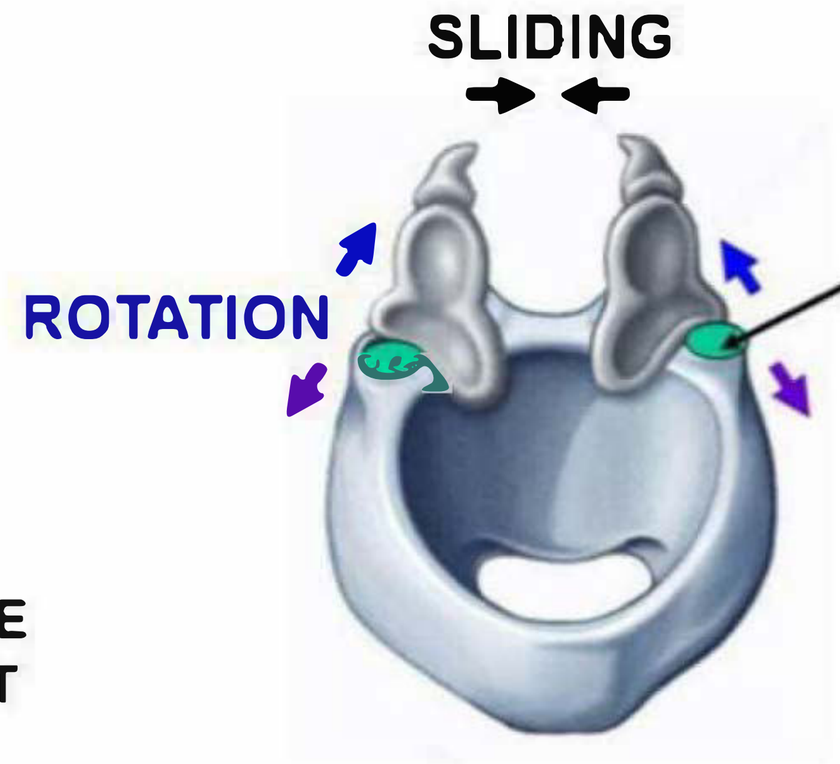

Freq: 173 Hz
Amp: 75 dB
Video-Stroboscopy
25.826
A
B

LARYNX REVIEW: SYNOVIAL JOINTS

THYROID and CRICOID



ARYTENOID and CRICOID



DAMS

LARYNGOMALACIA
GLOTTIC WEBS
STENOSIS

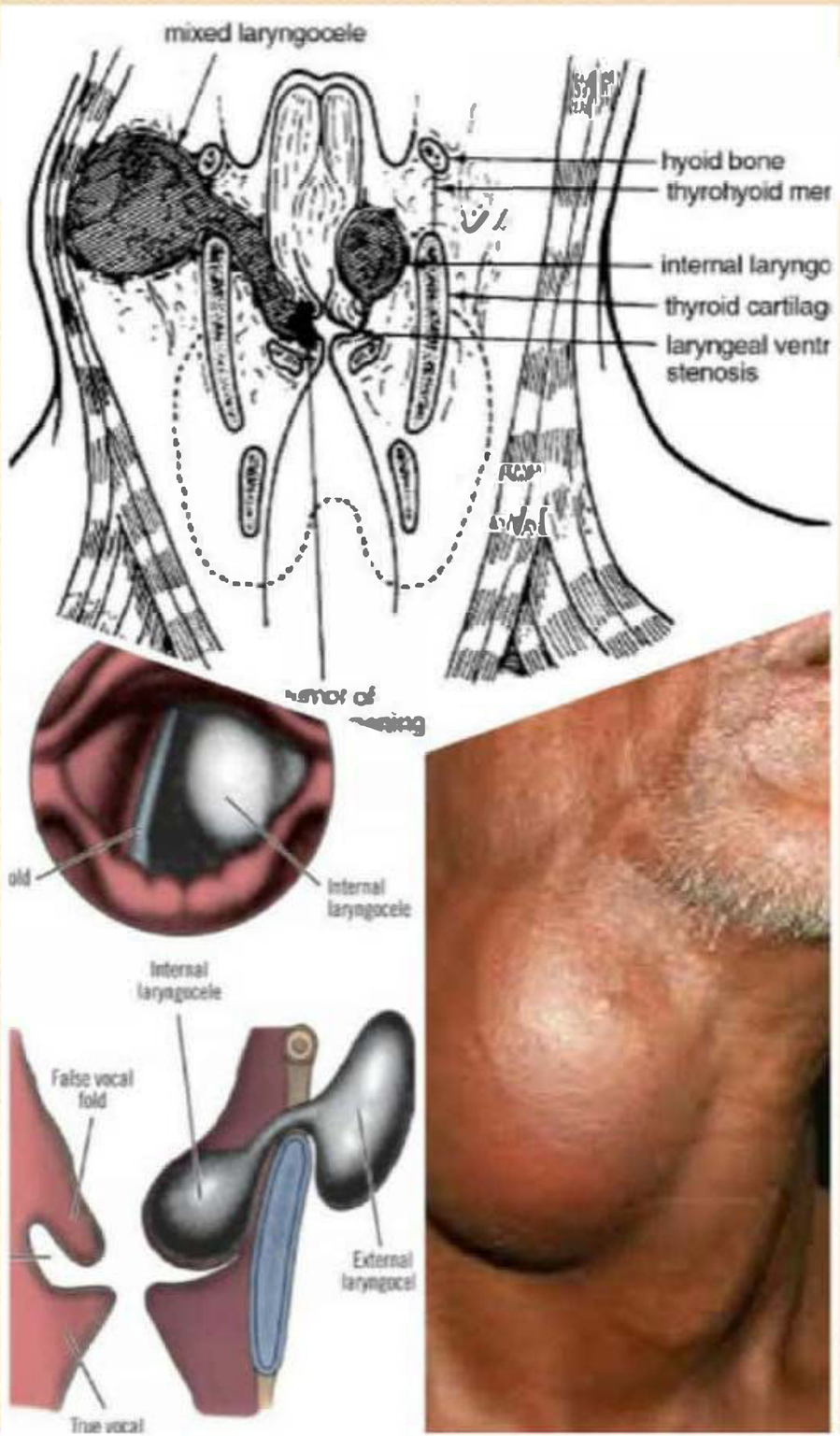
Grade 1
Stenosis 0-50%

Grade 2
Stenosis 51-70%

Grade 3
Stenosis 71-99%

Grade 4
Stenosis 100%

Cotton-Myer classification


mixed laryngocele
hyoid bone
thyrohyoid mer
internal laryngo
thyroid cartilag
laryngeal ventr
stenosis
old
Internal laryngocele
Internal laryngocele
False vocal fold
External laryngocel
True vocal

Which of the following is **incorrect statement** regarding **Laryngeal histoplasmosis**:

a.Commonly seen in eastern india.

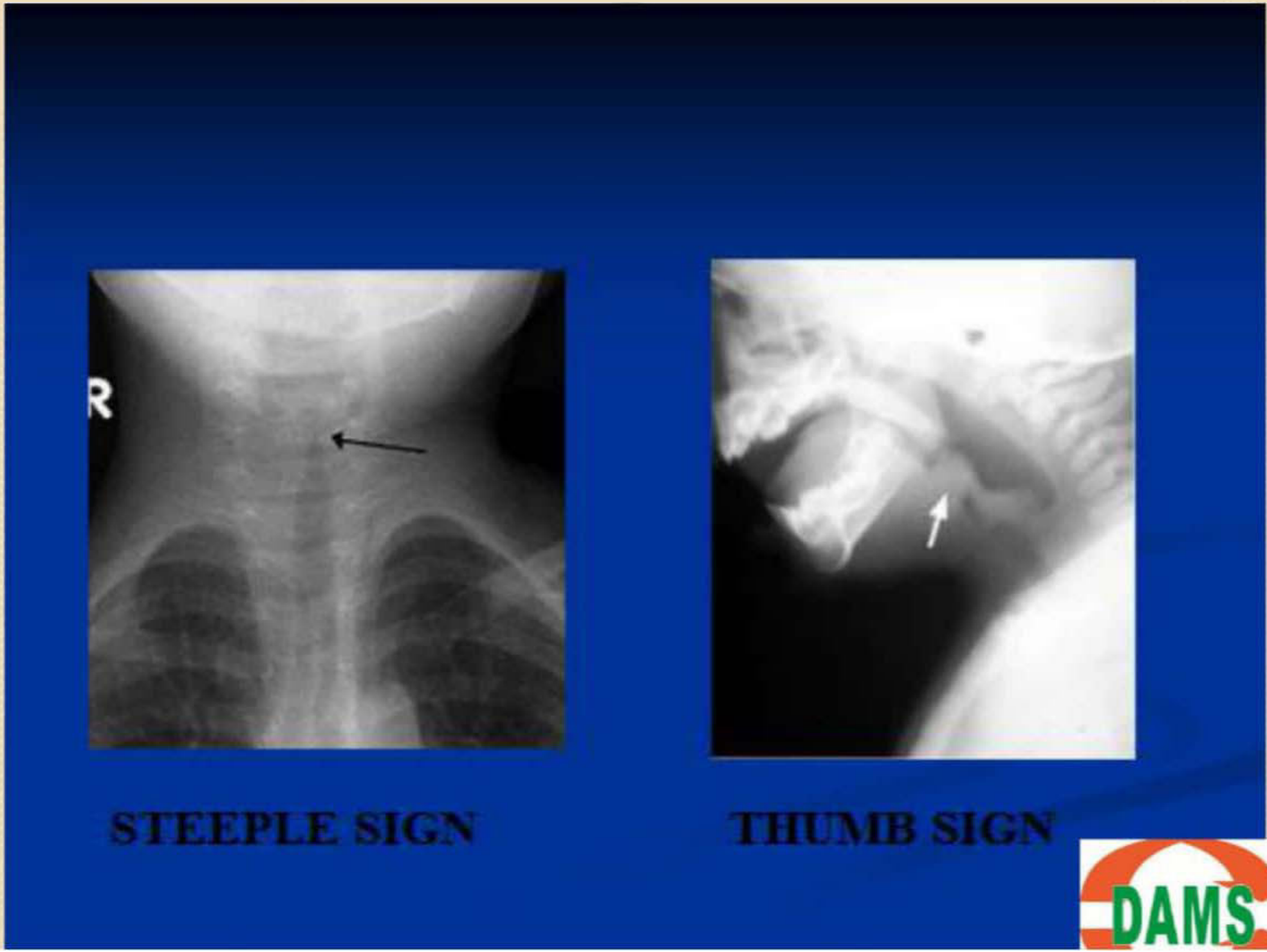
b.Laryngeal involvement occurs in chronic disseminated form of histoplasmosis.

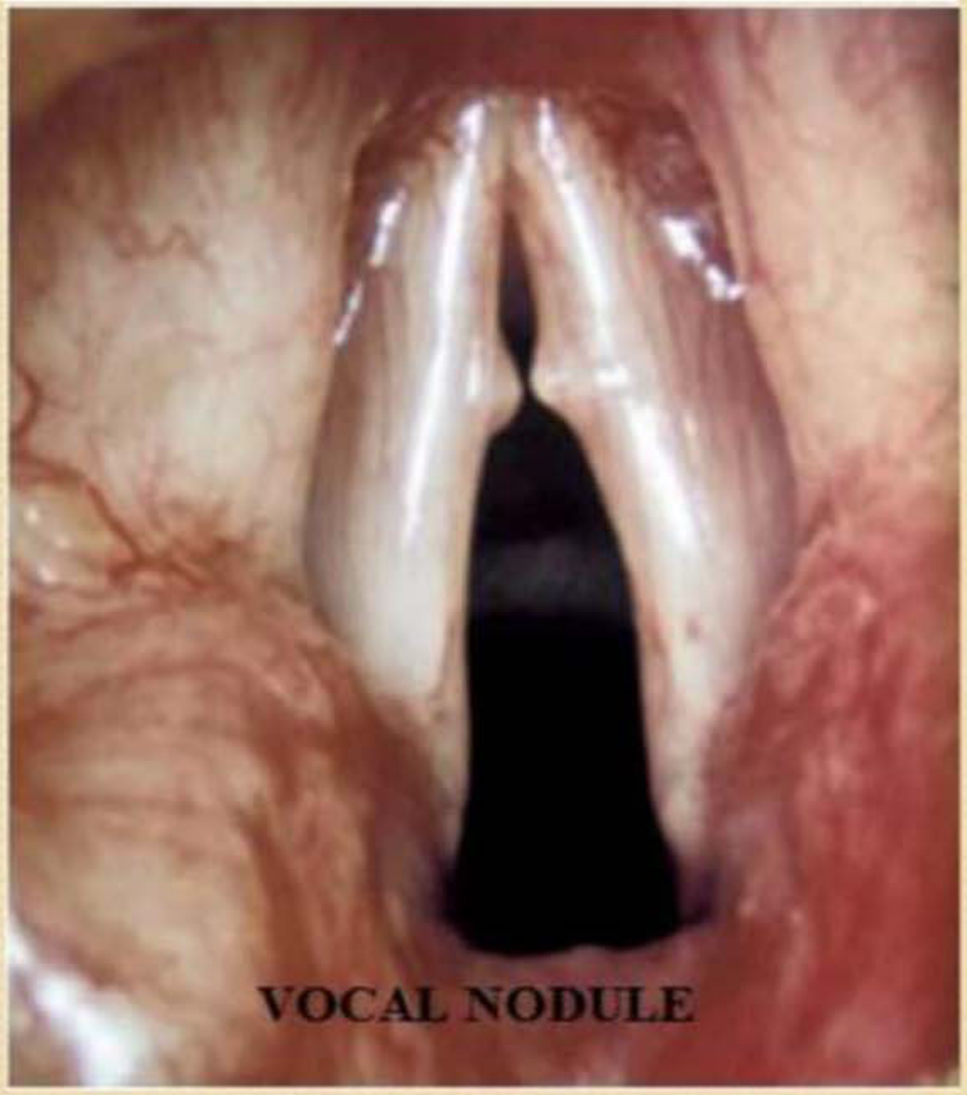
c.Microlarngeal excision is the treatment of choice.

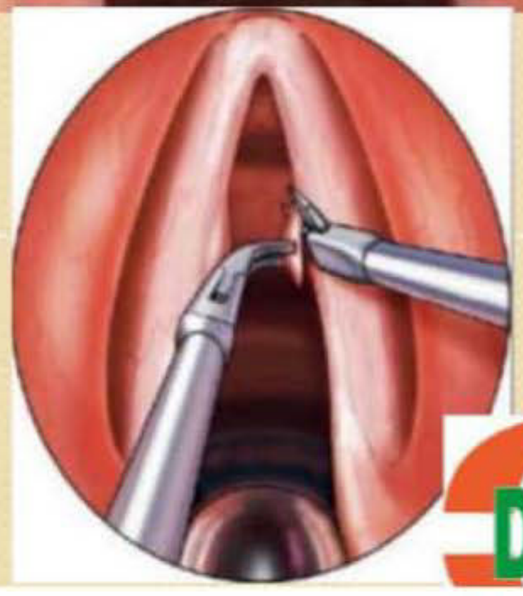
d.Leishmaniasis is the closest differential diagnosis

Acute epiglottitis: swan neck

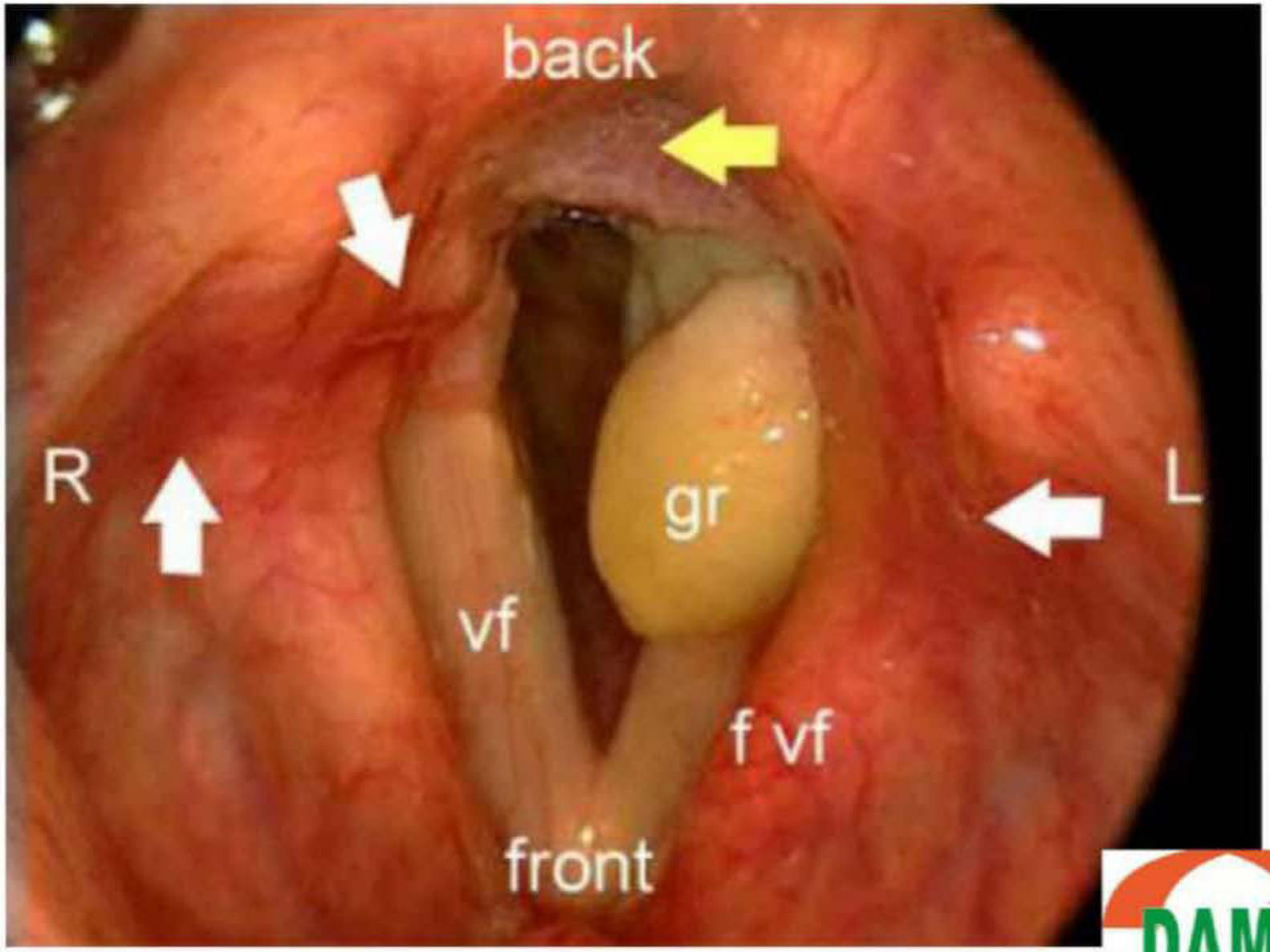
FIG 20–4.
Typical positioning of a person with partial airway obstruction who is trying to straighten the upper airway in

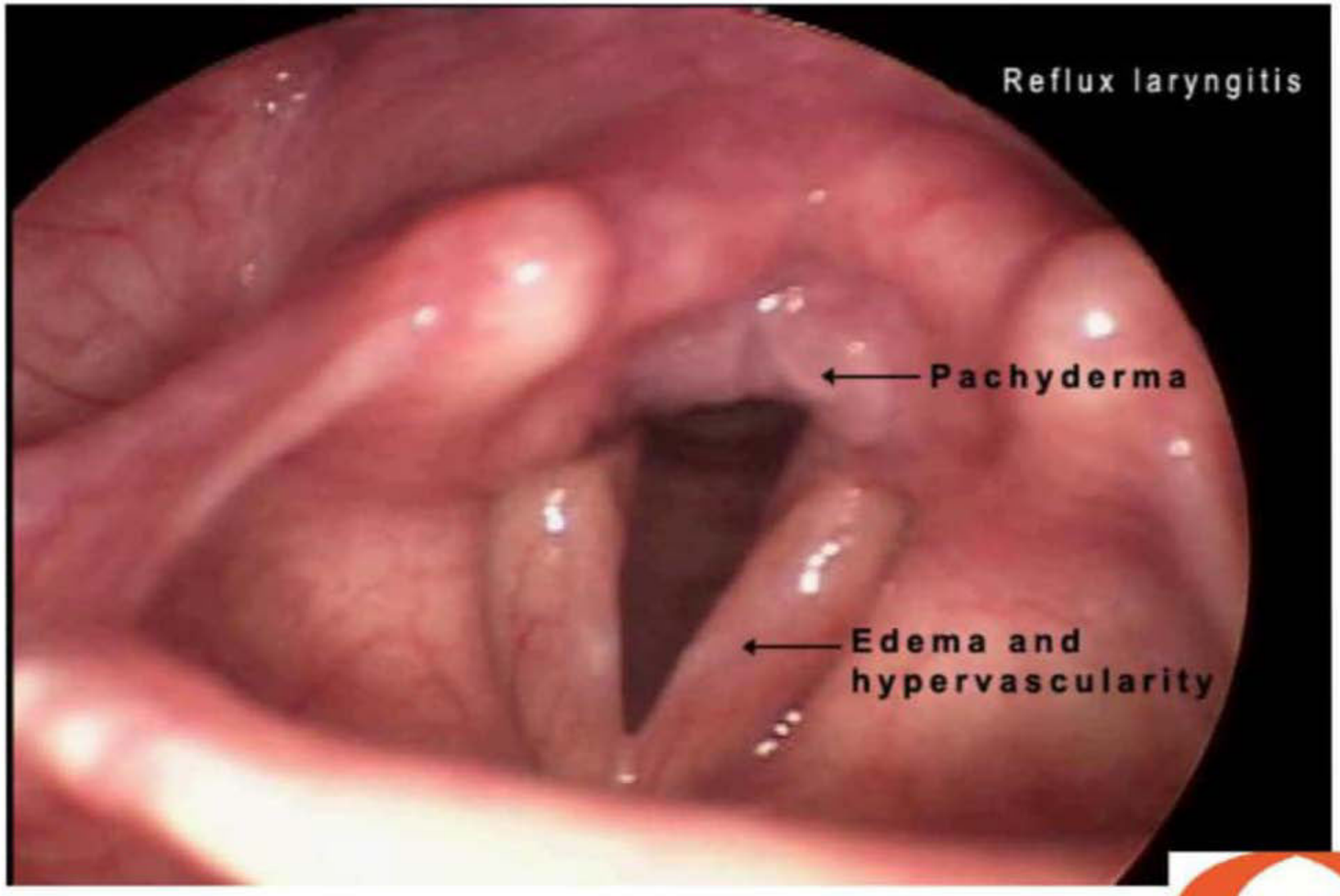

R
STEEPLE SIGN
THUMB SIGN
DAMS

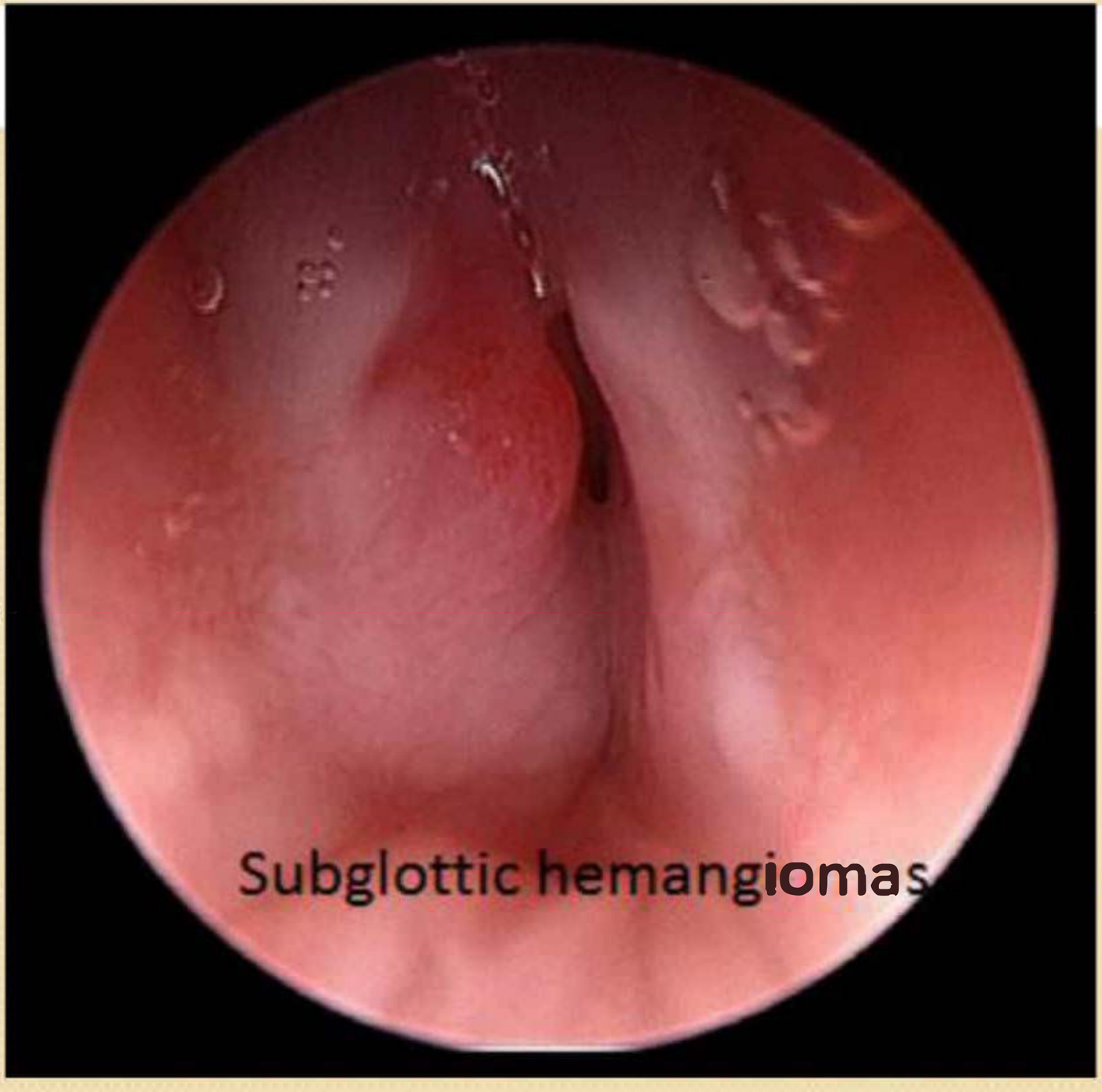

VOCAL NODULE


DAMS

INTUBATION GRANULOMA




Reflux laryngitis
Pachyderma
Edema and
hypervascularity


Subglottic hemangiomas

Papilloma obstructing anterior commissure. Microsurgical excision resulted in complete resolution of disease with restoration of normal vocal function.

CHONDROMA

Chondroma of Laryngeal cartilage is a rare, benign neoplasm

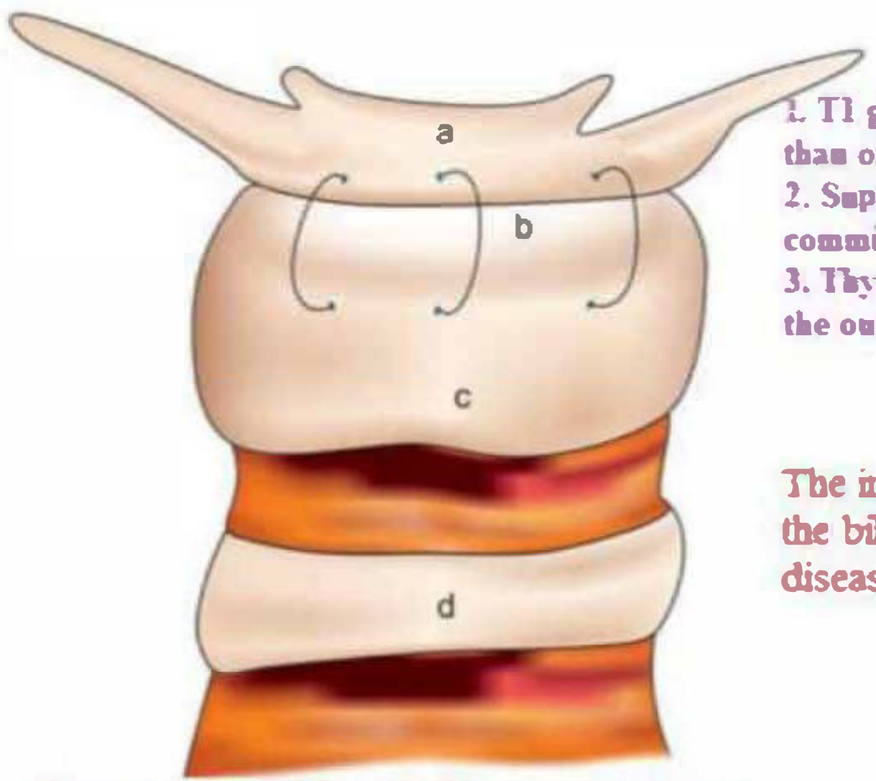
Posterior lamina of the cricoid cartilage > Thyroid > Arytenoid > Epiglottis

HPL

PFT MANDATORY

BOTH VOCAL CORDS NORMAL

Indications:



Supracricoid Partial Laryngectomy

INDICATIONS

1. T1 glottic carcinoma (a horse-shoe lesion) where more than one-third of the contralateral cord is involved
2. Supraglottic cancer with spread to the anterior commissure
3. Thyroid cartilage involvement but without invasion of the outer perichondrium

CONTRAINDICATIONS

The interarytenoid region, the postcricoid region, the bilateral pyriform sinuses should be free of disease.

Vertical Partial Laryngectomy:

Removal of:

- One vocal fold - from anterior commissure to vocal process
- ½ of opposite vocal fold may also be removed if involved
- Ipsilateral false vocal cord
- Ventricle
- Paraglottic space (and overlying thyroid cartilage)

Vertical Partial Laryngectomy: Contraindications

- Large T3 or any T4 lesion
- Intrarytenoid or cricoarytenoid joint involvement
- Bilateral arytenoid cartilage involvement or bilaterally diminished vocal cord mobility
- Thyroid cartilage penetration

DAMS

Advanced laryngopharyngeal cancers where the cord is fixed

Contraindications
Presence of edematous tissue in the interarytenoid region or the postcricoid region

NEAR TOTAL LARYNGECTOMY

DAMS

Neoadjuvant chemotherapy

- 2 cycles of Cisplatin(80-120mg/m2) + 5- FU(10-15mg/m2) given within 3 weeks interval

- Only those with > 50% tumour regression will receive radiation therapy

ELECTRO LARYNX

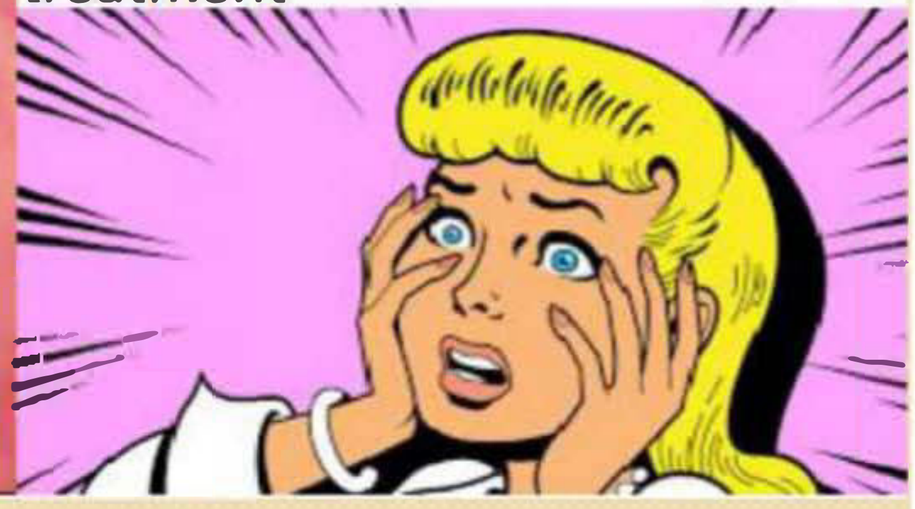
Study the **given image and choose most inappropriate statement**:

a.Usually caused by vocal abuse

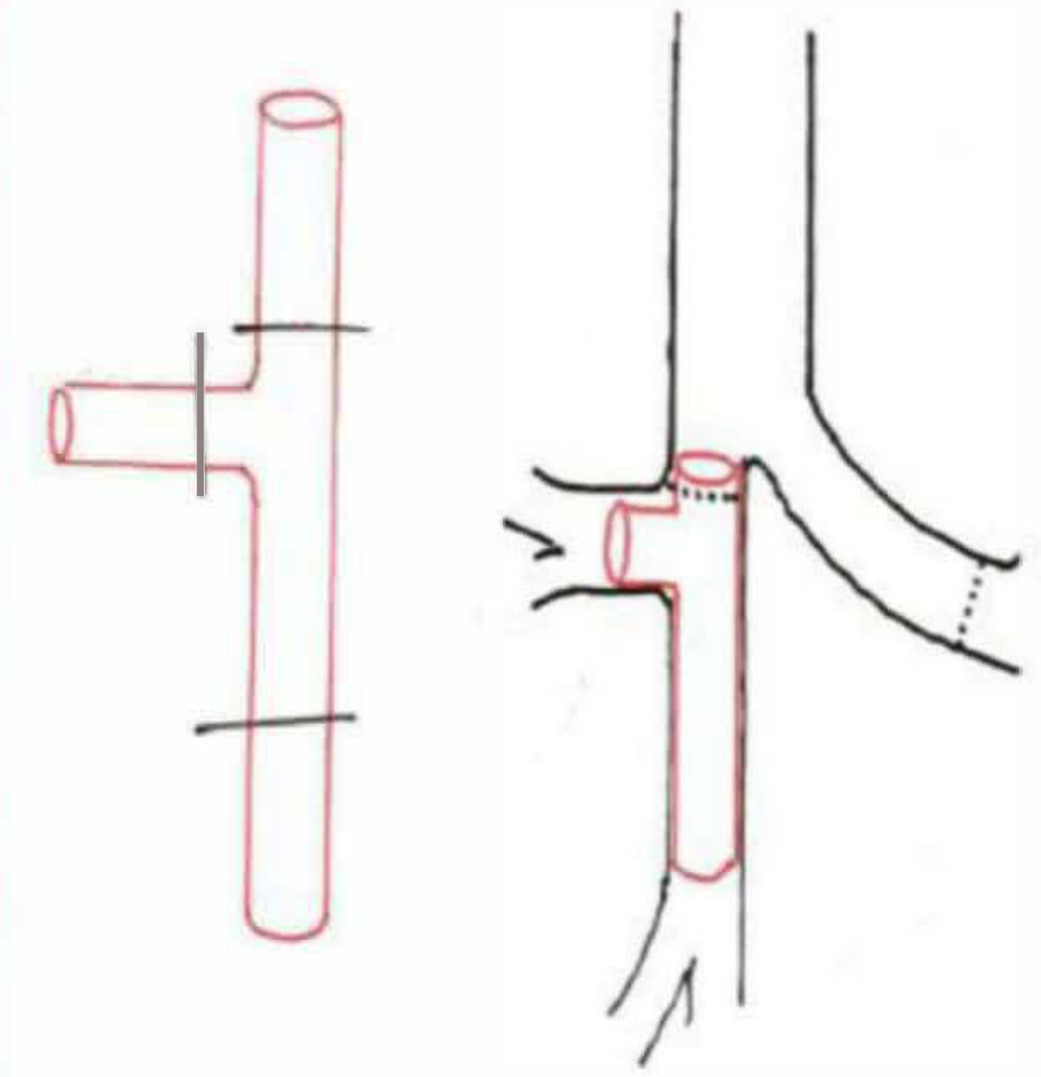
b.There is collection of oedema fluid in the subepithelial space

c.There is asymmetrical swelling of vocal cords

d.Vocal cord stripping is the treatment



DAMS



Montgomery T-tube
Bronchial Stenosis

DAMS

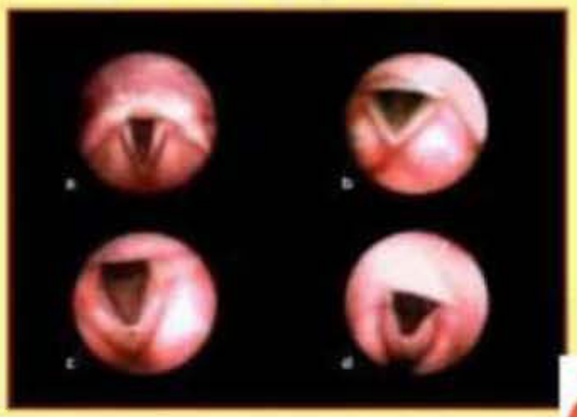
Findings of Laryngopharyngeal Reflux

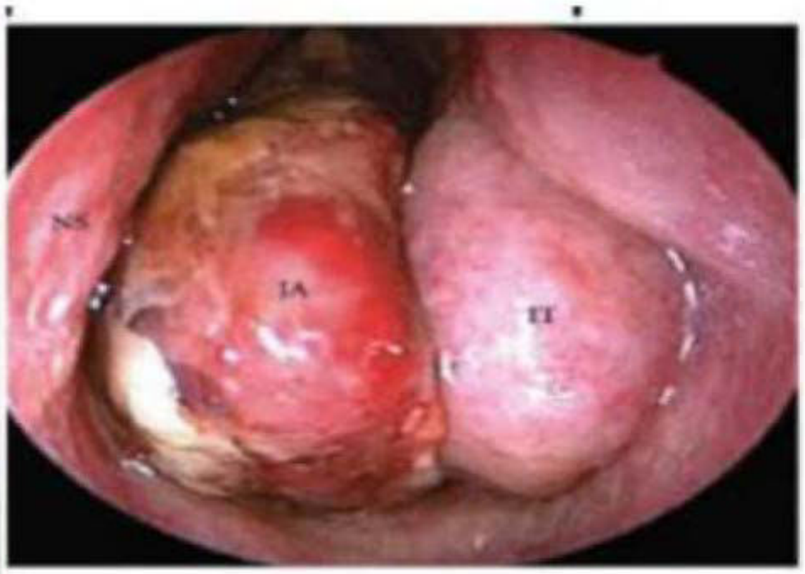
I. Subglottic Edema (Pseudosulcus Vocalis)

II. Ventricular Obliteration

IV. Vocal Fold Edema

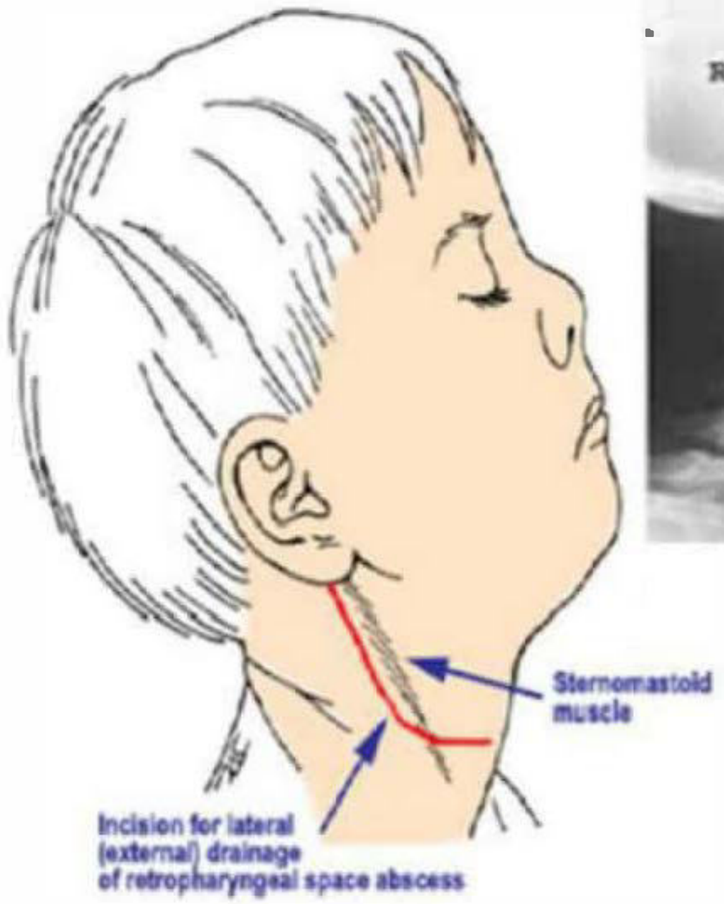
V. Posterior Commissure Hypertrophy (PCH)

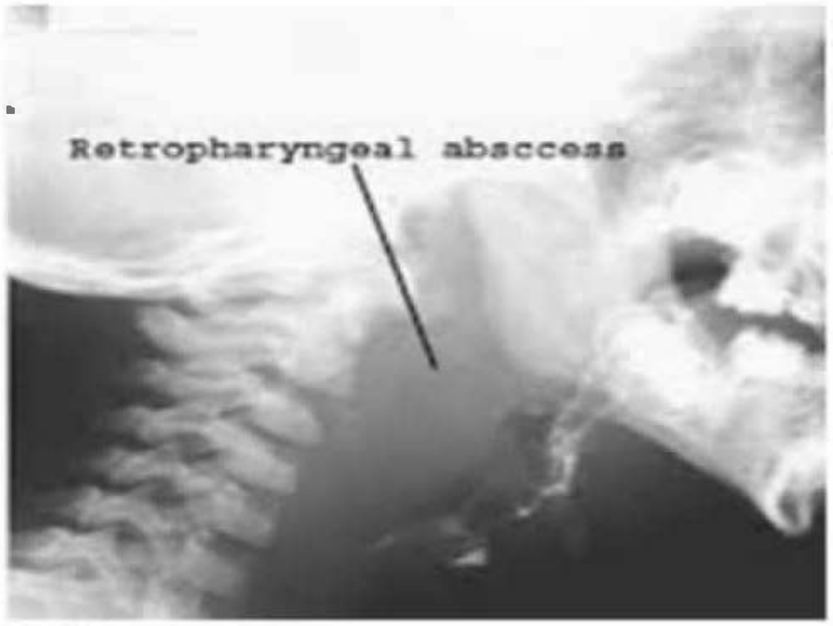


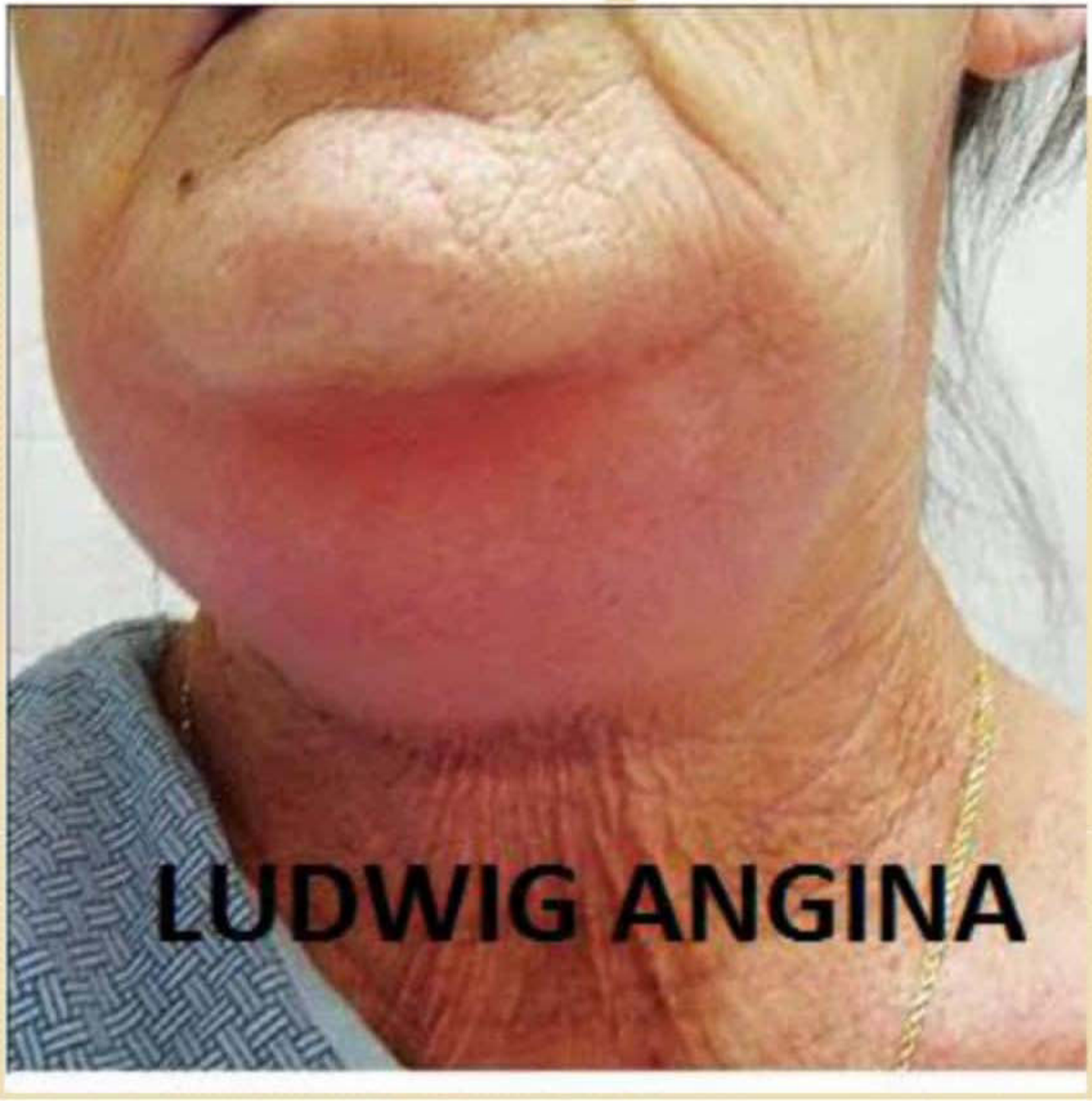


Transnasal endoscopic
Internal maxillary artery

Nasopharyngeal Cancer


Sternomastoid muscle
Incision for lateral (external) drainage of retropharyngeal space abscess


Retropharyngeal abscess


LUDWIG ANGINA

What is Moure's sign?
Hold the larynx between index and thumbfingers of your right hand and move it from side to side over the vertebral column

Normally you feel click
In postcricoid carcinoma there is absence of click

ABSENT
MALIGNANCY
ABSCESS

DAMS

Tangier Disease

Tonsillectomy

- **Dissection and snare method**
- Electrocautery Heat of electrocautery (400 °C)
- **Radiofrequency ablation**
- **Coblation tonsillectomy**

  Use plasma by combines radiofrequency energy and saline to create a plasma field
- **Harmonic Scapel**

  Ultrasonic energy to vibrate its blade at 55 kHz
- **Thermal Welding**

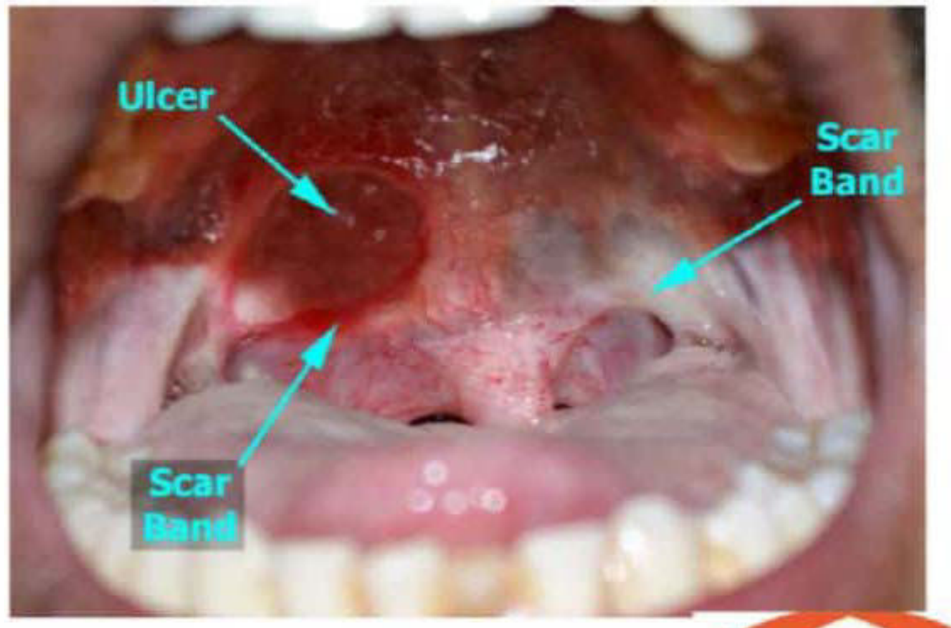
  Tissue Weld/Painless/ 2-3 °C than body temp.
- **Carbon dioxide laser**
- **Microdebrider**

SUBMUCOUS FIBROSIS

Age → 20 – 40 years Sex → F > M

Is PREMALIGNANT

Steroids



SECTION VIII

COLOR PLATES

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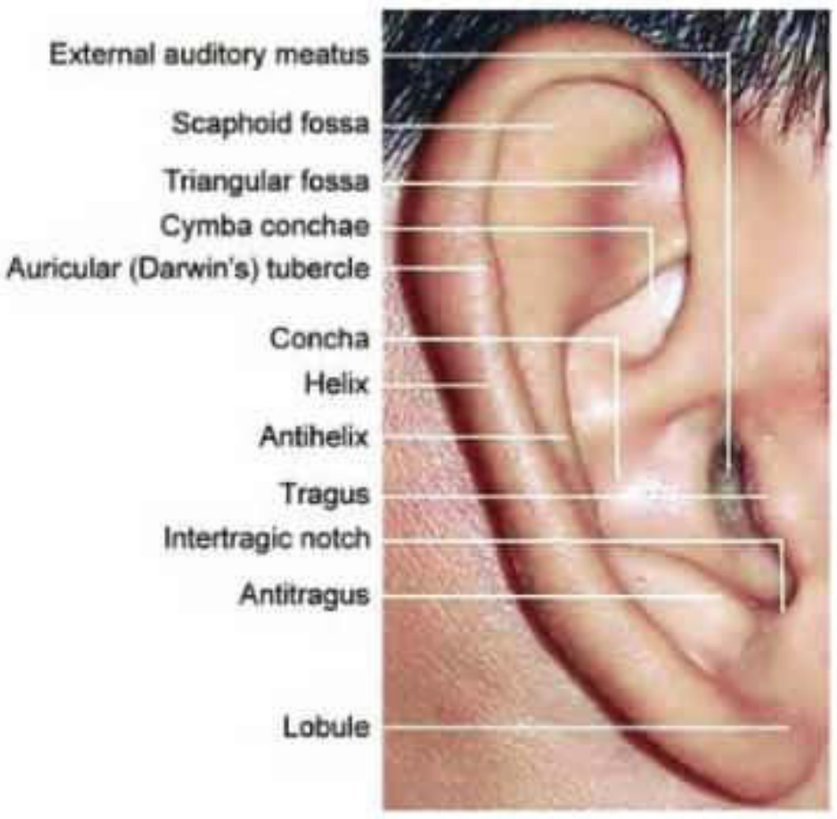
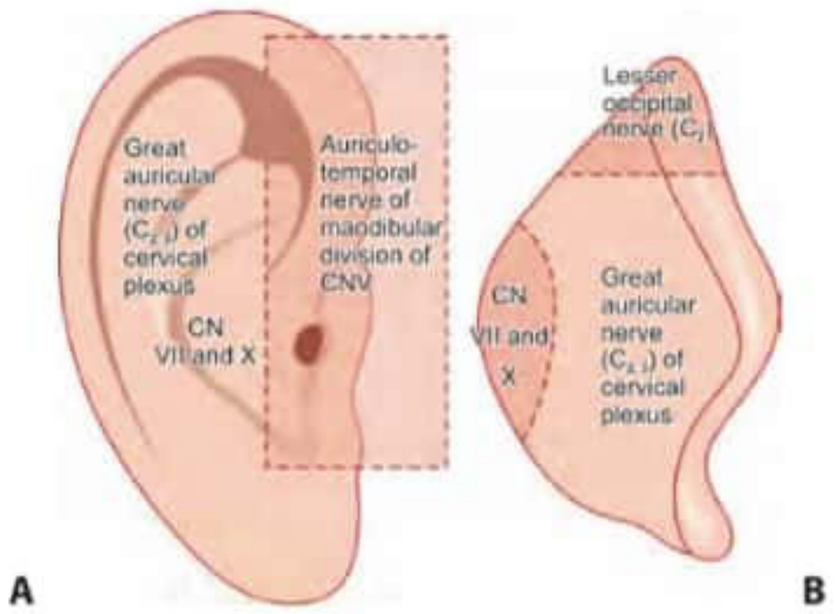
IMPORTANT PICTURES FOR PICTORIAL QUESTIONS

EAR

1. Auricular cartilage: external features

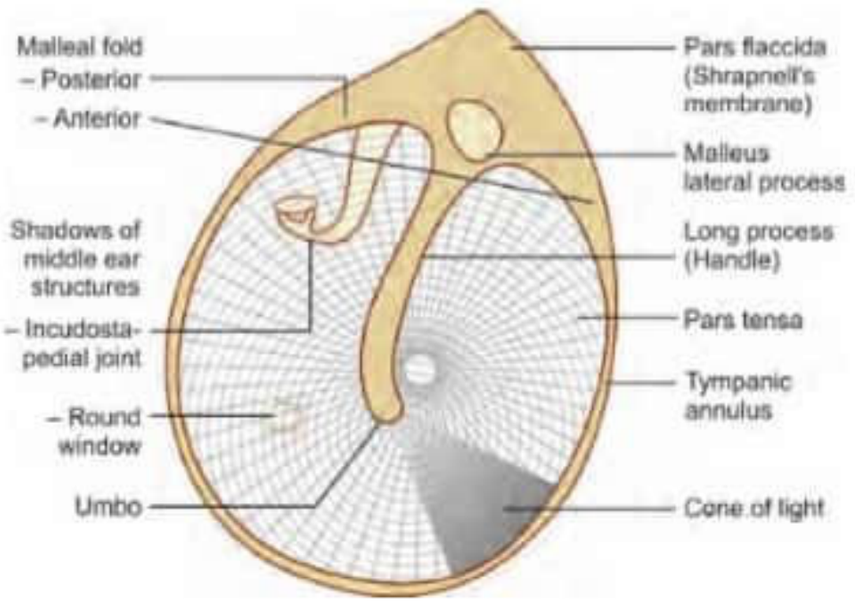


2. Nerve supply of Pinna

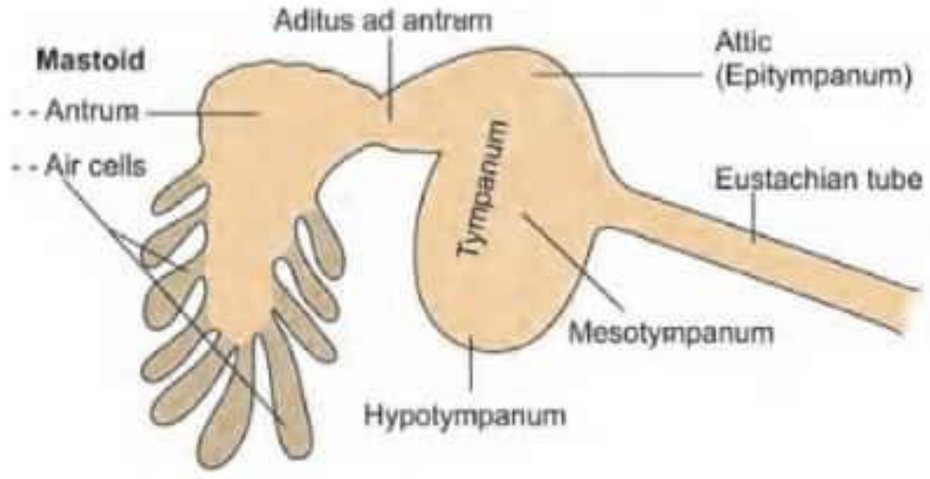


A. Lateral surface **B.** Medial surface

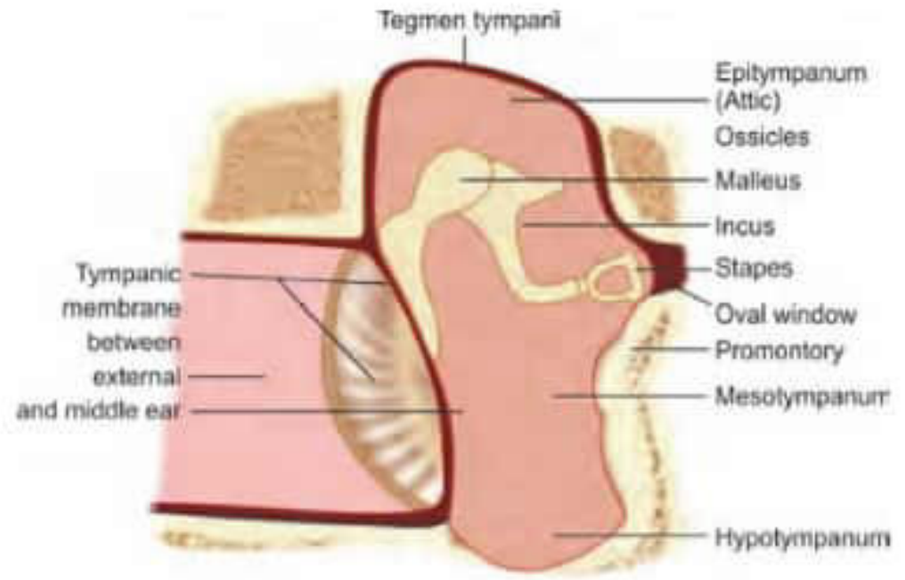
3. Tympanic membrane as seen on otoscopy



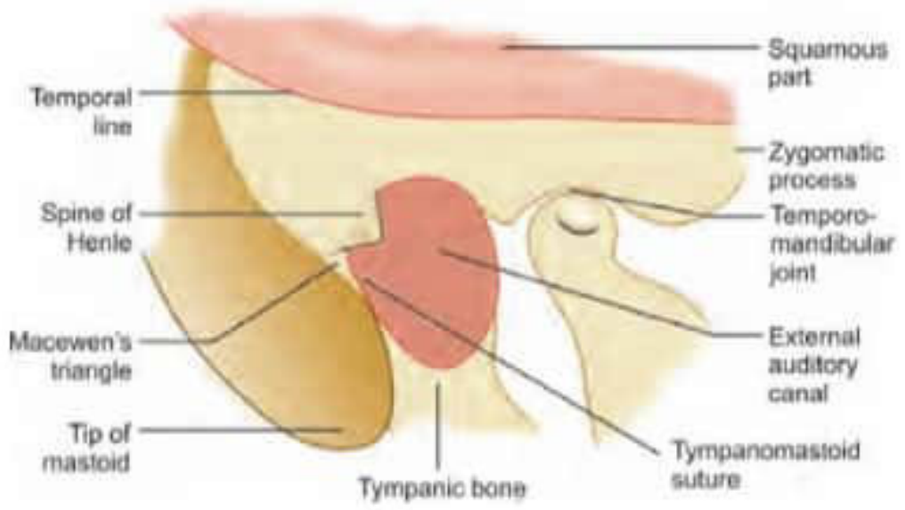
4. Parts of middle ear cleft



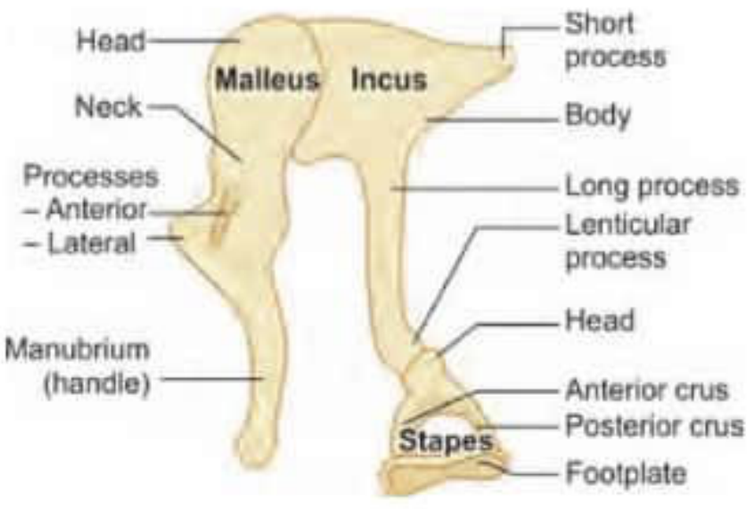
5. Parts of middle ear as in seen on coronal section



6. Mc-Ewan triangle: Surface landmark for mastoid antrum



7. Middle ear ossicles



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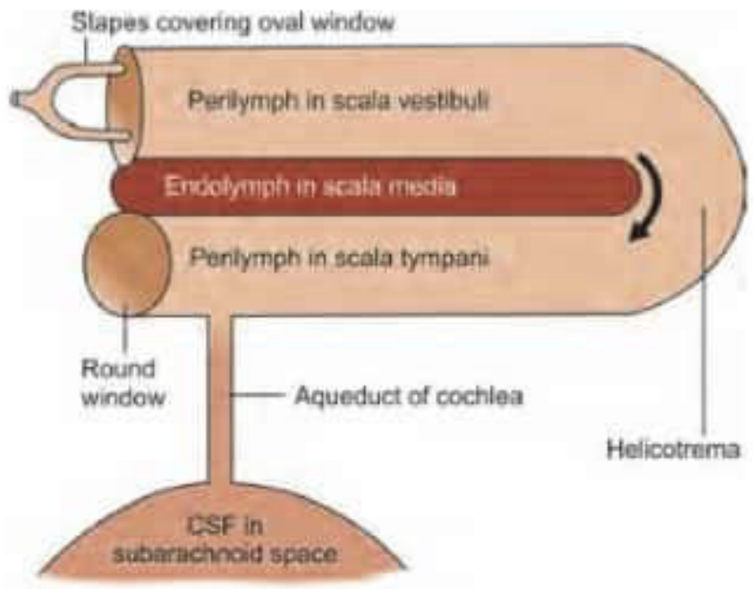
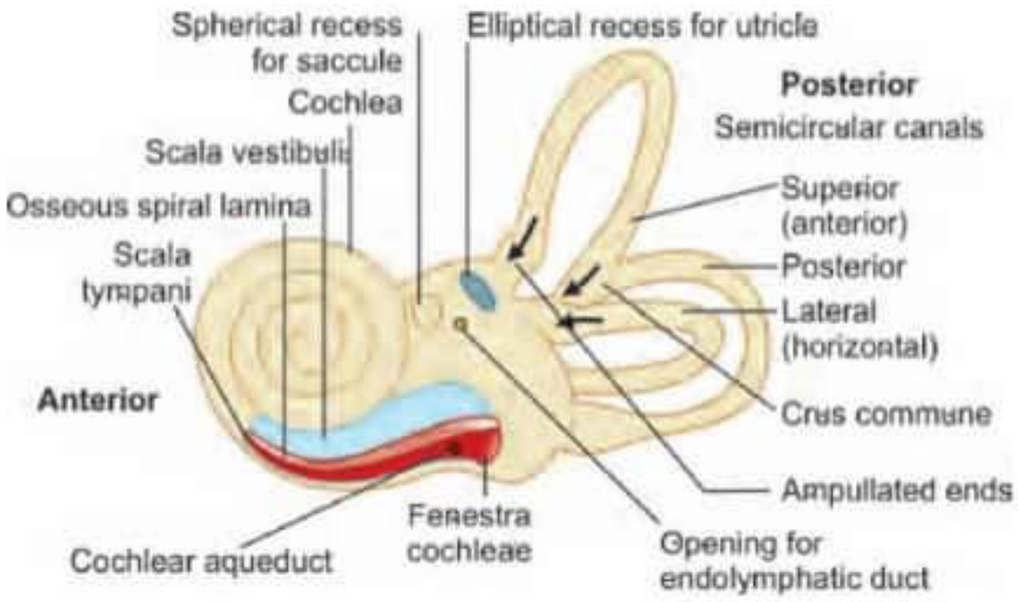
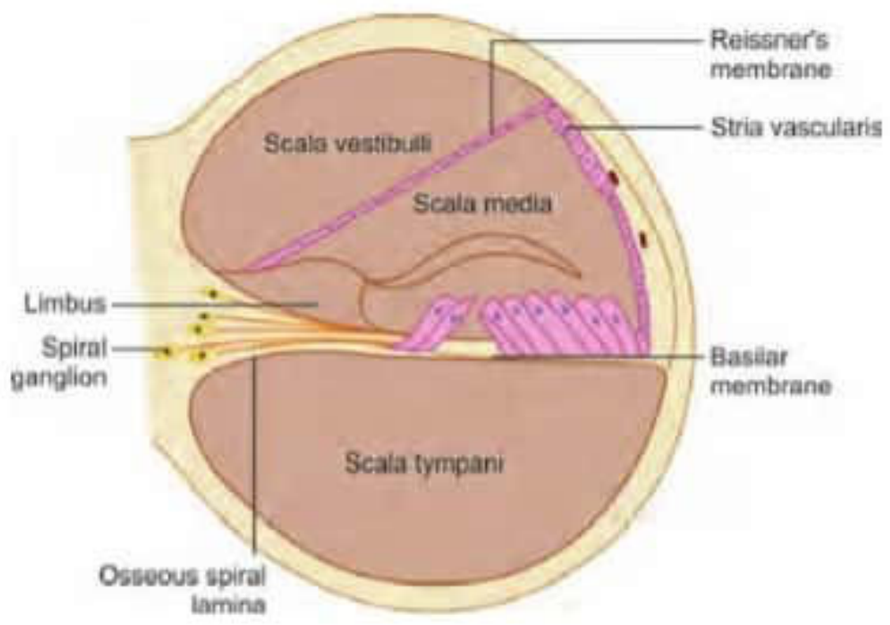
8. Cochlea: Peri and endolymphatic systems: relation



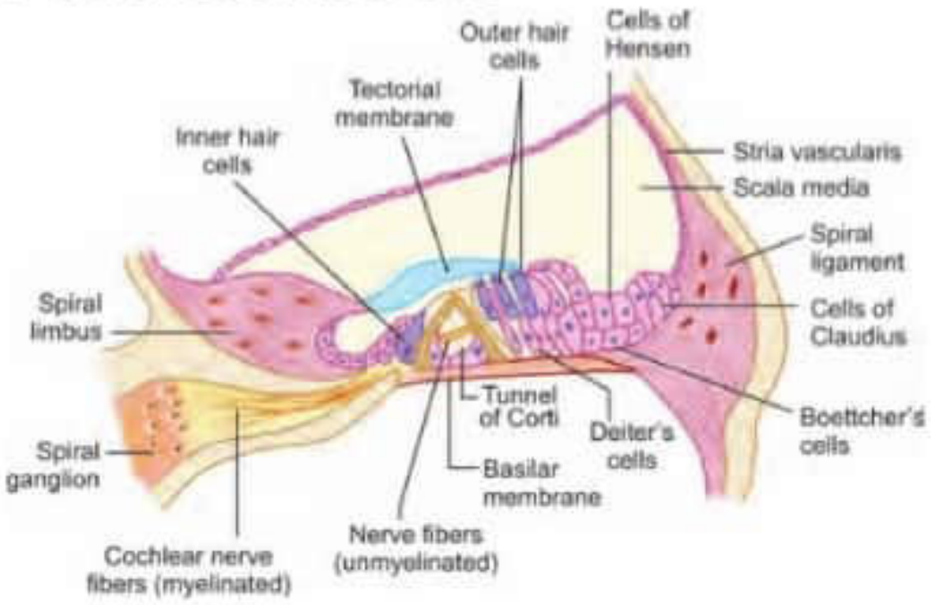
9. Medial wall of left bony labyrinth seen from lateral side after the removal of its lateral wall



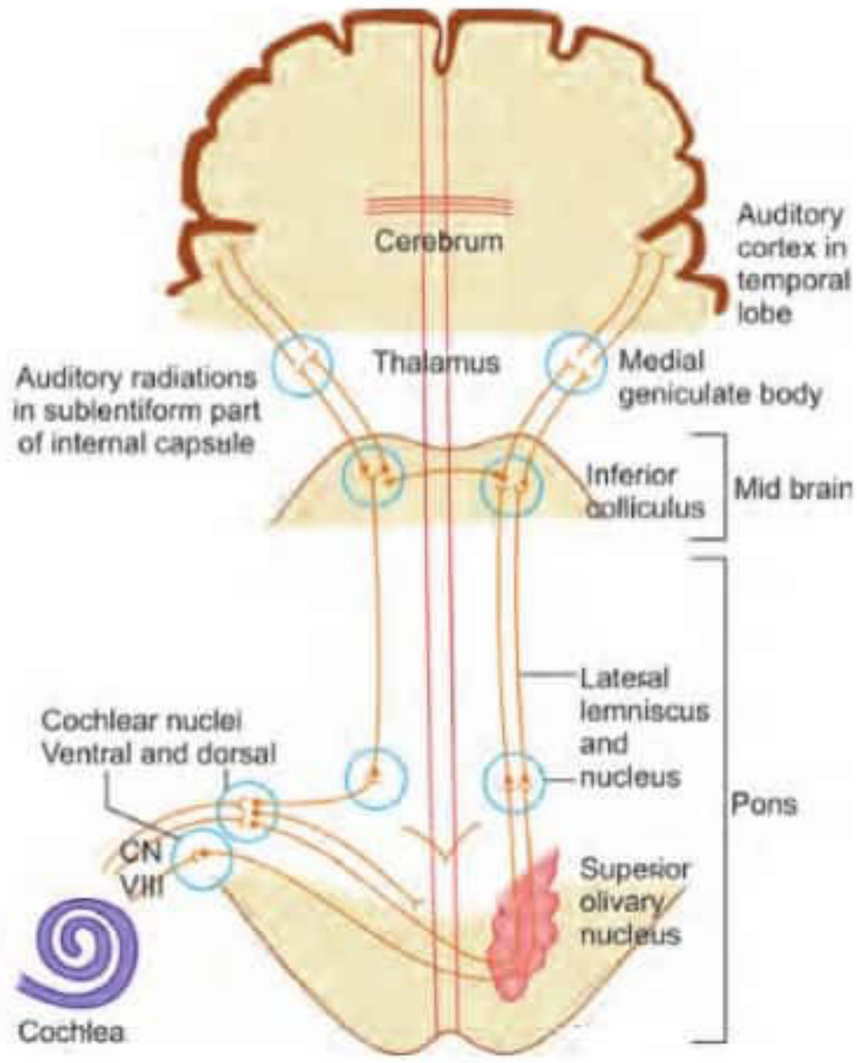
10. Structure of cochlear canal after its cut section



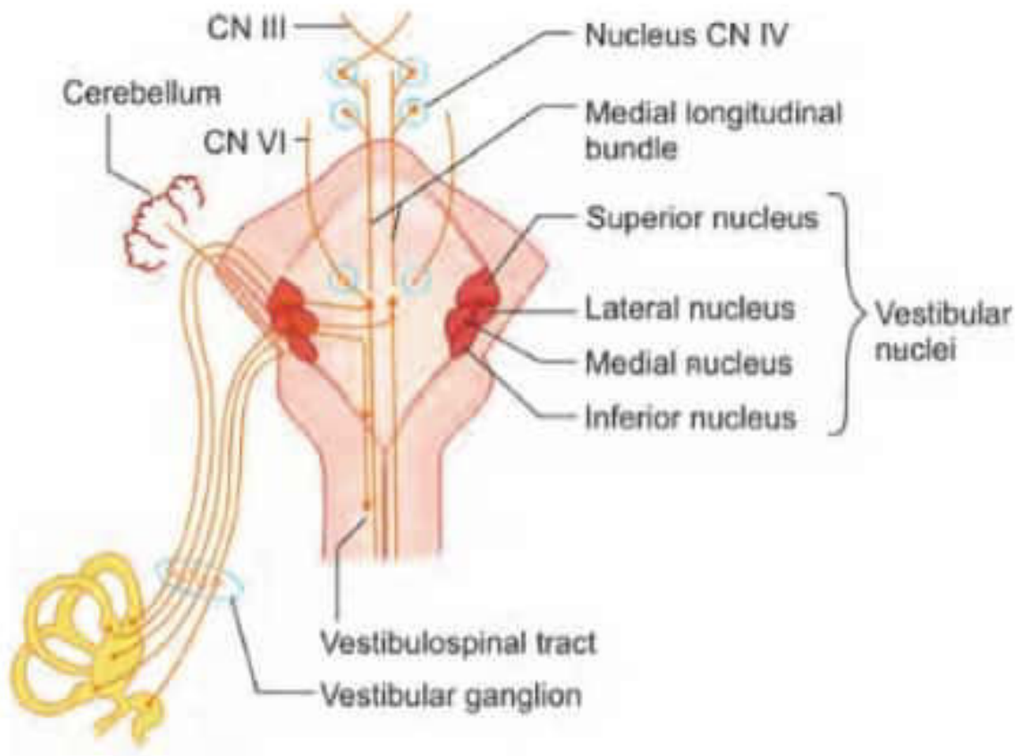
11. Structure of Organ of Corti



12. Central auditory pathway



13. Vestibular pathway



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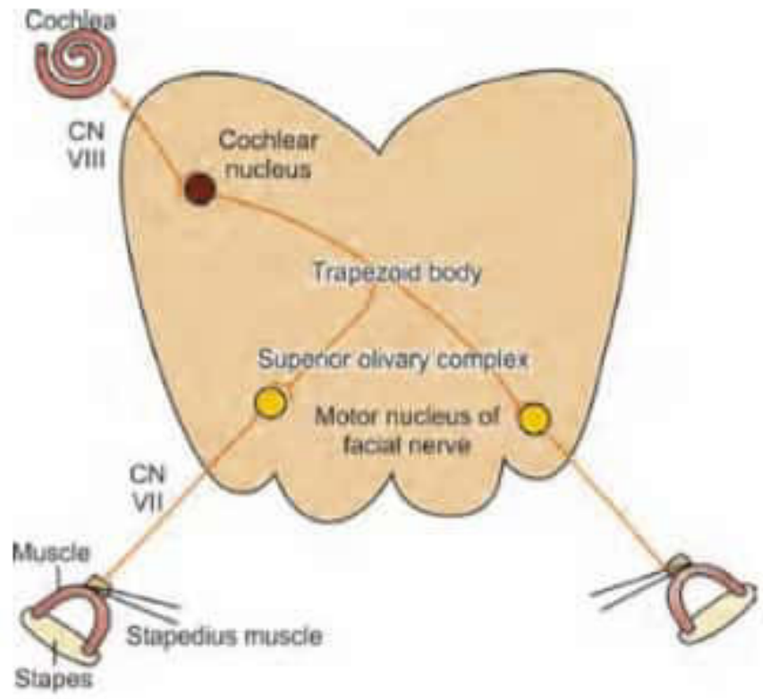
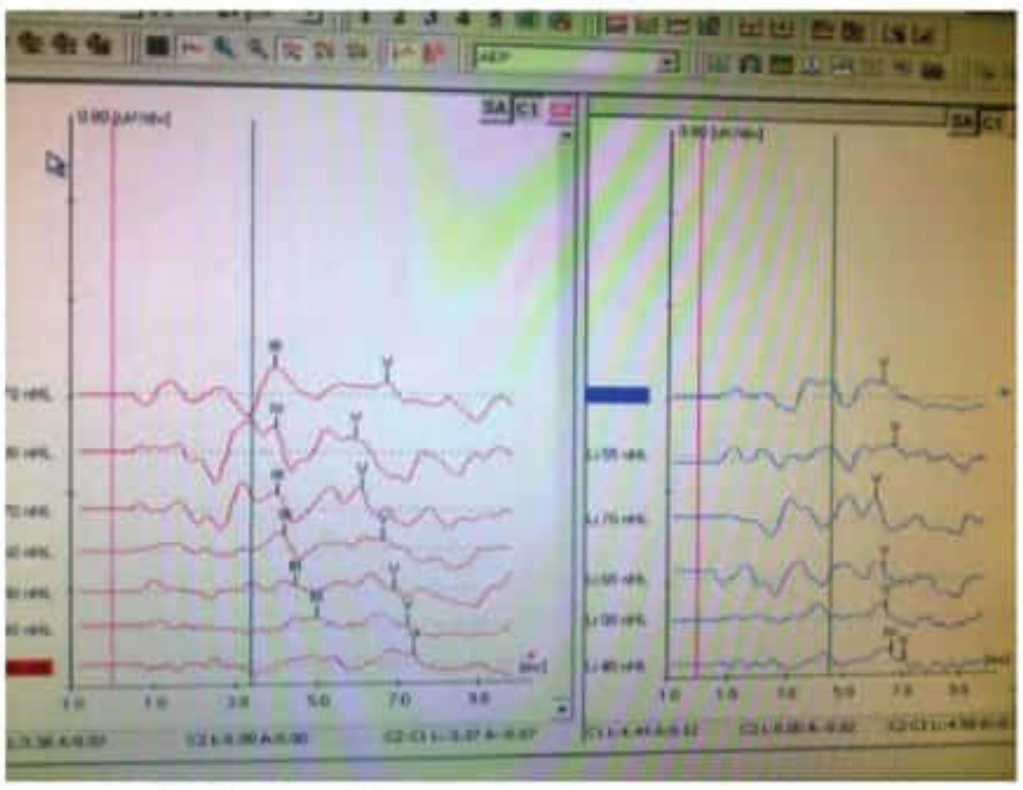
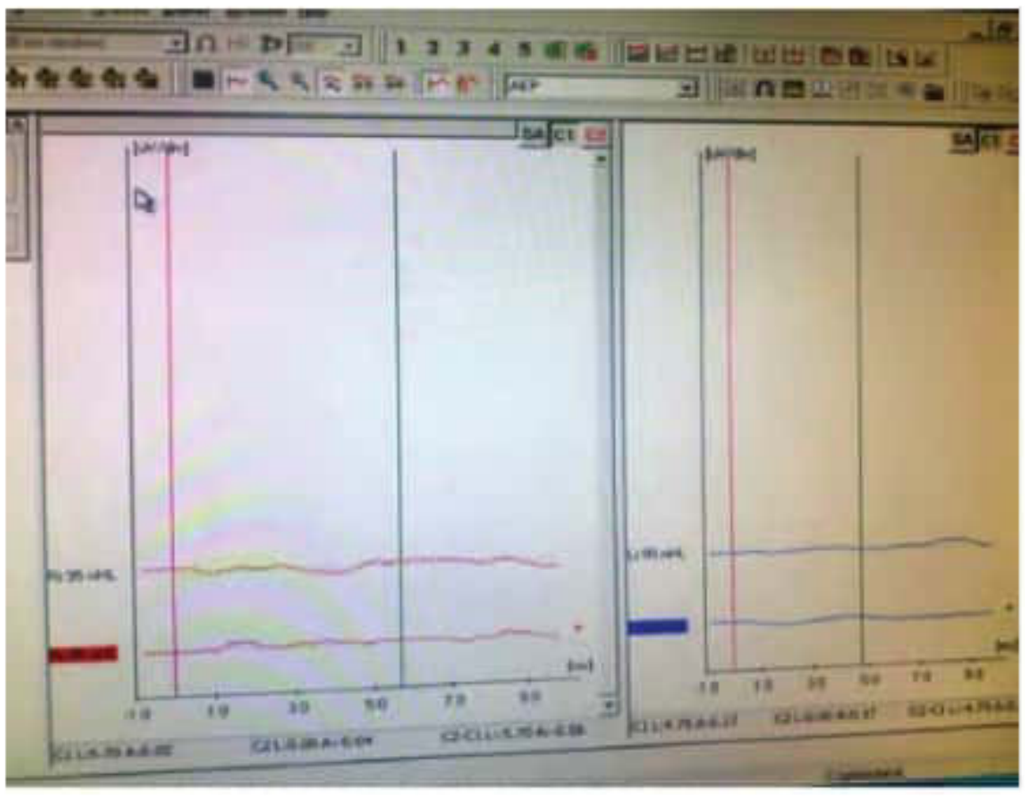
14. Acoustic reflex pathway



15. BERA



A. Normal with normal latency

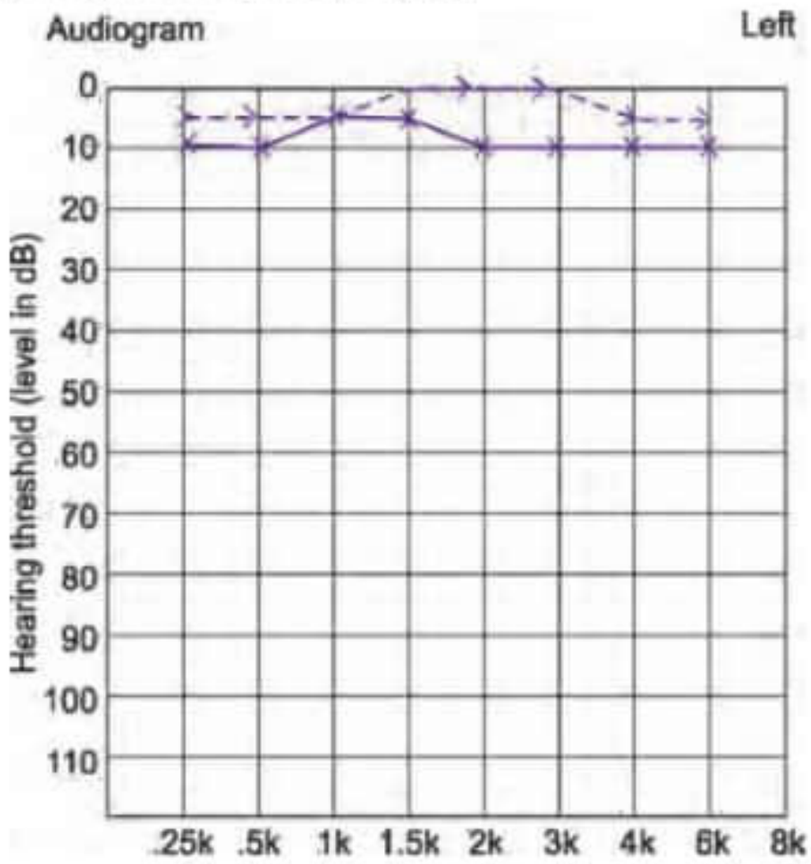


B. BERA in severe hearing less. Note: No peaks seen

16. Symbols used in audiogram charting

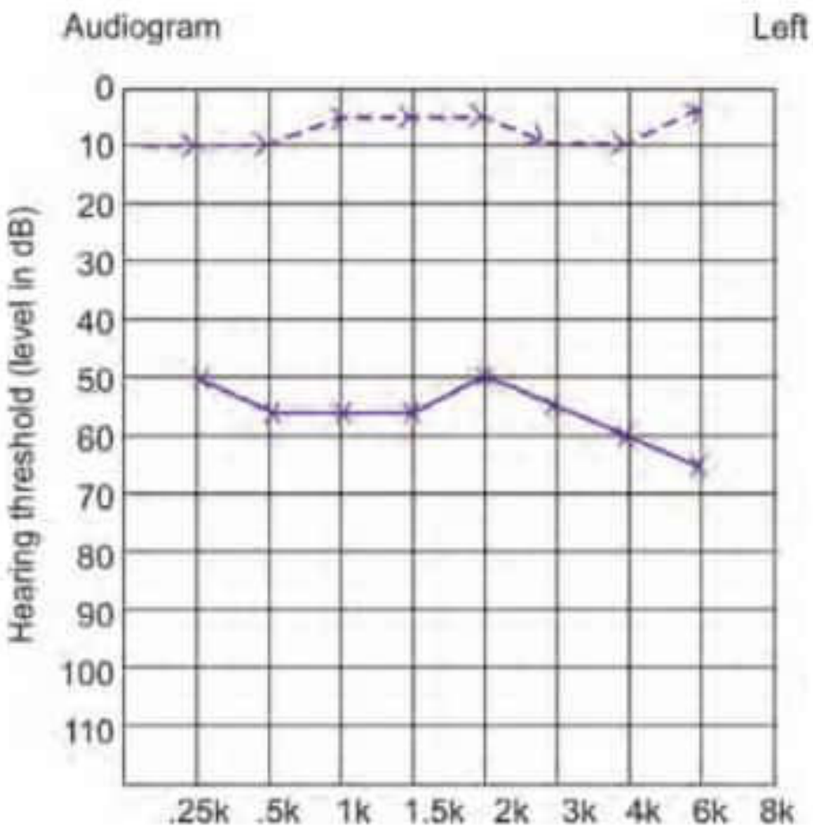
Modality	Ear	
	Right	Left
AC unmasked	○	×
AC masked	△	□
BC unmasked	<	>
BC masked	[	]
No response		

17. Audiogram of left normal ear



In normal persons, hearing threshold values with both air and bone remain between 0 and 10 dB

18. Audiogram of left ear with conductive hearing loss



In this graph, bone-air gap is seen which means a patient can hear by bone under 10-20 dB, while with air hearing is much below, depending on the severity, indicating conductive hearing loss.

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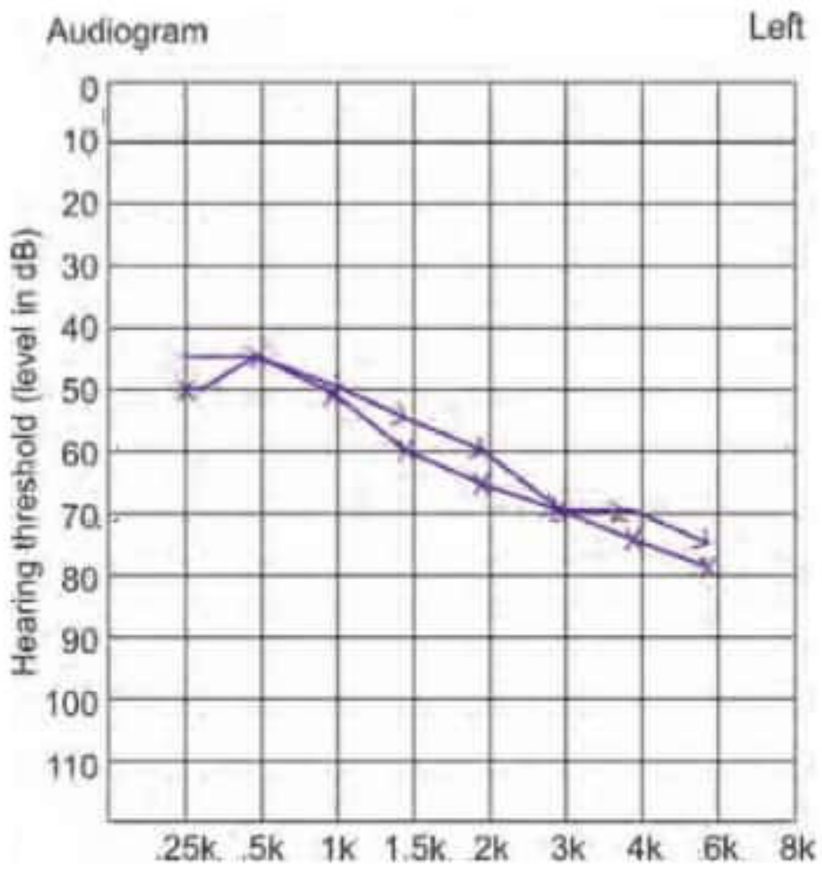
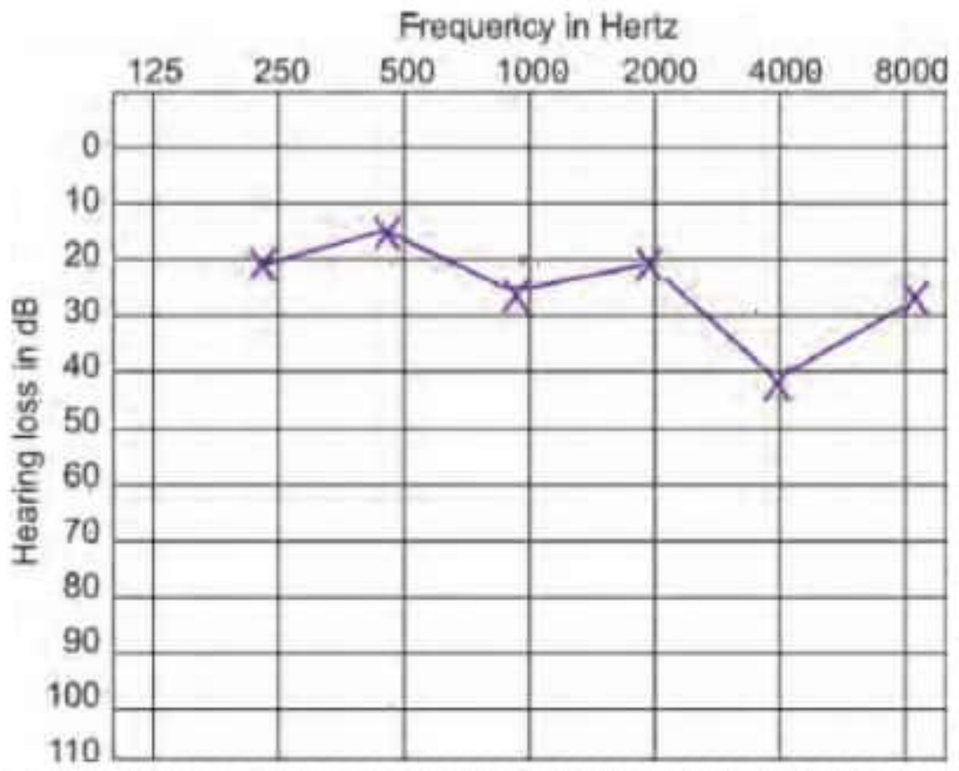
19. Audiogram of left ear with SNHL



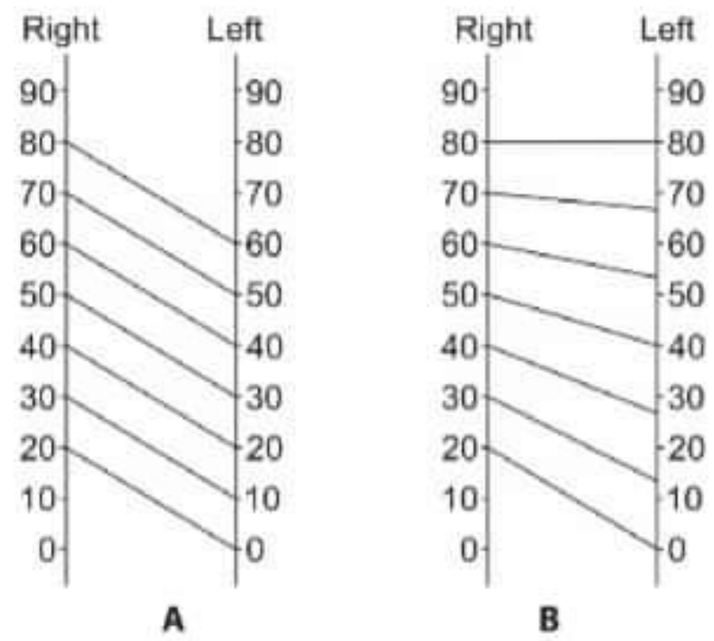
In SNHL, both bone and air conduction values are decreased and may even overlap each other.

20. Audiogram in Early case of noise-induced hearing loss.



In acoustic trauma, there is a sudden dip at 4000 Hz both in air and bone conduction values

21. Alternate binaural loudness balance test



(A) Nonrecruiting ear. The initial difference of 20 dB between the right and left ear is maintained at all intensity levels.
(B) Recruiting ear right side. At 80 dB loudness perceived by right ear is as good as left ear though there was difference of 30 dB initially

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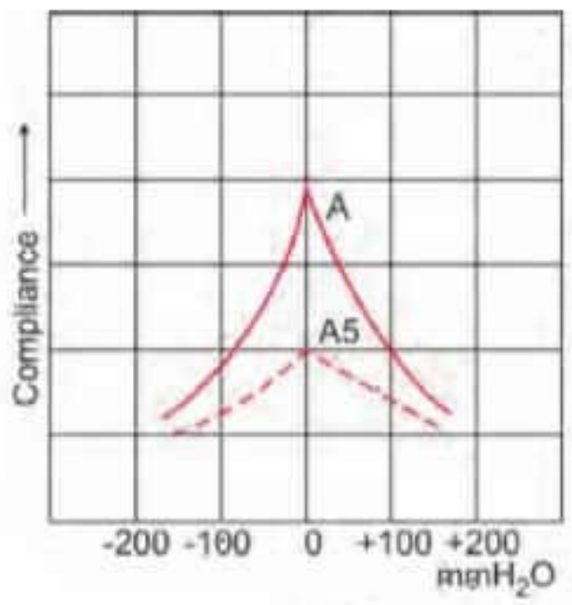
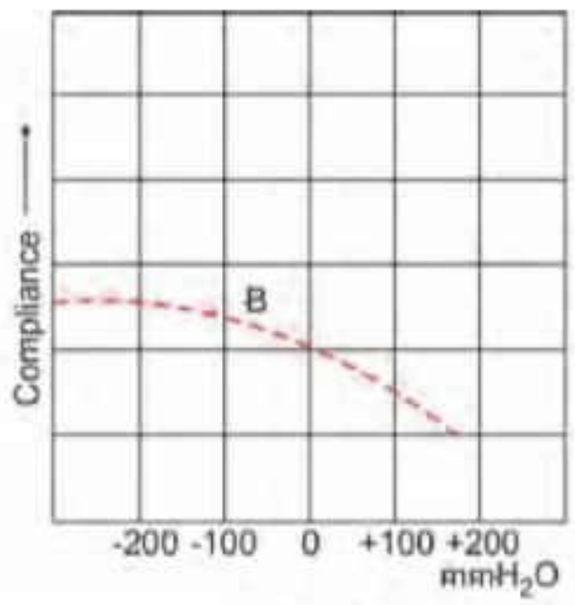
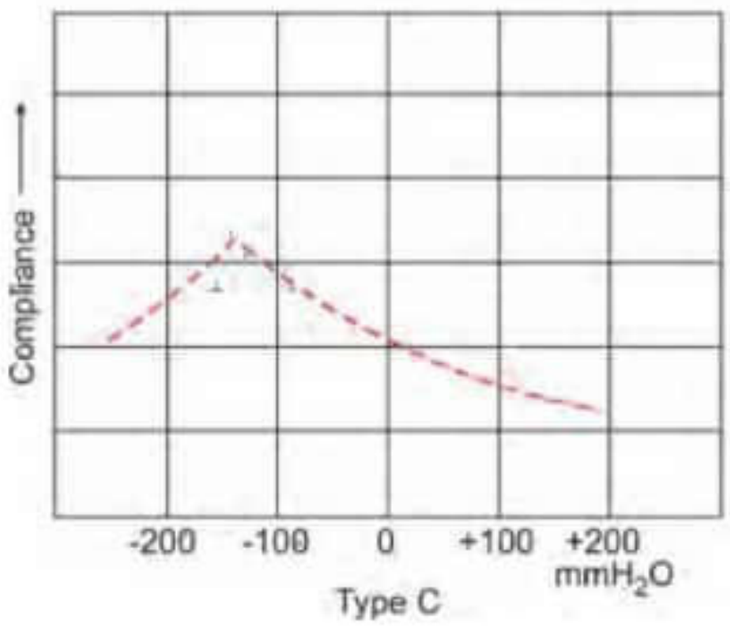
22. Types of Tympanogram: Impedene Audiometry Curves:



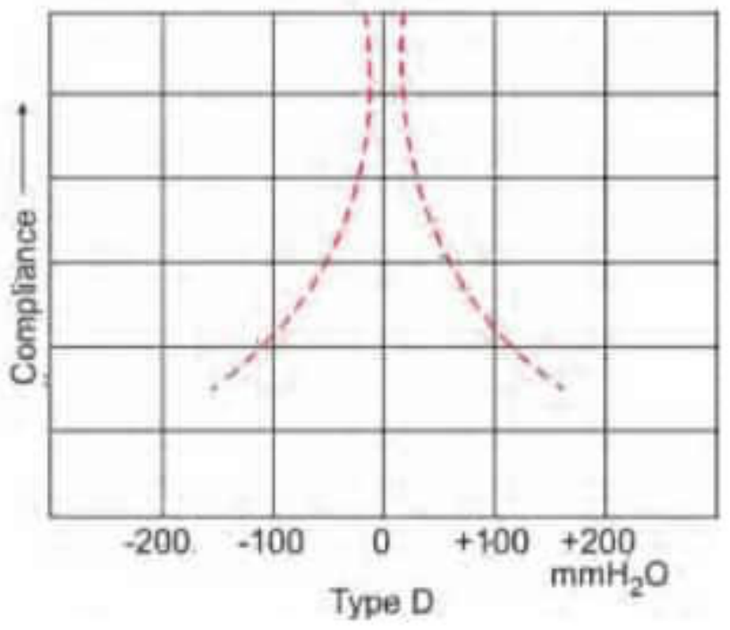
Type A
A- Normal



Type B
B-flat or dome shaped audiogram
(middle ear fluid)



Type C



Type D

Types of curve	Conditions seen in
A curve (Normal peak height and pressure	Normal Eustachian tube obstruction
As curve[Q] (It is also a variant of normal tympanogram but may be shallow	Otosclerosis[Q] Tumors of middle ear Fixed malleus syndrome Tympanosclerosis
Ad curve (Variant of normal with high peak)	Ossicular discontinuity Post stapedectomy Monometric ear drum
B curve (Flat or dome-shaped curve)[Q] Indicating lack of compliance	Fluid on middle ear[Q] Secretory otitis media[Q] Tym+anic membrane perforation[Q] Grommet in ear[Q]
C curve (negataive peak pressure)	Retracted tympanic membrane Faulty function of Eustachian tube/ Eustachian tube obstruction

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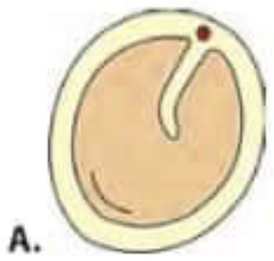
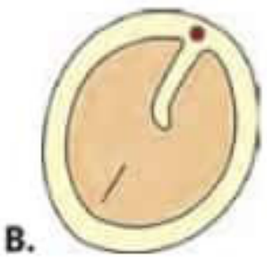
23. Incisions for myringotomy

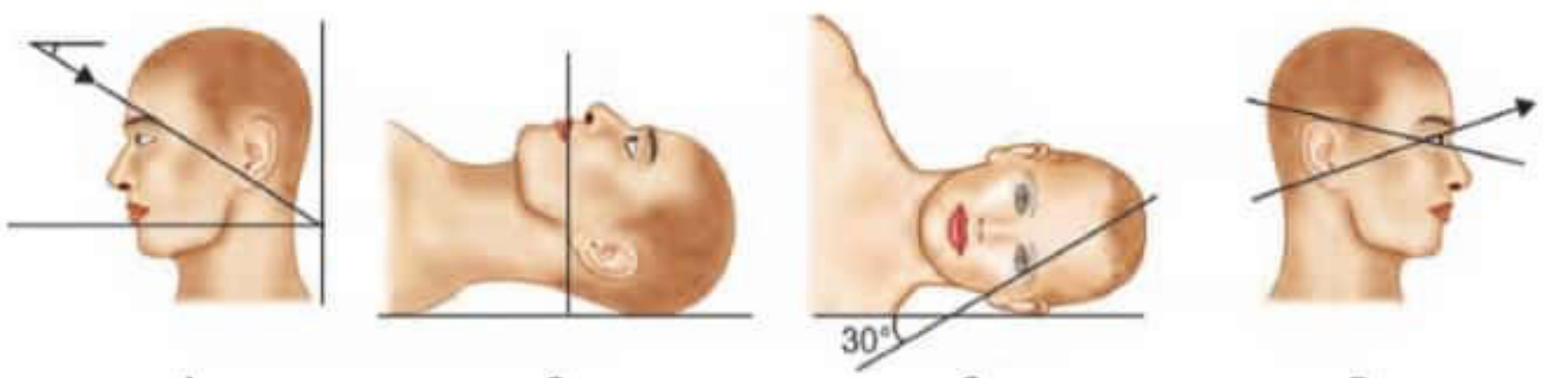


A. In case of Acute Suppurative Otitis Media (ASOM)



B. In case of Serous Otitis Media ± grommet insertion

24. Different view of X-ray for diseases of the ear

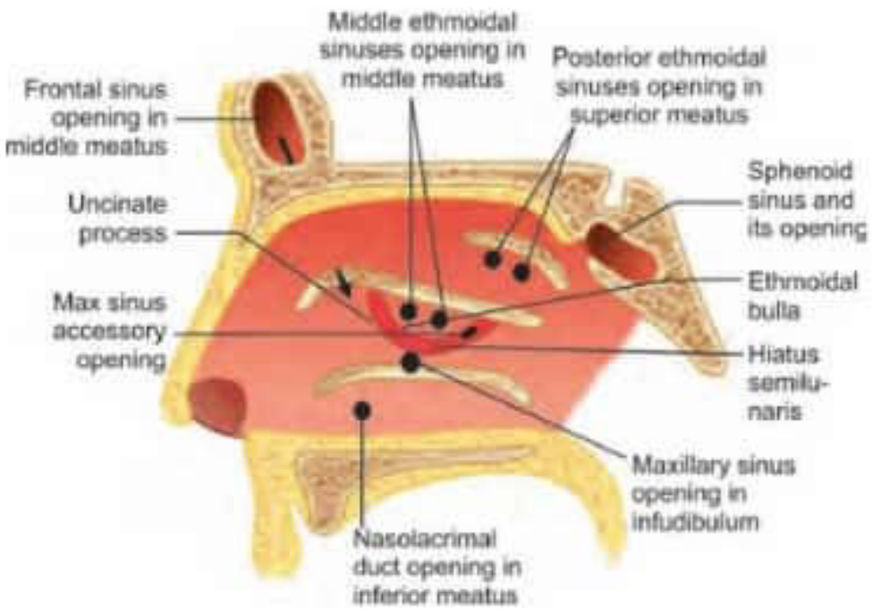


A. B. C. D.

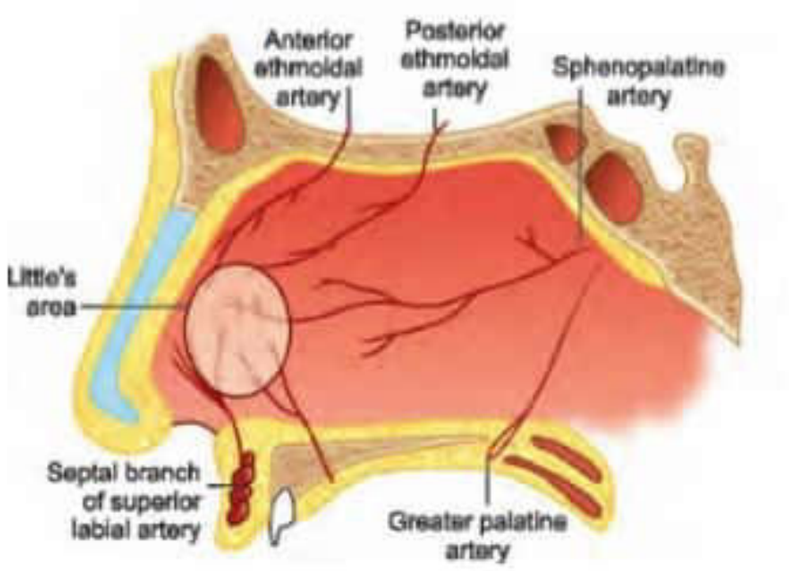
(A) Towne's (Fronto-occipital): (B) Submento-vertical view, (C) Stockholm-B view (Lateral-oblique): (D) Stenvers view (Oblique-posterior anterior)

NOSE

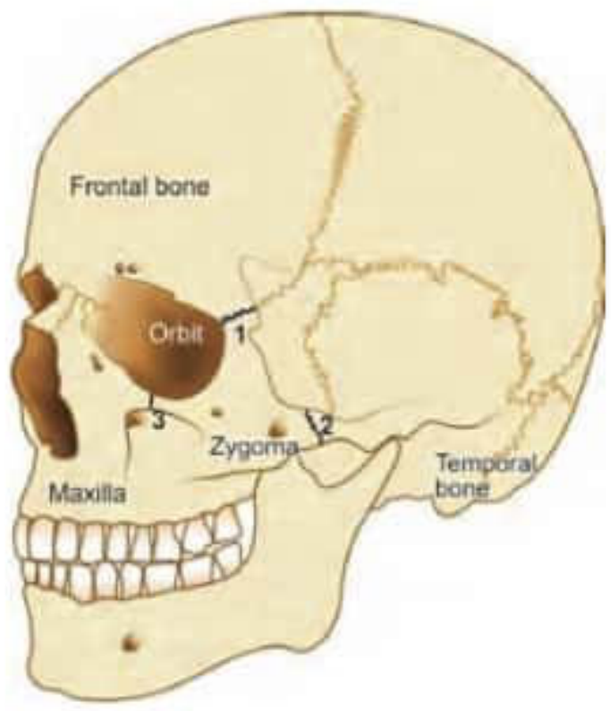
1. Openings of paranasal sinus as in lateral wall of nose after removal of turbinates



2. Blood supply of nasal septum



3. Tripod fracture



Left zygoma (tripod) fracture showing three sites of fracture. (1) Zygomaticofrontal: (2) Zygomaticotemporal; (3) Infraorbital

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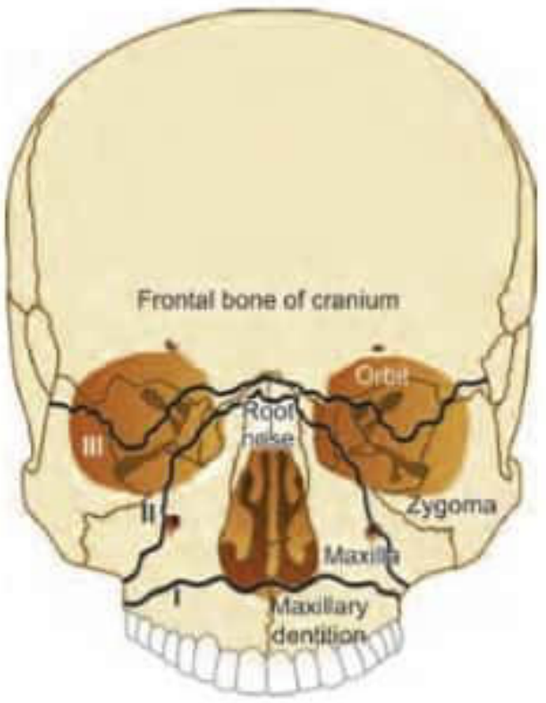
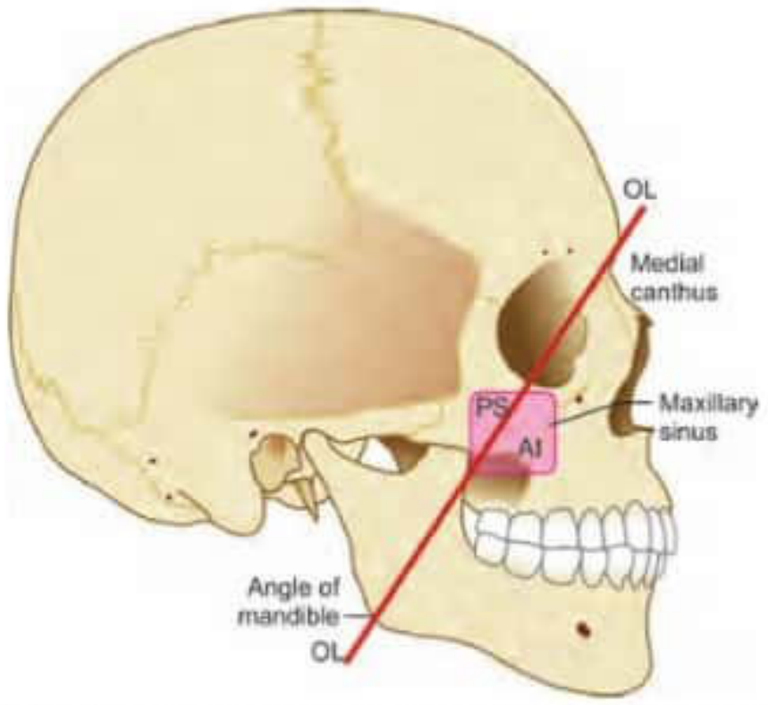
4. Le fort classification of fracture of nasomaxillary complex



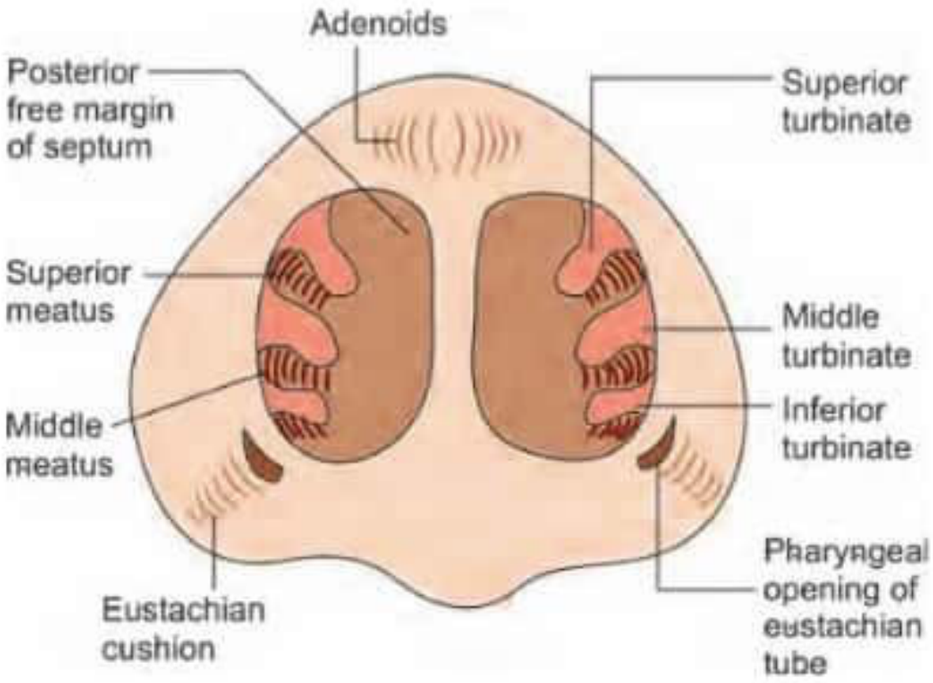
Le Fort classification of fractures of nasomaxillary complex crossing nasal septum and pterygoid plates. (I) Transverse (separating maxillary dentition); (II) Pyramidal (fracture of root of nose, medial wall and floor of orbit and maxilla), (III) Craniofacial disjunction (separating face from the cranium)

5. Ohngrens classification for malignant neoplasm of PNS

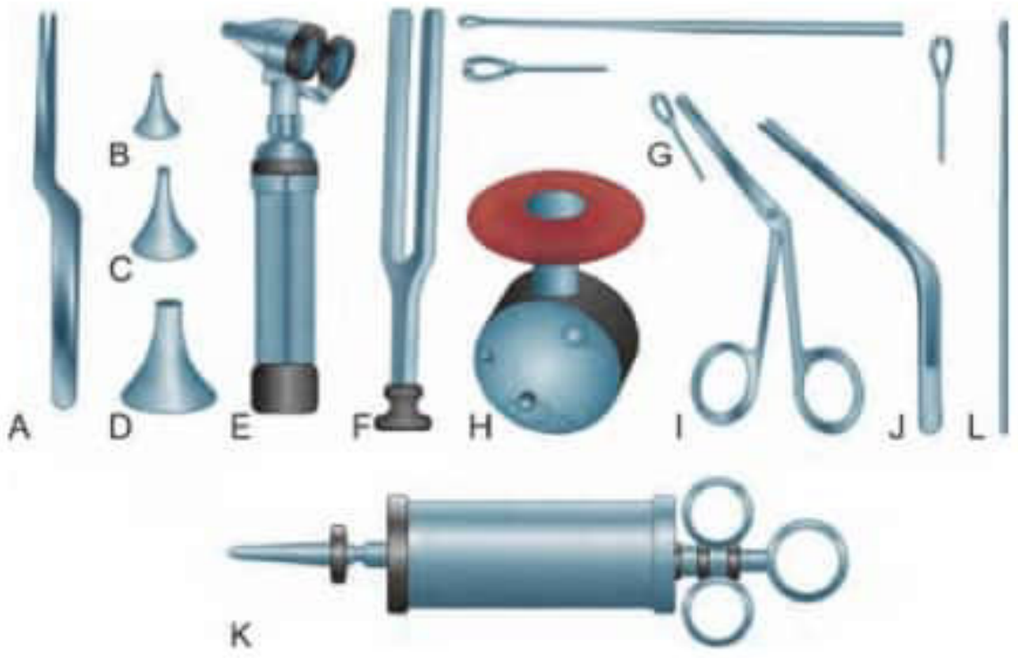


Ohngren's classification: Ohngren's line is an imaginary line (OL), which extends between medial canthus and the angle of mandible, divides the maxilla into two regions anteroinferior (AI) and posterosuperior (PA). AI growths are easy to manage and have better prognosis than PS tumors

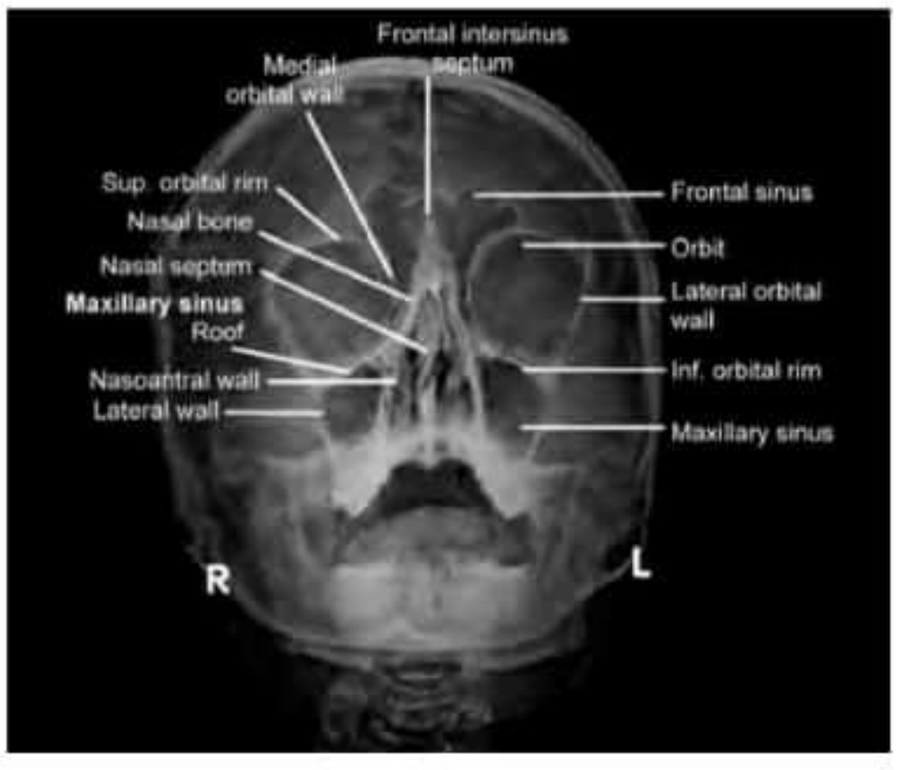
6. Structures seen an posterior rhinoscopy



7. A radiopaque foreign body in the nose of a child



8. X-ray: PNS, Water's view



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9. View for the paranasal sinuses

A. B. C.

Radiology of nasal structures: (A) Occipitomental view: (B) Occipitofrontal view: (C) Submentovertical view

It is difficult to examine all the paranasal sinuses on one projection, so the examination of individual sinus requires many views. The few standard views that are taken, which give an adequate idea about the condition of paranasal sinuses are as follow:

- **Occipitomental view (Waters view):** The X-ray is taken in the nose-chin position with an open mouth. The film demonstrates mainly the maxillary sinuses, nasal cavity, septum, frontal sinuses and few cells of the ethmoids. The view taken in the standing position may show fluid level in the antrum (Fig. A)
- **Occipitofrontal view (Caldwell view):** The patient's forehead and tip of the nose are kept in contact with the film. This view is particularly useful for fontal sinuses. A portion of the maxillary antrum and nasal cavity are also shown (Fig. B)
- **X-ray, the base of the skull (Submentovertical view):** The neck and head are fully extended so that vertex faces the film and the rays are directed beneath the mandible. The view is useful for demonstrating sphenoid sinuses, ethmoids, nasopharynx, petrous apex, posterior wall of the maxillary sinus and fracture of the zygomatic arch (Fig. C)
- **Lateral view:** The patient's head is placed in a lateral position against the film and the ray is directed behind the outer canthus of the eye towards the film.

The maxillary, ethmoidal and frontal sinuses superimpose each other but this film is useful for the following purposes:

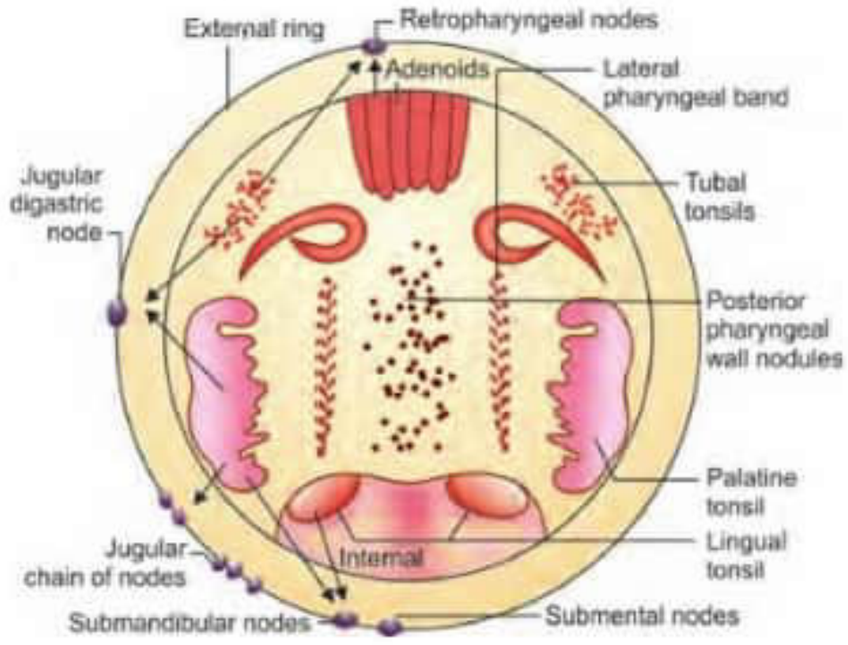
- To demonstrate the extent of pneumatization of the sphenoid and frontal sinuses.
- To demonstrate the position of a radiopaque foreign body in the nasal cavity or nasopharynx.
- To demonstrate the thickness of soft tissues of the nasopharynx which should not normally be more than 5 mm.
- To show the nasopharyngeal airway.
- To demonstrate the adenoid mass or a tumor in the nasopharynx.

- **Lateral oblique view for ethmoids:** If the disease involves the ethmoids, a special lateral oblique view provides an idea about the ethmoidal air cells, being relatively free of superimposition by other structures.

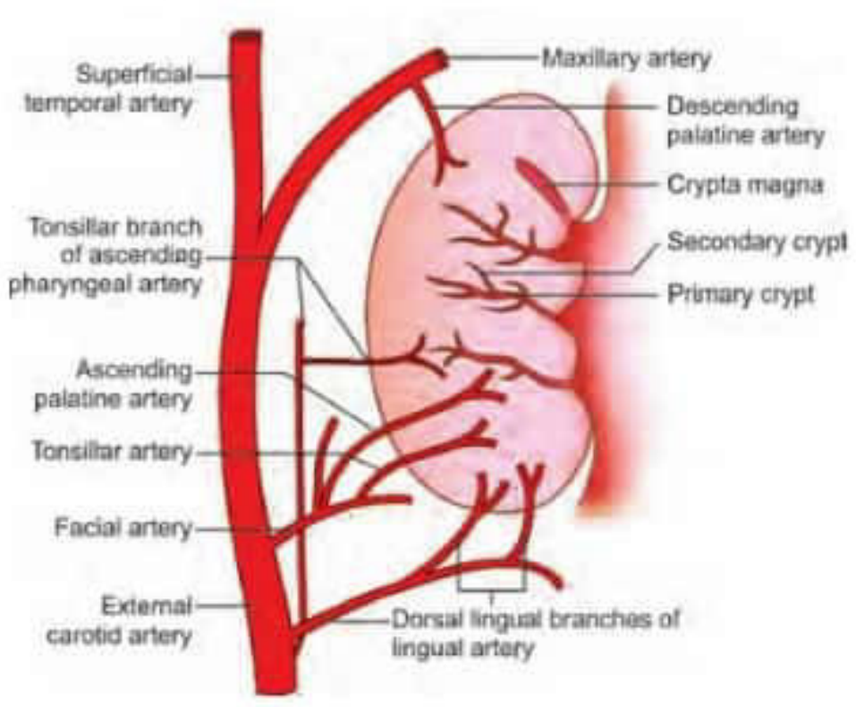
On plain radiography, the normal sinuses appear as air filled translucent cavities. Opacity of the sinuses can be caused by fluid, thickened mucosa or tumors. Bony erosion can occur because of tumors, osteomyelitis or mucoceles.

PHARYNX

1. Waldeyers ring



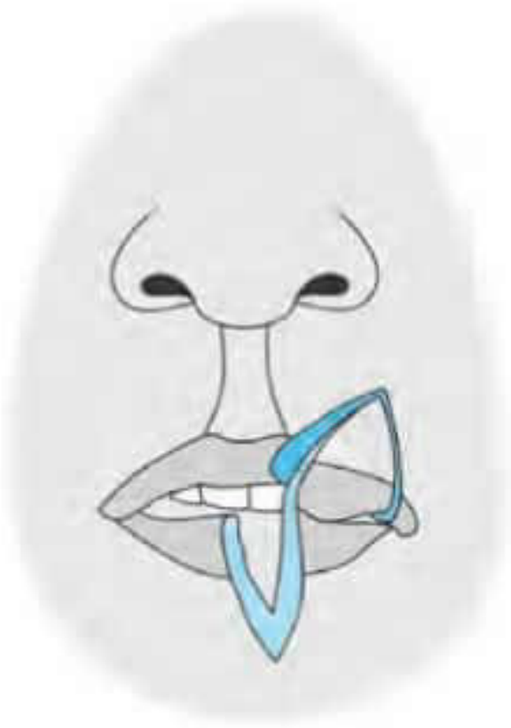
2. Blood supply of tonsil



Scanned by CamScanner

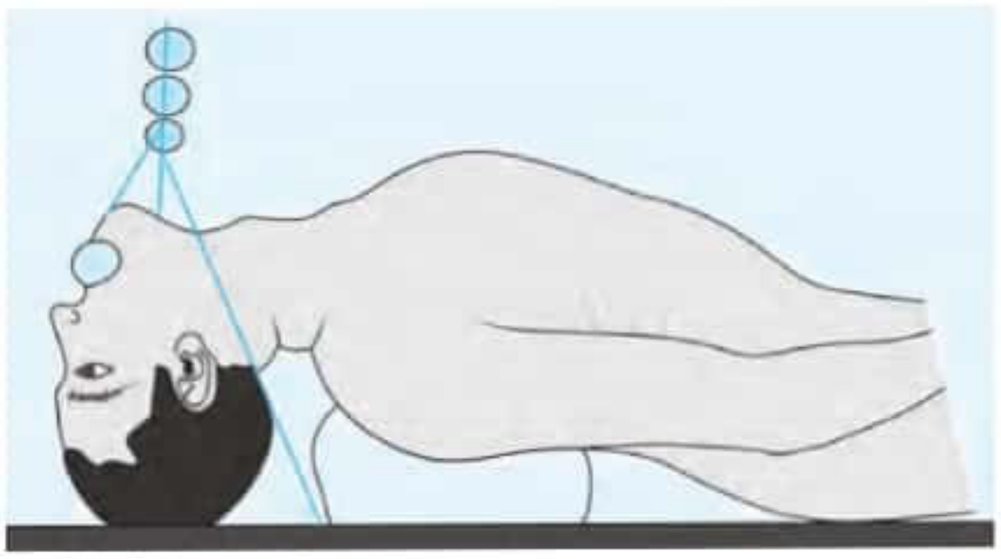
IMPORTANT INCISIONS AND POSITION IN ENT SURGERY

1. **Abbe estander flap**



Used for lip reconstruction

2. **Rose Position**

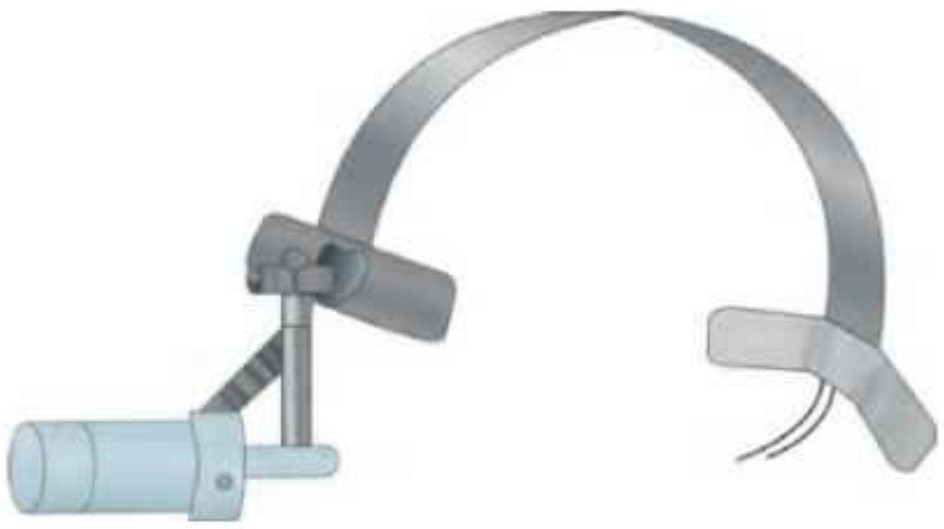


Used during I. Tonsillectomy II. Abenoidectomy III. Tracheostomy

INSTRUMENTS

1. **Head mirror**

2. **Head light**



3. **Aural speculum**

4. **Electrical otoscope**

5. **Jobson's aural probe**

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6. Tuning fork

7. Aural syringe

8. Eustachian catheter

9. Siegel's pneumatic speculum

10. Politzer bag

11. Myringotome

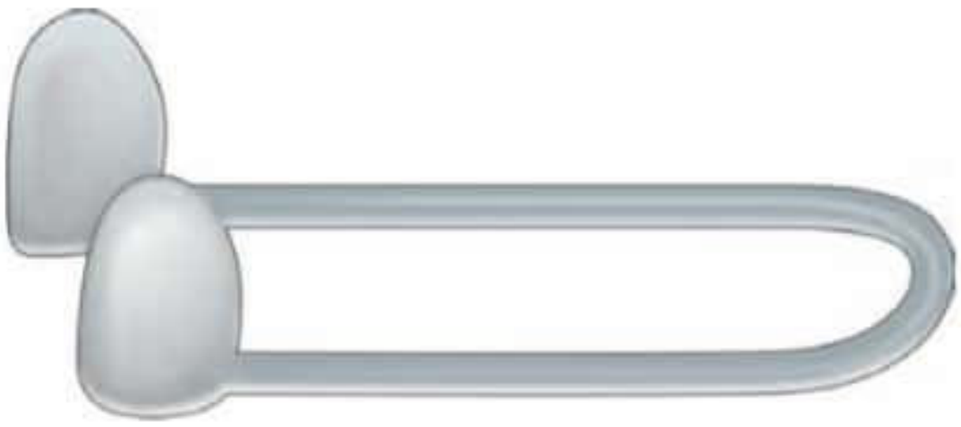
12. Mastoid retractor

13. Mastoid gouge

14. Mallet

15. Mastoid cell seeker with scoop

16. Thudicum's nasal speculum



17. Correct method of holding Thudicum's nasal speculum

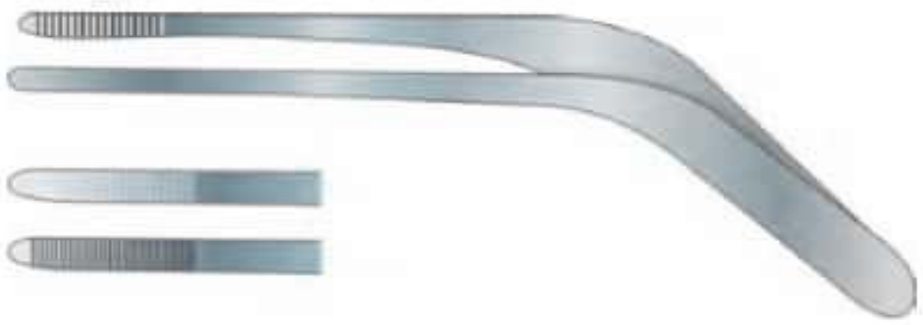
Scanned by CamScanner

18. St Clair-Thompson's nasal speculum

19. Posterior rhinoscopy mirror

20. Nasal foreign body hook

21. Nasal packing, forceps



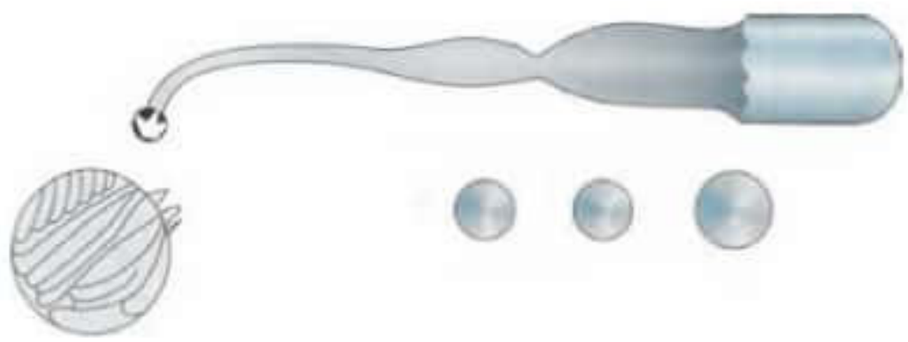
22. Antral trocar and cannula

23. Antral cannula

24. Antral perforator

25. Myle's nasoantral perforator

26. Antral burr



27. Antral wash cannula

28. Luc's forceps

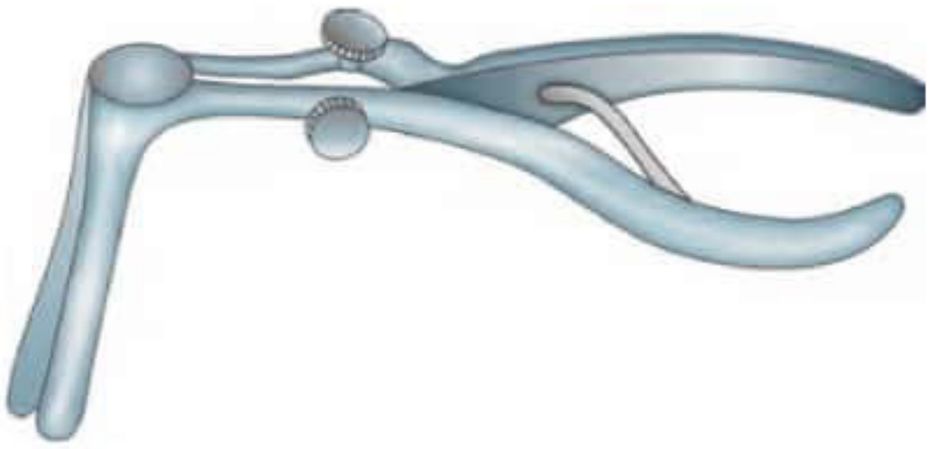
29. Nasal snare

30. Freer's septal knife

31. Long-bladed nasal speculum

Scanned by CamScanner

32. Killian's nasal speculum



33. Ballinger's swivel knife

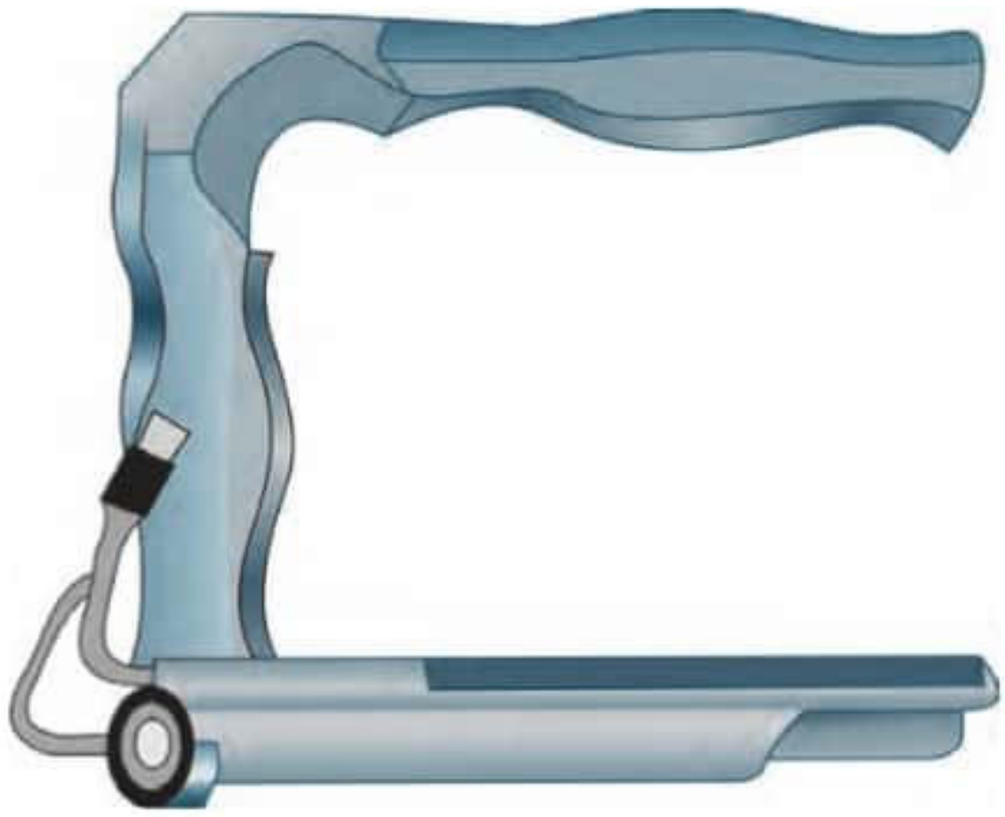
34. Bayonet-shaped gouge

35. Walsham's forceps

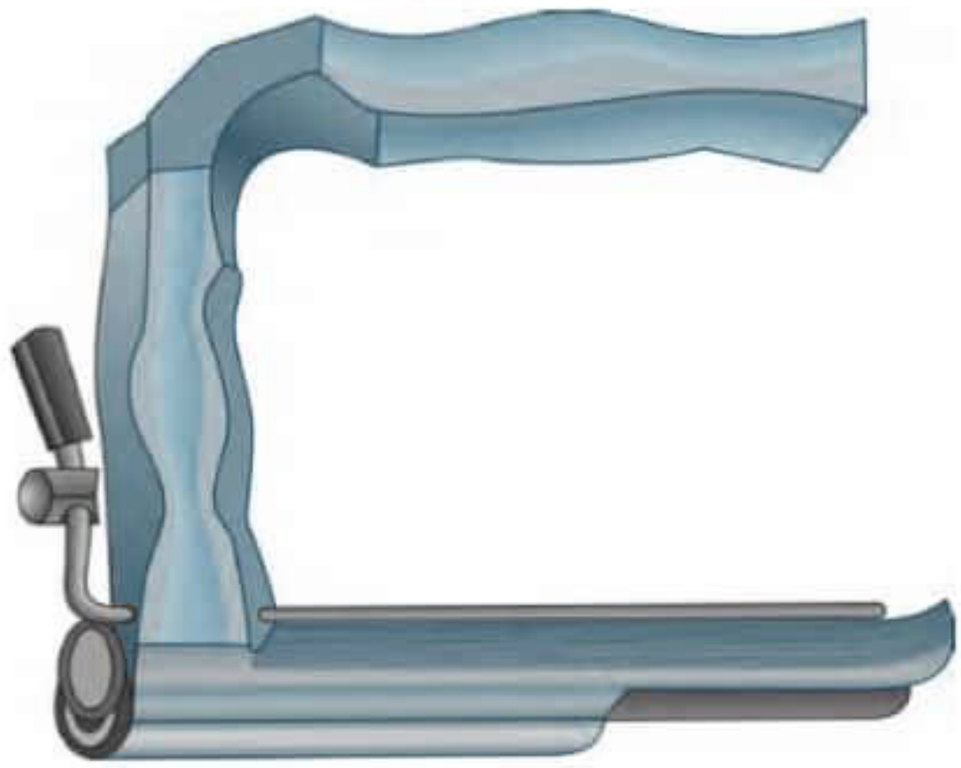
36. Lackj's spatula

37. Laryngeal mirror

38. Direct laryngoscope



39. Chevalier-Jackson laryngoscope with removable slide



40. Distal light arrangement

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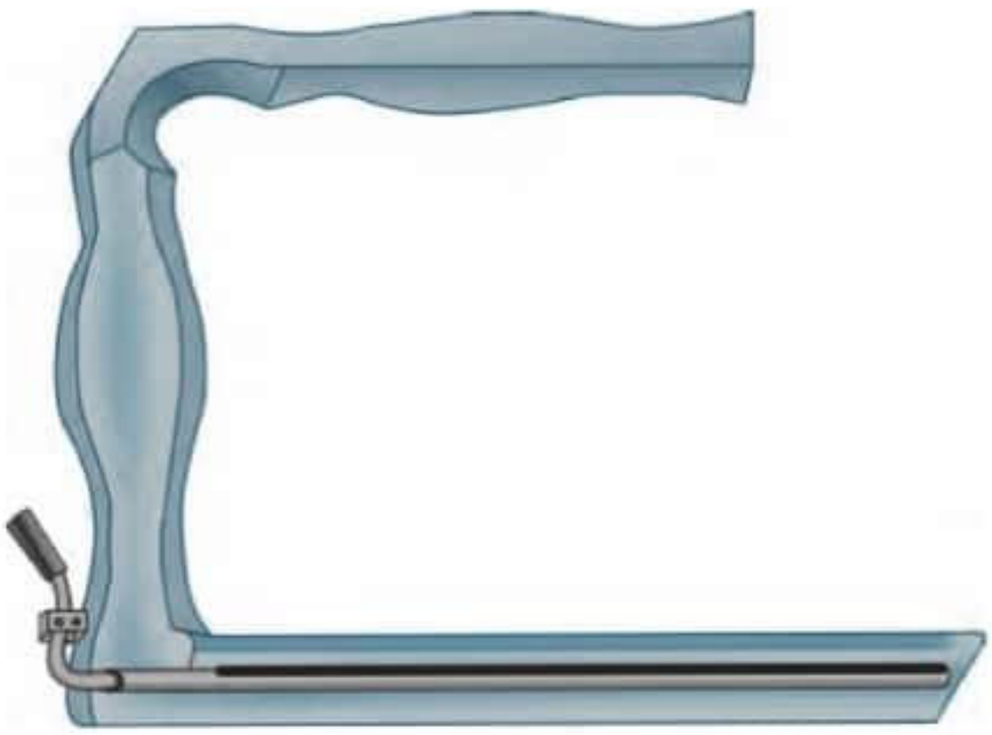
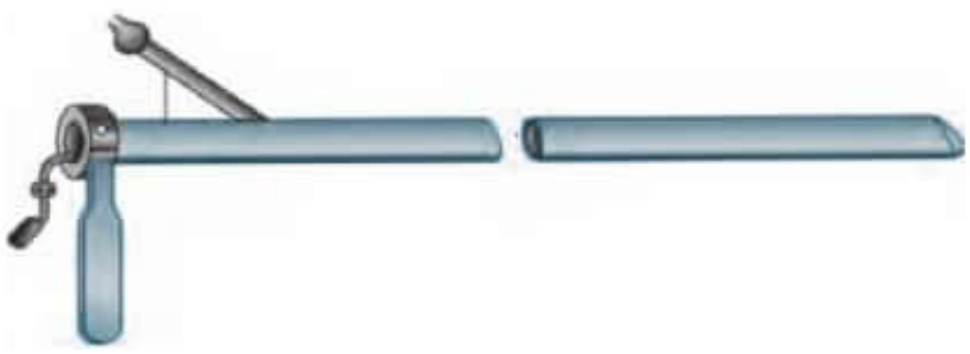
41. **Anterior commissure larynogoscope**

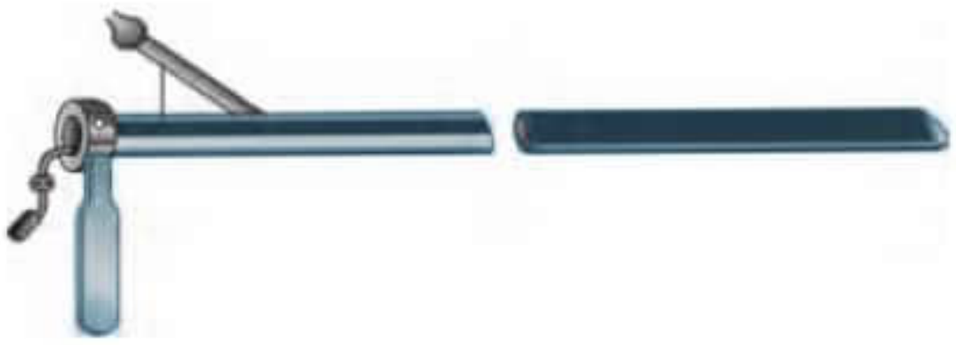


42. **Negus bronchoscope**

43. **Chevalier-Jackson bronchoscope**



44. **Chevalier-Jackson esophagoscope**



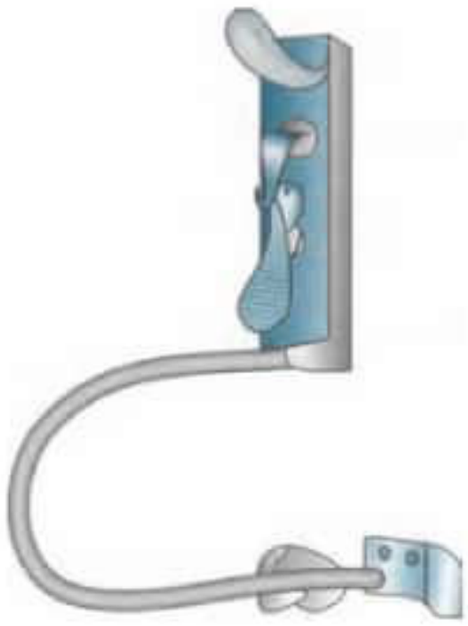
45. **Negus esophagoscope**

46. **Esophageal speculum**

47. **Laryneal forceps**

48. **Crocodile punch biopsy foreps**

49. **Boyle-Davis mouth gag**



50. **Tongue plate with throat suction**

Scanned by CamScanner

51. Tonsil holding forceps

52. Tonsillar suction

53. Tonsil pillar retractor and dissector

54. Tonsillar snare

55. Guillotine

56. Adenoid curette with cage

57. Peritonsillar abscess drainage forceps

58. Fuller's tracheostomy tube

59. Jackson's tracheostomy tube

60. Blunt tracheal hook

61. Sharp tracheal hook

62. Draffin bipod stand with plate

Scanned by CamScanner

ENT INSTRUMENTS

Head mirror

It is used to reflect the light for the examination of nasal and oral cavity

Thudichum Nasal Speculum

1) Used to see floor of the nasal cavity

Seigles pneumatic speculum

1) Used to open the ear canal and give a magnification
2) To test the mobility of tympanic membrane
3) To see a magnified image of small perforations

Killian nasal Speculum

Used in SMR or Septoplasty to keep the mucoperiosteal flaps away

Negus knot tier (Negus ligature slipper)

Used with Negus or Wilson artery forceps to help tie sutures; help to slip the ligature over the tip of Negus or Wilson forceps during ligation of vessels following tonsillectomy

Myringotome

It is used for incising the tympanic membrane in certain cases of otitis media

Posterior rhinoscopy mirror

1) For posterior rhinoscopy (seeing the inner parts of the nose like the choanae)
2) To see the eustachian tube and adenoid

Bayonet shaped gouge

1) It is used to remove the bony spur from the nasal septum during septoplasty and SMR
2) The curved edges of the instrument allows easy cutting of the sleeves of the bone

Suction Tube

Used to provide suction and remove unwanted blood, fluid and debris from the operative field

Laryngeal Mirror

Used in indirect laryngoscopy to see the base of the tongue, vallecula,

HURD TONSIL DISSECTOR and PILLAR RETRACTOR

The sharp end is used to dissect the tonsil during tonsillectomy and the retractor end is used to retractor the anterior pillor

Draffin Bipod

It is used to vary the height of mouth gag during operation

Periosteal Elevator

Used to elevate the periostium during surgery

Tilley's Harpoon

1) It is used to create an artificial passage into the maxillary sinus through the nose
2) It is used to puncture medial wall of inferior meati

Laryngoscope

Used for the direct laryngoscope
during operation and aneshtesia

Auriscope (Otoscope)

1) To examine the external auditory canal and ear drum
2) Used during aural toileting, removal of wax, myringotomy, stapedectomy and to dilate the stenosis of canal

Negus tonsil artery forcep

1) As a haemostat replaces tonsil artery forceps
2) Used to tie ligature at a depth and ligature wont slip due to its curve tip

Self Retaining Mastoid Retractor

It is used in mastoidectomy to retract the skin, subcutaneous tissue and periostium

Antral Trocar

1) Used in nasal sinus surgery
2) To conform presence of puss in maxillary sinus
3) Cytological examination of antral wash out fluid
4) Lavage of the maxillary sinus

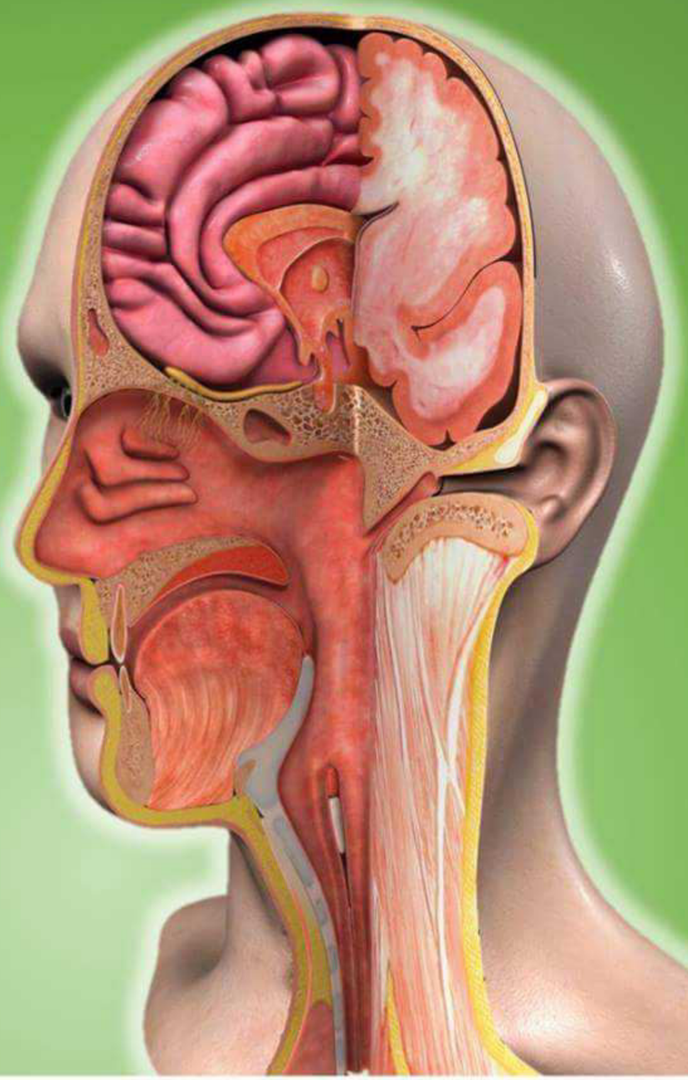
Terbinectomy Scissor

Used in terbinectomy

Boyle-Davis mouth gag and Tongue plate

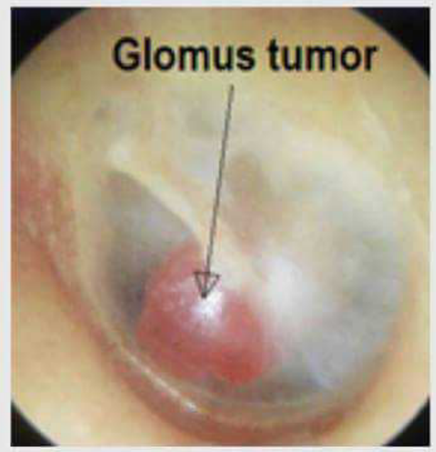
Tilley nasal dressing forcep

1) Used in the anterior part of the nasal cavity most importantly, anterior nasal packing
2) Used for removal of foriegn bodies from nasal cavity

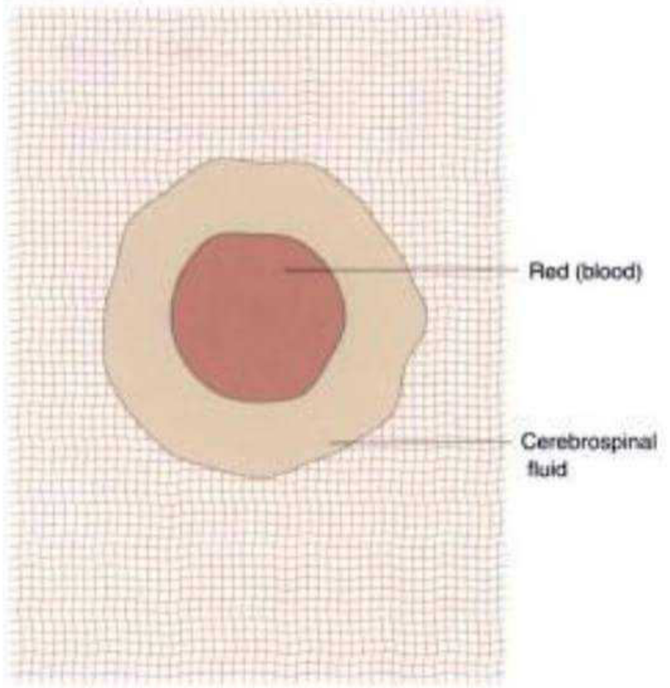


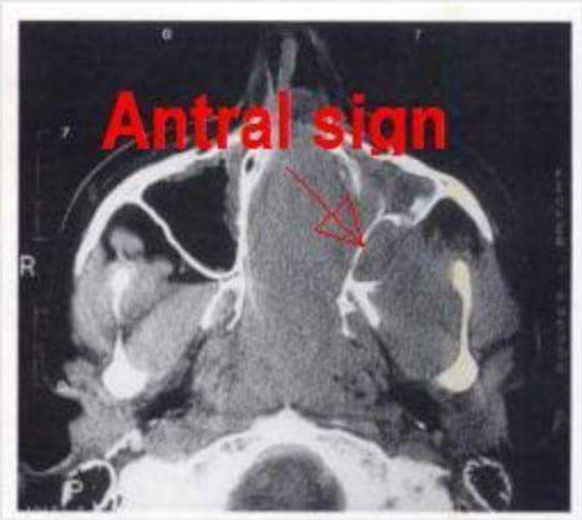
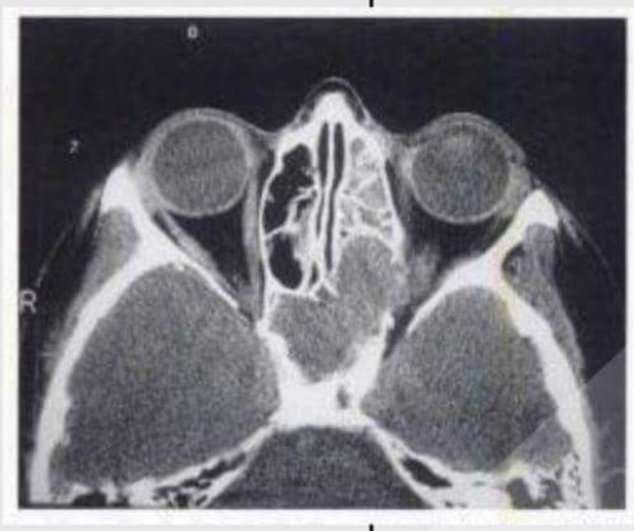
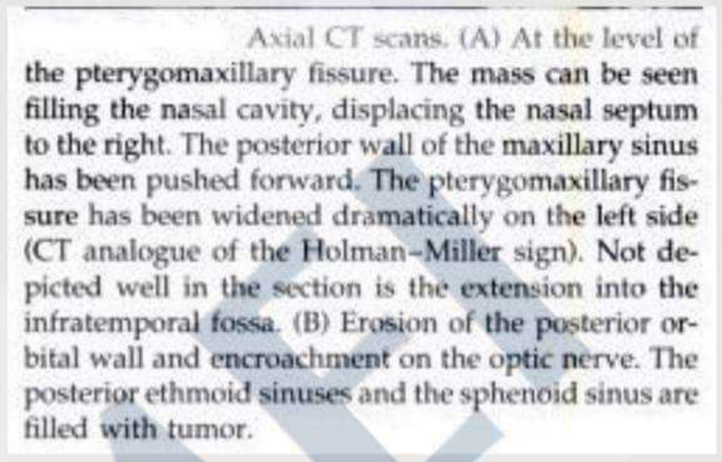
Eponymous Signs in ENT

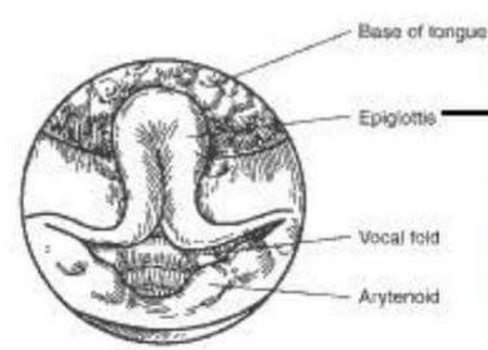
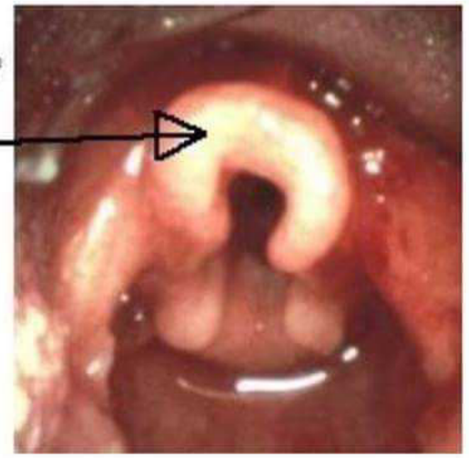
PGMEI

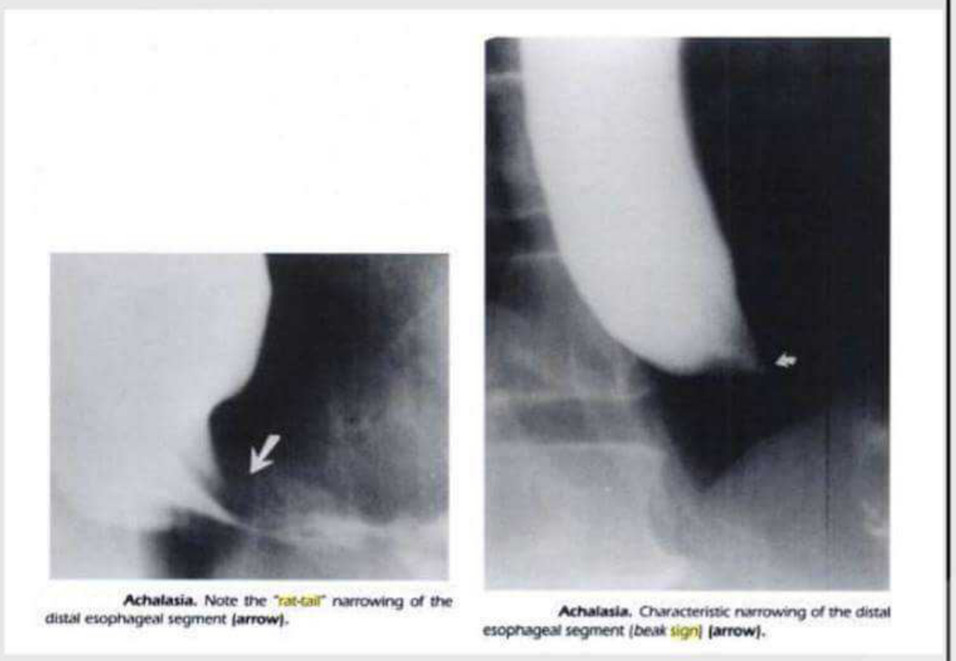
SIGN	SEEN IN	WHATS IT
AQUINO'S SIGN	Glomus tumors	blanching of the tympanic mass with gentle pressure on the carotid artery
BATTLE SIGN	Petrous temporal bone fracture (middle fossa #)	Bruising behind ear at mastoid region, due to petrous temporal bone fracture (middle fossa #)
BEZOLD'S SIGN	mastoiditis	Inflammatory edema at the tip of the mastoid process in mastoiditis
BOCCA'S SIGN	Ca Larynx	Absence of post cricoid crackle(Muir's crackle) in Ca post cricoid
BROWNE'S SIGN	Glomus tumor	blanching noted when applying positive pressure{with Siegel's speculum} to the tympanic membrane 

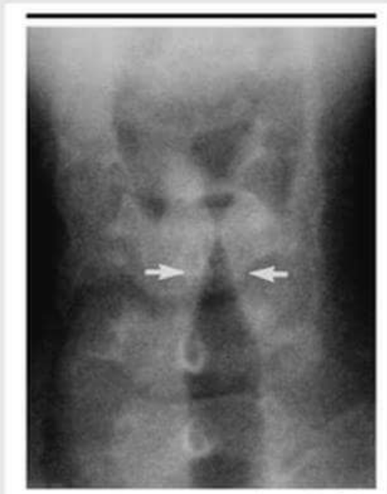
SIGN	SEEN IN	WHATS IT
BRYCE SIGN	combined laryngocele & external laryngocele	If combined laryngocele & external laryngocele is presenting as a neck mass, compression will cause a hissing sound as the air escapes from it into the larynx. This test is fraught with danger in cases of combined laryngoceles because air from the external component may get forced into the internal component causing acute airway obstruction.
DELTA SIGN	Lateral sinus thrombosis	CT or MRI with contrast shows an empty triangle appearance of the thrombosed sinus surrounded by contrast enhanced dura (empty triangle sign)
DODD'S SIGN / CRESCENT SIGN	positive in AC ployp Negative in Angiofibroma	X-ray finding-Crescent of air between the mass and posterior pharyngeal wall
FURSTENBERG'S SIGN	Positive in Encephaloceles	Owing to the intracranial connection, there is pulsation and expansion of the mass with crying, straining, or compression of the jugular vein (Furstenberg test).This is used to differentiate Nasal Encephaloceles from other congenital midline nasal masses like Nasal Gliomas

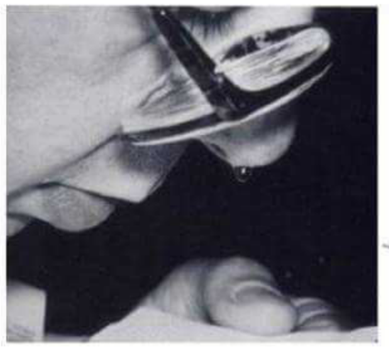
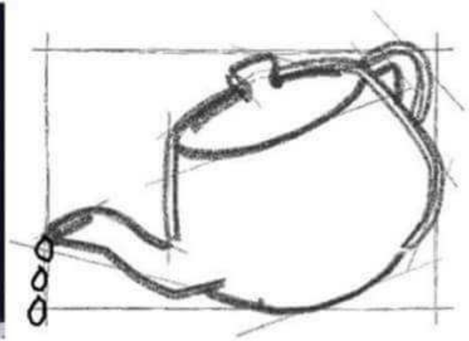
SIGN	SEEN IN	WHATS IT
GRIESINGER'S SIGN	lateral sinus thrombosis	Erythema and oedema posterior to the mastoid process resulting from septic thrombosis of the mastoid emissary vein
HALO SIGN/ HANDKERCHIEF SIGN	CSF rhinorrhea when CSF is mixed with Blood.	In patients with head trauma, a mixture of blood and CSF may make the diagnosis difficult. CSF separates from blood when it is placed on filter paper, and it produces a clinically detectable sign: the ring sign, double-ring sign, or halo sign.  Halo sign. Clear drainage that separates from bloody drainage suggests the presence of cerebrospinal fluid.
HITSELBERGER'S SIGN	Acoustic neuroma	loss of sensation in the postero-superior part of external auditory meatus supplied by Arnold's nerve(branch of Vagus nerve to ear)
HONDOUSA SIGN	X-ray finding in Angiofibroma	indicating infratemporal fossa involvement characterised by widening of gap between ramus of mandible and maxillary body

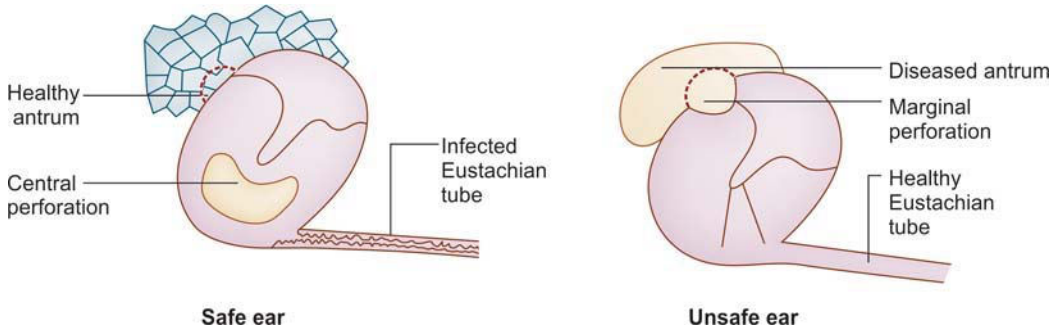
SIGN	SEEN IN	WHATS IT
HOLMAN MILLER SIGN, ANTRAL SIGN	 	The anterior bowing of the posterior wall of the antrum seen on lateral skull film .**Pathognomic for juvenile nasopharyngeal angiofibroma**.  Axial CT scans. (A) At the level of the pterygomaxillary fissure. The mass can be seen filling the nasal cavity, displacing the nasal septum to the right. The posterior wall of the maxillary sinus has been pushed forward. The pterygomaxillary fissure has been widened dramatically on the left side (CT analogue of the Holman–Miller sign). Not depicted well in the section is the extension into the infratemporal fossa. (B) Erosion of the posterior orbital wall and encroachment on the optic nerve. The posterior ethmoid sinuses and the sphenoid sinus are filled with tumor.
IRWIN MOORE'S SIGN	**chronic tonsillitis**	positive squeeze test in **chronic tonsillitis**
LAUGIER'S SIGN	**basilar skull fracture**	Blood behind the eardrum suggests **basilar skull fracture**
LEUDET'S SIGN		Inflammation of the eustachian tube can produce a bright clicking sound heard by the examiner through the otoscope while the patient experiences it as tinnitus.caused by reflex spasm of the tensor palati muscle.
LIGHT HOUSE SIGN	**Acute suppurative otitis media**	A small pin hole perforation with a pulsatile ear discharge is seen in **Acute suppurative otitis media.**

SIGN	SEEN IN	WHATS IT
MILIAN'S EAR SIGN	**Erysipelas**	➢can spread to pinna(cuticular affection), where as cellulitis cannot ➢erysipelas involves the upper dermis and superficial lymphatics, whereas cellulitis involves the deeper dermis and subcutaneous fat **Involvement of the ear (Milian's ear sign) is a distinguishing feature for erysipelas since this region does not contain deeper dermis tissue.**
OMEGA SIGN	**LARINGOMALACIA**	INFANTILE OMEGA SHAPED EPIGLOTTIS SEEN IN **LARINGOMALACIA**.  Appearance of the infantile larynx in laryngomalacia. 
Paul Dudley White's winking ear lobe sign	**tricuspid insufficiency**.	Movement of the ear lobe coincident with the pulse suggests **tricuspid insufficiency**.
PHELP'S SIGN	**Glomus jugulare**	loss of crest of bone (as seen in CT-scan) between carotid canal and jugular canal in **glomus jugulare**.

SIGN	SEEN IN	WHATS IT
RAT TAIL SIGN /"Bird-beak" sign-	**barium swallow of achalasia**  **Achalasia.** Note the "rat-tail" narrowing of the distal esophageal segment **(arrow).** **Achalasia.** Characteristic narrowing of the distal esophageal segment (beak sign) **(arrow).**	he oesophagus is dilated, and contrast material passes slowly into the stomach as the sphincter opens intermittently. The distal oesophagus has a narrow segment and the image resembles a bird's beak
RISING SUN SIGN	**glomus tumour, high jugular bulb and aberant carotid artery in the floor of middle ear**	There is red vascular hue seen behind the intact tympanic membrane. it is seen in **glomus tumour, high jugular bulb and aberant carotid artery in the floor of middle ear**.
RACCOON SIGN		Indicate subgaleal hemorrhage,and not necessarly base of skull .
SCHWARTZ SIGN	**active phase of otosclerosis**	It is also called flamingo flush sign. it is seen because of increased vascularity in submucous layer of promontory in **active phase of otosclerosis**(otospongiosis).
STANKIEWICK'S SIGN	**orbital injury during FESS**	indicate **orbital injury during FESS**. fat protrude in to nasal cavity on compression of eye ball from outside .

SIGN	SEEN IN	WHATS IT
STEEPLE SIGN	**Acute laryngotracheobronchitis** Normal anteroposterior radiograph of the upper airway, with the normal appearance of the subglottic region (arrows).  Anteroposterior radiograph of the upper airway of a patient with croup. The subglottic tracheal narrowing produces an inverted V appearance known as the steeple sign (arrows).	X-ray finding in **Acute laryngotracheobronchitis** (CROUP).The **steeple sign** is produced by the presence of edema in the trachea, which results in elevation of the tracheal mucosa and loss of the normal shouldering (lateral convexities) of the air column.
TEAR DROP SIGN	**Orbital floor fracture** 	Seen in **Orbital floor fracture**. It is defined as tear drop shaped opacification seen hanging from the roof of the maxillary sinus on water's view. The floor of the orbit is the most common portion of the orbit to sustain fracture. A classic radiographic finding in blow-out fractures is the presence of a polypoid mass (the tear-drop) protruding from the floor of the orbit into the maxillary antrum The tear-drop represents the herniated orbital contents, periorbital fat and inferior rectus muscle.
THUMB SIGN		It is a thumb like impression (due to enlarged epiglottis) seen on X-ray lateral view neck in patients with acute **epiglottitis**.Direct visualization of the epiglottis by laryngoscope, if attempted, reveals a beefy red, edematous epiglottis.

SIGN	SEEN IN	WHATS IT
TRAGUS SIGN	**acute otitis externa**	In **acute otitis externa** there is marked tenderness when tragus is pressed against the pinna.
TEA POT SIGN	**CSF rhinorrhoea**  	is seen in **CSF rhinorrhoea**.This could be related to the relationship of the sphenoid ostium to the sinus floor. The sphenoid ostium lies at an appreciable distance anterosupe-rior from the sinus floor. An increase in the CSF rhinorrhea therefore occurs in a case of sphenoid sinus leak when the patient bends forward as an increasing amount of CSF gains access to the ostium "teapot" sign.
Uvula pointing sign		seen in rhinoscleroma .when scleroma involve nasopharynx ,uvula point towards roof of nasopharynx.
WOODS SIGN		palpable jugulodigastric lymphnodes



Figs 5.58A and B: CSOM safe and unsafe

The tympanic membrane is likely to get perforated either by trauma or by vascular necrosis as frequently seen in cases of chronic otitis media. The atticoantral disease may cause tympanic membrane to retract and result in retraction pocket which ultimately turns into a perforation. In safe Chronic suppurative otitis media (CSOM), a central perforation occurs, while in unsafe CSOM, a marginal or an attic perforation results.

Secretory otitis media is collection of nonpurulent fluid in the tympanum and may present as a hair line or air bubbles behind the intact tympanic membrane.

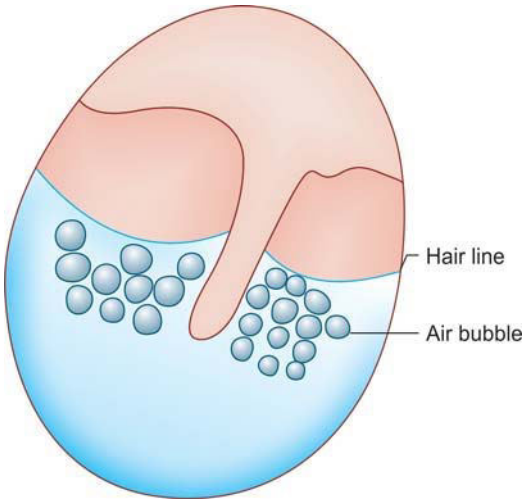


Fig. 5.59: Secretory otitis media

Secretory otitis media is usually bilateral. In a case of unilateral secretory otitis media in an adult, nasopharynx must be examined to rule out any growth or tumor.

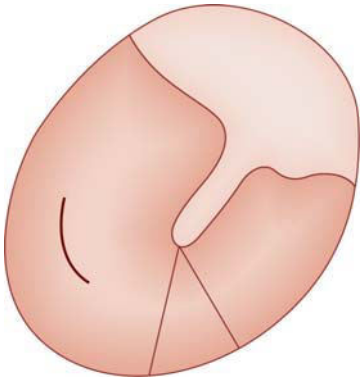


Fig. 5.60: Myringotomy–incision

Myringotomy is done in secretory otitis media when medical treatment fails. Myringotomy is necessary to relieve the pus from the middle ear in case of acute otitis media with complications like facial weakness. The incision and instruments needed for myringotomy as shown in Figures 5.60 to 5.62.

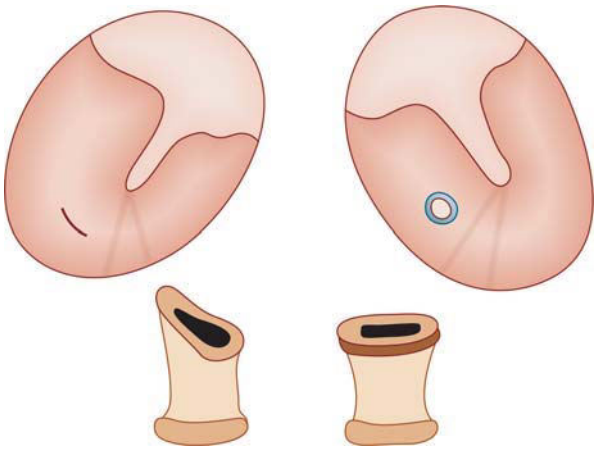


Fig. 5.61: Ventilation tube

Secretory otitis media is treated by myringotomy and insertion of ventilation tubes (grommet) when medical treatment fails.

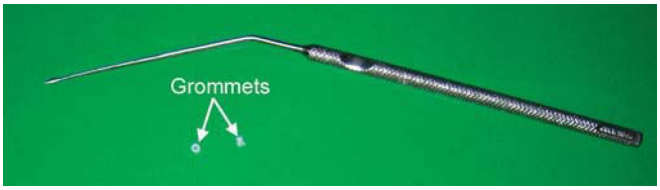


Fig. 5.62: Myringotomy knife and grommet

Table 5.11: Classification of chronic otitis media
Nonsuppurative
• Serous otitis media
• Glue ear or secretory otitis media
Suppurative
• Tubotympanic
• Atticoantral
Tuberculous

Chronic otitis media is infection of the middle ear of long duration. Classification of chronic otitis media is shown in the chart.

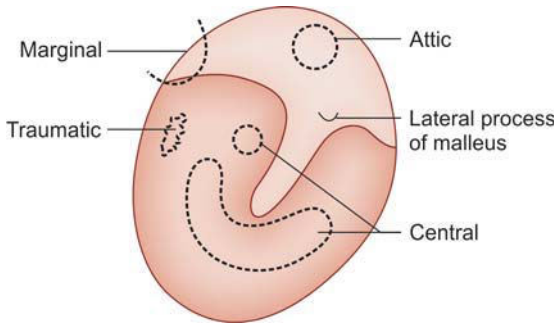


Fig. 5.63: Various perforations of tympanic membrane

Chronic suppurative otitis media due to the tubotympanic pathology produces central perforation and due to atticoantral pathology it causes posterosuperior or an attic perforation and is associated with cholesteatoma.

AURAL POLYP

Fig. 5.64: Aural polyp

The atticoantral disease produces osteitis. This initiates formation of granulation tissue which may present as ear polyp.

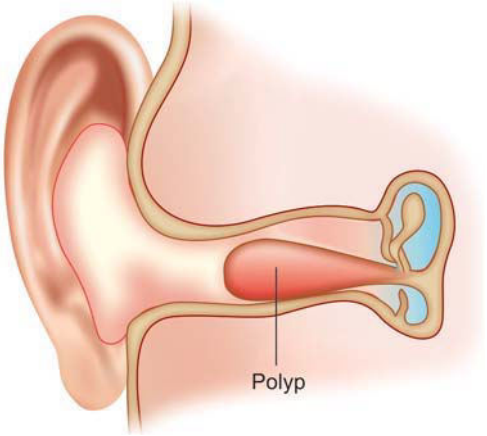


Fig. 5.65: Diagrammatic representation aural polyp

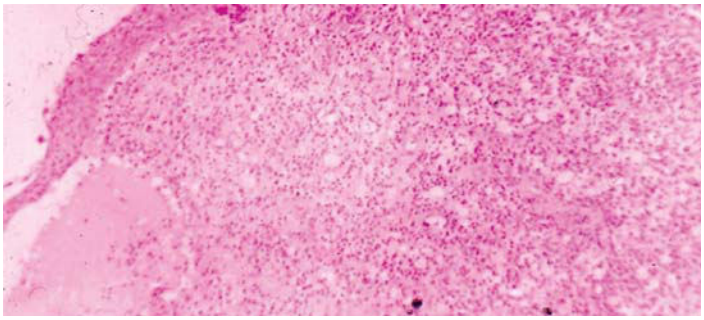


Fig. 5.66: Histopathological examination of aural polyp

Photomicrograph of the ear polyp shows marked inflammation of the connective tissue covered with mucosal lining.

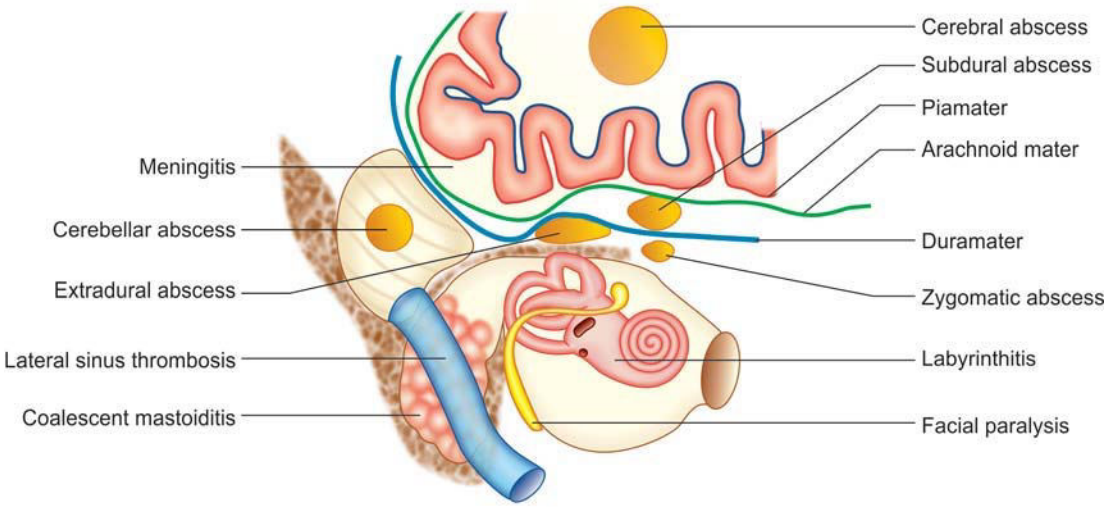


Fig. 5.67: Complications of CSOM

Chronic suppurative otitis media associated with cholesteatoma results in several complications through thrombophlebitis and osteitis. Figures 5.68 and 5.69 showing the various complications of otitis media is shown.

Fig. 5.68: Zygomatic abscess

Fig. 5.69: Postaural abscess

A zygomatic abscess (Fig. 5.68) or a postaural abscess (Fig. 5.69) can form as a complication of acute otitis media or acute exacerbation of the chronic otitis media.

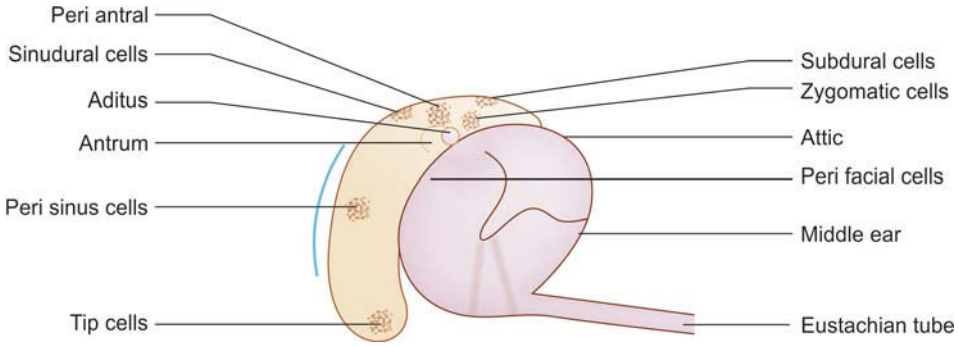
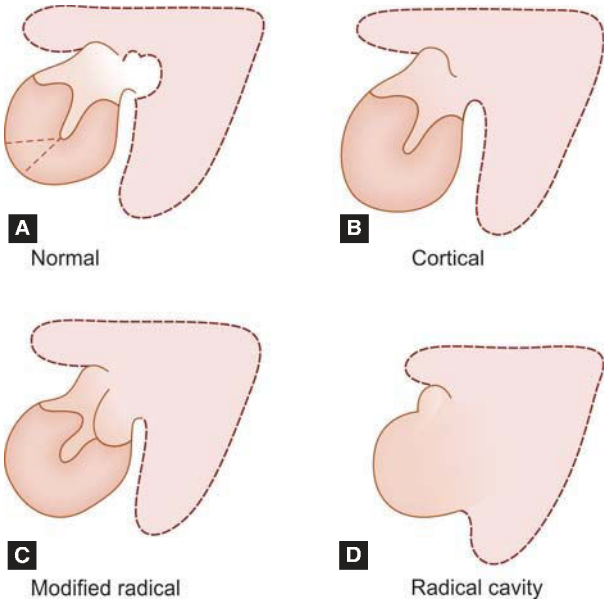


Fig. 5.70: Middle ear cleft and mastoid air cells



Figs 5.71A to D: Mastoidectomy

Mastoidectomy is an operation to provide relief in case of mastoiditis. Surgery is classified into three categories, simple or cortical, radical and modified radical mastoidectomy according to the extent of the surgical removal.

Fig. 5.72: Mastoidectomy instruments

The instruments required for the operation of mastoidectomy are shown. A bone drill makes it precise as well as minimizes the time for surgery.

Table 5.12: Prerequisites for tympanoplasty

• Good cochlear reserve
• Good Eustachian tube function
• Dry ear
• Healthy middle ear mucosa
• Sound transmission to oval window and sound protection to round window.

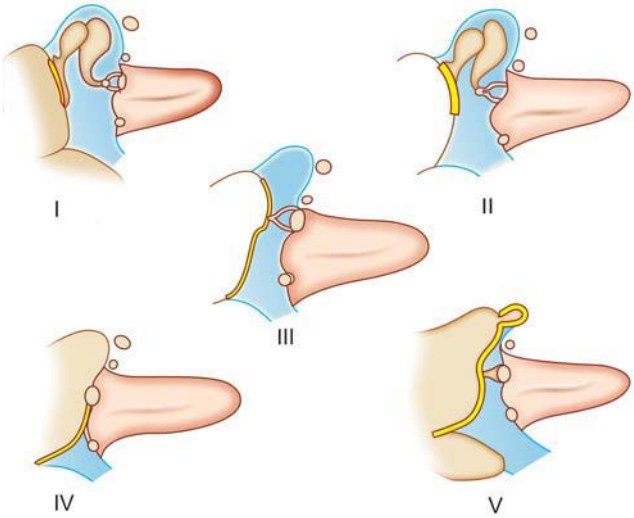


Fig. 5.73: Types of tympanoplasty

Various types of tympanoplasty with repair of the hearing mechanism in the ear are shown.

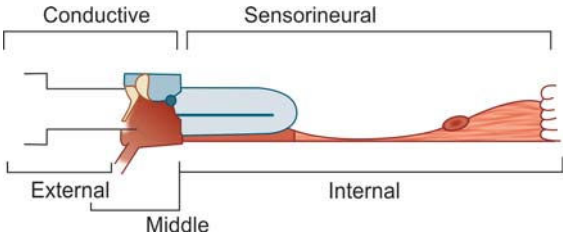


Fig. 5.74: Causes of deafness

Deafness is due to defective transmission or abnormal perception of the sound waves. The cause may be in the external, middle or inner ear mechanism.

Table 5.13: Deafness: External, middle and inner ear causes

External ear	*Middle ear*	*Internal ear*
Atresia	Congenital	Congenital
Wax	Otitis media	Viral
Foreign body	Otosclerosis	Presbycusis
Otitis externa	Trauma	Ménière's syndrome
Trauma	Tumor	Noise trauma
Tumor		Ototoxic drugs
		Head injury
		Labyrinthitis
		Vascular
		Idiopathic

Table 5.14: Management of deafness

Medical
• Antibiotics
• Antihistaminics and decongestants
• Suction, aural toilet, topical antibiotics
Surgical
• Adenoidectomy
• Myringotomy and grommet insertion
• Tympanoplasty
• Stapedotomy
• Bone anchored hearing aid (single sided deafness, congenital deafness, deafness with cavity problems)
• Cochlear implants
• Brainstem implants after acoustic neuroma excision in NF2
Rehabilitation
• Hearing aid
• Lip reading

In majority of cases deafness can be treated. Complete deafness before the speech development (prelingual) needs an approach different from cases where deafness is moderate and speech has developed (postlingual).

SUDDEN SENSORINEURAL HEARING LOSS: AN OTOLOGIC EMERGENCY

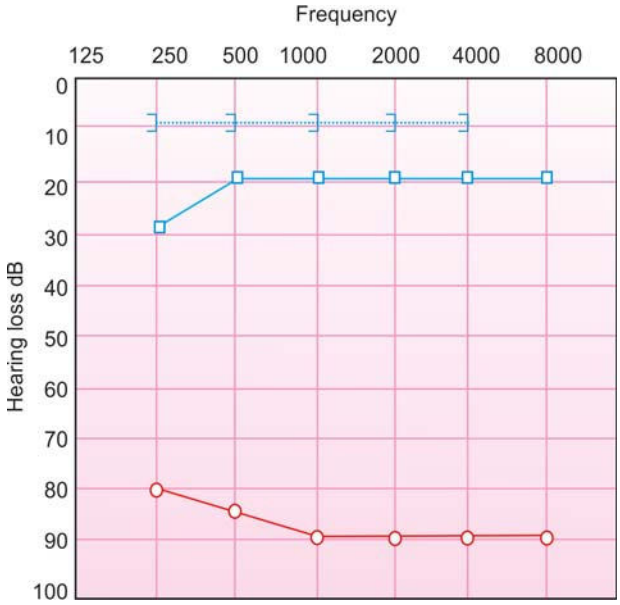


Fig. 5.75: Sudden sensorineural hearing loss in a telecaller

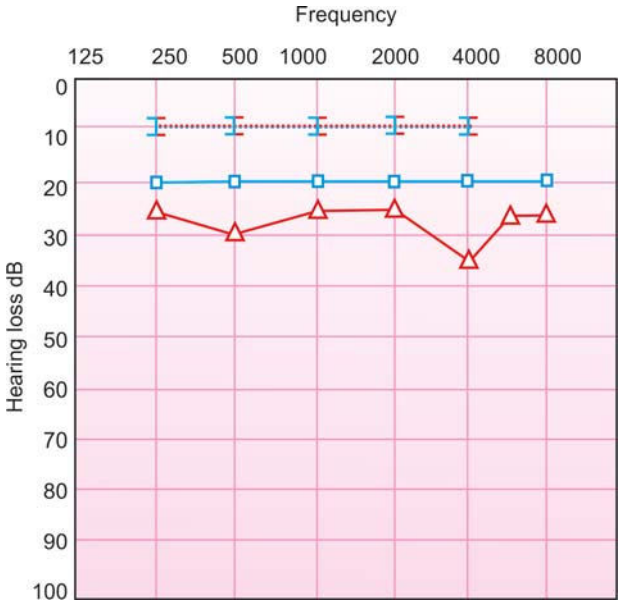


Fig. 5.76: Complete recovery of same patient as in 5.75 after 3 weeks [treated with oral steroids (prednisolone, 1 mg/kg/day), acyclovir, cinnarizine, Vitamin B12, low salt diet]

COCHLEAR IMPLANT

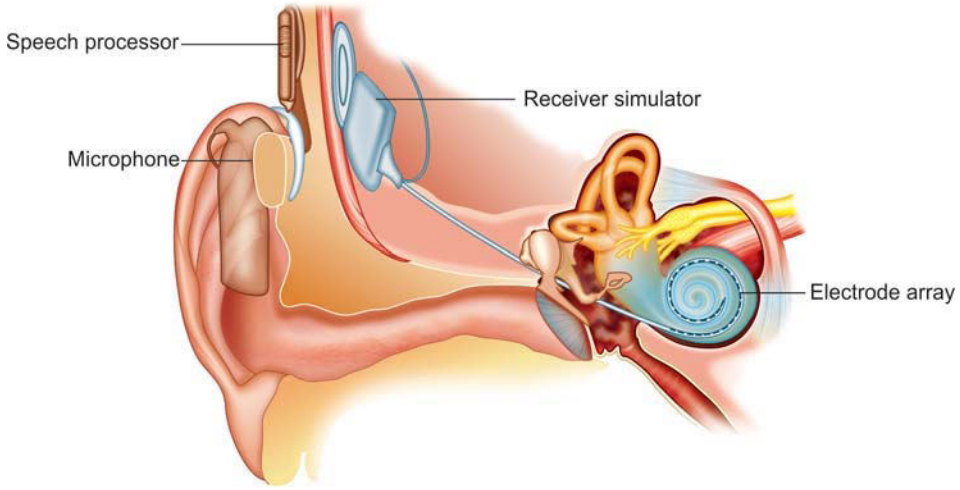


Fig. 5.77: Cochlear implant

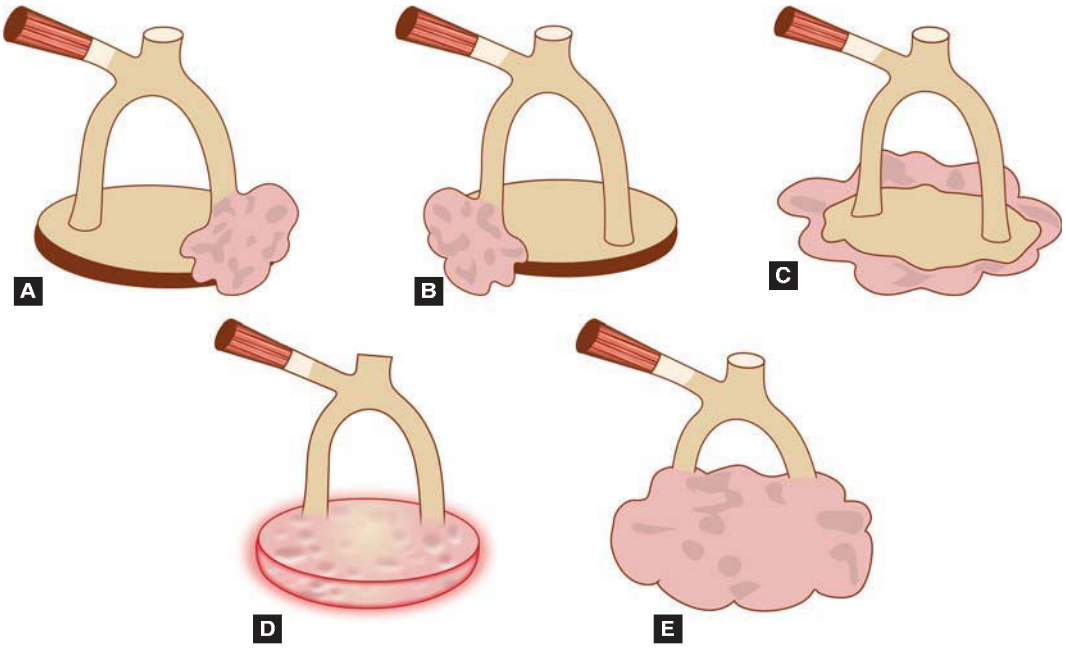
Table 5.15: How does a cochlear implant work?
• Sound is received by microphone
• Sound is sent from microphone to speech processor
• Speech processor selects and codes useful sounds
• Code is sent to transmitting coil
• Transmitting coil sends code across skin to receiver/stimulator
• Receiver/stimulator converts code to electrical signals
• Electrical signals are sent to electrodes to stimulate VIII nerve fibers
• Signals are recognized as sounds by the brain, producing a hearing sensation

Table 5.16: Cochlear implant rehabilitation
Postsurgery
• Switch on—first tune up and program the implant
• Regular mapping over next few weeks
Listening training
• An individual program is worked out for each recipient
• Focus is on listening
• For children: Listening training follows normal child speech and language developmental milestones (hearing age vs chronological age)
• Family must be involved with children. The parents are trained to carry over listening to home in a natural environment

OTOSCLEROSIS

Table 5.17: Otosclerosis

Clinical features
• Progressive conductive deafness
• Age 20–40 years
• Females more than males (2:1)
• Familial tendency present (50%)
• Commonly bilateral (70%)
• Paracusis Wilsii (patient hears better in noisy environment)
• Normal tympanic membrane
• Advanced case may have sensorineural deafness (Due to cochlear involvement)
• Flamingo red sign in active otosclerosis (Schwartz sign)
Pathology
• Replacement of normal lamellar bone by spongy vascular bone of increased cellularity and density which fixes the footplate of stapes
Treatment
• Stapedotomy
• Hearing aid



Figs 5.78A to C: Types of otosclerotic focus (A) Anterior; (B) Posterior; (C) Annular; (D) Biscuit; (E) Obliterative

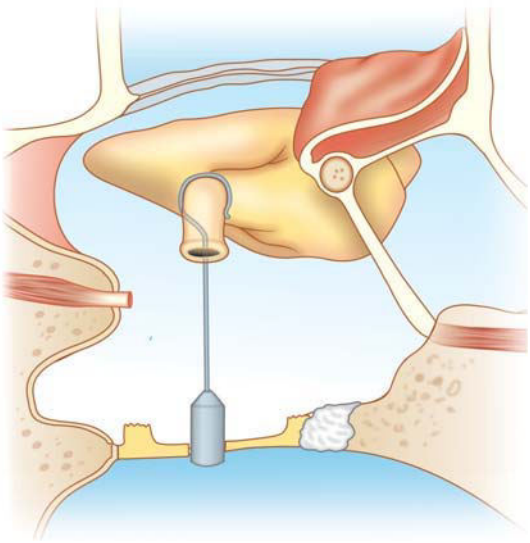


Fig. 5.79: Stapes piston

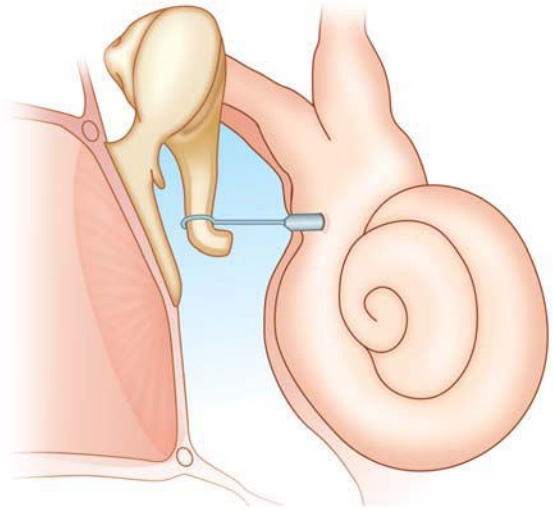


Fig. 5.80: Stapes piston *in situ* in coronal view

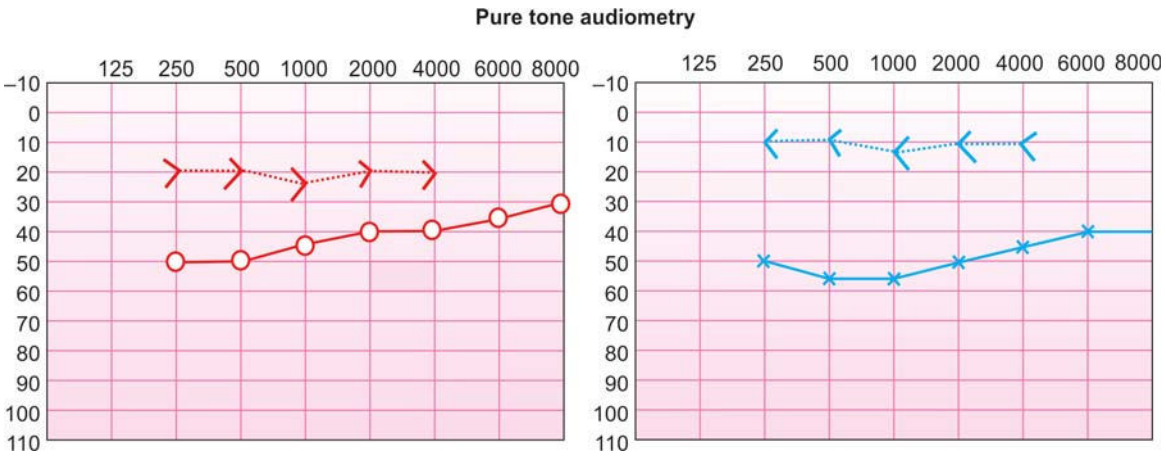


Fig. 5.81: Audiogram of a patient with otosclerosis showing bilateral conductive hearing loss

A bilateral progressive conductive deafness in a young adult female who has history of familial tendency is generally due to otosclerosis. The deafness aggravates during pregnancy.

Tympanometry usually shows type As curve with reduced compliance. Stapedial reflex is absent in late stages of the disease.

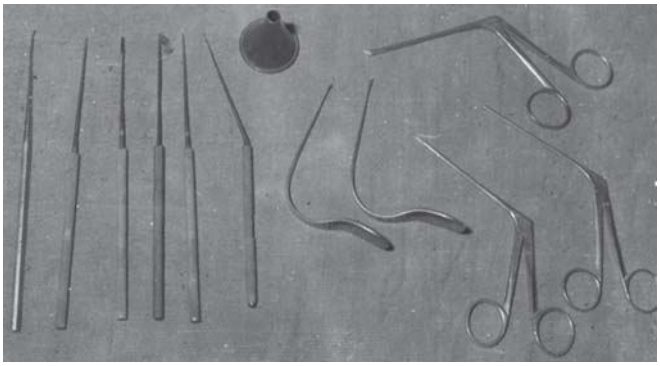


Fig. 5.82: Instruments for stapedotomy

In otosclerosis stapes gets fixed by new bone formation, stapedotomy is the recommended treatment. A set of instrument used for stapedotomy is shown. One may be benefited by using a hearing aid if the operation is not undertaken.

Table 5.18: Causes of vertigo

Ear	*Central*	*Vascular*	*Miscellaneous*
Labyrinthitis	Migraine	Cervical vertigo	Orthostatic hypertension
Vestibular neuronitis	Epilepsy	Vertebrobasilar insufficiency	Hypoglycemia
Cupulolithiasis	Head injury	Vascular accidents	Hyperventilation syndrome
Perilymphatic fistula	Multiple sclerosis		Hypothyroidism
Ménière's disease	Drugs		Hormonal imbalance
Syphilitic vertigo			
Acoustic neuroma			
Ototoxic drugs			
Herpes oticus			
Eustachian tube dysfunction			
Wax			

Maintenance of satisfactory equilibrium depends on the coordination between labyrinth, muscles, tendons, joint sense, skin receptors and visual centers. Any interference with coordination of these components leads to a sense of vertigo. A detailed history and thorough examination of the ear is of paramount importance to find out the nature and thereby the causes of vertigo. Causes of vertigo are listed in the Table 5.18. Treatment of the detected cause and suppression of the vertigo by antivertigo drugs cinnarizine, betahistine or prochlorperazine followed by rehabilitation is the line of management.

MÉNIÈRE'S DISEASE

Table 5.19: Ménière's disease

Cardinal symptoms
• Episodic vertigo
• Fluctuating deafness
• Tinnitus.
Other symptoms
• Aural fullness
• Headache
• Nystagmus
• Anxiety state.
Pathology
• Endolymphatic hydrops.

Ménière's disease is characterized by acute vestibular and cochlear symptoms. Its attacks come in bouts.

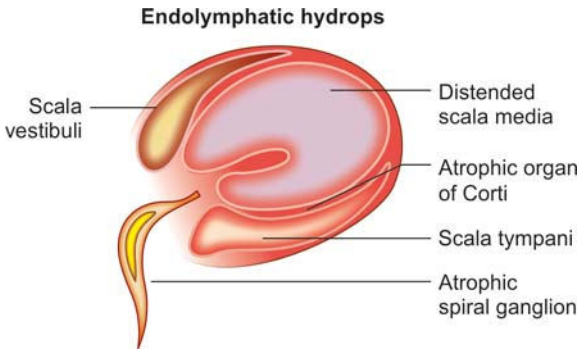


Fig. 5.83: Pathogenesis of Ménière's disease

Figure 5.83 shows the endolymphatic hydrops. This is a characteristic finding of Ménière's disease.

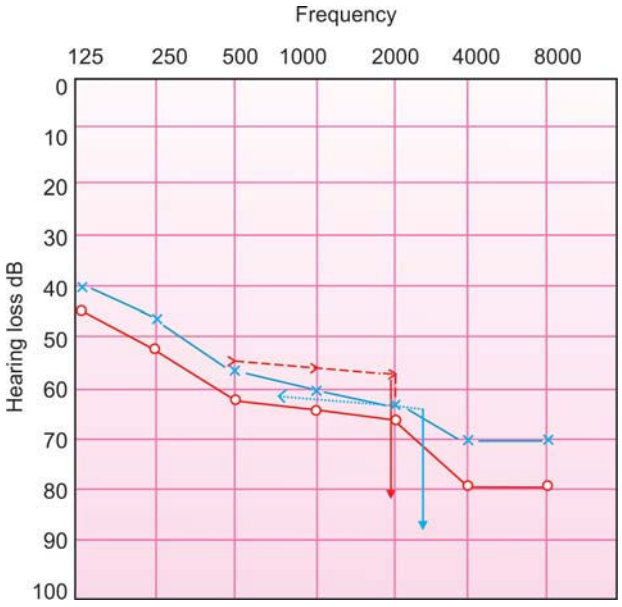


Fig. 5.84: Audiogram of patient with Ménière's disease

Audiogram reveals perceptive deafness as seen in Ménière's disease. Initially only lower frequencies are affected. As disease progresses, all frequencies get involved.

Table 5.20: Ménière's disease treatment

Acute phase
• Reassurance
• Complete bed rest
• Sedation
• Stellate ganglion block.
Remission phase
Medical • Salt restricted diet • Diuretics (Thiazide) • Antihistamines (Cinnarizine)
Surgical • Sac decompression • Sac shunt • Vestibular neurectomy • Labyrinthectomy (Chemical/Surgical) when hearing is poor

Treatment for Ménière's disease depends on whether it is in acute phase or in remission phase.

Fig. 5.85: Malignancy ear: Fibrosarcoma. Note ipsilateral facial weakness

Tumors of the ear are uncommon. However, squamous cell carcinoma is the most common malignancy of the middle ear. Sarcomas are more common in children. A child with fibrosarcoma involving the ear and producing facial palsy is shown in the photograph.

Fig. 5.86: X-ray skull

Radiograph shows complete destruction of the petrous temporal bone by malignancy of the ear. CT scanning of the temporal bone has brought in revolutionary changes in the diagnosis and treatment planning for these cases.

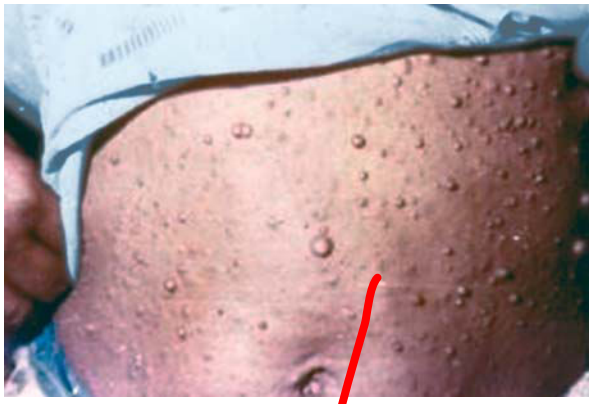
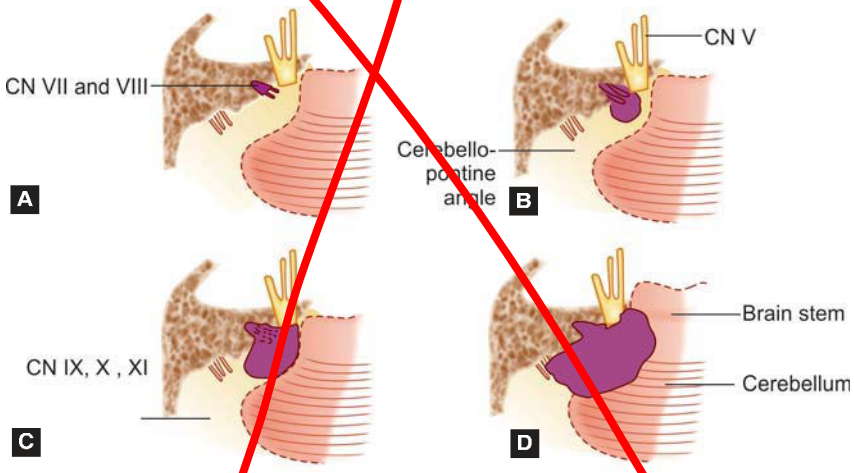


Fig. 5.87: Neurofibromatosis (NF2) can also present as acoustic neuroma

Acoustic neuroma is a tumor of the eighth cranial nerve. It may occur as a solitary tumor or be a part of generalized neurofibromatosis. Patients with unilateral sensorineural deafness, tinnitus or deafness and/or absence of corneal reflex should be fully investigated.



Figs 5.88A to D: Acoustic neuroma and its expansion: (A) Intracanalicular; (B) Tumor extending into cerebellopontine angle; (C) Tumor pressing on CNV; (D) Very large tumor pressing on CN V, IX X, XI, and brainstem and cerebellum

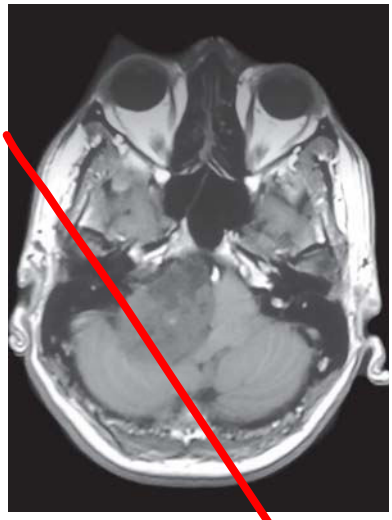


Fig. 5.89: T1 weighted axial MRI showing giant tumor arising from vestibulocochlear nerve spreading into cerebellopontine angle. *Note:* The relation of the tumor to the temporal bone (*Source:* Photo by Dr Bhavin Parikh-Vadodara)

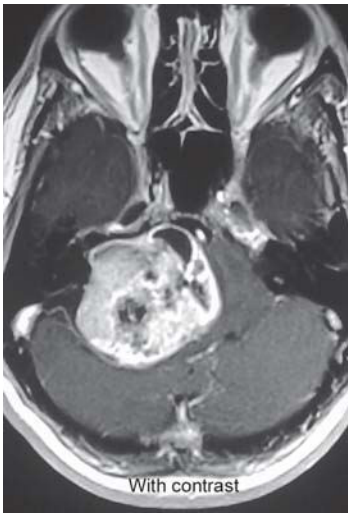


Fig. 5.90: Gadolinium enhanced T1 weighted axial MRI showing brightly enhancing acoustic neuroma (*Source:* Photo by Dr Bhavin Parikh)

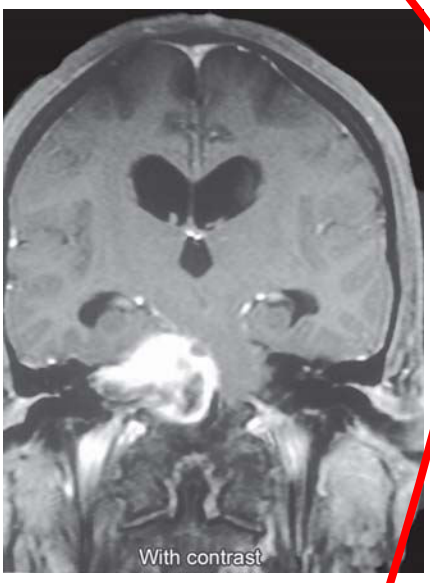


Fig. 5.91: Gadolinium enhanced T1 weighted coronal MRI showing brightly enhancing acoustic neuroma tumor causing marked deviation of the brainstem to opposite side with hydrocephalus. *Note:* The extension of the tumor in internal auditory meatus (*Source:* Photo by Dr Bhavin Parikh)

Fig. 5.92: Gadolinium enhanced T1 weighted sagittal MRI showing brightly enhancing acoustic neuroma tumor with hydrocephalus causing compression of the cerebellum with tonsillar herniation into foramen magnum with hydrocephalus (*Source:* Photo by Dr Bhavin Parikh)

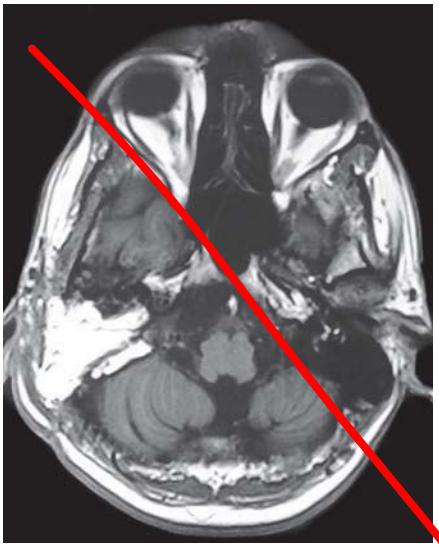


Fig. 5.93: Postoperative T2 weighted axial MRI picture after 10 days of removal of tumor by translabyrinthine approach. *Note:* The abdominal fat used for filling the defect in mastoid cavity and cerebellopontine angle to prevent CSF leak (*Source:* Photo by Dr Bhavin Parikh)

Fig. 5.94: Postoperative T1 weighted coronal MRI picture after 10 days of removal of tumor by translabyrinthine approach. *Note:* The hydrocephalus, compression of cerebellum and deviation of brainstem has resolved (*Source:* Photo by Dr Bhavin Parikh)

Fig. 5.95: Glomus jugulare

Glomus jugulare (nonchromaffin paraganglioma) is the most common benign tumor of the ear. It produces deafness, tinnitus or presents as a bleeding ear polyp. It may also involve other related structures; patient shown is having a right glomus jugulare tumor which had extended intracranially.

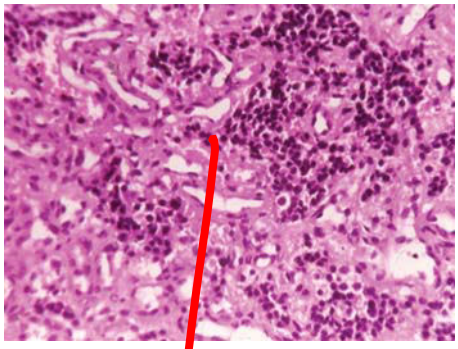


Fig. 5.96: Histopathological examination of glomus jugulare

Microscopically, the tumor consists of monomorphic cells arranged in groups which are separated by vascular connective tissue septae.

FACIAL NERVE

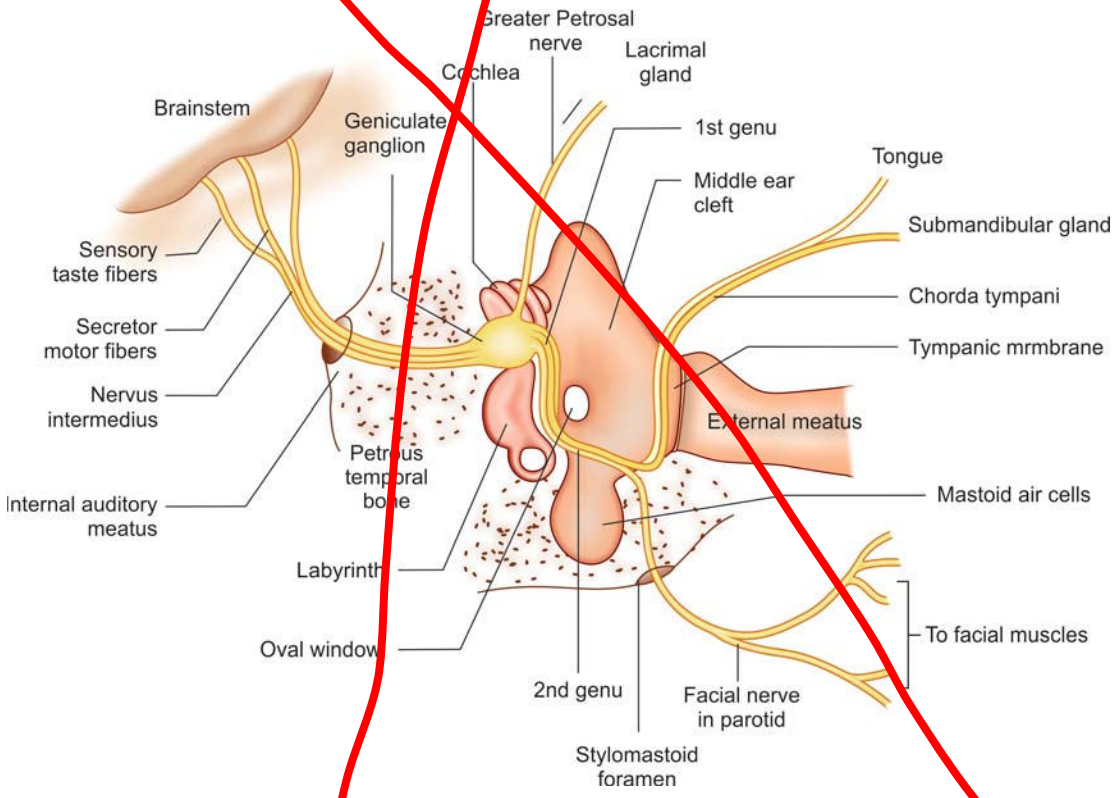


Fig. 5.97: Facial nerve

Facial paralysis can be caused damage to the facial nerve of different levels as shown.

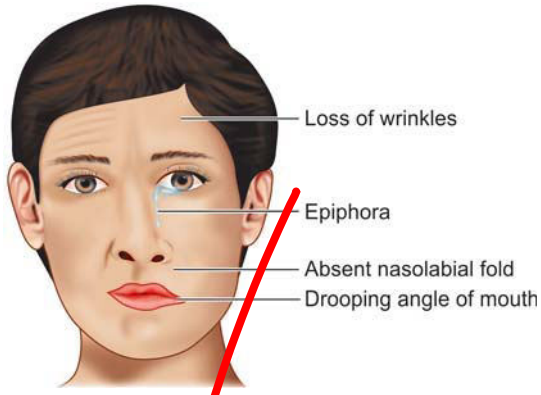


Fig. 5.98: Bell's palsy

Bell's palsy is a unilateral, idiopathic, rapid onset facial palsy. It may be partial or complete and is usually self limiting. It is the commonest cause of facial paralysis. It is treated with antivirals, steroids and physiotherapy.

Table 5.21: Causes of facial paralysis

Intracranial
Brainstem lesions • Vascular accidents • Poliomyelitis • Multiple sclerosis • Tumors
CP angle lesions • Acoustic neuroma • Primary cholesteatoma • Arachnoid cysts • Meningitis
Intratemporal • Otitis • Trauma • Tumors • Ramsay Hunt syndrome • Bell's palsy
Infratemporal • Parotid tumors • Trauma
Miscellaneous • Sarcoidosis • Polyneuritis • Melkersson syndrome • Leukemia

TEMPORAL BONE TRAUMA

Fig. 5.99: CSF otorrhea with Battles sign in a patient with temporal bone fracture following road traffic accident

HRCT of patient in Figure 5.99 showed longitudinal fracture of temporal bone. Facial canal and labyrinth had escaped. Otorrhea ceased after conservative management.

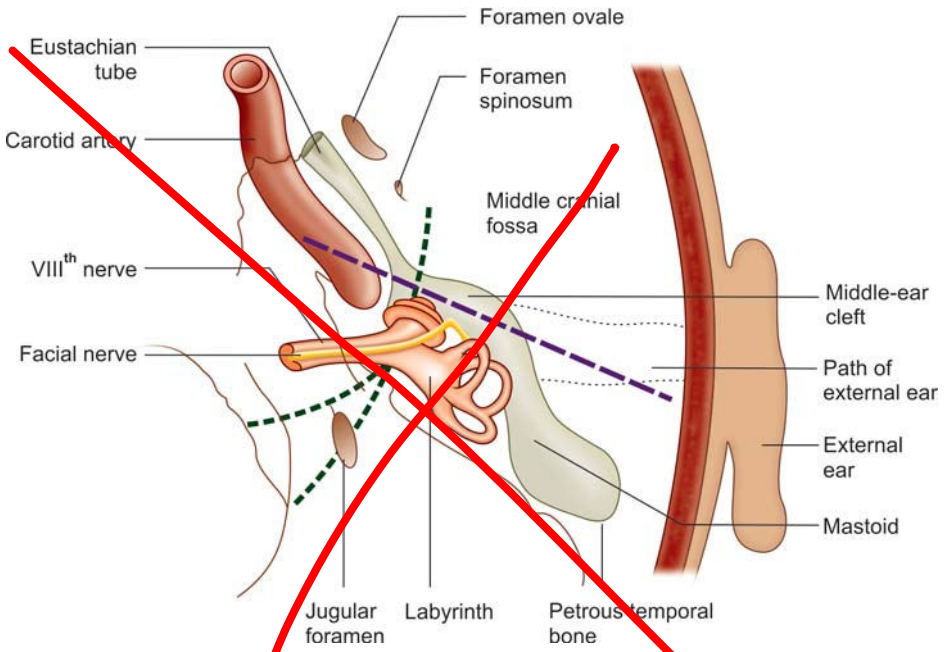


Fig. 5.100: Temporal bone fractures: Longitudinal (big dash) and transverse (small dot)

Transverse temporal bone fractures are more likely to involve the facial canal or labyrinth. Incidence of longitudinal temporal bone fractures is more common as compared to transverse fractures.

RAMSAY HUNT SYNDROME

Fig. 5.101: Herpes zoster oticus

Herpes zoster oticus presents with herpetic eruptions of the external ear and produces infranuclear facial paralysis. It is treated with antivirals (systemic and topical Acyclovir) steroids and analgesics.

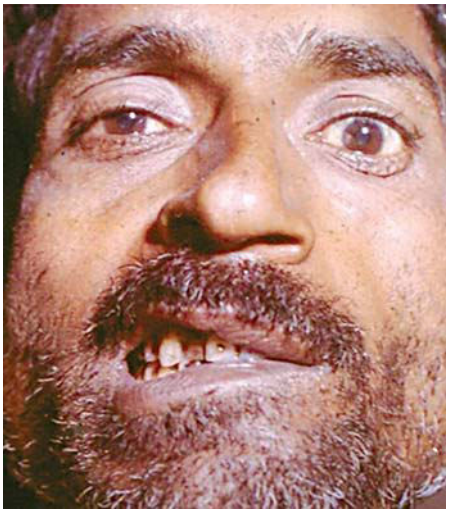


Fig. 5.102: Left facial paralysis in herpes zoster oticus

Recovery rates of facial paralysis in herpes zoster oticus are poorer than Bell's Palsy.

ESSENTIAL BLEPHAROSPASM

Fig. 5.103: Hyperkinetic disorder of facial nerve: Essential blepharospasm–patient unable to open eyes (*Source:* Photo by Dr Raj Anand, Oculoplastic Surgeon, Delhi)

Fig. 5.104: Essential blepharospasm relieved by Botox (*Source:* Photo by Dr Raj Anand)

Differental Diagnosis of Referred Otalgia

Table 5.22: Referred otalgia

Via V cranial nerve	*Via 9th and 10th cranial nerve*
Sinusitis	Acute tonsillitis
Nasopharyngeal malignancy	Retropharyngeal abscess
Nasopharyngitis	Post-tonsillectomy
Carious tooth	Laryngopharyngeal lesions
Costen syndrome	Glossopharyngeal neuralgia
Deviated nasal septum	Styloid process neuralgia
Via 2nd and 3rd cervical nerve	*Via 7th nerve*
Cervical disc lesions	Herpes zoster oticus
Arthritis	
Temporomandibular	
Fibrositis of sternomastoid muscle	
Herpetic lesions	

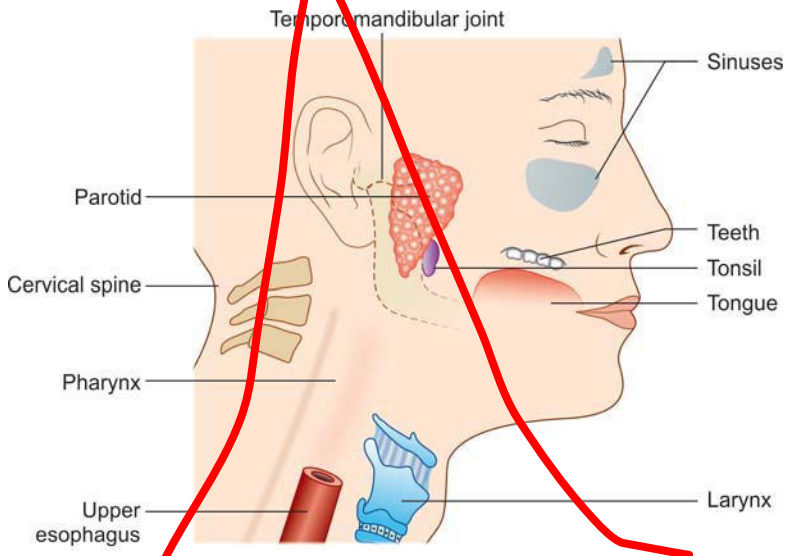


Fig. 5.105: Referred otalgia

Fig. 5.106: OPG of patient with styalgia (Eagle's syndrome)
Elongated styloid process marked by arrows on right side

Palpate the tonsillar fossa in patients with unexplained otalgia.

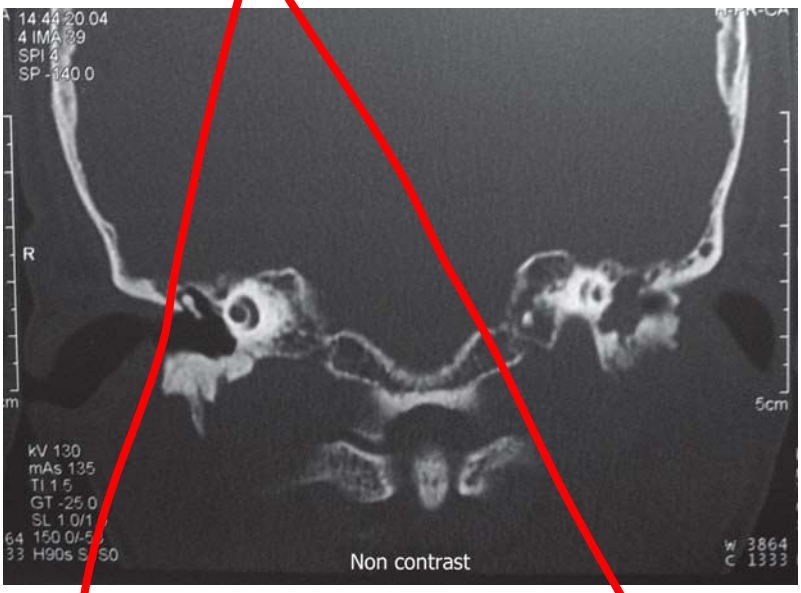


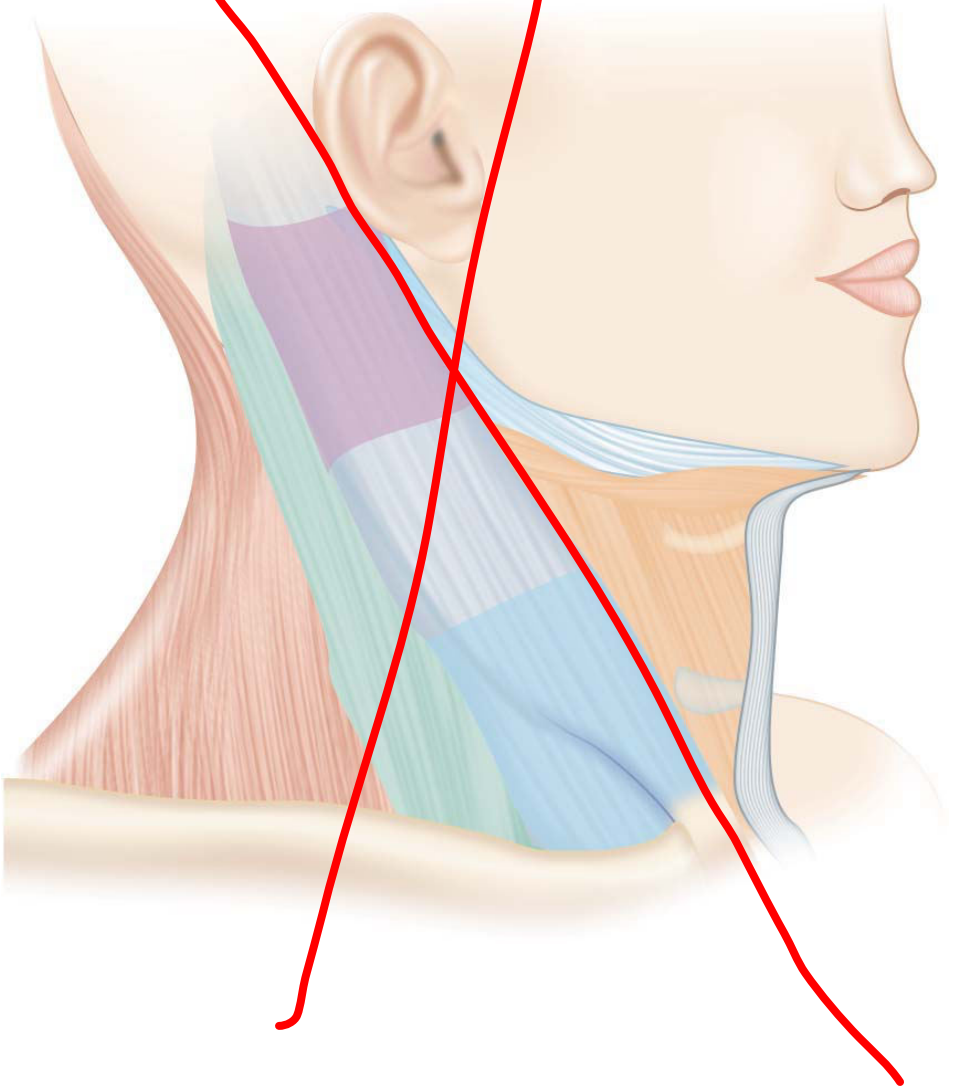
Fig. 5.107: HRCT temporal bone of patient with left malignant otitis externa

'Malignant' otitis externa is an aggressive disease termed 'malignant' because of poor treatment outcome and mortality. It occurs typically in diabetic or immunocompromised individuals. It is usually caused by *Pseudomonas aeruginosa*. It begins as a non healing external otitis which progresses into an osteomyelitis of the temporal bone and spreads along the skull base.

6

PS Saharia, Deepti Sinha

Neck



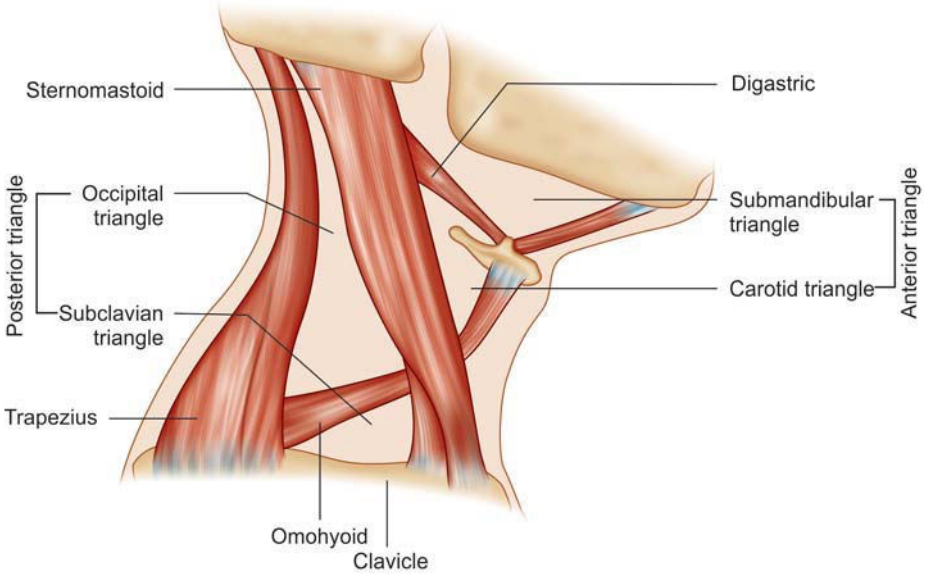


Fig. 6.1: Triangles of the neck

Fig. 6.2: Palpation of neck nodes

Flex the patient's neck slightly and stand behind the patient while palpating the neck.

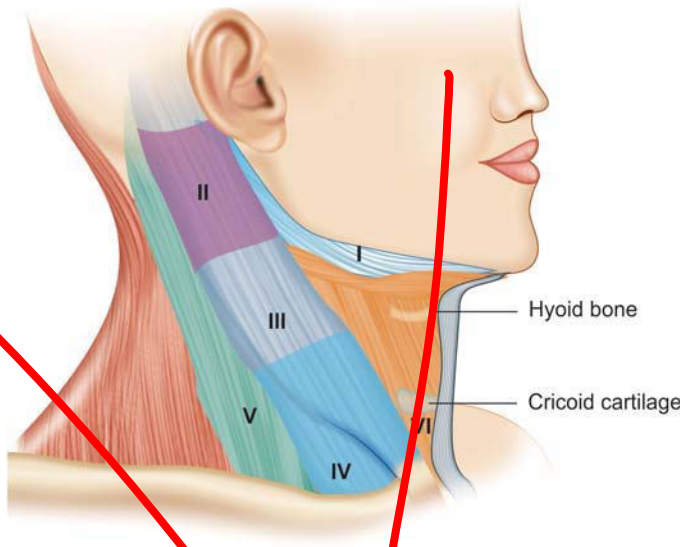


Fig. 6.3: Levels of neck nodes

Table 6.1: Levels of lymph nodes in neck

Level I A	:	Submental nodes
Level I B	:	Submandibular nodes
Level II, III and IV	:	Upper, mid and lower jugulodigastric nodes
Level V	:	Posterior triangle
Level VI	:	Below the hyoid in the anterior neck
Level VII	:	Anterior mediastinum

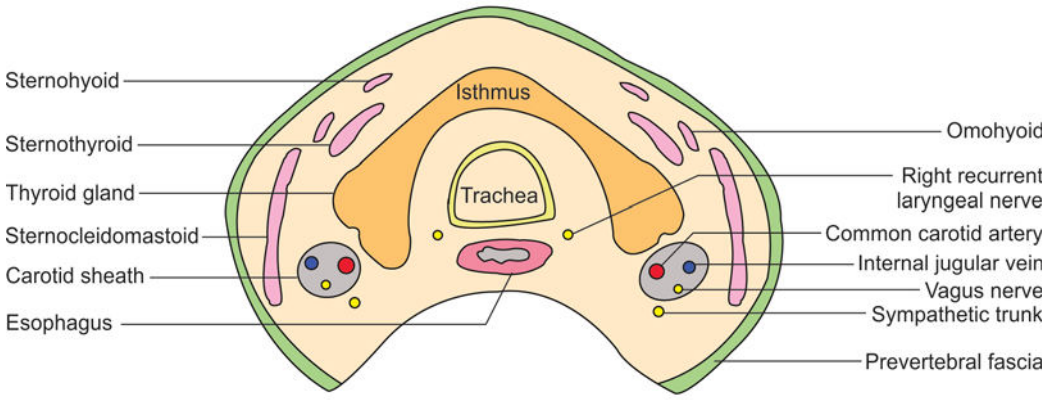


Fig. 6.4: Axial section of neck at level of thyroid isthmus

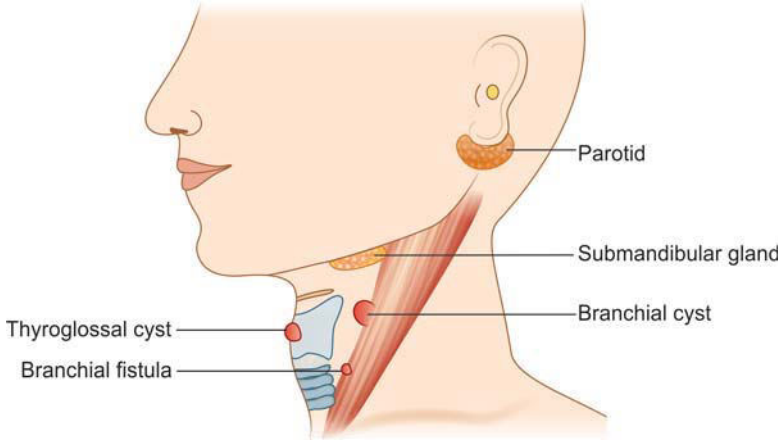


Fig. 6.5: Sites of various cysts, fistulae and tumors in neck

Neck swellings may be midline or lateral. Thyroglossal cyst is commonly encountered as a midline cyst. Thyroglossal cyst is the only midline neck swelling which moves up on protrusion of tongue.

TUBERCULAR LYMPHADENITIS

Tubercular lymphadenitis commonly affects upper deep jugular nodes in the neck. Initial stage is of enlargement of multiple nodes followed by matting due to periadenitis. This may further progress to caseation followed by rupture and cold abscess formation. Cold absceses are termed cold because the swelling is without heat, redness, pain or fever. All patients with tubercular lymphadenitis are screened for pulmonary tuberculosis. WHO Multidrug regimen for TB is followed (2 months treatment with isoniazid, rifampicin, pyrizinamide, ethambutol + 4 months continuation therapy with isoniazid and rifampicin). Nonresponders or MDR cases are referred to dedicated TB centers for treatment.

Fig. 6.6: Tubercular cold abscess

Microscopically, tuberculous lymph node is characterized by granulomas consisting of lymphocytes, epitheloid cells and Langhans giant cells as shown in low and high power view.

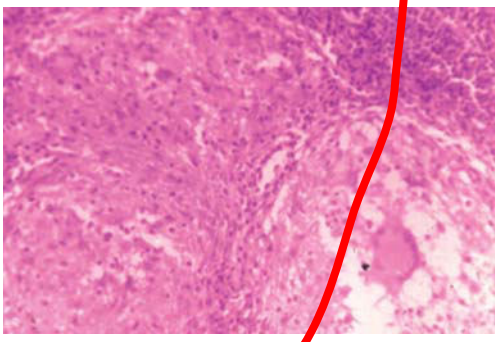


Fig. 6.7: Langhans cells

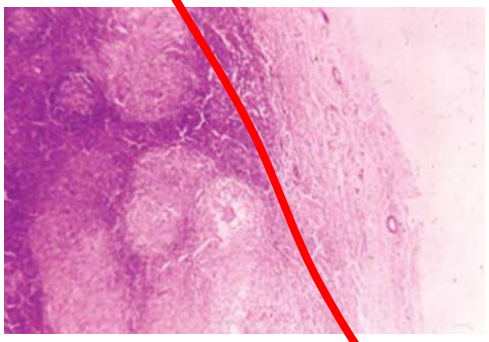


Fig. 6.8: Histopathology of tubercular lymphadenitis

METASTATIC NECK NODE

Fig. 6.9: Hard and fixed cervical lymph node

The cervical lymph nodes may be involved by malignant lesions in the tongue, pharynx or nasopharynx. Large fixed nodes as in Figure 6.9 mostly represent advanced malignant disease and are often unresectable. After establishing diagnosis appropriate curative or palliative treatment is offered.

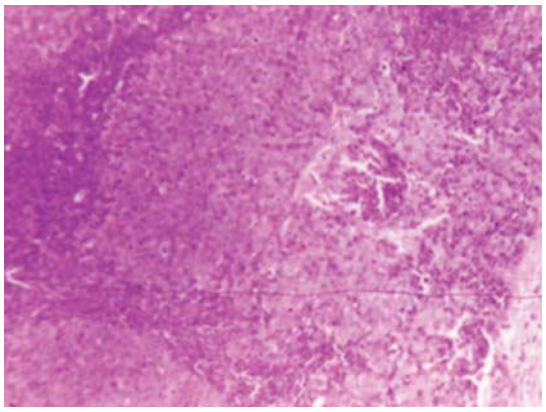
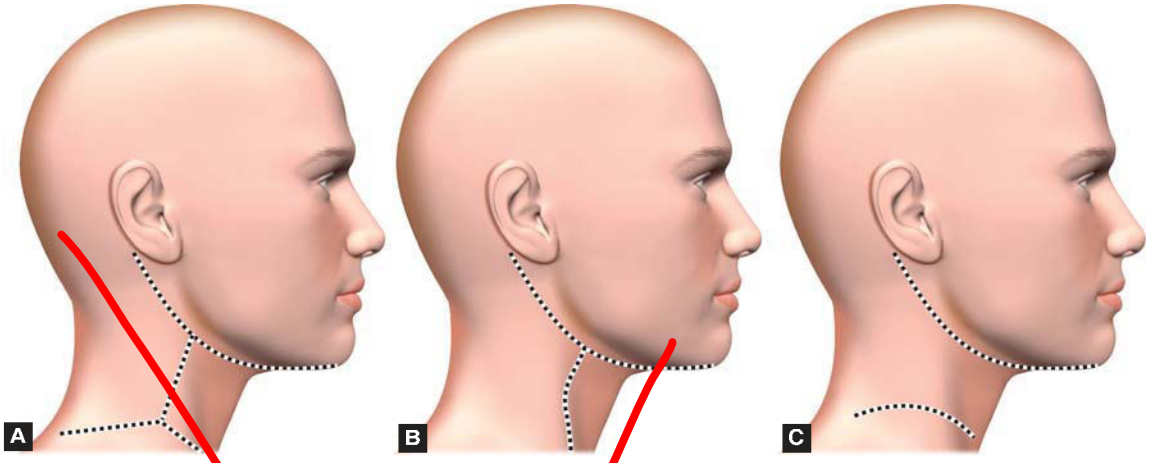


Fig. 6.10: Histopathological examination of metastatic cervical node



Figs 6.11A to C: Incisions for block dissection: Double Y,Y, Mc Fee

Block dissection of the cervical lymph nodes is indicated for the metastatic cancer. A double Y incision for the block dissection is shown. With the development of the newer diagnostic facilities, better pre- and postoperative management and improved tissue transfer technique, surgical procedures dealing with a lesion in the base of the skull have evolved in coordination with neurosurgical colleagues.

PAROTID GLAND

Parotid glands are the largest salivary glands. The parotid duct exits the gland medially, crosses the superficial border of the masseter, pierces the buccinator, and enters the oral cavity through the buccal mucosa opposite the second maxillary molar.

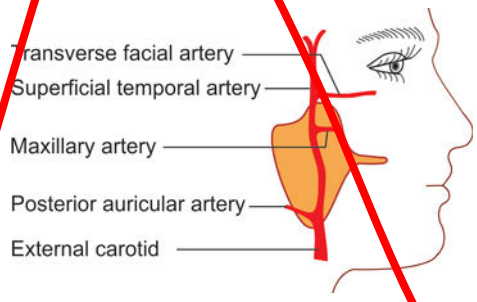


Fig. 6.12: Arterial supply of parotid gland

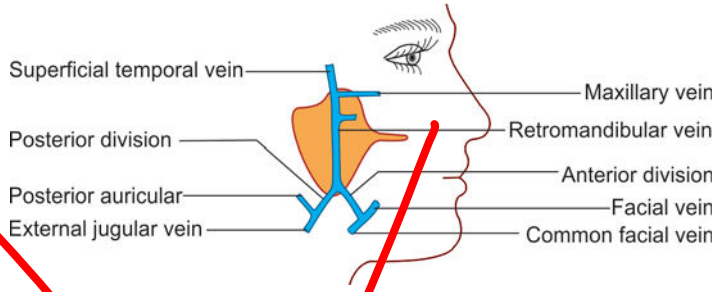


Fig. 6.13: Venous drainage of parotid gland

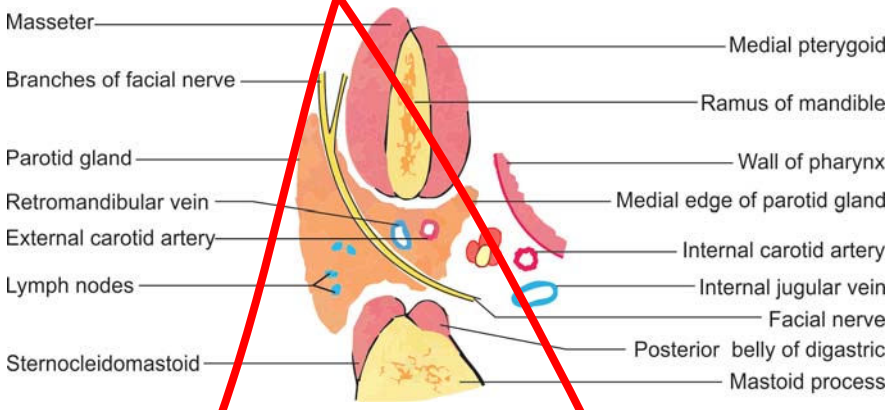


Fig. 6.14: Relations of parotid gland: coronal section

Parotid gland is divided into a superficial and deep portion by the facial nerve, which passes through the gland.

Fig. 6.15: Pleomorphic adenoma

Pleomorphic adenoma of the parotid gland is a common tumor. It is slow growing and can attain a big size if neglected. Treatment is superficial parotidectomy with preservation of the facial nerve. Recurrence may occur if removal is not complete.

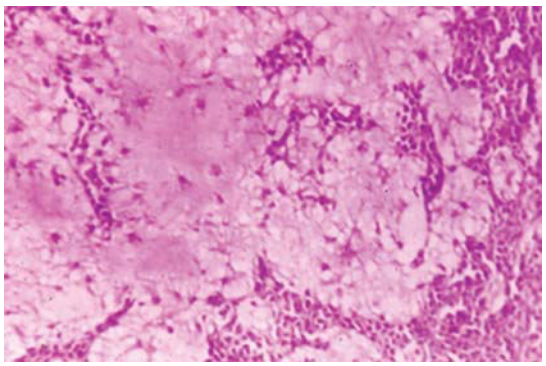


Fig. 6.16: Histopathological examination of pleomorphic adenoma

Photomicrograph of pleomorphic adenoma showing epithelial component merging imperceptibly with mesenchymal tissue. Pseudocartilaginous change is seen in the mesenchyme.

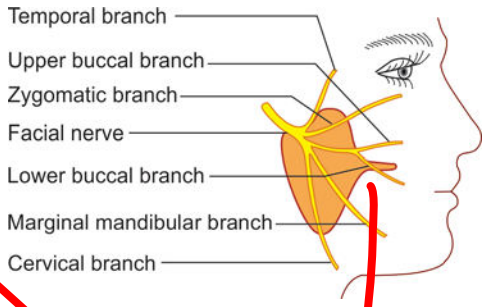


Fig. 6.17: Facial nerve branches in the parotid

The facial nerve emerges from the stylomastoid foramen and courses through the substance of the parotid dividing it into superficial lobe (superficial or lateral to the facial nerve) and deep lobe (deep or medial to the facial nerve). The facial nerve branching pattern within the parotid, can be highly variable. The main trunk bifurcates into the zygomaticotemporal branch and the cervicofacial branch at the pes anserinus and further into the temporal, zygomatic, buccal, marginal, and cervical branches. Pes is located 1.3 cm from the stylomastoid foramen.

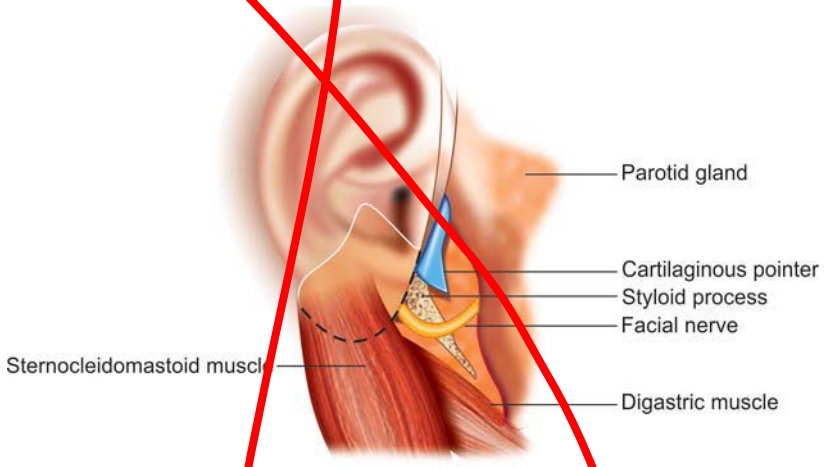
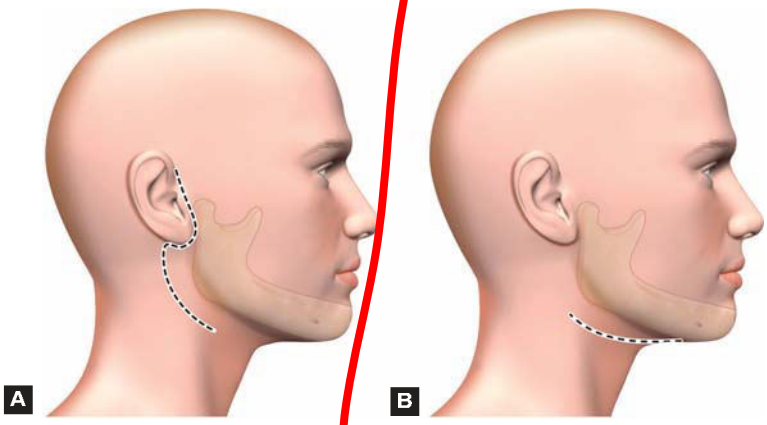


Fig. 6.18: Surgical landmarks of facial nerve in parotid surgery



Figs 6.19A and B: Incisions for parotid and submandibular gland surgery respectively

SUBMANDIBULAR GLAND

Submandibular gland lies in the submandibular triangle. It consists of a larger superficial and smaller and more posterior deep lobe which connect around the posterior border of mylohyoid muscle.

Fig. 6.20: Submandibular duct stones on X-ray

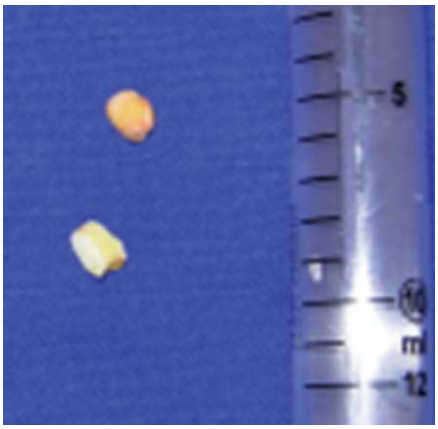


Fig. 6.21: Submandibular duct stones specimen

Ninety percent of salivary calculi are found in submandibular salivary glands due to the nature of secretions and antigravity drainage.

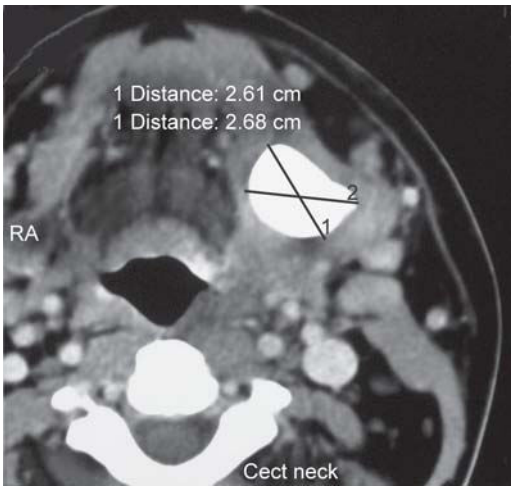


Fig. 6.22: Large submandibular gland stone on CT

Fig. 6.23: Large submandibular gland stone on X-ray

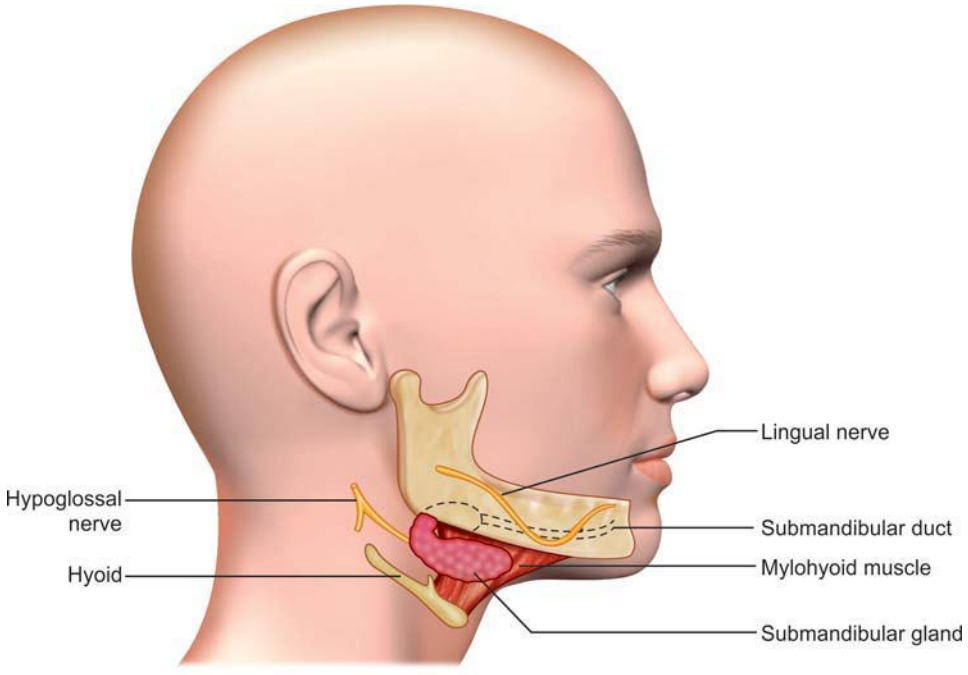


Fig. 6.24: Relations of submandibular duct during gland surgery

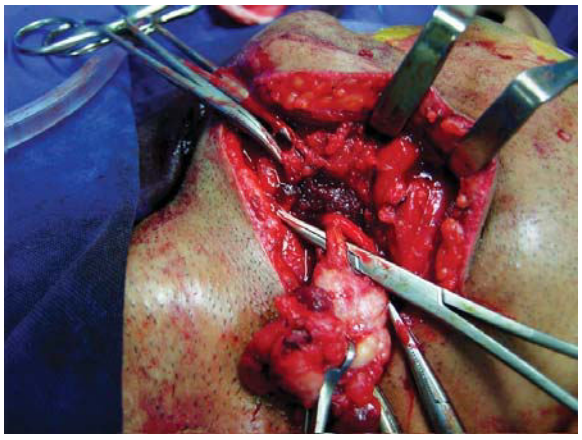


Fig. 6.25: Intraoperative photograph of submandibular gland excision

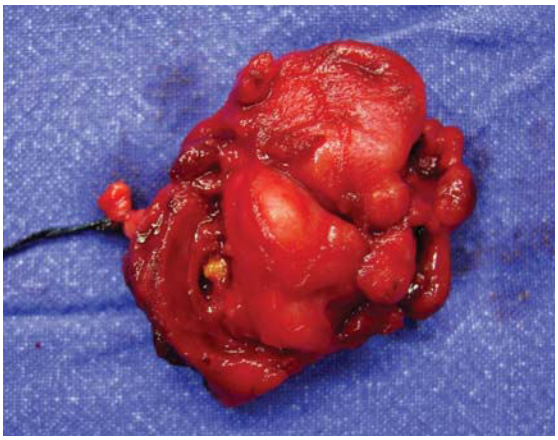


Fig. 6.26: Submandibular gland specimen with stone

The salivary duct prior to being cut. Note the lingual nerve crossing over it (Fig. 6.25).

THYROID

Thyroid gland occupies the center of neck, in front of and close to the trachea just above the thoracic inlet. It has two lobes which are connected across midline by the isthmus. Because of fascial attachments the gland moves upwards with swallowing and slides under the examining fingers.

Fig. 6.27: Thyroglossal cyst (moves with deglutition and is the only midline neck swelling which moves up on tongue protrusion)

A thyroid scan should be done prior to excision of thyroglossal cyst to confirm that it is not the only thyroid tissue. Since the thyroglossal tract is closely related to the hyoid bone, the central part of the hyoid must be excised together with the cyst; to prevent recurrence. Also the tract is traced upwards to a foramen caecum, where it is excised along with a core of lingual muscle.

Fig. 6.28: Goiter

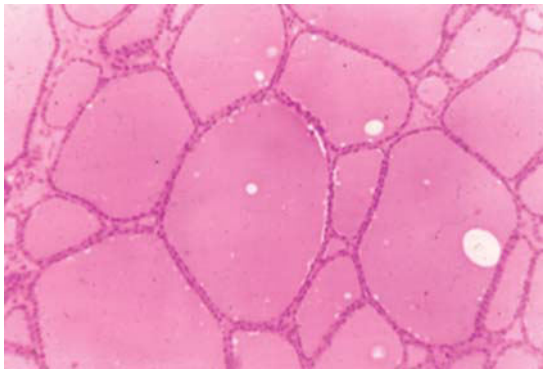


Fig. 6.29: Histopathological examination of colloid goiter

The term goiter is used to describe generalized enlargement of thyroid gland. A discrete swelling in one lobe with no palpable abnormality elsewhere is termed as solitary thyroid nodule. Photomicrograph of goiter shows dilated thyroid acini filled with pink colloid material.

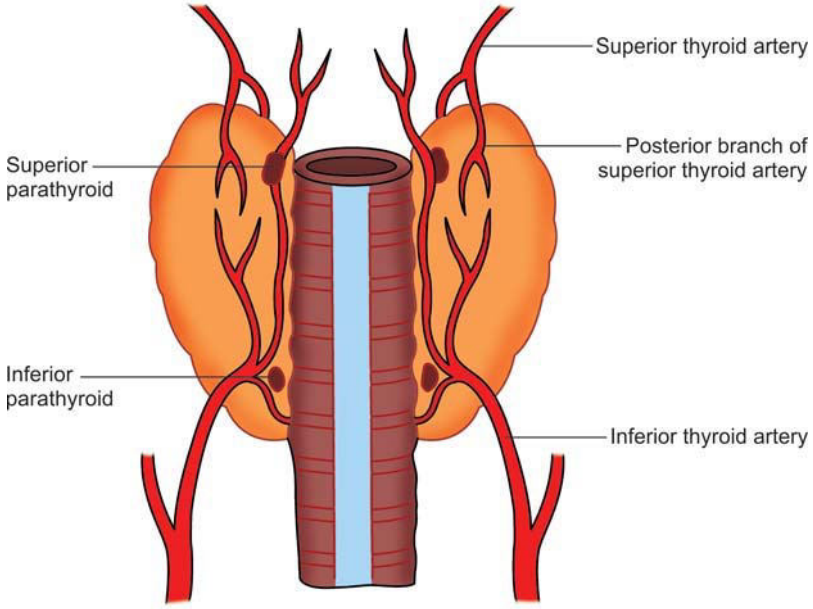


Fig. 6.30: Arterial supply of thyroid gland

Each thyroid lobe is supplied by a superior and an inferior thyroid artery and drained by three veins.

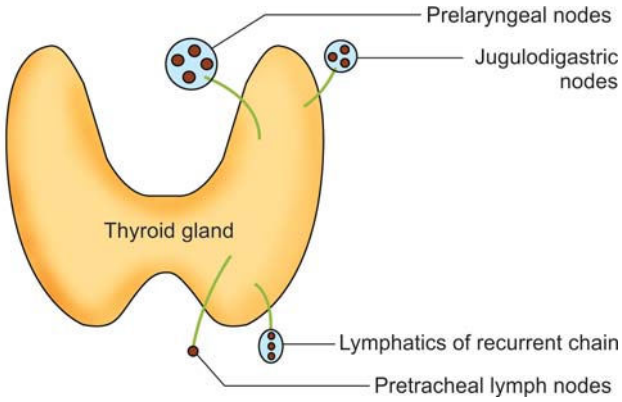
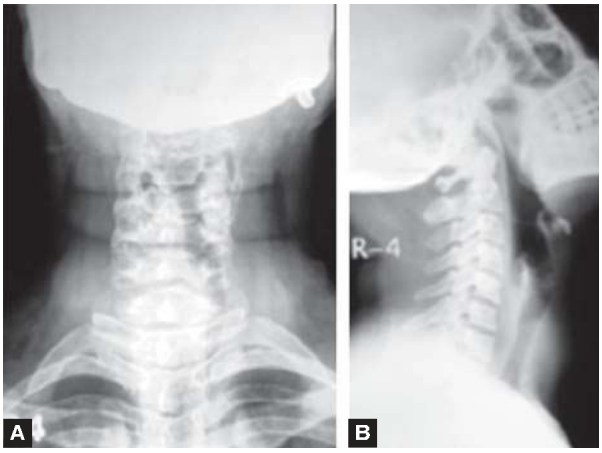


Fig. 6.31: Lymphatic drainage of thyroid gland

Fig. 6.32: Right thyroid lump



Figs 6.33A and B: X-ray showing tracheal shift and compression by thyroid lump

Fig. 6.34: Thyroid scan showing cold nodule on the right side
(Same patient as in Figures 6.32 and 6.33)

Cold nodules are more likely to be malignant as compared to warm nodules.

CAROTID BODY TUMOR

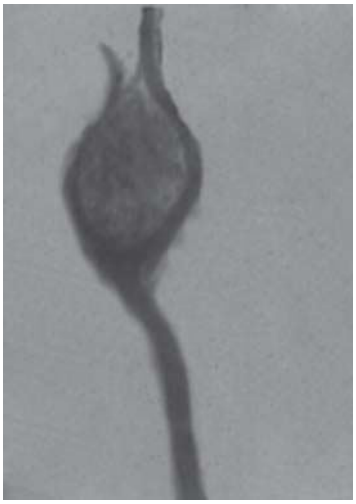


Fig. 6.35: CT with 3D reconstruction: Carotid body tumor (*Source:* Carotid Paraganglioma AK. Monga, Deepti Sinha; Indian Journal of Otolaryngology and Head and Neck Surgery. Special issue 2005;217-20)

A 50-year-male had presented with a pulsatile swelling in left carotid triangle. Carotid body tumor (CBT) are treated by surgery or radiotherapy. The patient in figure 6.35 underwent successful surgery.

Fig. 6.36: Carotid body tumor in Axial CT scan *(Carotid Paraganglioma. AK Monga, Deepti Sinha. Indian Journal of Otolaryngology and Head and Neck Surgery. Special issue 2005;217-20)*

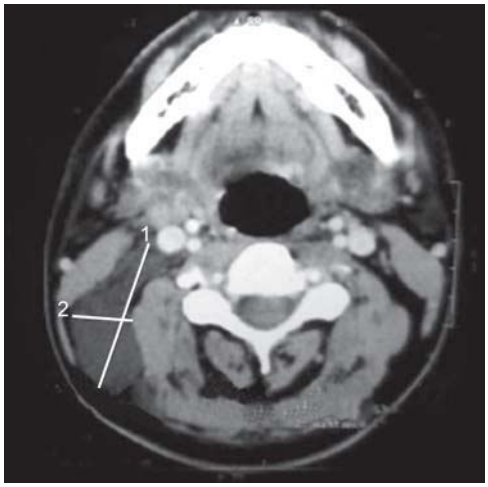


Fig. 6.37: CT: Cervical lymphangioma (*Source:* Photo by Dr Mukul Gupta)

7

PS Saharia, Deepti Sinha

Endoscopy

DYSPHAGIA

Common complaints of swallowing are difficulty in swallowing (Dysphagia), pain on swallowing (Odynophagia), lump or sticking in throat (globus). Dysphagia may affect the oral, pharyngeal or esophageal phases of swallowing. Thorough history and physical examination are important in the diagnosis and treatment of dysphagia.

Table 7.1: Dysphagia

- A detailed history often suggests the cause
- Look for cranial nerve palsies
- Barium swallow is diagnostic in most cases
- Remember that dysphagia and aspiration often occur together
- Aspiration is best assessed with videofluoroscopy
- Endoscopy is mandatory if malignancy is suspected

Fig. 7.1: Drink test

Make the patient drink water to see at what stage of swallowing the patient has dysphagia. Direct observation of lip closure, jaw closure, chewing and mastication, tongue mobility and strength, palatal and laryngeal elevation, salivation, and oral sensitivity is necessary. Cranial nerves V, VII-XII should be examined. Gag reflex should be tested.

Table 7.2: Causes of dysphagia

Oral cavity			
Pharynx and larynx			
	Lumen	*Wall*	*Outside*
Stomatitis	Impacted FB	Pharyngitis	Retropharyngeal abscess
Quinsy		Laryngitis	Malignancy
Tonsillitis		Growth	
Palatal palsy		Hysterical	
		Plummer Vinson syndrome	
		Bulbar palsy	
Esophagus			
Lumen	*Wall*	*Outside*	
Foreign body	Stricture	Retrosternal goiter	
	Cardiospasm	Pharyngeal diverticulum	
	Bulbar poliomyelitis	Aneurysm of aorta	
	Malignancy	Mediastinal mass	
		Dysphagia lusoria	

Fig. 7.2: Patient with Patterson Brown-Kelly syndrome

Angular stomatitis (white arrow in figure above), glossitis, koilonychia and hypochromic microcytic anemia constitute Patterson Brown-Kelly syndrome. The patient has increasing dysphagia. It is a precancerous condition, commonly seen in female adults.

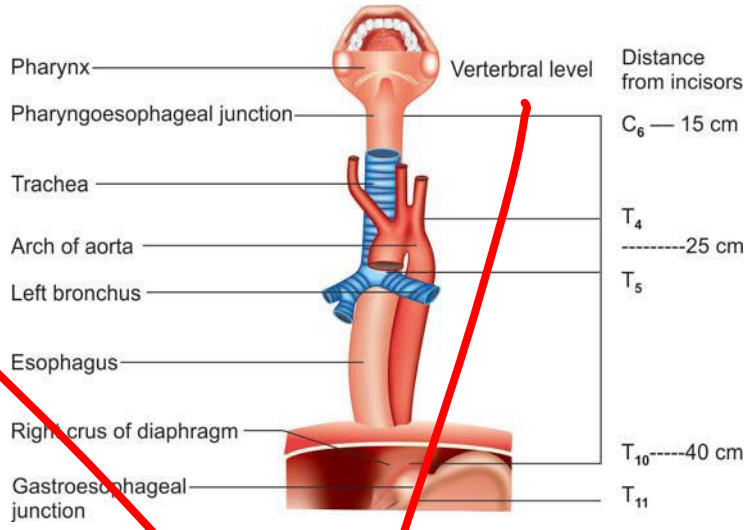


Fig. 7.3: Levels of constrictions of esophagus with distance from incisors

Table 7.3: Key points in management of esophageal foreign bodies

Esophageal foreign bodies
• A carefully taken history is important
• Lateral soft-tissue X-ray of the neck is often helpful. Coins, bones, metallic part of dentures are often seen. Air in the upper esophagus is also a sign of an impacted foreign body. Sharp foreign bodies should be removed at the earliest opportunity
• Ruling out esophageal perforation is important
• Soft foreign bodies may be treated conservatively for a short period
• Esophageal foreign bodies can perforate the esophagus

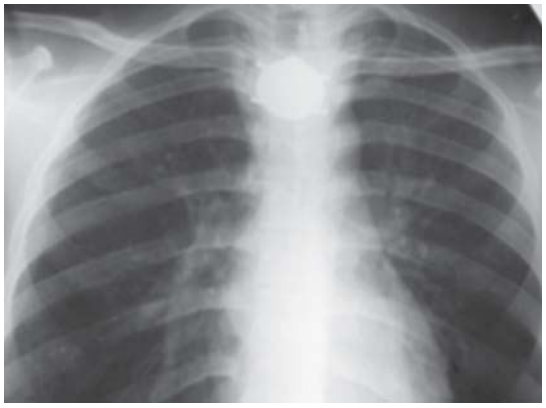


Fig. 7.4: Wrist watch dial in esophagus (*Source:* An amazing case of working wrist watch in the esophagus. Nitin Aggarwal, Deepti Sinha. Indian Journal of Otolaryngology and Head and Neck Surgery 2006;58(1);105-106)

Fig. 7.5: Wrist watch removed (*Source:* An amazing case of working wrist watch in the esophagus. Nitin Aggarwal, Deepti Sinha. Indian Journal of Otolaryngology and Head and Neck Surgery 2006;58(1);105-106)

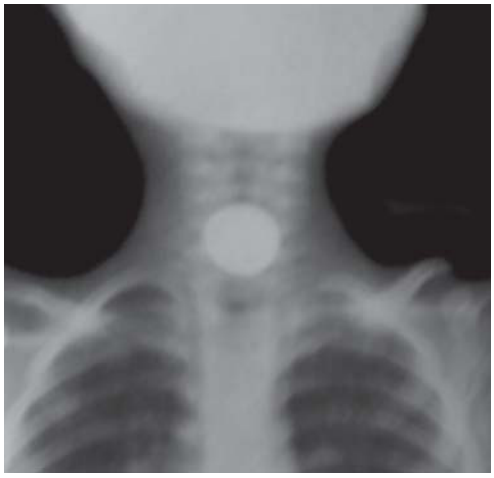


Fig. 7.6: Foreign body (coin) at level of cricopharynx

Fig. 7.7: Ring in esophagus (*Source:* Photo by Dr AK Khurana)

A foreign body in the food passages and bronchi is more common in children than in adults. The size of the foreign body is important to produce the relevant symptoms. Coin is a frequent foreign body to get stuck at the cricopharynx of a child. A denture with or without hooks is common in adults. A sharp foreign body may injure the soft tissue and produce complications.

Fig. 7.8: Foreign body esophagus (*Source:* Photo by Dr AK Khurana)

Fig. 7.9: Foreign body seen in esophagus (*Source:* Photo by Dr AK Khurana)

Mucosal injury may be caused by drugs like potassium chloride, nonsteroidal anti-inflammatory drugs (NSAIDs), or antibiotics (doxycycline, azithromycin, clindamycin, trimethoprim-sulfamethoxazole). Xerostomia may be caused by antihistamines, anticholinergics, alpha-adrenergic blockers or angiotensin-converting enzyme (ACE) inhibitors.

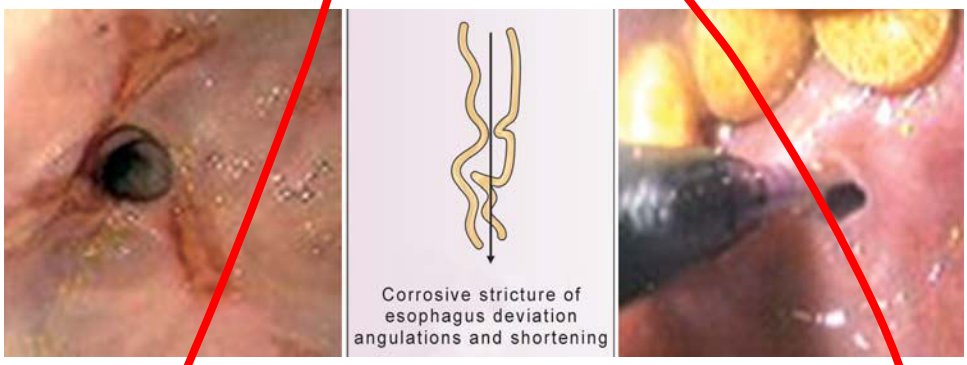


Fig. 7.10: Stricture of esophagus (*Source:* Photo by Dr AK Khurana)

Stricture of the food passage is invariably because of corrosive poisoning. Stricture may be partial or complete. Stricture may distort the shape and the size of the esophagus. A passable stricture is gradually dilated. Retrograde dilation is better.

Fig. 7.11: Pharyngeal diverticulum (*Source:* Photo by Dr AK Khurana)

A pharyngeal diverticulum may be central or lateral. It is the herniation of the pharyngeal mucosa through Killian's dehiscence. Regurgitation of food and dysphagia are the common symptoms. A visible swelling may be felt when the pouch is big. Barium swallow helps in the diagnosis. Diathermy or surgical excision provides the best treatment for the symptomatic cases.

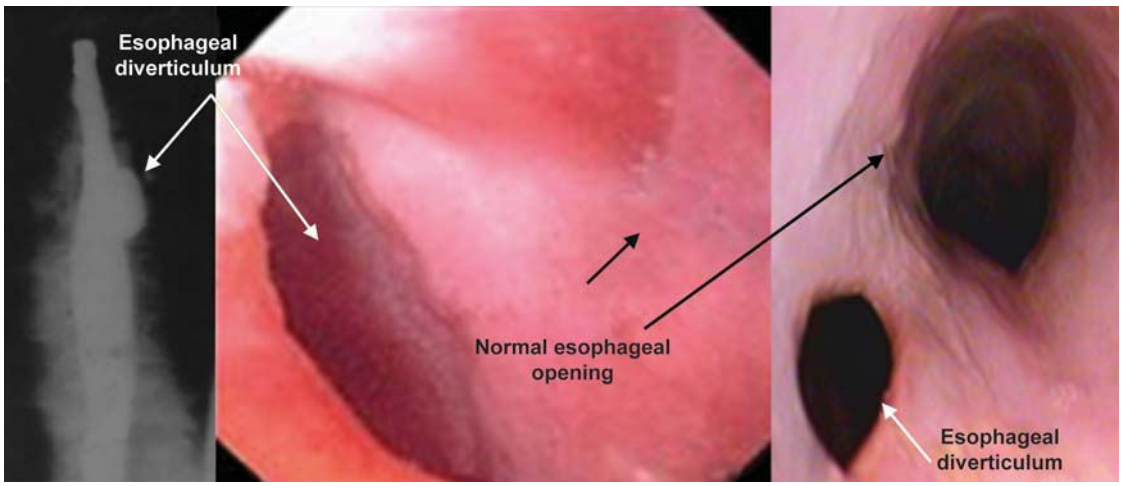


Fig. 7.12: Esophageal diverticulum (*Source:* Photo by Dr AK Khurana)

Diverticulas in the esophagus are rare and usually asymptomatic and are accidentally discovered by barium swallow done for some other reasons.

Fig. 7.13: Barium swallow

Malignancy of the esophagus is not uncommon. The diagnosis is often delayed. A barium swallow showing rat tailing appearance of the esophagus.

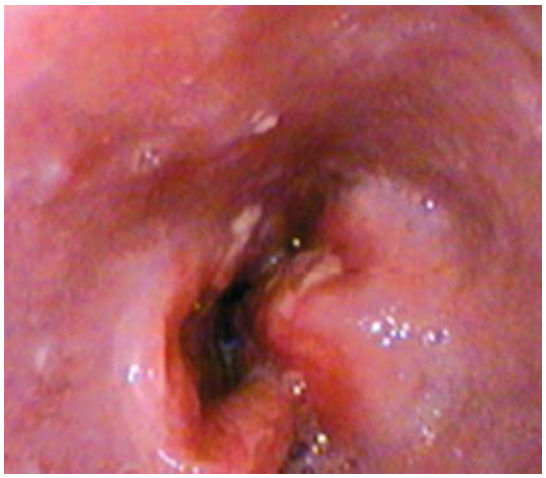


Fig. 7.14: Endoscopic view of esophagus showing the obstruction to lumen

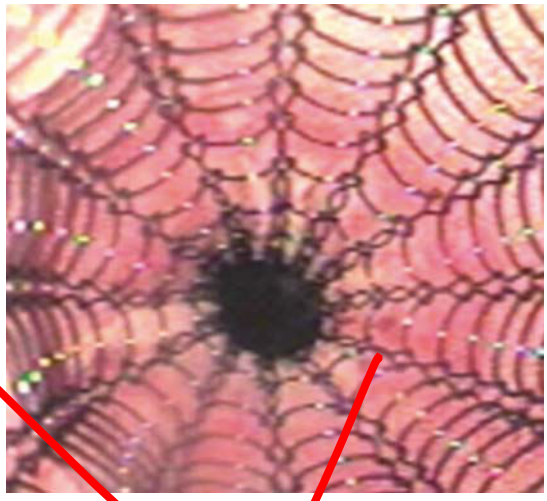


Fig. 7.15: Stenting—a stent in the esophagus

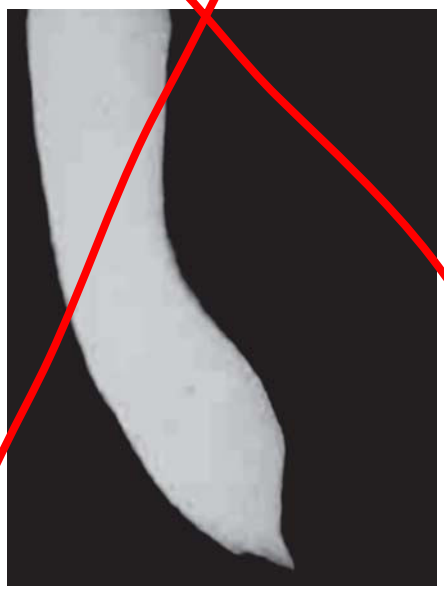


Fig. 7.16: Achalasia cardia

Achalasia cardia is failure of the cardiac end of the esophagus to open to allow the food to go into the stomach. Subsequently, esophagus gets enlarged. Patient develops progressive dysphagia and regurgitation of food. Repeated esophageal dilatation is effective. Heller's operation is indicated when dilation falls.

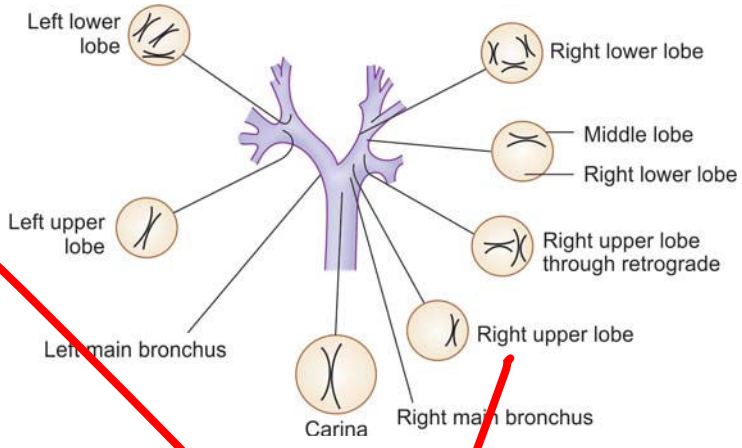


Fig. 7.17: Bronchial tree

Direct examination of the bronchial tree is termed as bronchoscopy. It is performed for diagnostic and therapeutic purposes such as biopsy or foreign body removal in India. Peanuts and other nuts are the most common airway foreign bodies in children.

Table 7.4: Size of bronchoscope

Age group	*Lumen (mm)*		*Length (cm)*
Infant	4	X	30
Small child	5	X	30
Large child	6	X	35
Adolescent	7	X	40
Small adult	8	X	40
Large adult	9	X	40

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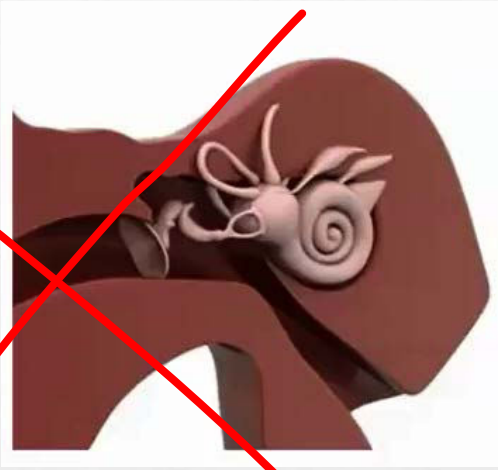
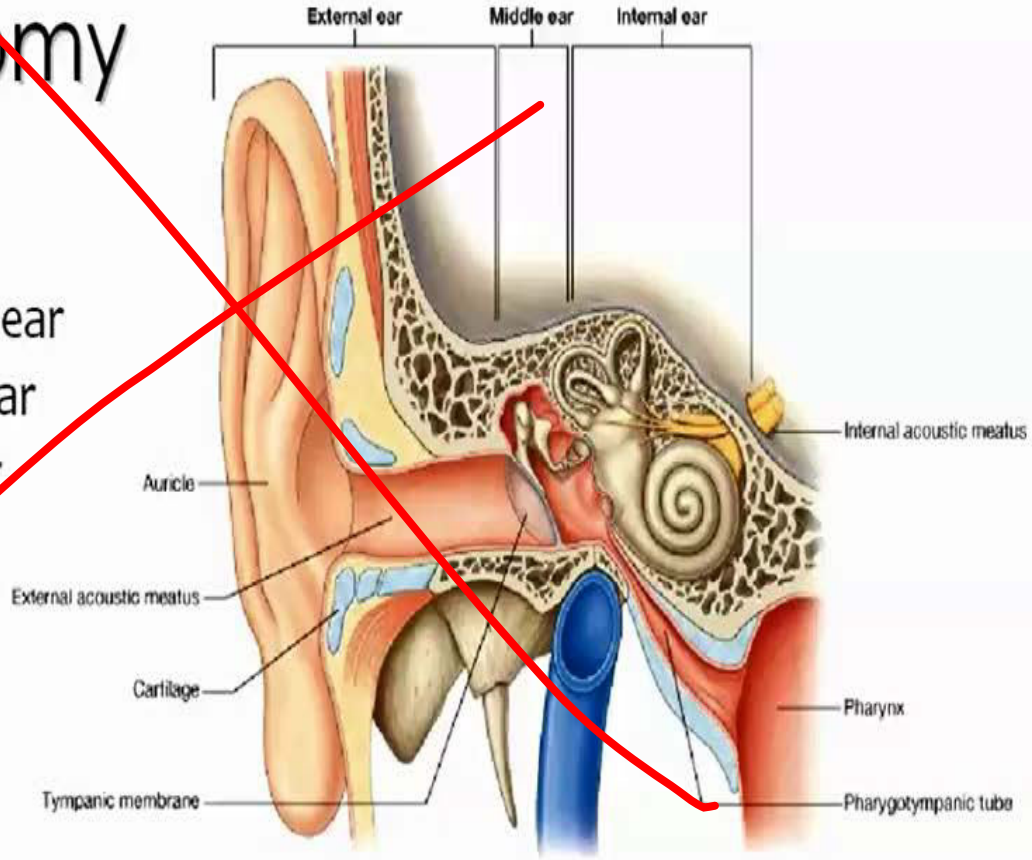
Imaging Modalities of the ear (CT)



Dr.Murad Al-Naqshbandi
3rd Year Otolaryngology resident
Kurdish board of Medical specialty

Anatomy

- External ear
- Middle ear
- Inner ear

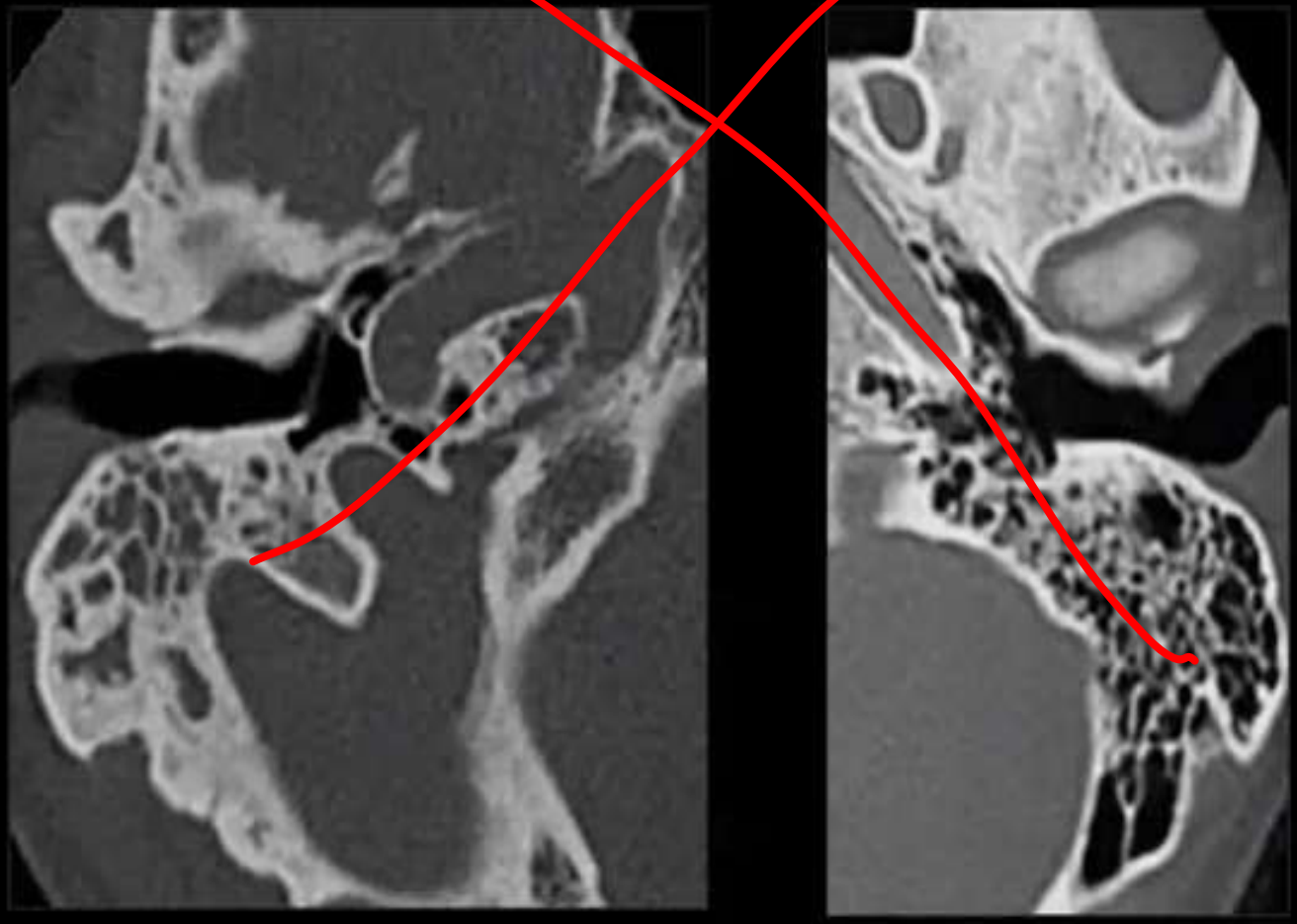


Examination technique

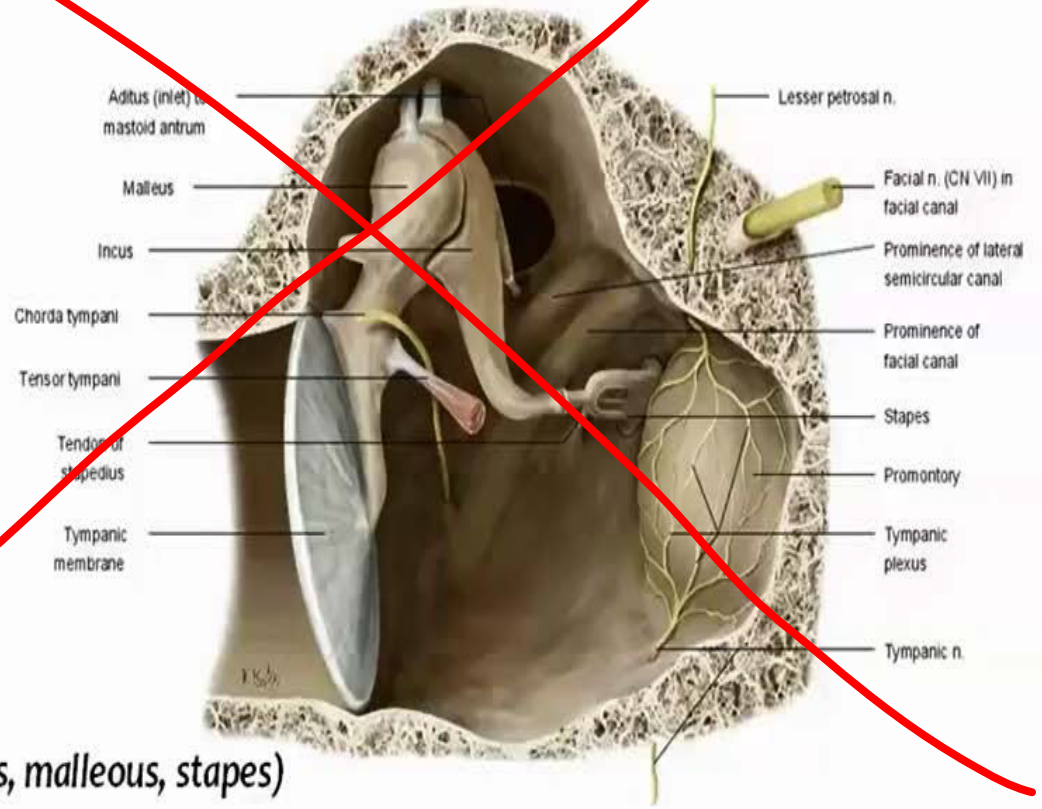
- 2 mm slice thickness
- Suitable field of view
- Bone and soft tissue windows
- Symmetric patient position
- Contrast injection ?!

External ear

- Cartilaginous and bony parts
- Ends by the tympanic membrane which is attached to the tympanic annulus
- The drum The anterior and superior part of tympanic annulus = scutum

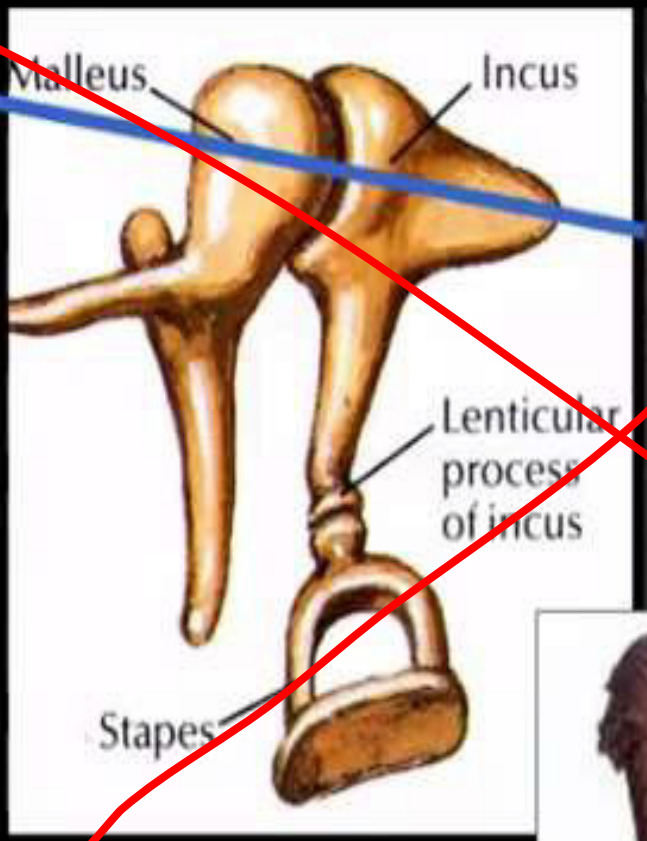


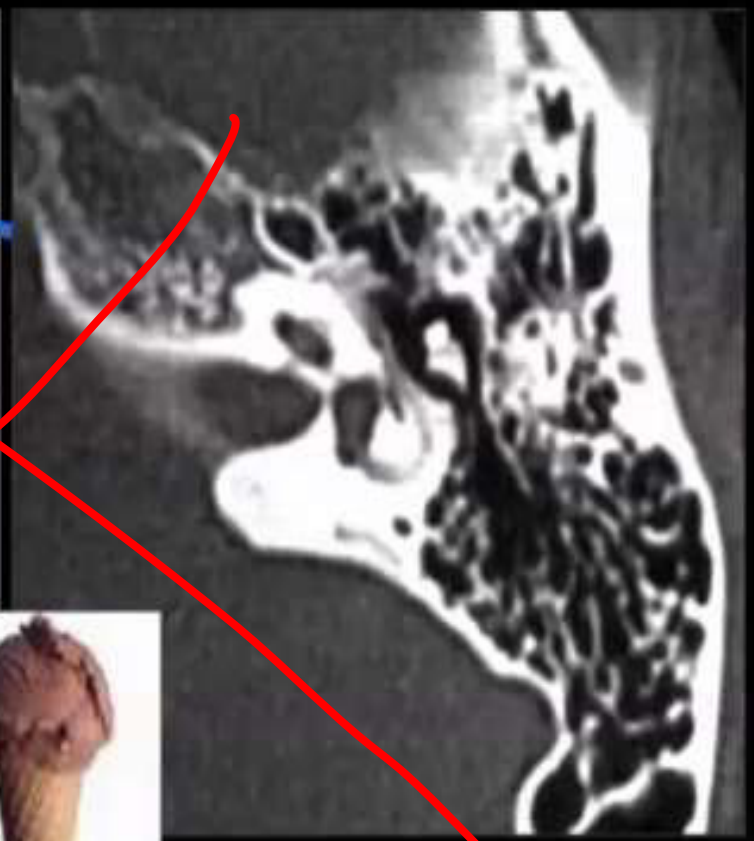
Middle ear anatomy



- Ossicles (Incus, malleous, stapes)
- Muscles and nerves

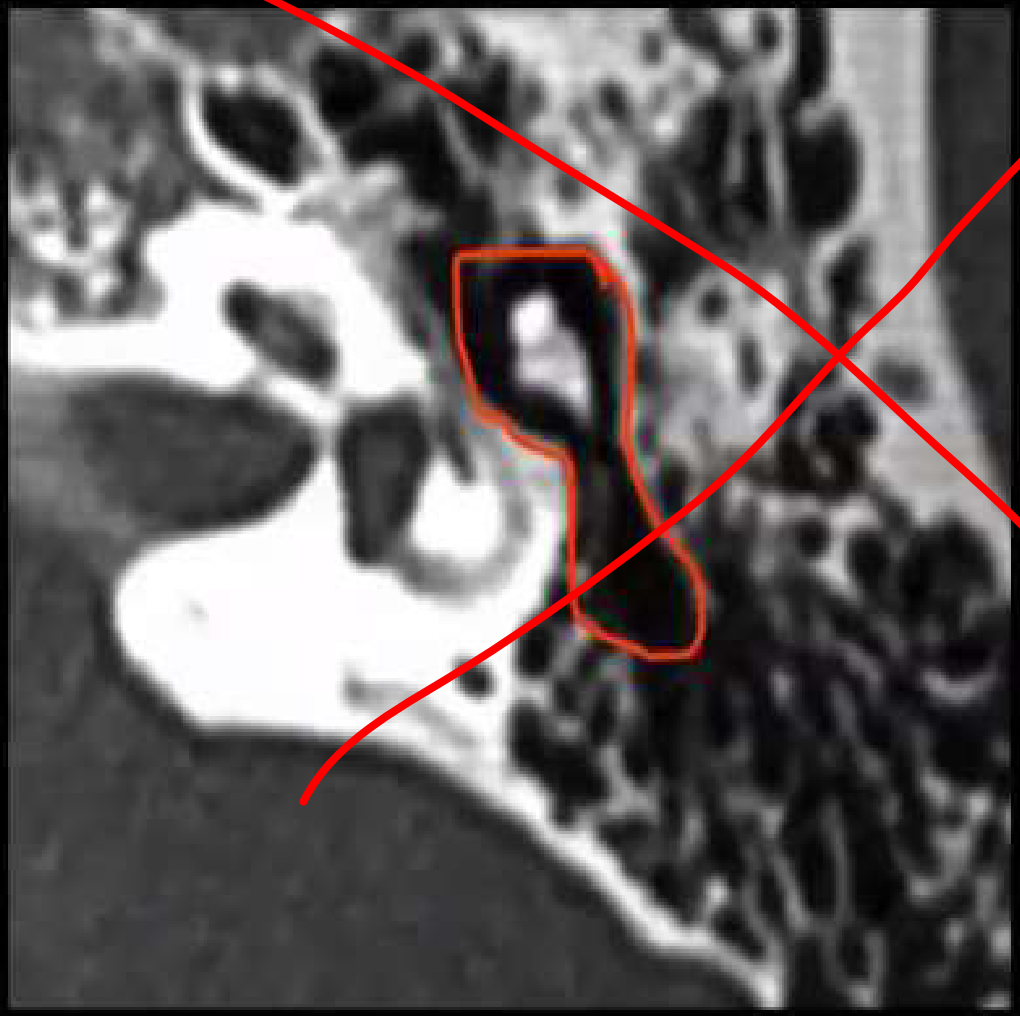
Middle ear anatomic land mark



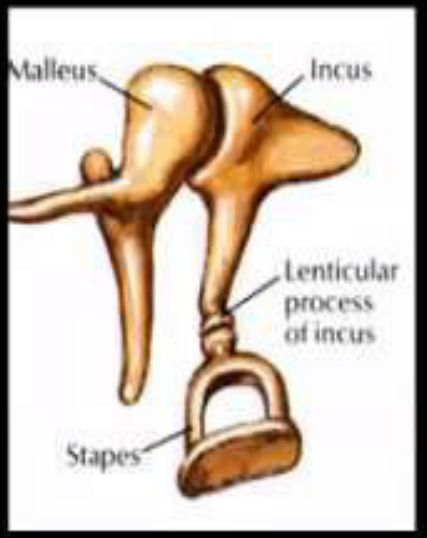


Ice cream cone

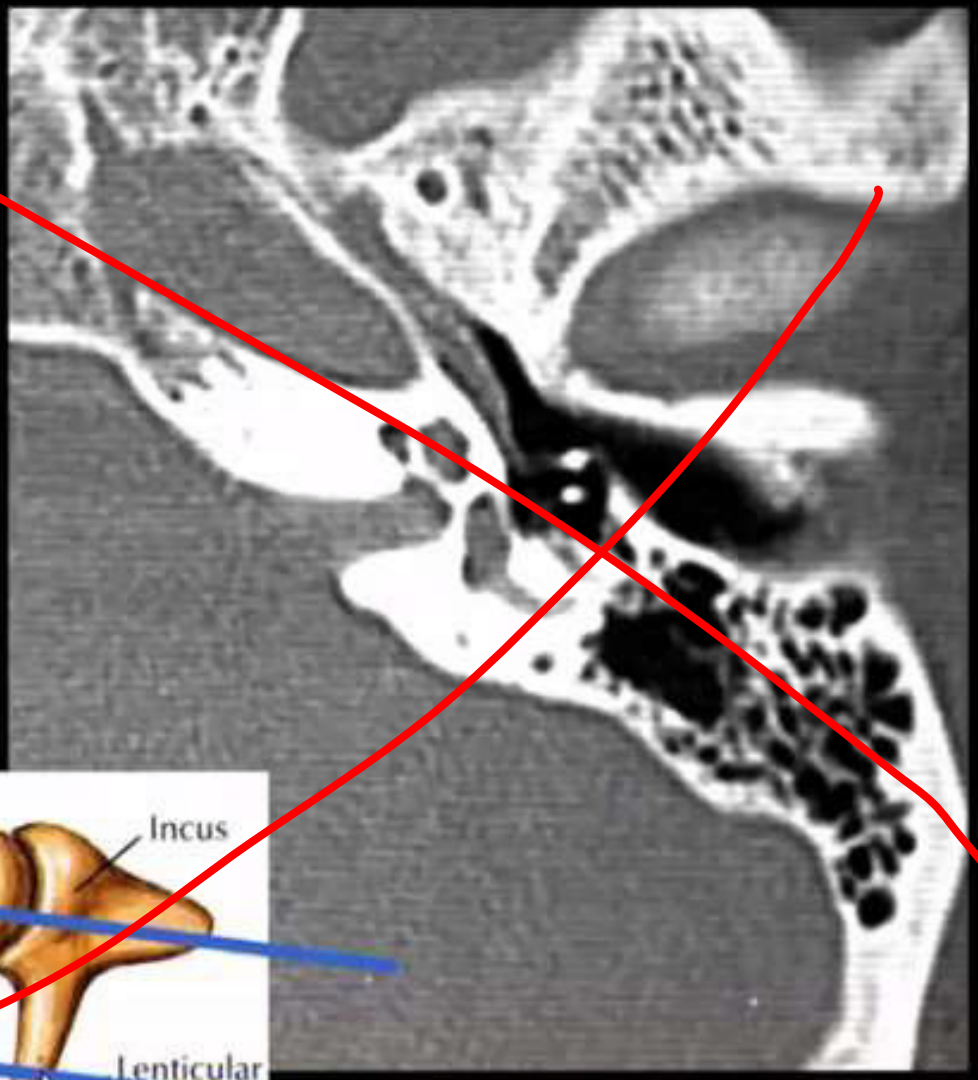
Middle ear cavity



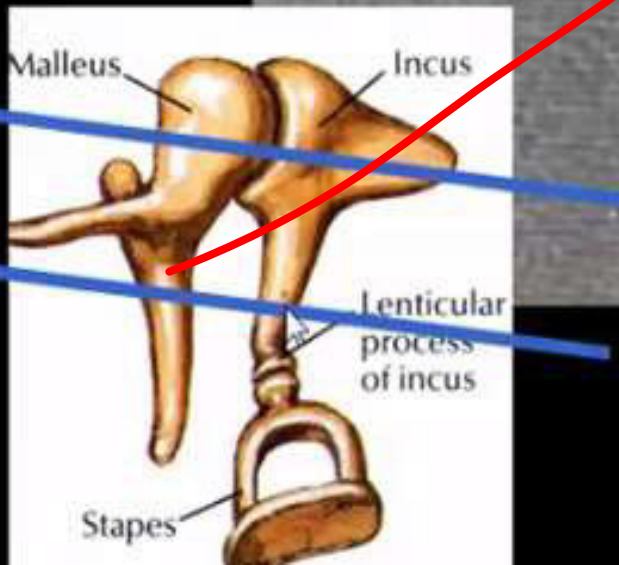
- Ice cream cone
- Incus
- Malleus
- Aditus ad antrum
- Mastoid antrum
- Incudo - malleal joint

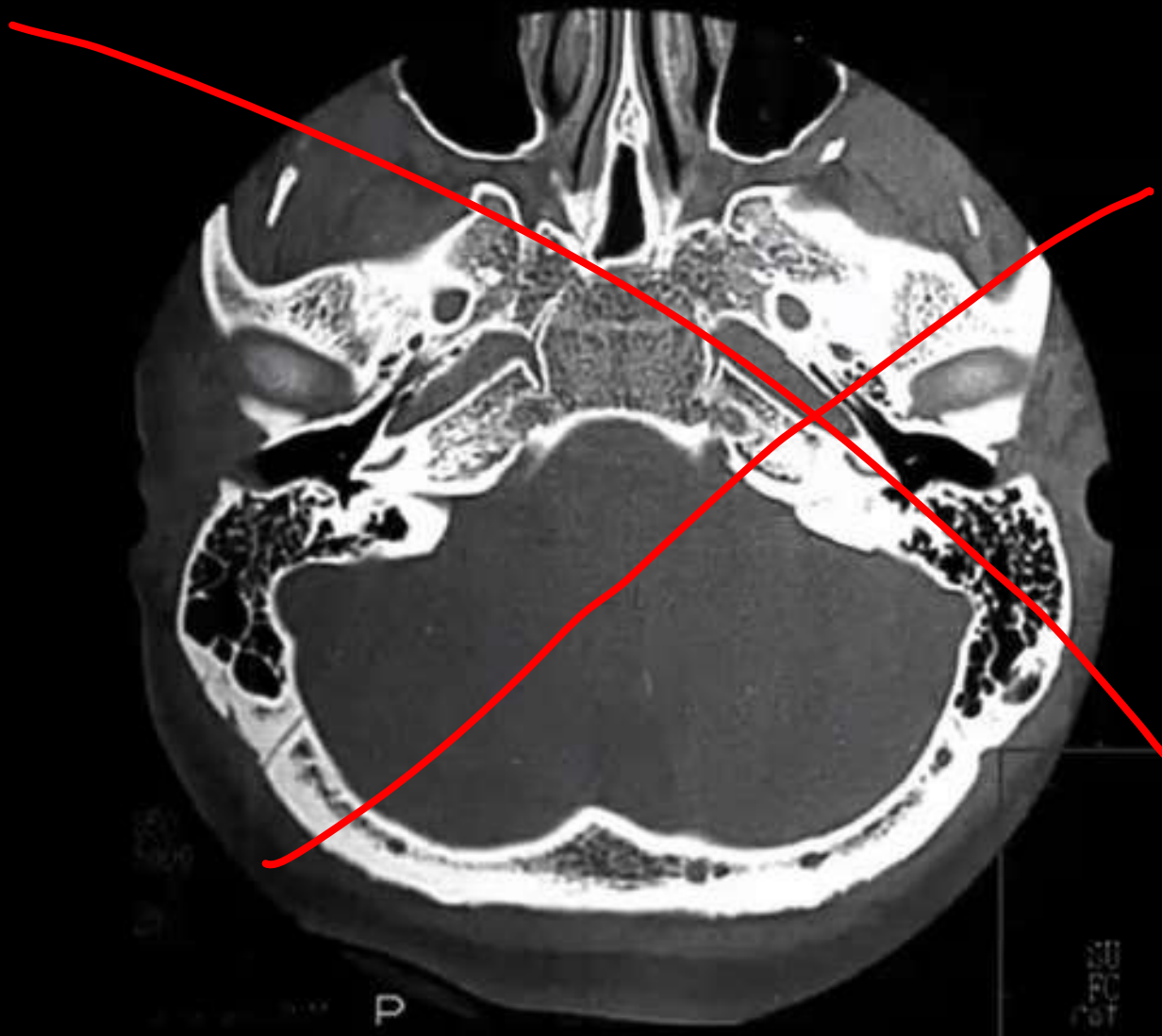


- Incus
- Malleus
- Stapes
- Mastoid



- Incus
- Malleus
- Eustachian tube

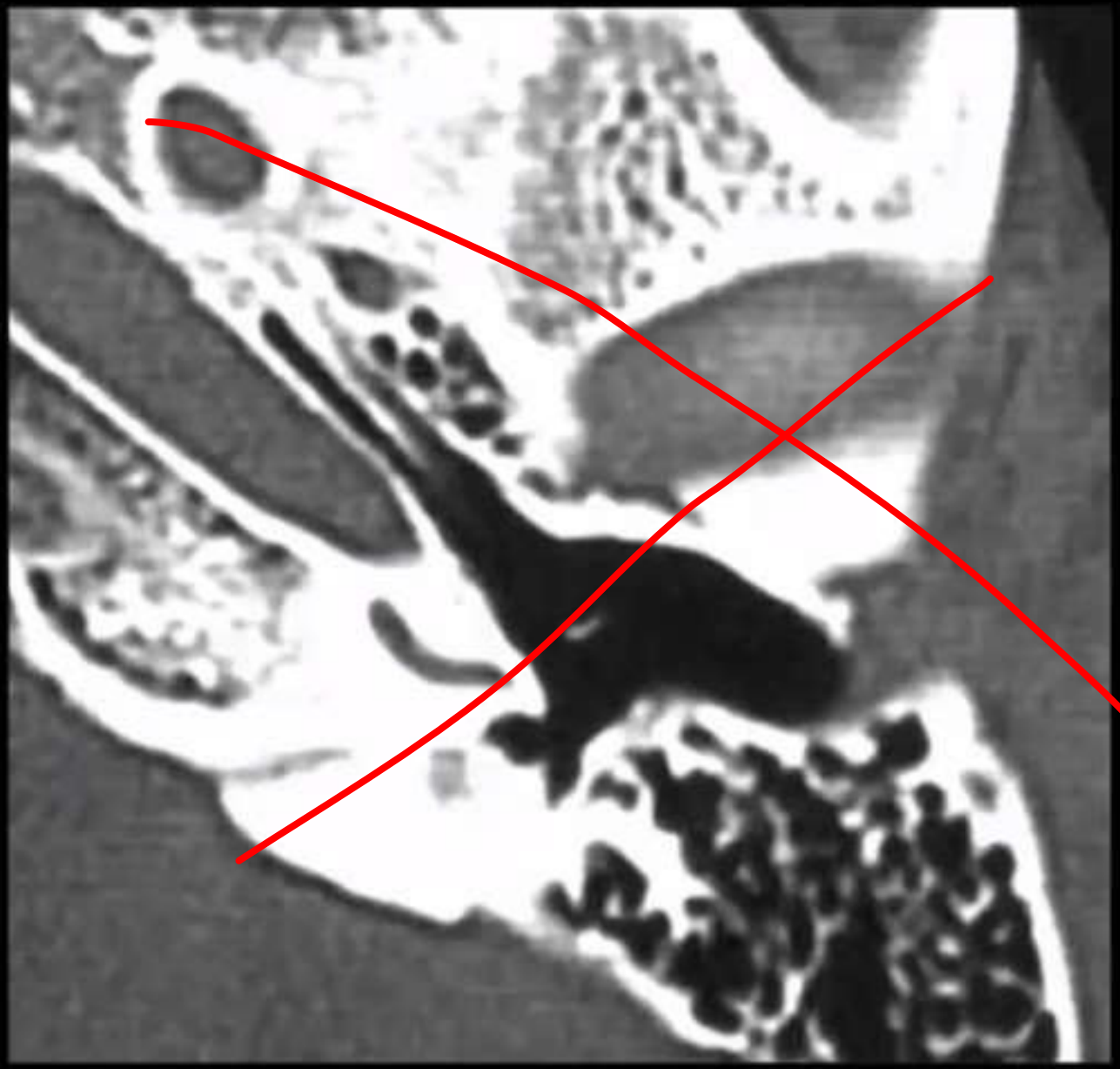




- Sinus tympani
- Facial recess
- Tympanic P.
- Eustachian tube
- Mastoid

Sea loin

Hypo tympanum

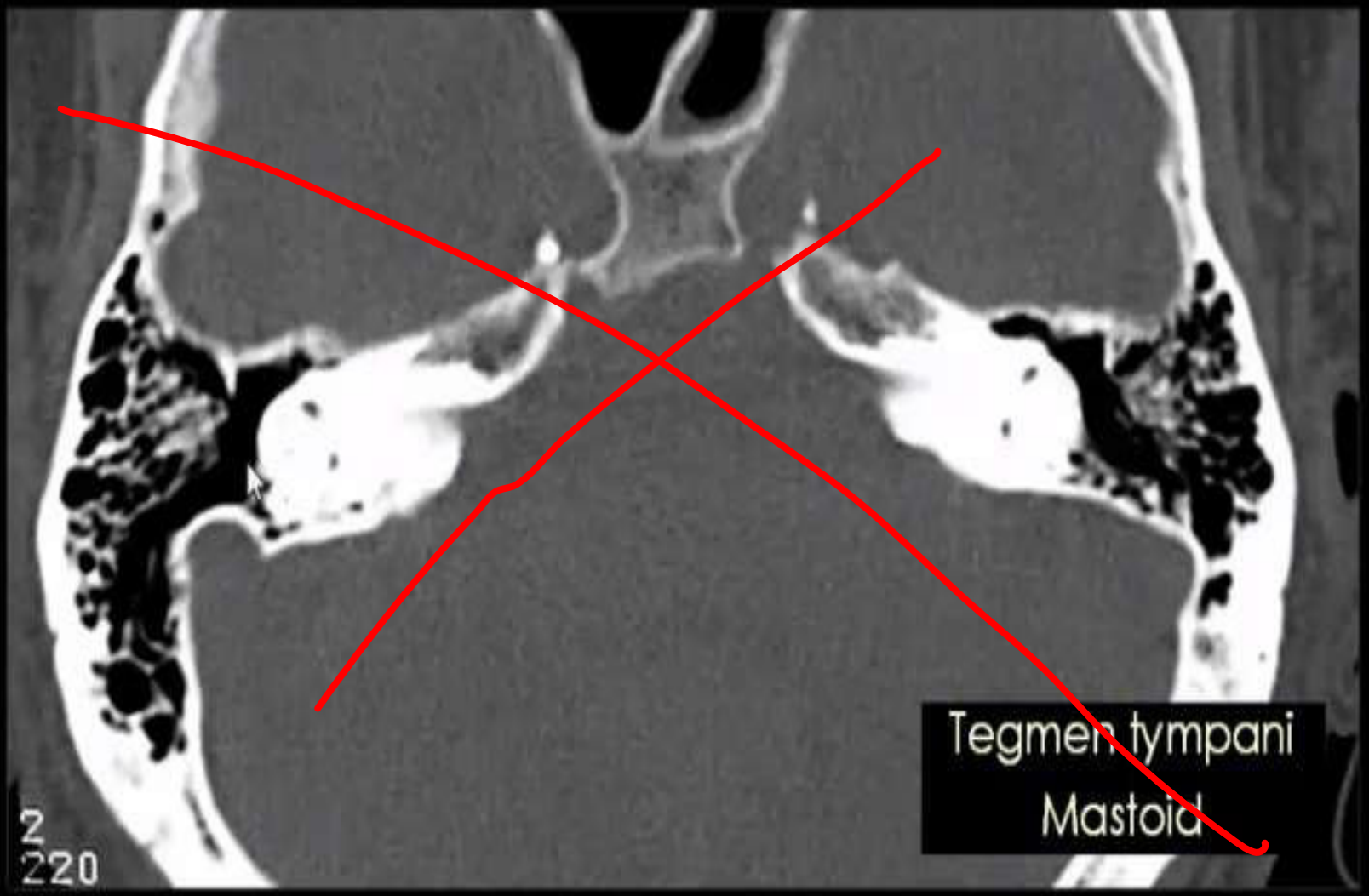


- Sinus tympani
- Facial recess
- Tympanic P.
- Eustachian tube
- Mastoid

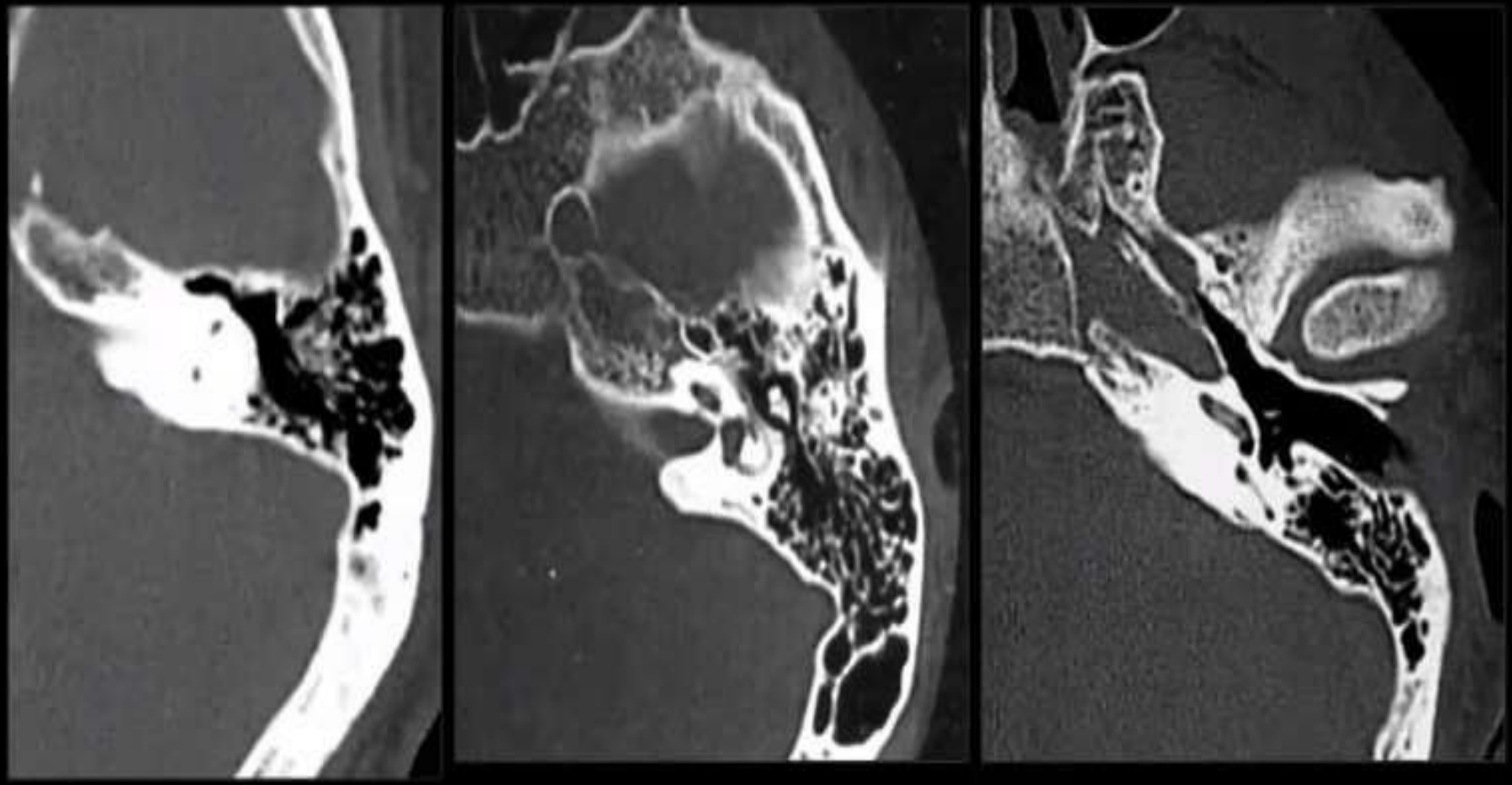
Sea loin

Hypo tympanum

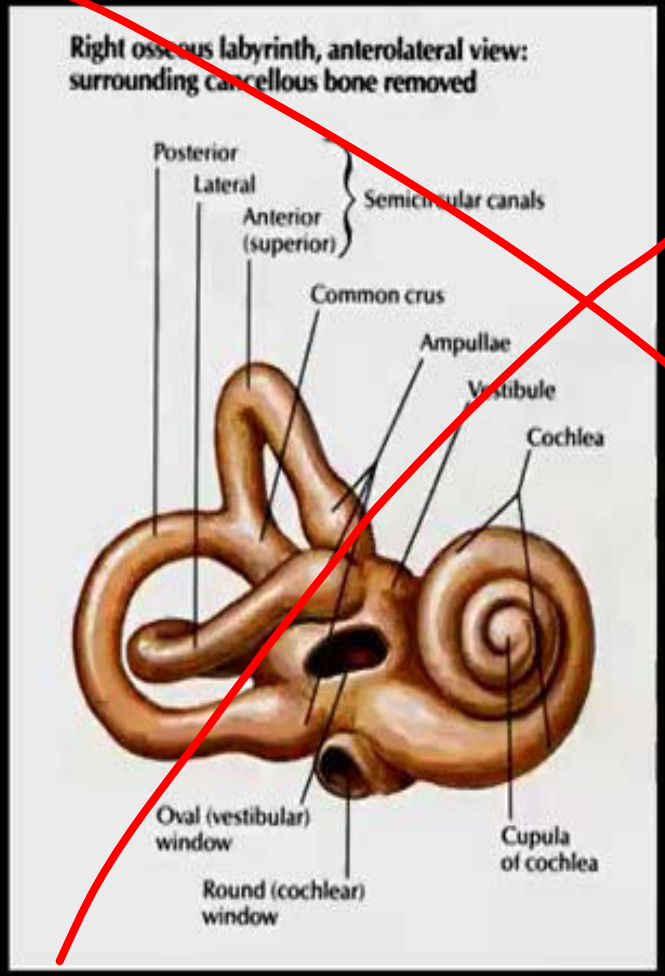
Epitympanum

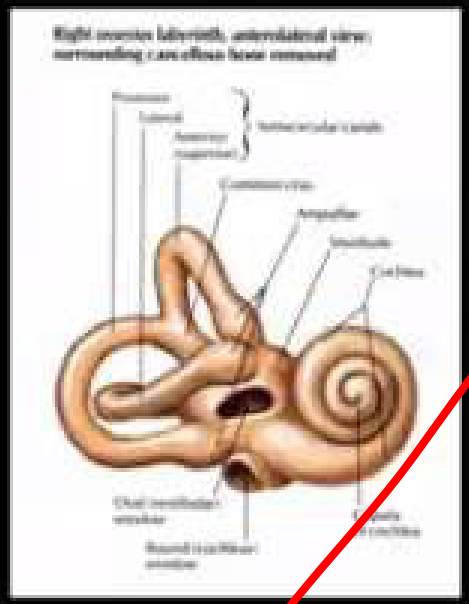


Middle ear



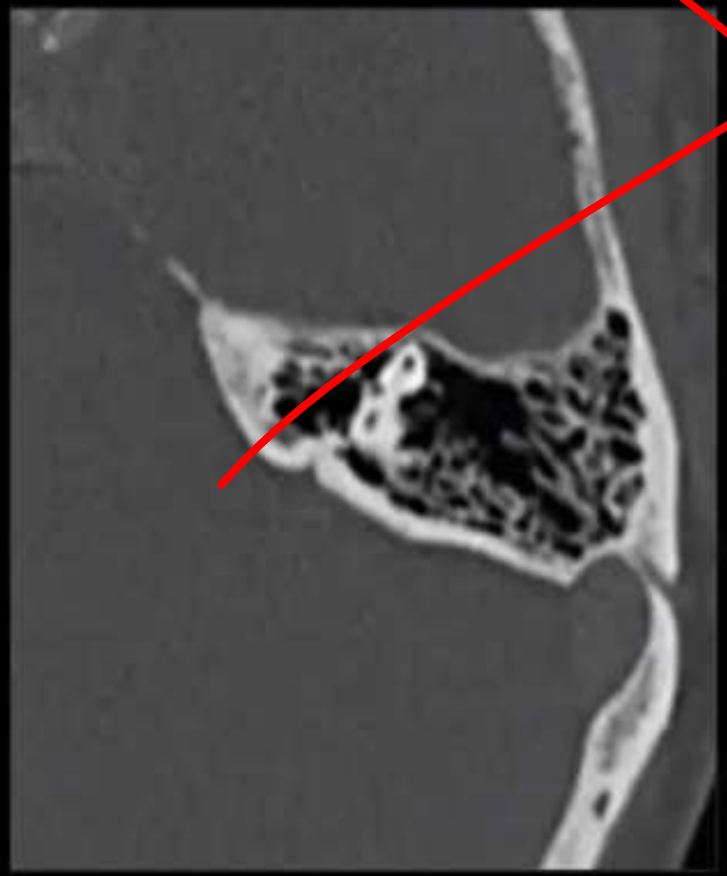
Inner ear anatomic landmark

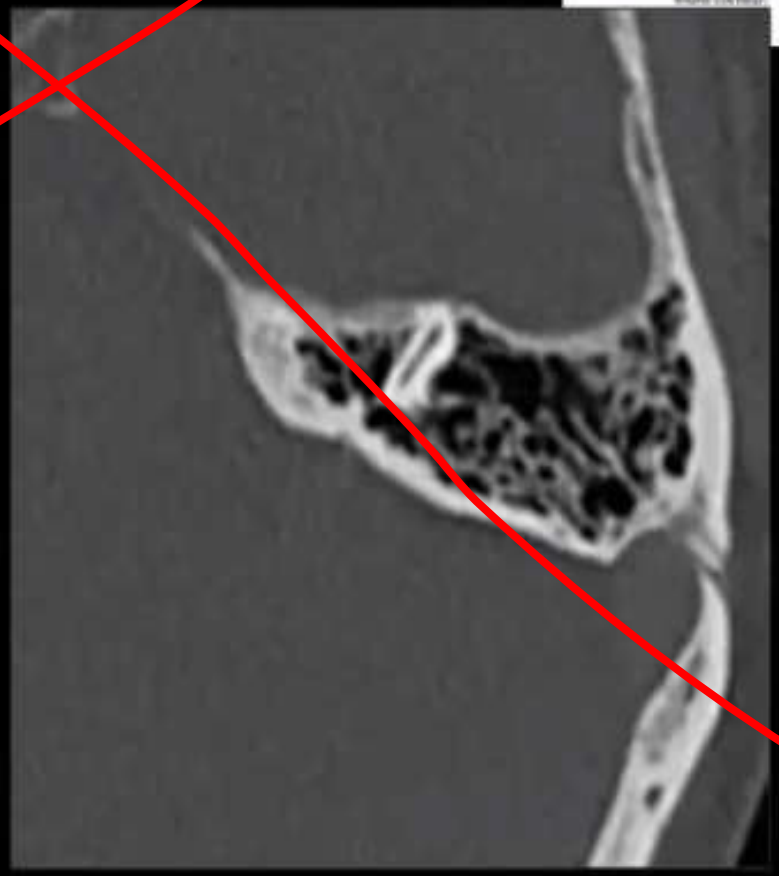


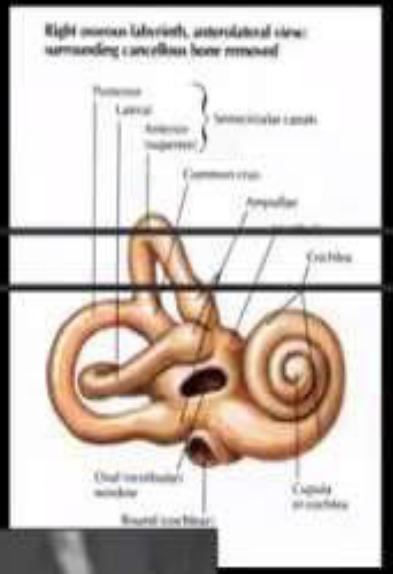


The lateral SCC is seen almost at the level of the ice cream cone if the scan angle is well adjusted

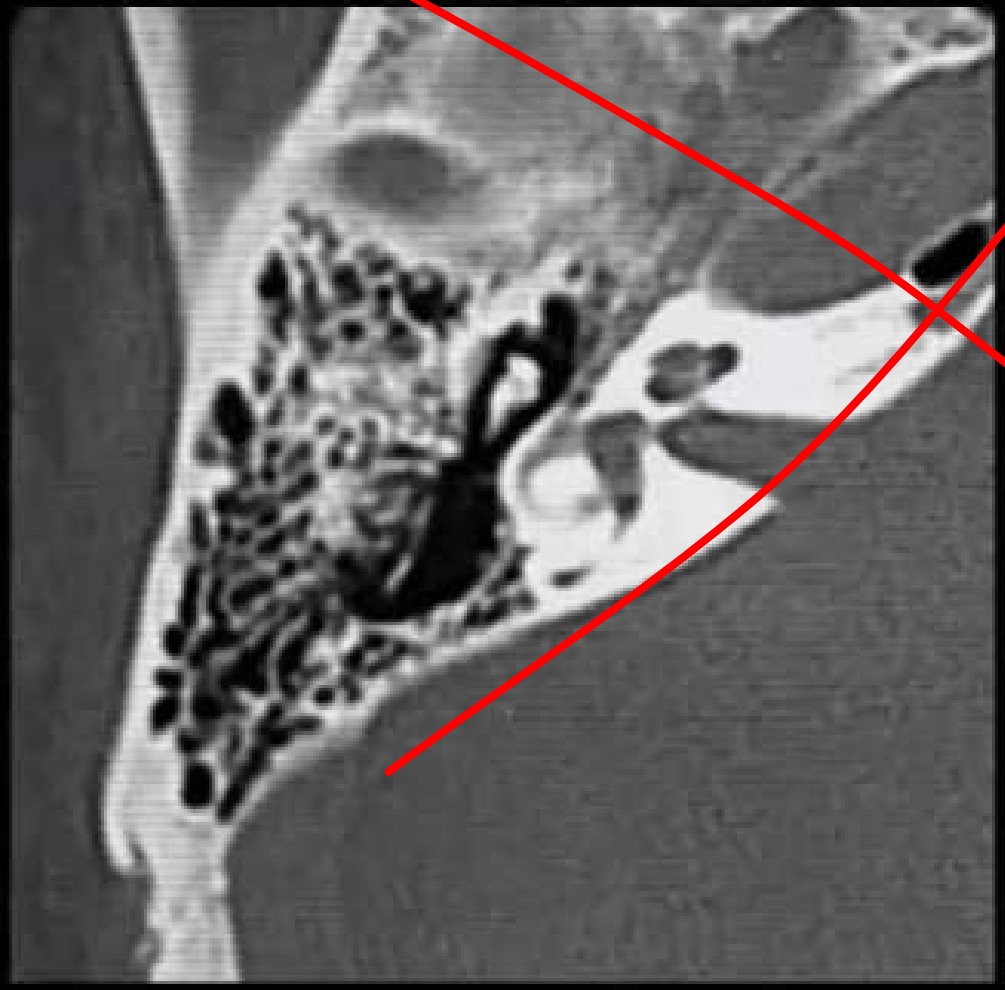
Superior SCC

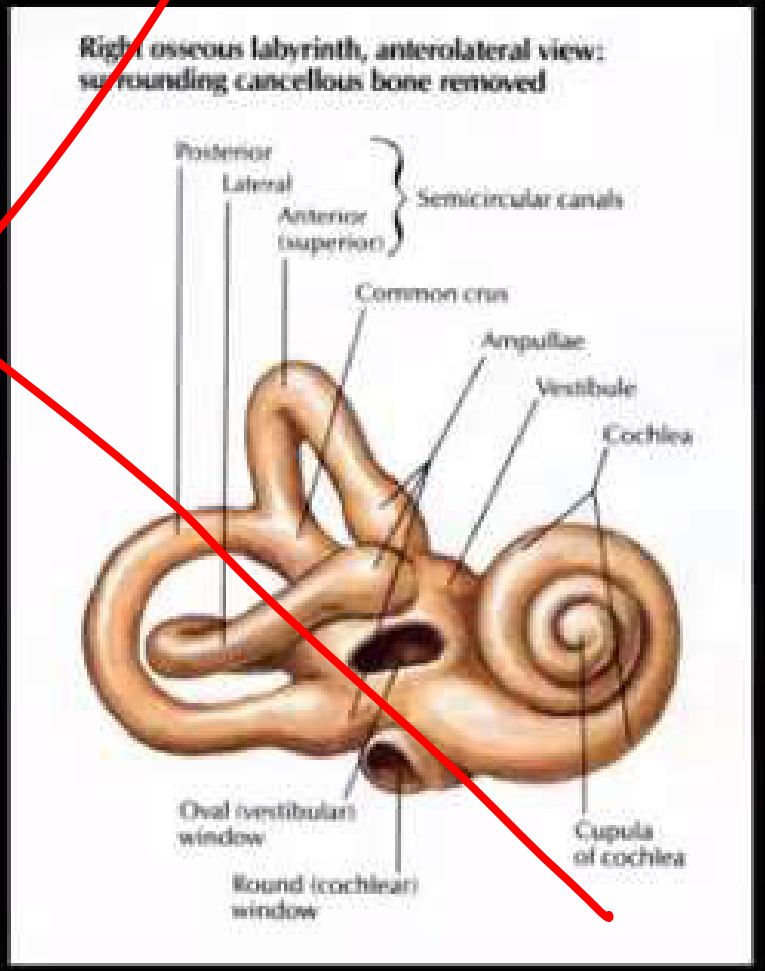






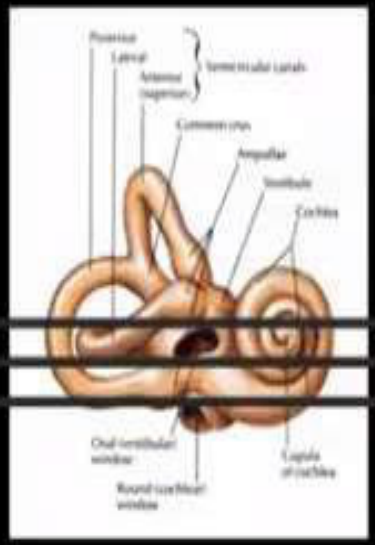
Cochlea

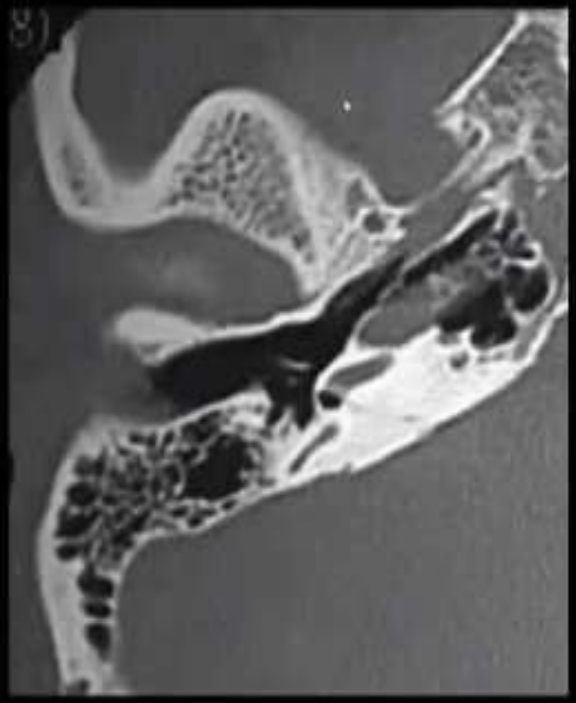


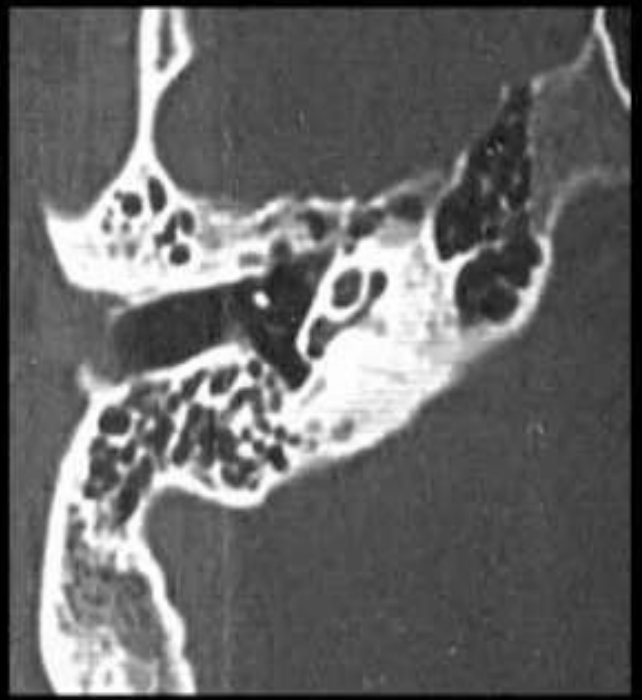

Right osseous labyrinth, anterolateral view:
surrounding cancellous bone removed
Posterior
Lateral
Anterior
(superior)
Semicircular canals
Common crus
Ampullae
Vestibule
Cochlea
Oval (vestibular)
window
Round (cochlear)
window
Cupula
of cochlea

- Middle ear cavity
- Ice cream cone
- Mastoid
- Lateral SCC
- Posterior SCC
- Dome of cochlea
- IAC
- Vestibule

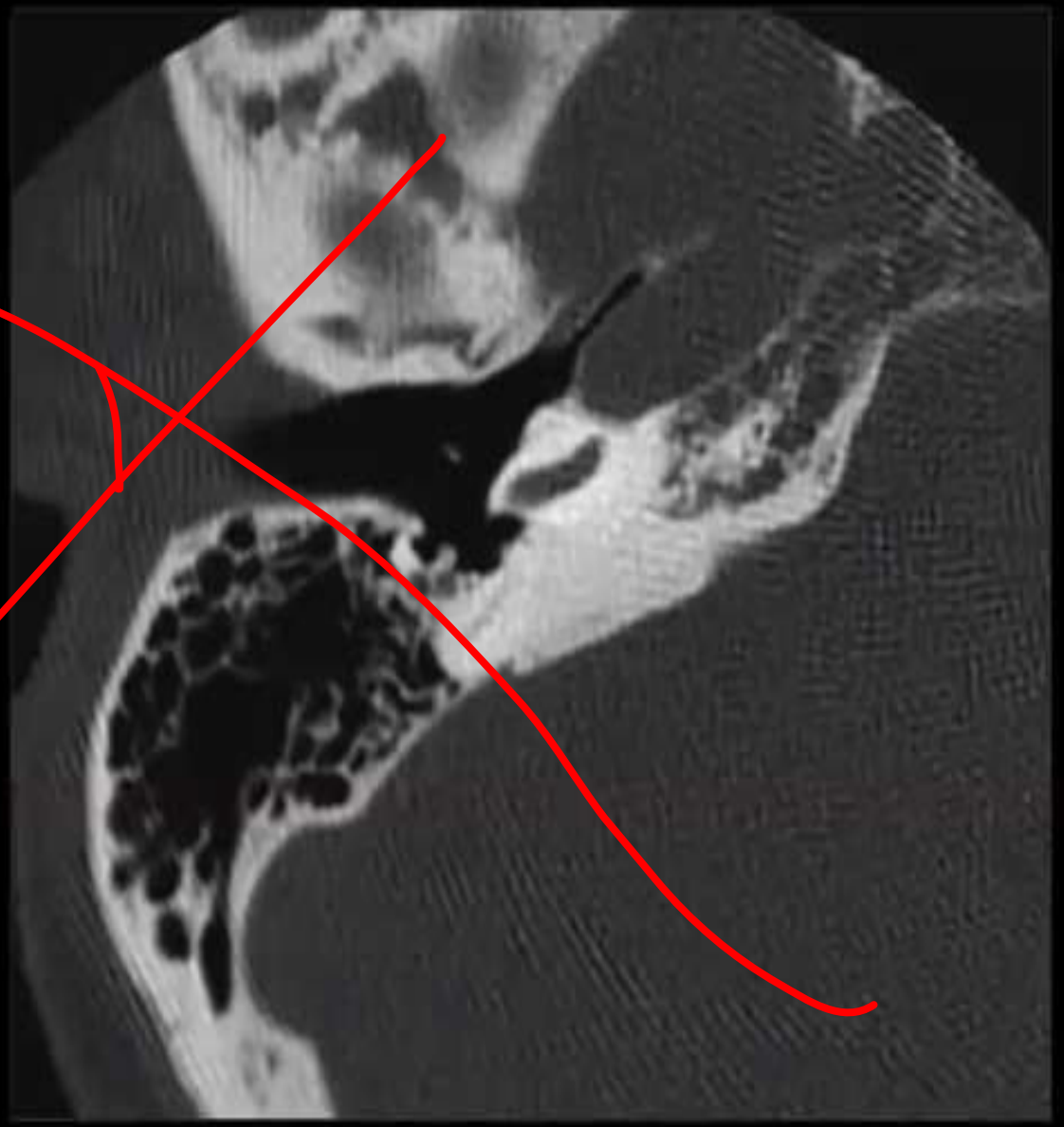
Cochlea


Posterior
Lateral
Anterior (superior)
Semicircular canals
Common crus
Ampullae
Vestibule
Cochlea
Oval (vestibular) window
Cupola of cochlea
Round (cochlear) window

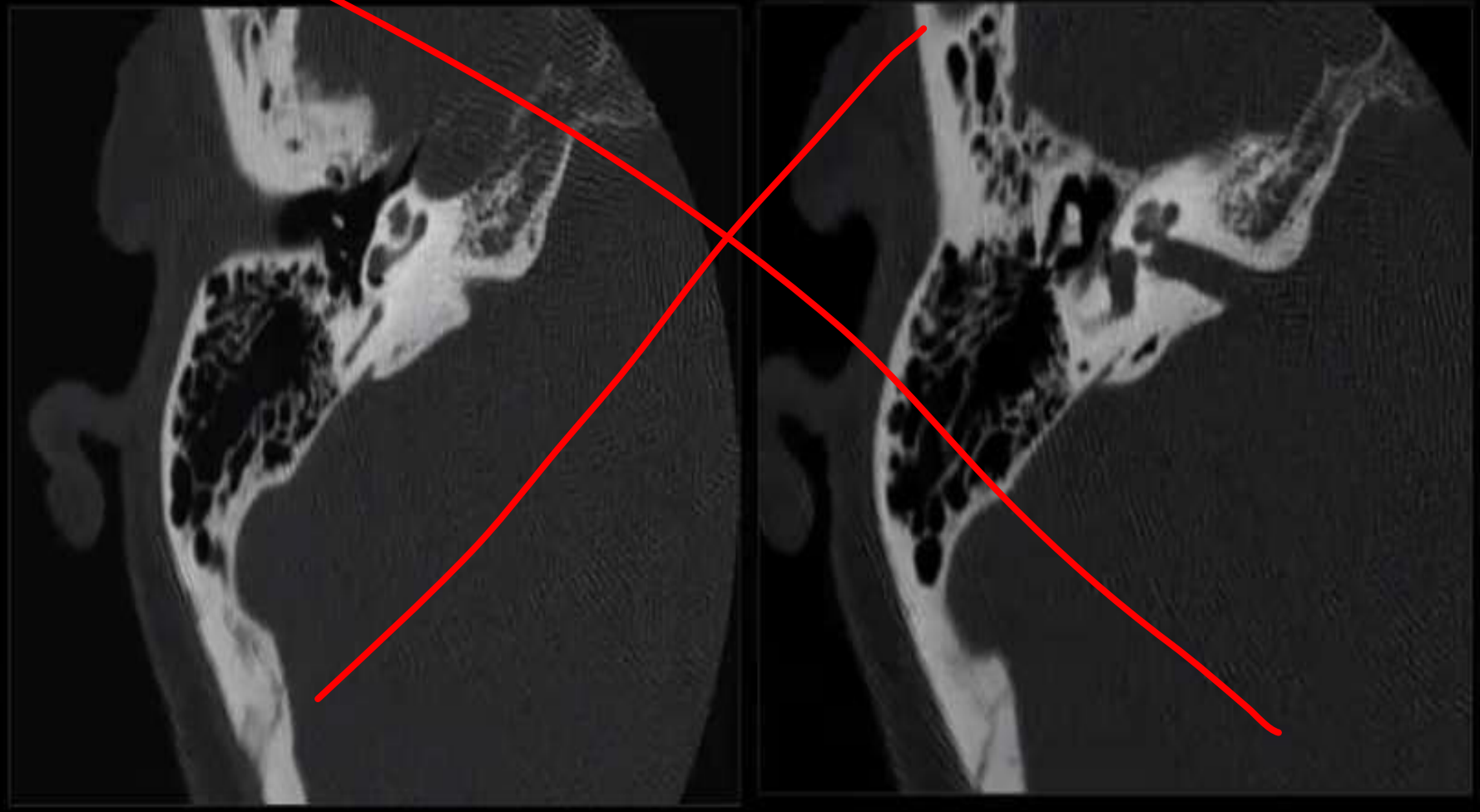




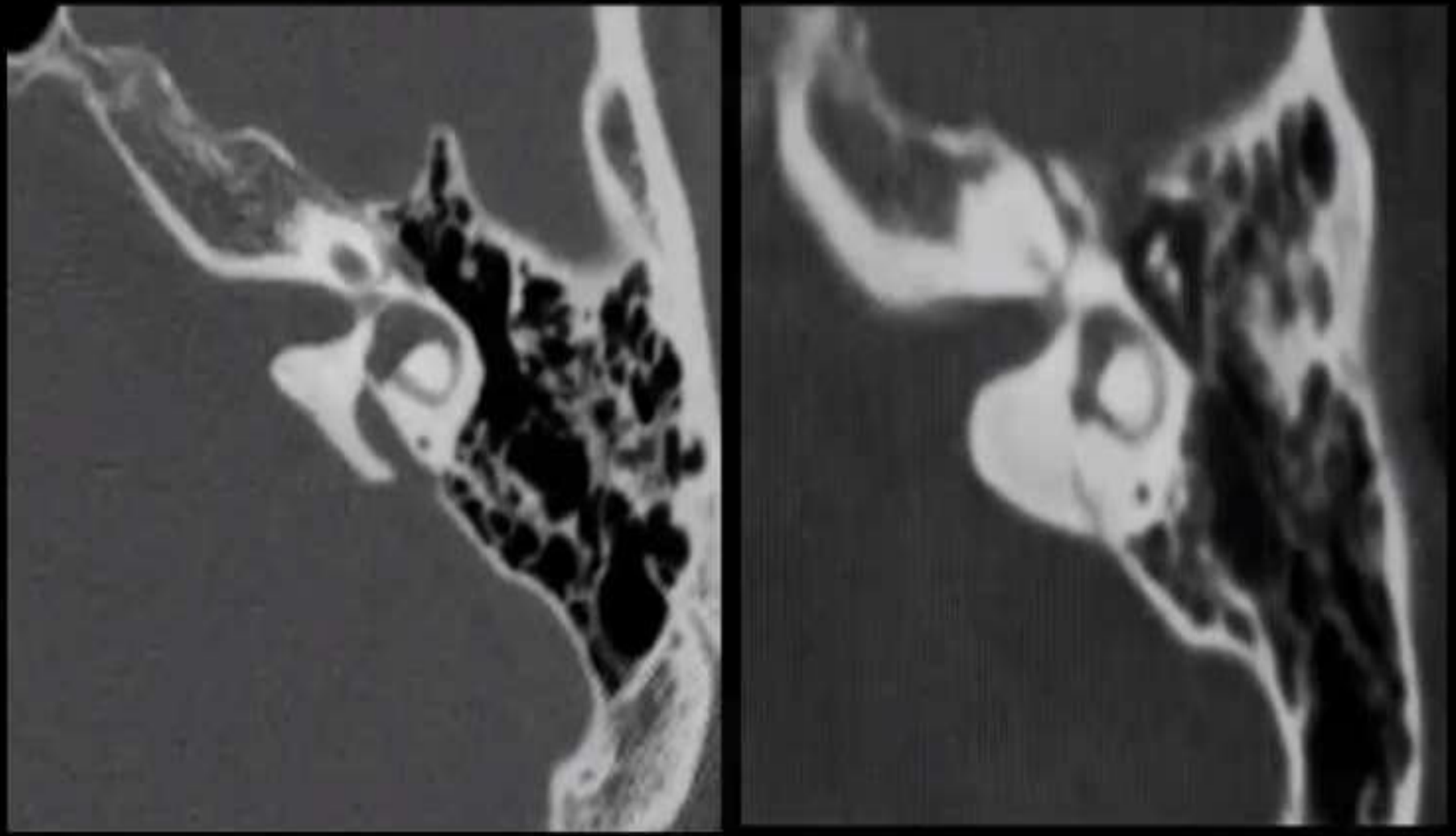
- Sinus tympani
- Facial recess
- Tympanic pyramid
- Basal turn of cochlea
- Eustachian tube
- Niche of round window



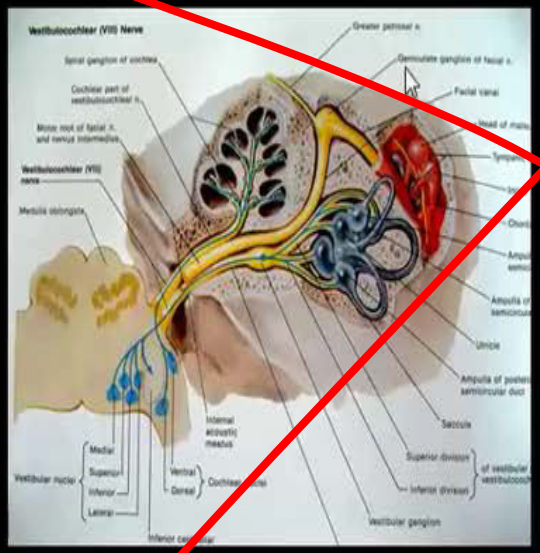
Cochlear aqueduct



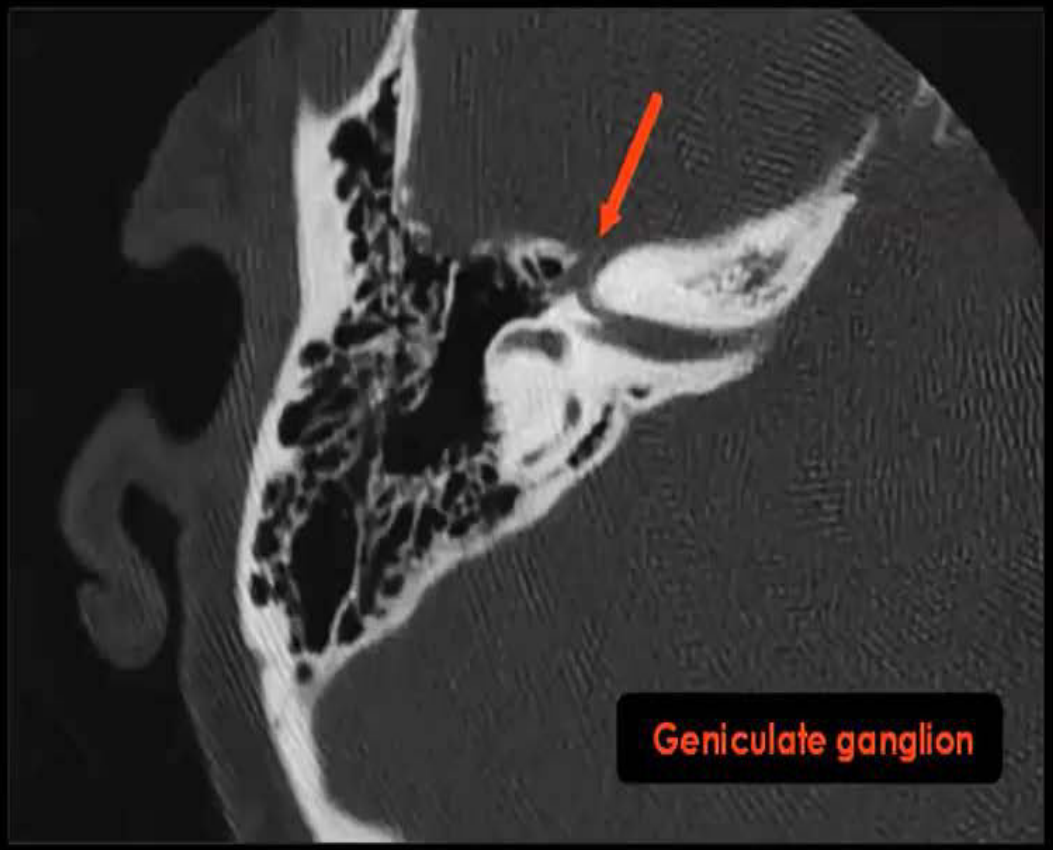
Vestibular aqueduct



Facial nerve anatomy

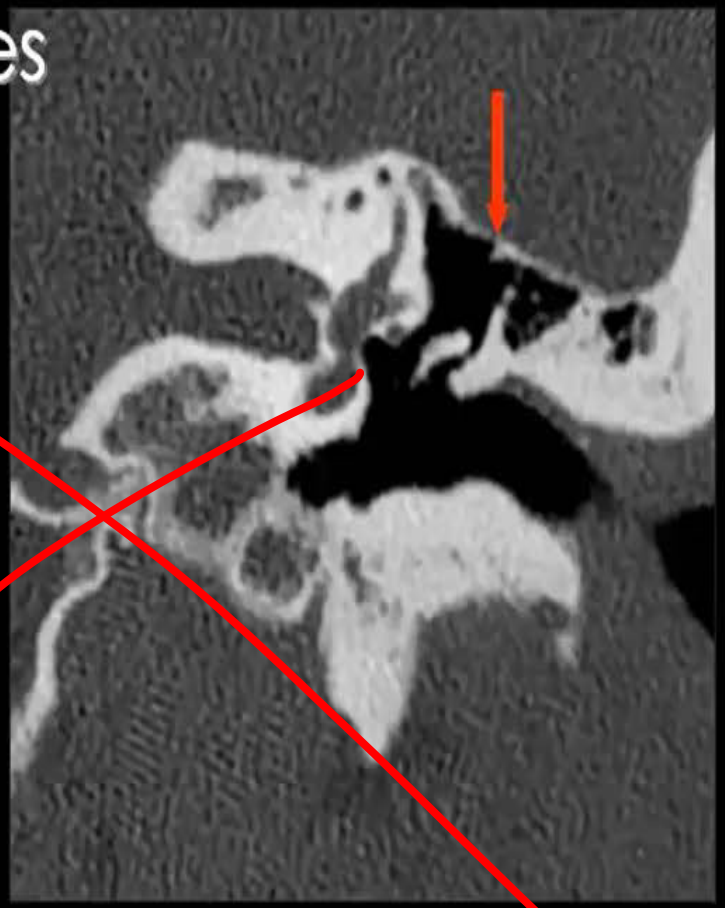


Facial nerve anatomy

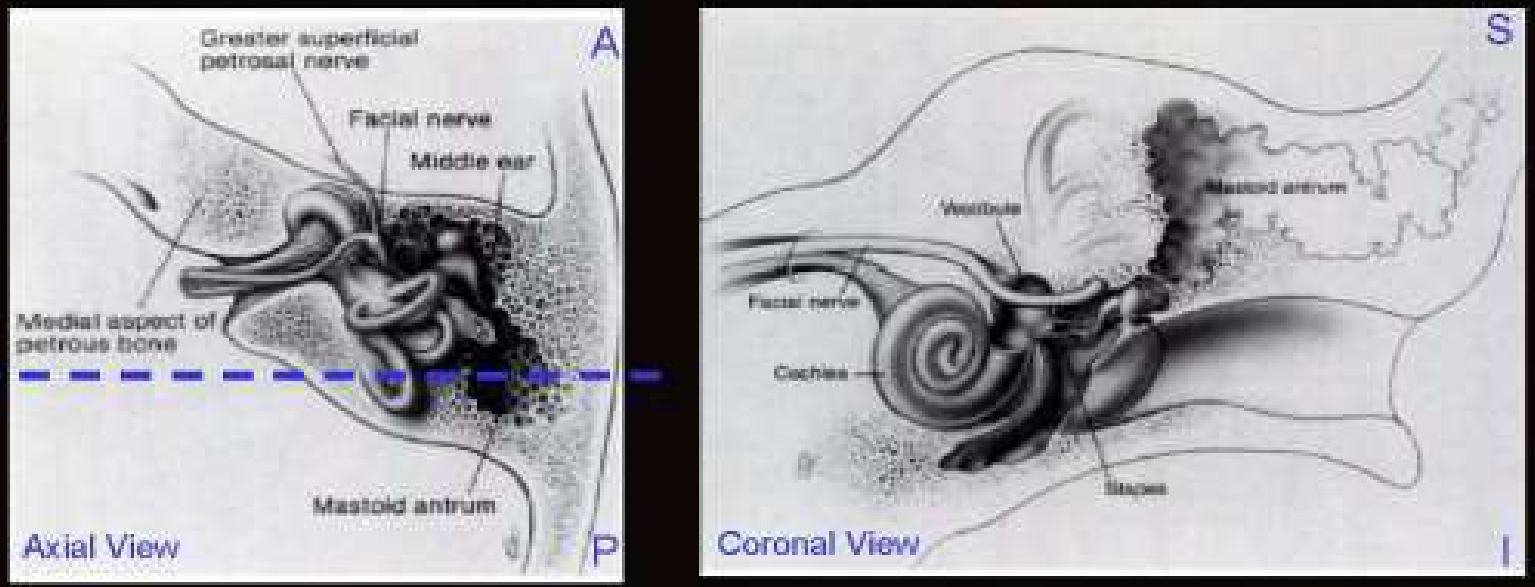


Coronal images

- Tegmen tympani
- Oval window
- Facial nerve



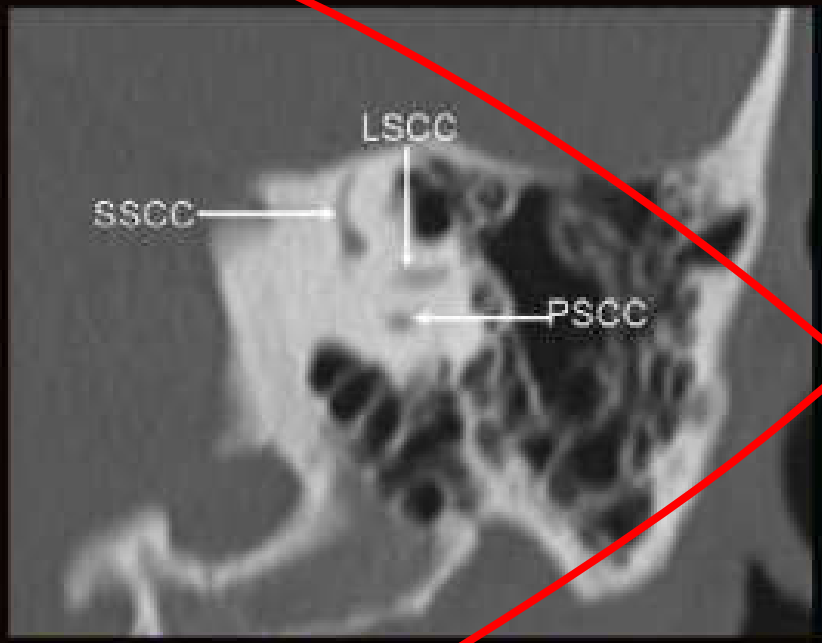
Coronal Slice (Posterior to Anterior)

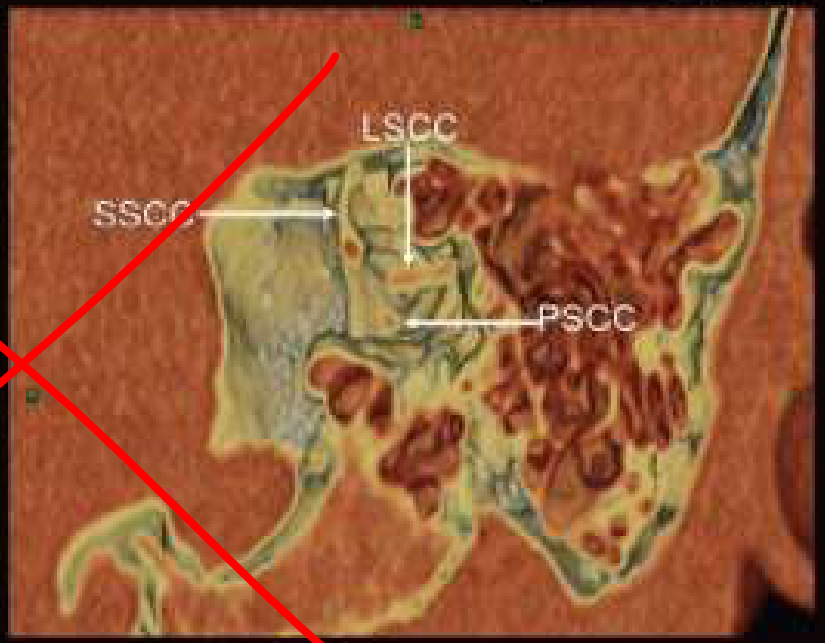


- The following slides are representative coronal slices through the normal temporal bone, proceeding anteriorly from the most posterior cut (through the semicircular canals).

Silver JA et al. 1987

Coronal: Semicircular Canals (posteriorly)

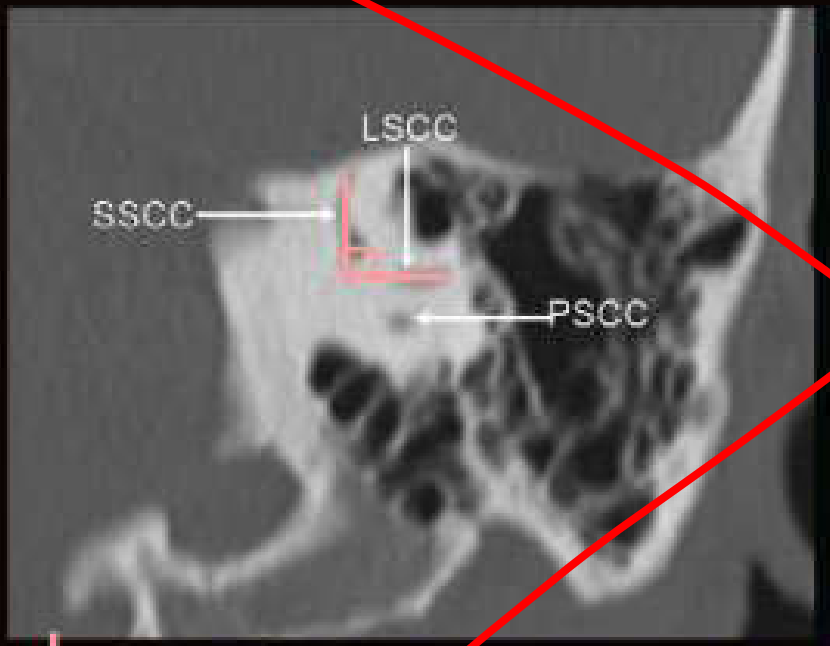


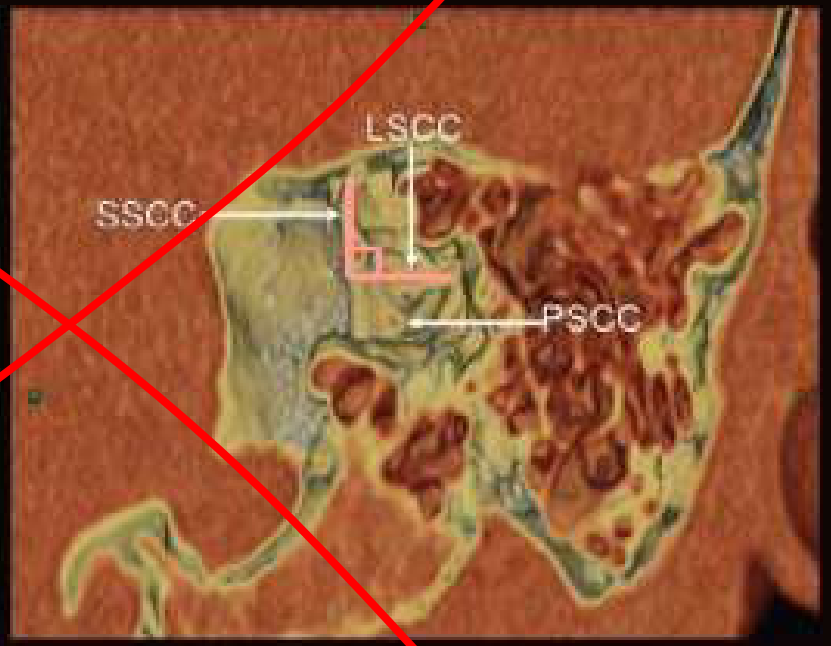


- Posterior, Lateral and Superior SCC

Fatterpekar et al., 3D CT of the temporal bone. www.terarecon.com

Coronal: Semicircular Canal Relationship

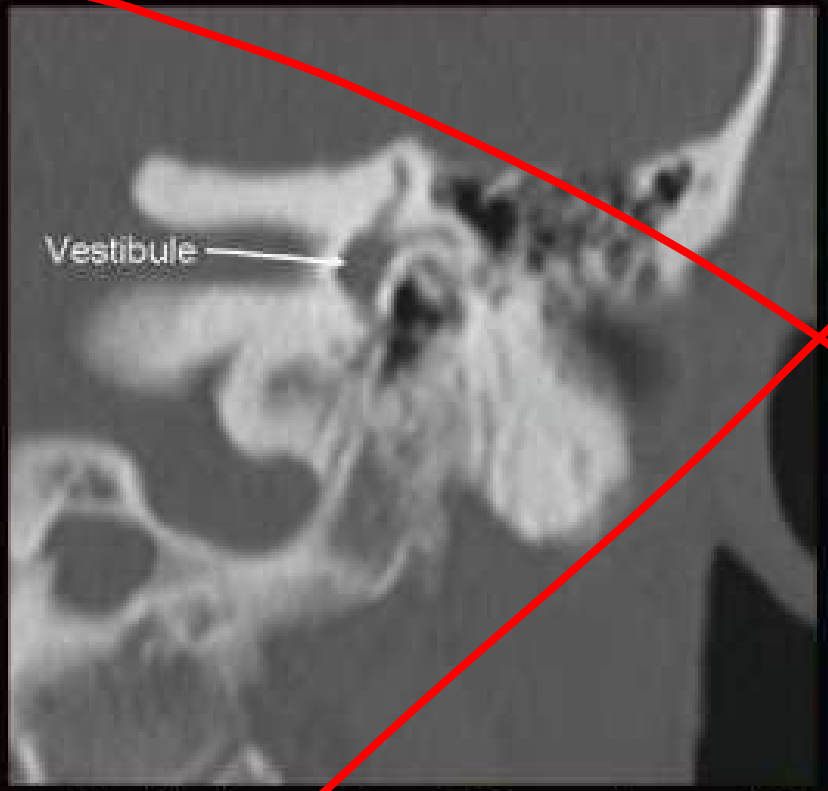


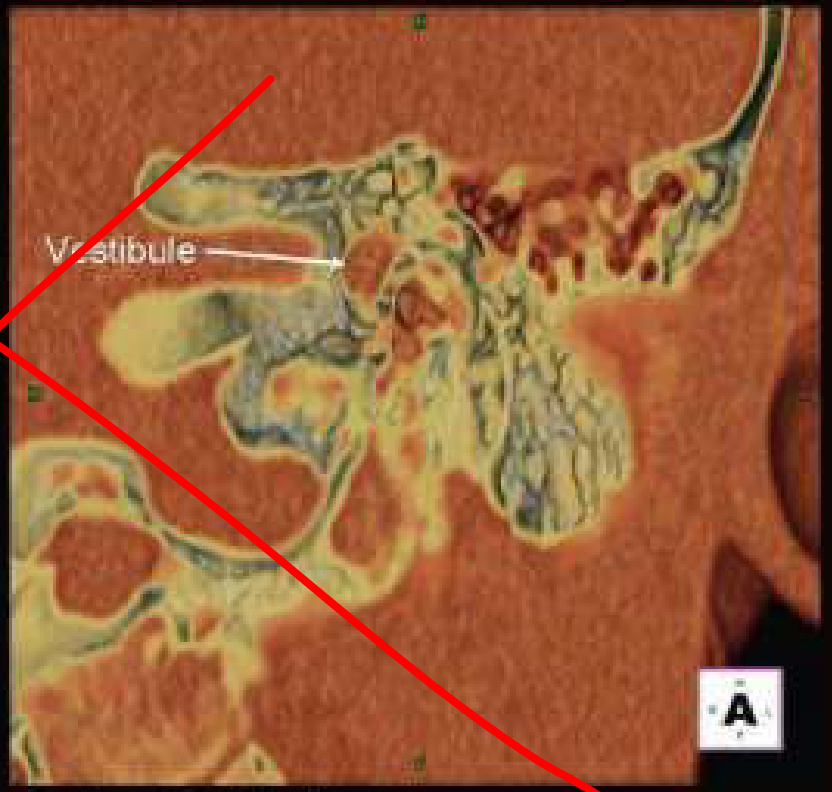


90° angle between Lateral and Superior SCC

Fatterpekar et al., 3D CT of the temporal bone. www.terarecon.com

Coronal: more anteriorly, the vestibule is visualized:

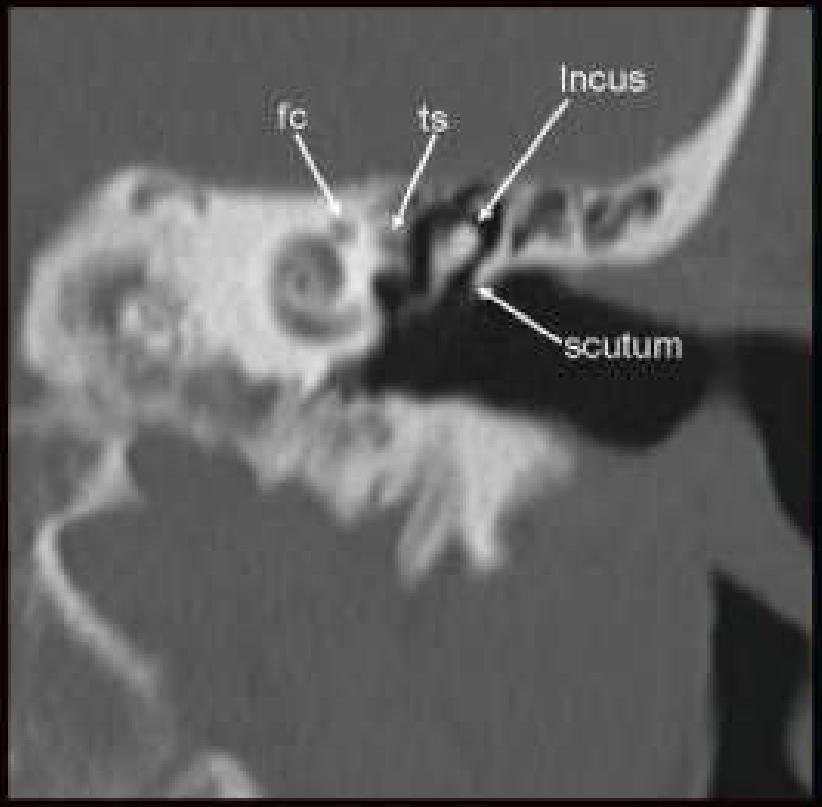


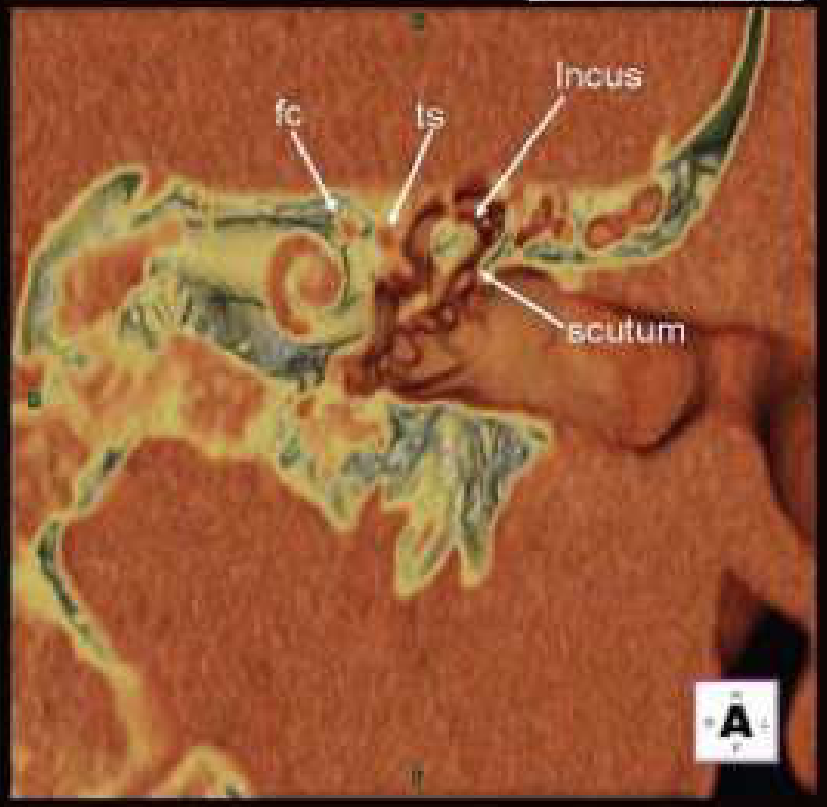


- Vestibule: union of Superior and Posterior SC ducts.

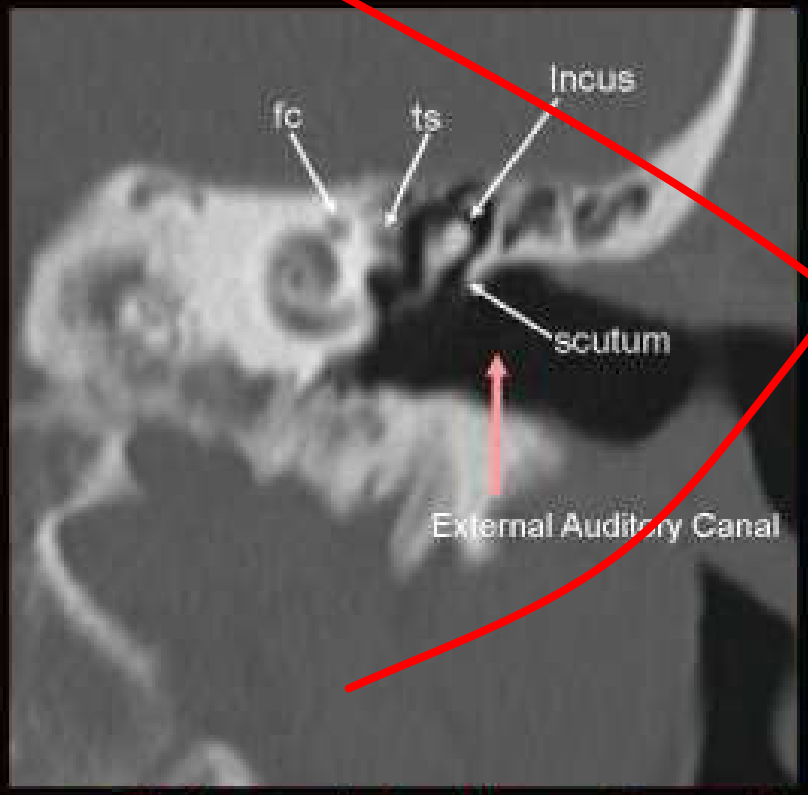
Fatterpekar et al., 3D CT of the temporal bone. www.terarecon.com

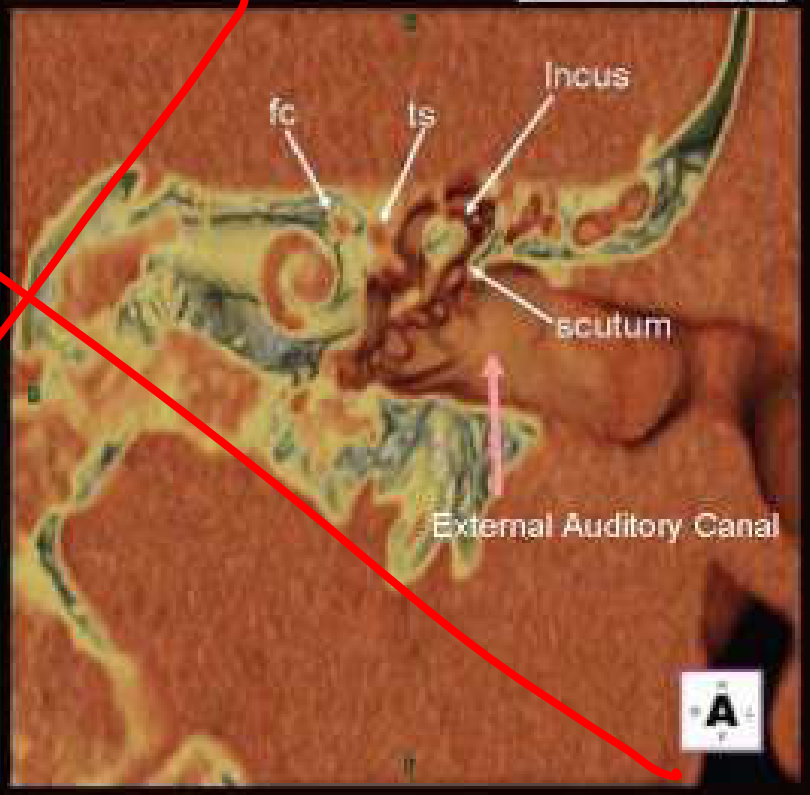
Coronal: continuing anteriorly





Coronal: External auditory canal

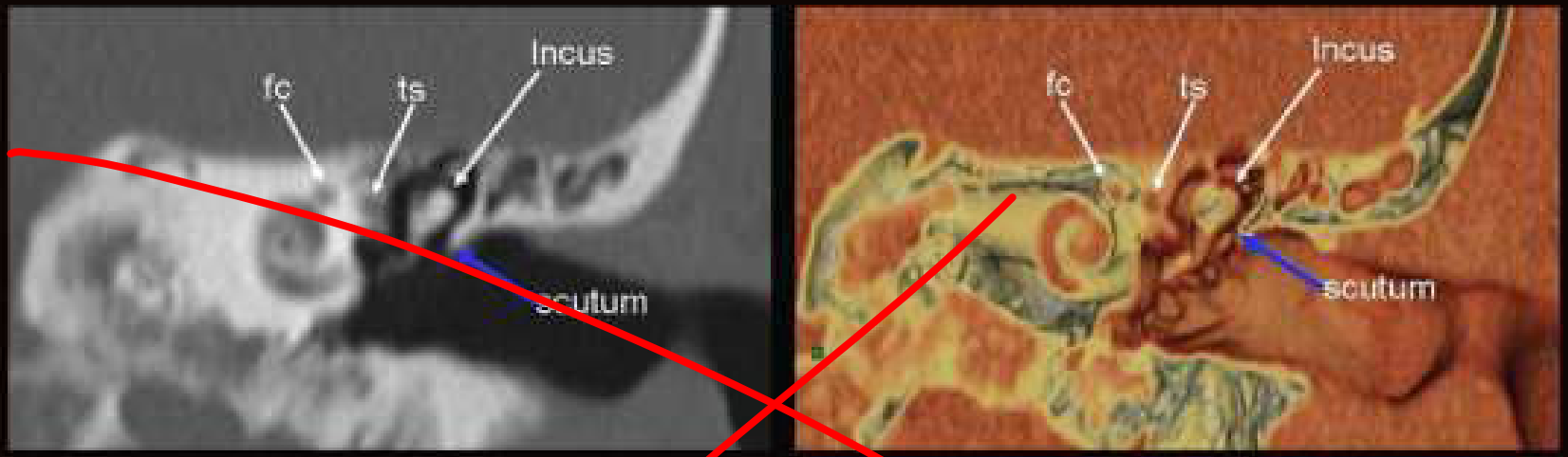




- External Auditory Canal

Fatterpekar et al , 3D CT of the temporal bone, www.terarecon.com

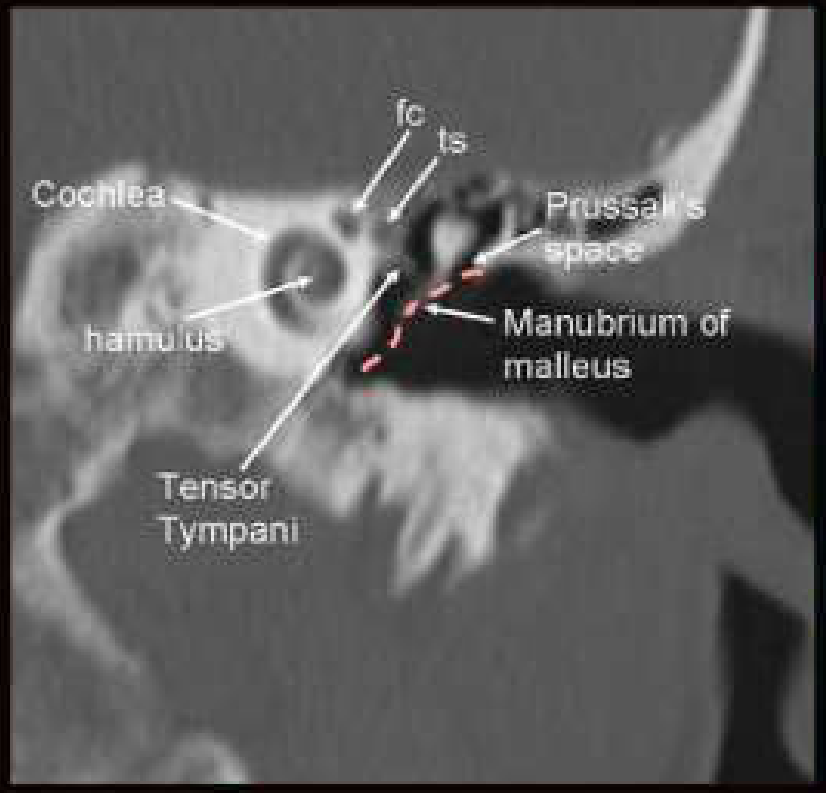
Coronal: Scutum

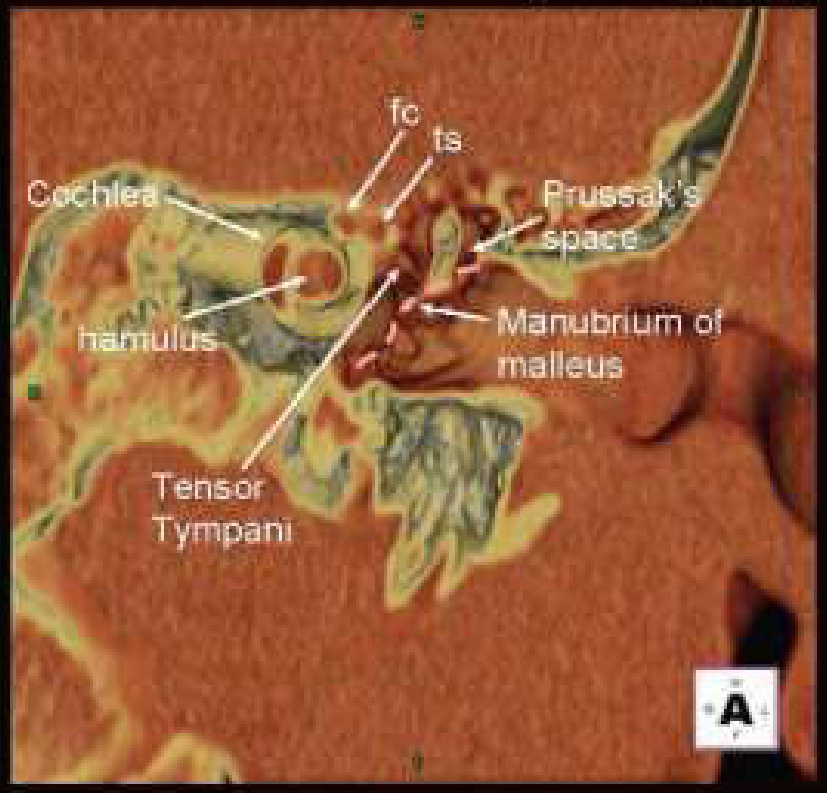


- Scutum: spur-shaped bony projection top of tympanic membrane
  - Often eroded by cholesteatoma

Fatterpekar et al , 3D CT of the temporal bone. www.terarecon.com

Coronal: Tympanic Membrane

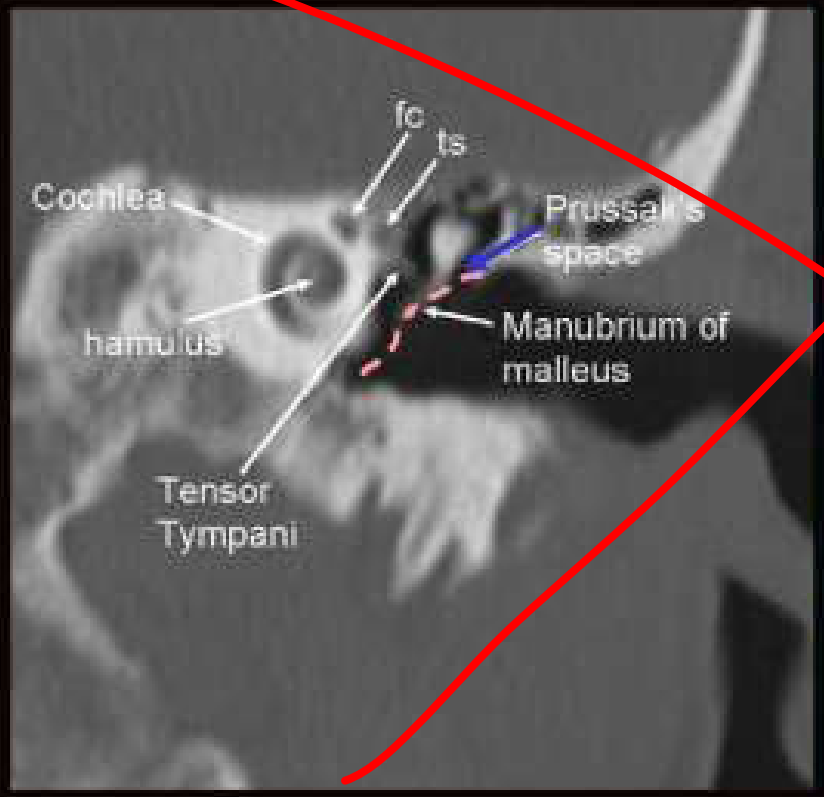


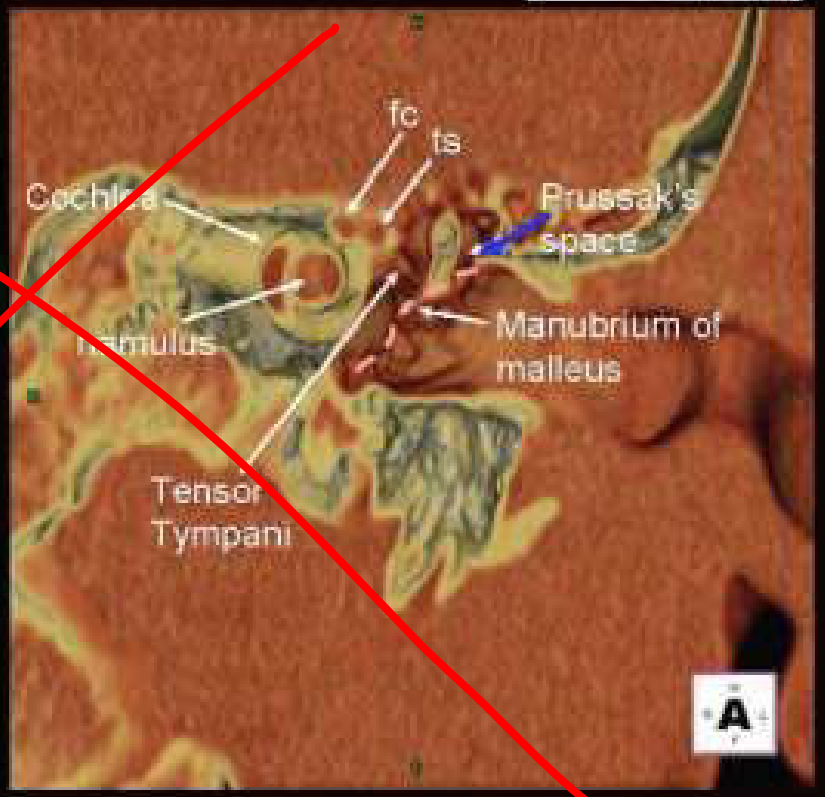


Outline of Tympanic Membrane

Fatterpekar et al , 3D CT of the temporal bone. www.terarecon.com

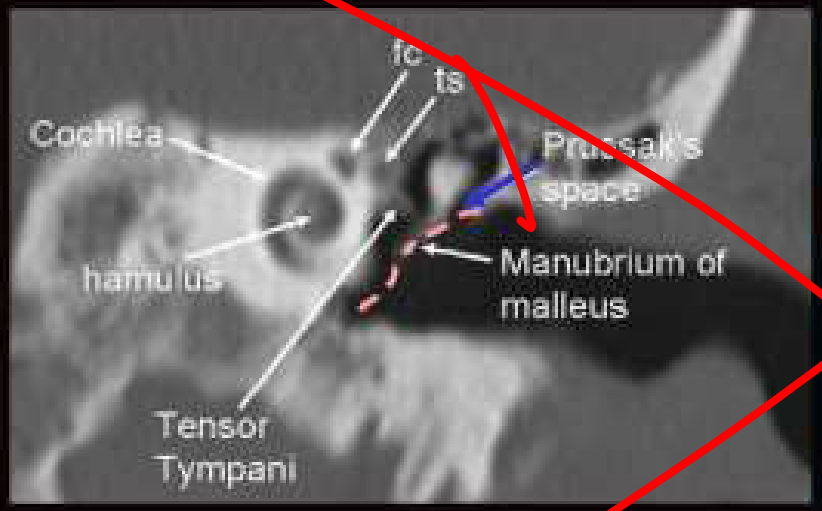
Coronal: Prussak's space

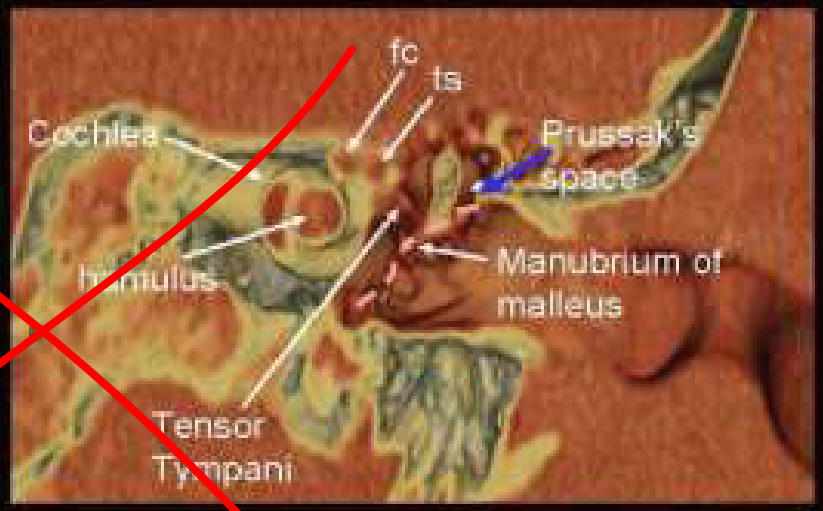




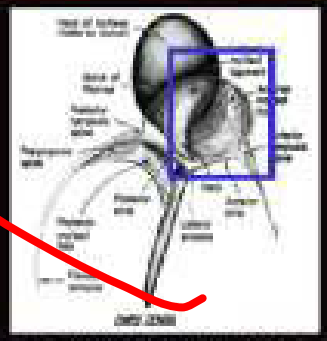
Fatterpekar et al , 3D CT of the temporal bone. www.terarecon.com

Coronal: Prussak's space





- Prussak's space: anterior recess of TM, between neck of malleus and pars flaccida of tympanic membrane
  - *most common site of acquired attic cholesteatoma.

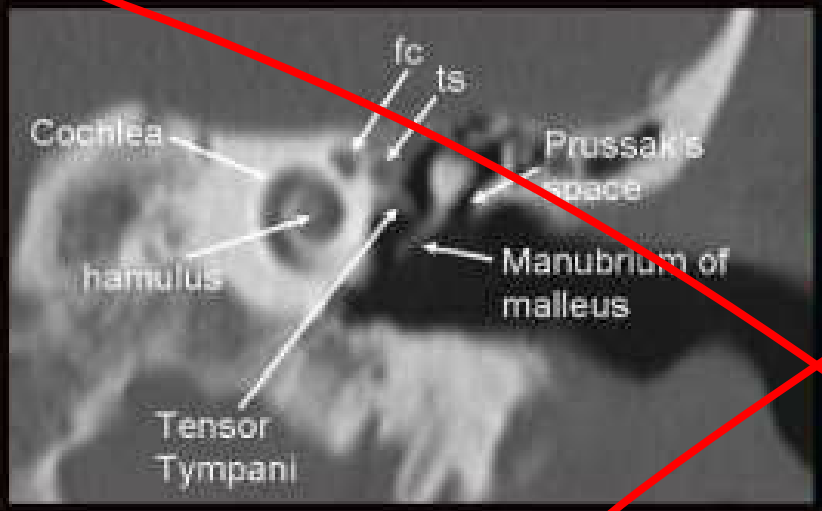


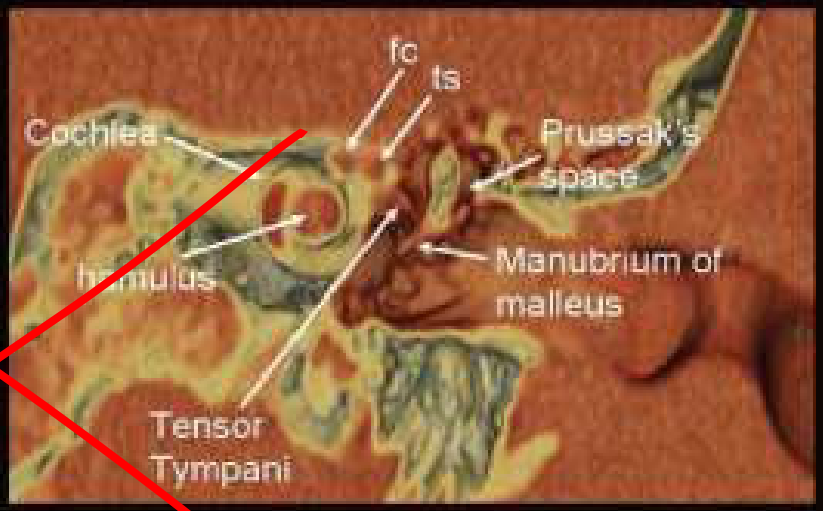
Prussak's space (pars flaccida (lateral border) removed)

Gulya JA, 1995.

Fatterpekar et al , 3D CT of the temporal bone, www.terarecon.com

Coronal

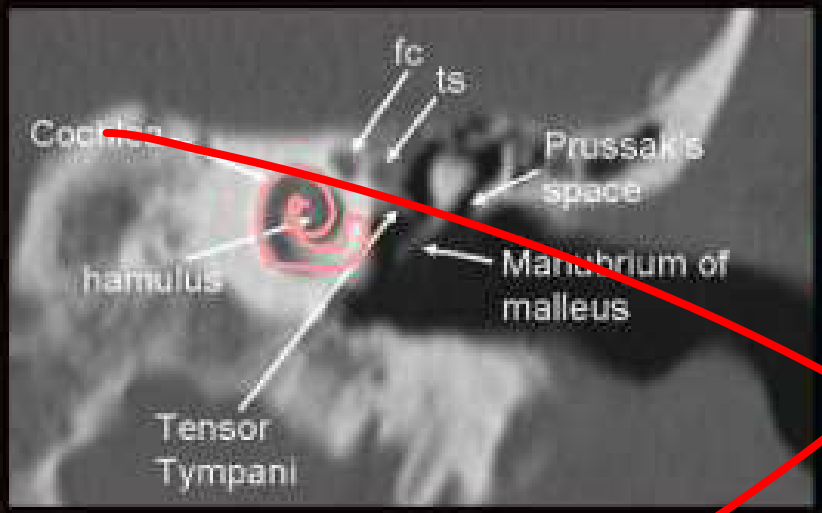


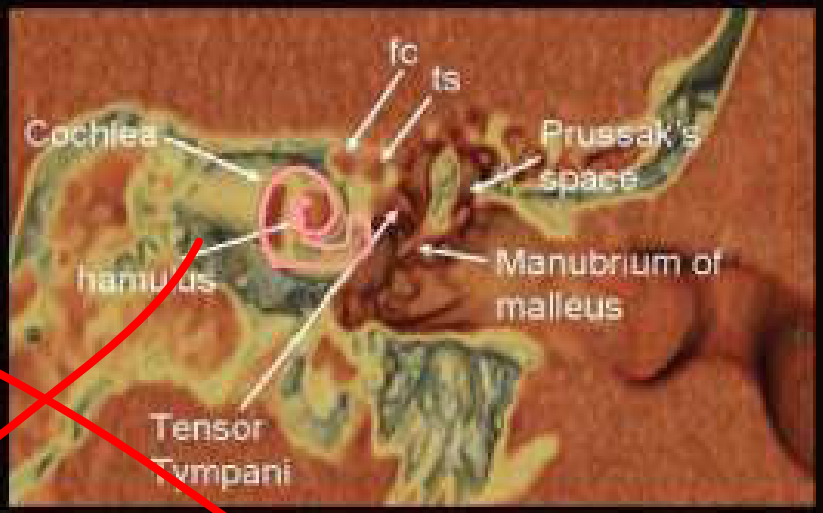


- Observe:
  - Cochlea "snail"

Anderson JF, 1883
Fatterpekar et al , 3D CT of the temporal bone, www.terarecon.com

Coronal





- Observe:
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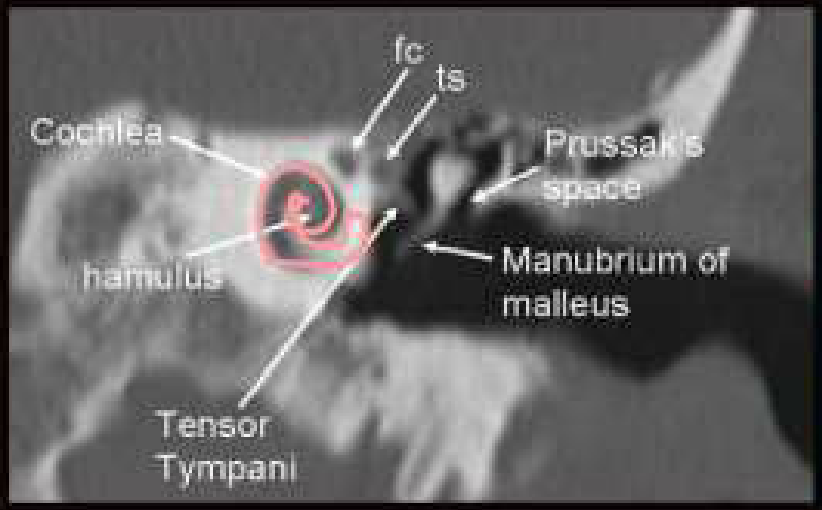
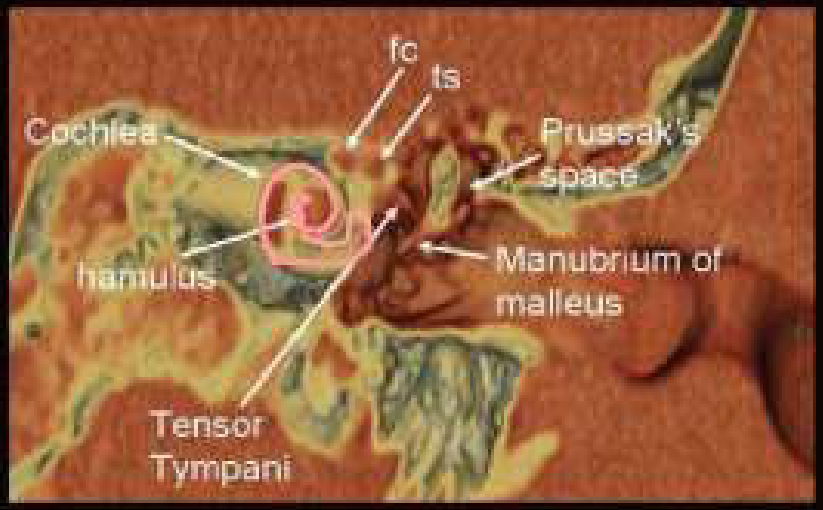
Anderson JF, 1883
Fatterpekar et al , 3D CT of the temporal bone. www.terarecon.com

Coronal

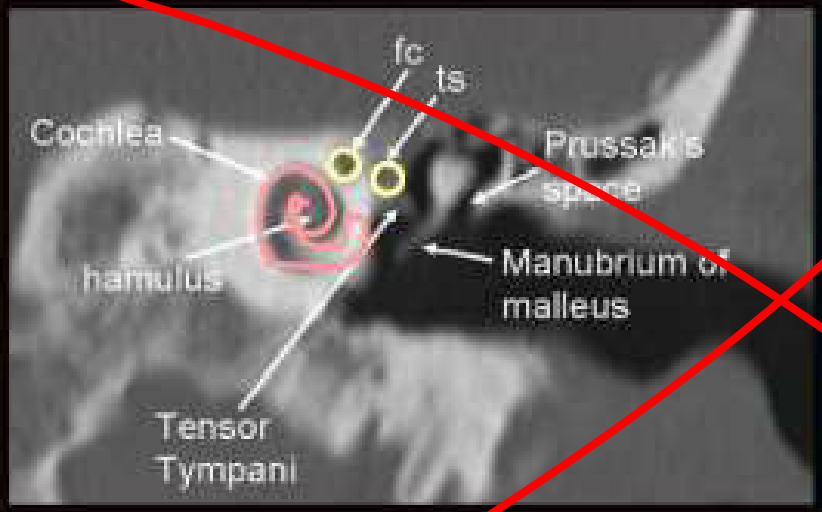


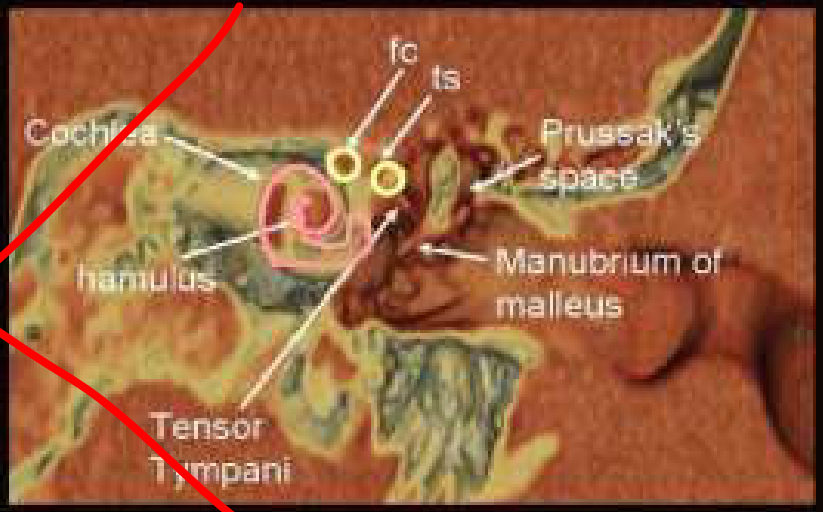


- Observe:
  - Cochlea "snail"
  - Two low attenuating areas above

Anderson JF, 1883
Fatterpekar et al , 3D CT of the temporal bone. www.teraricon.com

Coronal

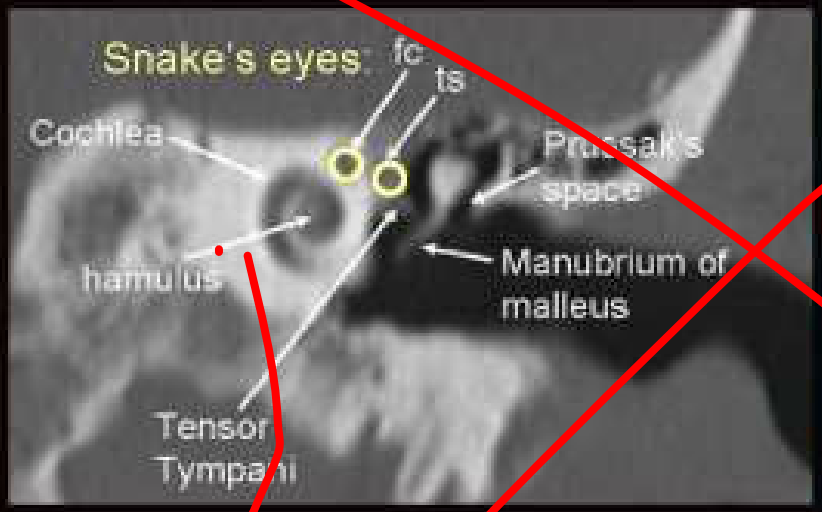


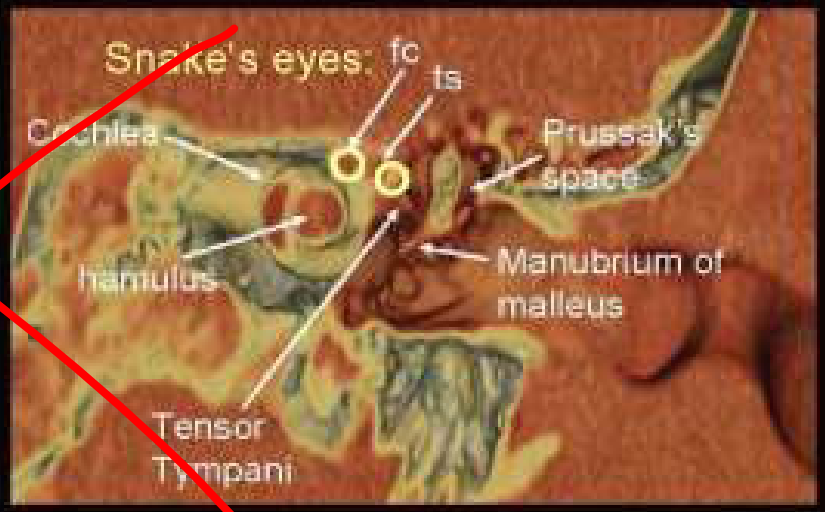


- Observe:
  - Cochlea "snail"
  - Two low attenuating areas above

Anderson JF, 1883
Fatterpekar et al , 3D CT of the temporal bone, www.terarecon.com

Coronal

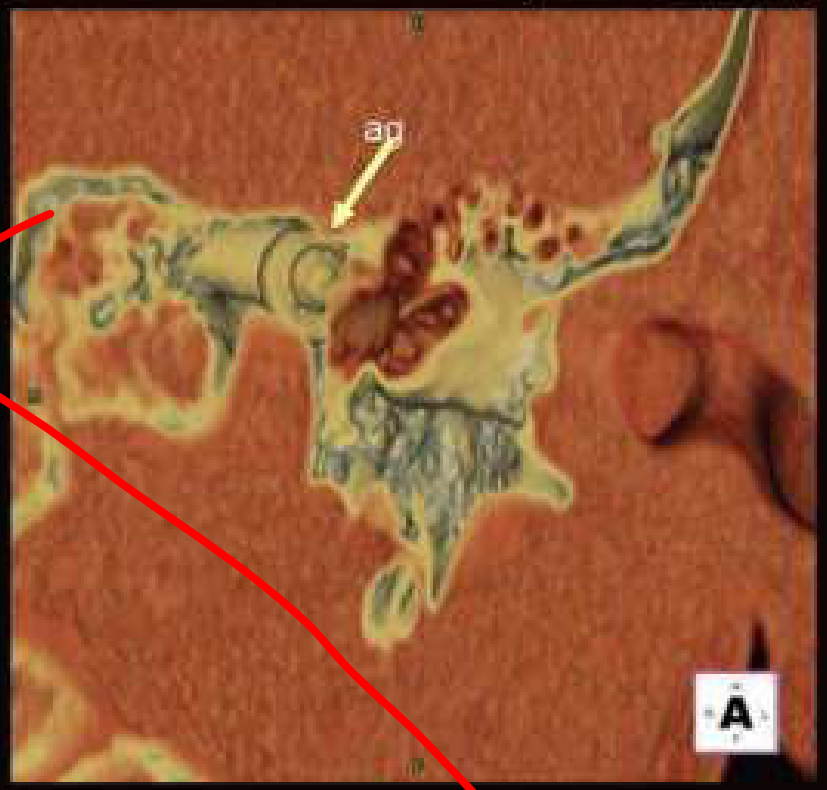




- "snake eyes": distal labyrinthine and proximal tympanic segments of facial nerve.
  - Locate anterior to the "snail" (cochlea)

Fatterpekar et al., 3D CT of the temporal bone. www.terarecon.com

Coronal: Anterior genu



- Anterior genu

ESKERRIK asko!
спасибо
Merci Beaucoup!
¡GRACIAS!
thank you!
Moc děkuji
ευχαριστώ
Danksagung!